



US012404237B2

(12) **United States Patent**
Hattori et al.(10) **Patent No.:** **US 12,404,237 B2**(45) **Date of Patent:** ***Sep. 2, 2025**(54) **METHOD OF TREATING EXCESSIVE SLEEPINESS AND/OR CATAPLEXY USING N-{2-((1,1'-BIPHENYL)-3-YL)METHYL}PYRROLIDIN-3-YL}SULFONAMIDE DERIVATIVES**(71) Applicant: **TAKEDA PHARMACEUTICAL COMPANY LIMITED**, Osaka (JP)(72) Inventors: **Yasushi Hattori**, Kanagawa (JP); **Yuhei Miyano**, Kanagawa (JP); **Yuichi Kajita**, Kanagawa (JP); **Tatsuki Koike**, Kanagawa (JP); **Yasutaka Hoashi**, Kanagawa (JP); **Norihito Tokunaga**, Kanagawa (JP); **Alexander Martin Pawliczek**, Kanagawa (JP); **Tsuneo Oda**, Kanagawa (JP); **Tohru Miyazaki**, Kanagawa (JP); **Yoshiteru Ito**, Kanagawa (JP); **Kohei Takeuchi**, Kanagawa (JP); **Keisuke Imamura**, Kanagawa (JP); **Takahiro Sugimoto**, Kanagawa (JP)(73) Assignee: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

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(21) Appl. No.: **19/079,199**(22) Filed: **Mar. 13, 2025**(65) **Prior Publication Data**

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Related U.S. Application Data

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Jan. 31, 2019 (JP) 2019-015488

(51) **Int. Cl.****C07D 207/14** (2006.01)**C07D 403/06** (2006.01)**C07D 407/08** (2006.01)**C07D 417/06** (2006.01)(52) **U.S. Cl.**CPC **C07D 207/14** (2013.01); **C07D 403/06** (2013.01); **C07D 407/08** (2013.01); **C07D 417/06** (2013.01)(58) **Field of Classification Search**

CPC C07D 207/14

See application file for complete search history.

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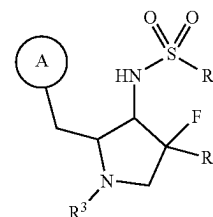
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(57) **ABSTRACT**

The present invention provides a heterocyclic compound having an orexin type 2 receptor agonist activity.

A compound represented by the formula (I):



(I)

wherein each symbol is as described in the specification, or a salt thereof has an orexin type 2 receptor agonist activity, and is useful as an agent for the prophylaxis or treatment of narcolepsy.

19 Claims, No Drawings

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**METHOD OF TREATING EXCESSIVE
SLEEPINESS AND/OR CATAPLEXY USING
N-{2-([(1,1'-BIPHENYL]-3-YL)METHYL)
PYRROLIDIN-3-YL}SULFONAMIDE
DERIVATIVES**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 18/175,215, filed Feb. 27, 2023, which is a continuation of U.S. application Ser. No. 17/239,191, filed Apr. 23, 2021, now abandoned, which is a continuation of U.S. application Ser. No. 16/777,273, filed Jan. 30, 2021, now issued as U.S. Pat. No. 11,028,048, which claims priority from JP 2019-015488, filed Jan. 31, 2019, all of which are incorporated by reference in their entireties.

TECHNICAL FIELD

The present invention relates to a heterocyclic compound, particularly, a heterocyclic compound having an orexin type 2 receptor agonist activity.

BACKGROUND OF THE INVENTION

Orexin is a neuropeptide specifically produced in particular neurons located sparsely in the lateral hypothalamus and its surrounding area, and consists of two subtypes, orexin A and orexin B. Both orexin A and orexin B are endogenous ligands of the orexin receptors, which are G protein-coupled receptors mainly present in the brain, and two types of subtypes, type 1 and type 2, are known for the orexin receptors (non-patent document 1).

Since orexin-producing neurons (orexin neurons) are localized in the vicinity of the feeding center, and intraventricular administration of orexin peptide results in an increase in food intake, orexin initially attracted attention as a neuropeptide having a feeding behavioral regulation. Thereafter, however, it was reported that the cause of dog narcolepsy is genetic variation of orexin type 2 receptor (non-patent document 2), and the role of orexin in controlling sleep and wakefulness has been also attracted.

From the studies using a transgenic mouse having denatured orexin neurons and a double transgenic mouse obtained by crossing this mouse with orexin overexpressing transgenic mouse, it was clarified that narcolepsy-like symptoms that appear by degeneration of orexin neurons disappear due to sustained expression of orexin. Similarly, when orexin peptide was intraventricularly administered to a transgenic mouse having denatured orexin neuron, improvement of narcolepsy-like symptoms was also observed (non-patent document 3). Studies of orexin type 2 receptor knockout mice have suggested that orexin type 2 receptor is important for maintaining arousal (non-patent document 4, non-patent document 5). Such background suggests that orexin type 2 receptor agonists become therapeutic drugs for narcolepsy or therapeutic drugs for other sleep disorders exhibiting excessive sleepiness (non-patent document 6).

In addition, it is suggested that a peptidic agonist that selectively acts on the orexin type 2 receptor improves obesity due to high fat diet load in mice (non-patent document 7).

In addition, it is suggested that intraventricular administration of orexin peptide shortens the systemic anesthetic time of rat (non-patent document 8).

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In addition, it is suggested that patients with sleep apnea syndrome show low orexin A concentration levels in plasma (non-patent document 9).

In addition, it is suggested that intraventricular administration of orexin peptide improves memory retention of senescence-accelerated model mouse (SAMP8) with cognitive dysfunction (non-patent document 10).

In addition, it is suggested that Orexin type 2 receptor agonist will be a therapeutic drug for cardiac failure (patent document 1, non-patent document 11).

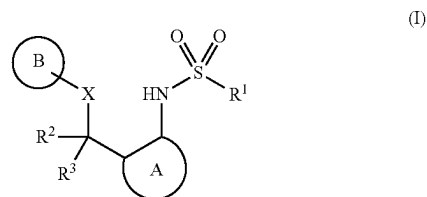
In addition, it is suggested that the daytime sleepiness of Parkinson's disease patients is caused by orexin nerve fallout (non-patent document 12).

In addition, it is suggested that orexin regulates bone formation and bone loss, and orexin type 2 receptor agonist will be a therapeutic drug for diseases related to bone loss such as osteoporosis, rheumatoid arthritis and the like (patent document 2).

In addition, it is suggested that orexin receptor agonist is useful for the prophylaxis or treatment of sepsis, severe sepsis and septic shock, since the mortality was significantly improved by mere continuous administration of orexin from the periphery in septic shock model mouse (patent document 3).

Therefore, a compound having an orexin type 2 receptor agonist activity is expected to be useful as a novel therapeutic drug for narcolepsy, idiopathic hypersomnia, hypersomnia, sleep apnea syndrome, disturbance of consciousness such as coma and the like, narcolepsy syndrome accompanied by narcolepsy-like symptoms, hypersomnia syndrome accompanied by daytime hypersomnia (e.g., Parkinson's disease, Guillain-Barre syndrome and Kleine Levin syndrome), Alzheimer, obesity, insulin resistance syndrome, cardiac failure, diseases related to bone loss, sepsis and the like, further, anesthetic antagonist, a prophylactic or therapeutic drug for side effects and complications due to anesthesia.

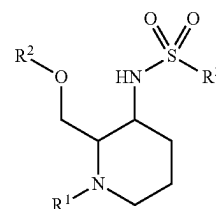
As sulfonamide derivatives, a compound represented by the formula



wherein each symbol is as described in the document (Patent Document 4) has been reported.

In addition, as compounds having an orexin type 2 receptor agonist activity, the following compounds have been reported.

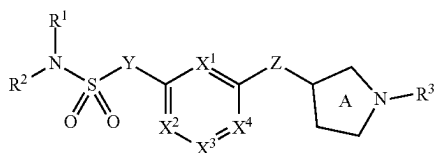
A compound represented by the formula



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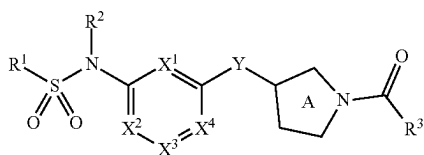
wherein each symbol is as described in the document (Patent Document 5).

A compound represented by the formula



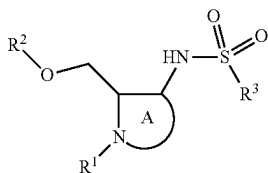
wherein each symbol is as described in the document (Patent Document 6).

A compound represented by the formula



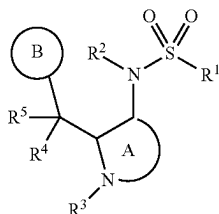
wherein each symbol is as described in the document (Patent Document 7).

A compound represented by the formula



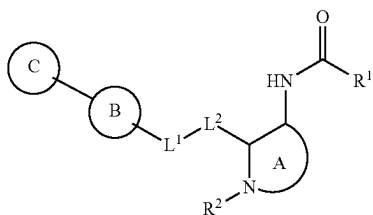
wherein each symbol is as described in the document (Patent Document 8).

A compound represented by the formula



wherein each symbol is as described in the document (Patent Document 9).

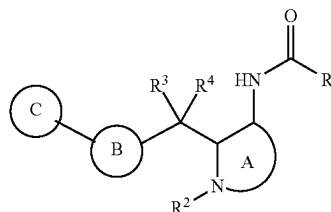
A compound represented by the formula



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wherein each symbol is as described in the document (Patent Document 10).

A compound represented by the formula



wherein each symbol is as described in the document (Patent Document 11).

Development of a novel compound having an orexin type 2 receptor agonist activity is desired.

DOCUMENT LIST

Patent Document

- [Patent Document 1] WO 2015/073707 A1
- [Patent Document 2] WO 2015/048091 A1
- [Patent Document 3] WO 2015/147240 A1
- [Patent Document 4] WO 2012/137982 A9
- [Patent Document 5] WO 2017/135306 A1
- [Patent Document 6] WO 2018/164191 A1
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- [Patent Document 8] WO 2019/027003 A1
- [Patent Document 9] WO 2019/027058 A1
- [Patent Document 10] WO 2020/004536 A1
- [Patent Document 11] WO 2020/004537 A1

Non-Patent Document

- [Non-Patent Document 1] Cell, Vol. 92, 573-585, 1998
- [Non-Patent Document 2] Cell, Vol. 98, 365-376, 1999
- [Non-Patent Document 3] Proc. Natl. Acad. Sci. USA, Vol. 101, 4649-4654, 2004
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- [Non-Patent Document 7] Cell Metabolism, Vol. 9, 64-76, 2009
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- [Non-Patent Document 12] Brain, Vol. 130, 2007, Pages 1586-1595

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

The present invention aims to provide a heterocyclic compound having an orexin type 2 receptor agonist activity.

Means of Solving the Problems

The present inventors have found that a compound represented by the following formula (I) or a salt thereof

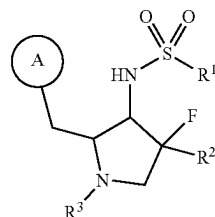
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(sometimes to be referred to as compound (I) in the present specification) has an orexin type 2 receptor agonist activity. As a result of further studies, they have completed the present invention.

Accordingly, the present invention relates to the following.

[1]

A compound represented by the formula (I):



wherein

R¹ is an optionally substituted C₁₋₆ alkyl group, an optionally substituted mono- or di-C₁₋₆ alkylamino group, an optionally substituted C₃₋₆ cycloalkyl group, or an optionally substituted 3- to 14-membered non-aromatic heterocyclic group;

R² is a hydrogen atom, a fluorine atom, an optionally substituted C₁₋₆ alkyl group, or an optionally substituted C₃₋₆ cycloalkyl group;

R³ is an acyl group, an optionally substituted C₁₋₆ alkyl group, an optionally substituted C₃₋₆ cycloalkyl group, an optionally substituted C₆₋₁₄ aryl group, an optionally substituted 3- to 14-membered non-aromatic heterocyclic group, or an optionally substituted 5- to 14-membered aromatic heterocyclic group; and Ring A is an optionally further substituted C₆₋₁₄ aromatic hydrocarbon ring, or an optionally further substituted 5- to 14-membered aromatic heterocycle, or a salt thereof.

[2]

The compound or salt of the above-mentioned [1], wherein

R¹ is

- (1) a C₁₋₆ alkyl group,
- (2) a mono- or di-C₁₋₆ alkylamino group, or
- (3) a C₃₋₆ cycloalkyl group;

R² is

- (1) a hydrogen atom,
- (2) a fluorine atom, or
- (3) a C₁₋₆ alkyl group;

R³ is

- (1) a C₁₋₆ alkyl-carbonyl group optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom,
 - (b) a hydroxy group, and
 - (c) a cyano group,
- (2) a C₁₋₆ alkoxy-carbonyl group,
- (3) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom,
 - (b) a hydroxy group,
 - (c) a cyano group, and
 - (d) a C₁₋₆ alkyl group,
- (4) a 3- to 14-membered non-aromatic heterocyclyl-carbonyl group optionally substituted by 1 to 3 substituents selected from

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(a) a halogen atom,

(b) a hydroxy group, and

(c) a C₁₋₆ alkyl group,

(5) a mono- or di-C₁₋₆ alkyl-carbamoyl group,

(6) a N—C₁₋₆ alkyl-N—C₁₋₆ alkoxy-carbamoyl group, or

(7) a N—C₁₋₆ alkyl-N',N'-di-C₁₋₆ alkylhydrazine-carbonyl group; and

Ring A is

(1) a benzene ring further substituted by one substituent selected from

(a) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom,

(ii) an optionally halogenated C₁₋₆ alkyl group, and

(iii) an optionally halogenated C₁₋₆ alkoxy group, and

(b) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from

(i) a C₁₋₆ alkyl group optionally substituted by 1 to 3 substituents selected from a halogen atom and a hydroxy group,

(ii) a C₁₋₆ alkoxy group,

(iii) a halogen atom, and

(iv) a C₁₋₆ alkoxy-carbonyl group, and

optionally further substituted by 1 to 3 halogen atoms, or

(2) a 5- or 6-membered aromatic heterocycle further substituted by one C₆₋₁₄ aryl group optionally substituted by 1 to 3 halogen atoms.

[3]

The compound or salt of the above-mentioned [1], wherein

R¹ is

- (1) a C₁₋₆ alkyl group,
- (2) a mono- or di-C₁₋₆ alkylamino group, or
- (3) a C₃₋₆ cycloalkyl group;

R² is

- (1) a hydrogen atom, or
- (2) a fluorine atom;

R³ is

- (1) a C₁₋₆ alkyl-carbonyl group optionally substituted by 1 to 3 hydroxy groups,
- (2) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom, and
 - (b) a hydroxy group,
- (3) a 3- to 8-membered monocyclic non-aromatic heterocyclyl-carbonyl group, or
- (4) a mono- or di-C₁₋₆ alkyl-carbamoyl group; and

Ring A is a benzene ring

further substituted by one C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom, and

(ii) an optionally halogenated C₁₋₆ alkyl group, and optionally further substituted by 1 to 3 halogen atoms.

[4]

The compound or salt of the above-mentioned [1], wherein

R¹ is

- (1) a C₁₋₆ alkyl group, or
- (2) a mono- or di-C₁₋₆ alkylamino group;

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R² is

- (1) a hydrogen atom, or
- (2) a fluorine atom;

R³ is

- (1) a C₁₋₆ alkyl-carbonyl group optionally substituted by 1 to 3 hydroxy groups,
- (2) a 3- to 8-membered monocyclic non-aromatic heterocyclic carbonyl group, or
- (3) a mono- or di-C₁₋₆ alkyl-carbamoyl group; and

Ring A is a benzene ring

further substituted by one C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from

- (i) a halogen atom, and
- (ii) a C₁₋₆ alkyl group, and

optionally further substituted by 1 to 3 halogen atoms.

[5] N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof.

[6] N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof.

[7] N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide or a salt thereof.

[8] A medicament comprising the compound or salt of any of the above-mentioned [1] to [7].

[9] The medicament of the above-mentioned [8], which is an orexin type 2 receptor agonist.

[10] The medicament of the above-mentioned [8], which is an agent for the prophylaxis or treatment of narcolepsy.

[11] The compound or salt of any of the above-mentioned [1] to [7] for use in the prophylaxis or treatment of narcolepsy.

[12] A method of activating an orexin type 2 receptor in a mammal, which comprises administering an effective amount of the compound or salt of any of the above-mentioned [1] to [7] to the mammal.

[13] A method for the prophylaxis or treatment of narcolepsy in a mammal, which comprises administering an effective amount of the compound or salt of any of the above-mentioned [1] to [7] to the mammal.

[14] Use of the compound or salt of any of the above-mentioned [1] to [7] for the manufacture of an agent for the prophylaxis or treatment of narcolepsy.

Effect of the Invention

The compound of the present invention has an orexin type 2 receptor agonist activity, and is useful as an agent for the prophylaxis or treatment of narcolepsy.

DETAILED DESCRIPTION OF THE INVENTION

The definition of each substituent used in the present specification is described in detail in the following. Unless otherwise specified, each substituent has the following definition.

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In the present specification, examples of the "halogen atom" include fluorine, chlorine, bromine and iodine.

In the present specification, examples of the "C₁₋₆ alkyl group" include methyl, ethyl, propyl, isopropyl, butyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, 1-ethylpropyl, hexyl, isohexyl, 1,1-dimethylbutyl, 2,2-dimethylbutyl, 3,3-dimethylbutyl and 2-ethylbutyl.

In the present specification, examples of the "optionally halogenated C₁₋₆ alkyl group" include a C₁₋₆ alkyl group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include methyl, chloromethyl, difluoromethyl, trichloromethyl, trifluoromethyl, ethyl, 2-bromoethyl, 2,2,2-trifluoroethyl, tetrafluoroethyl, pentafluoroethyl, propyl, 2,2-difluoropropyl, 3,3,3-trifluoropropyl, isopropyl, butyl, 4,4,4-trifluorobutyl, isobutyl, sec-butyl, tert-butyl, pentyl, isopentyl, neopentyl, 5,5,5-trifluoropentyl, hexyl and 6,6,6-trifluorohexyl.

In the present specification, examples of the "C₂₋₆ alkenyl group" include ethenyl, 1-propenyl, 2-propenyl, 2-methyl-1-propenyl, 1-butenyl, 2-butenyl, 3-butenyl, 3-methyl-2-butenyl, 1-pentenyl, 2-pentenyl, 3-pentenyl, 4-pentenyl, 4-methyl-3-pentenyl, 1-hexenyl, 3-hexenyl and 5-hexenyl.

In the present specification, examples of the "C₂₋₆ alkynyl group" include ethynyl, 1-propynyl, 2-propynyl, 1-butylnyl, 2-butylnyl, 3-butylnyl, 1-pentylnyl, 2-pentylnyl, 3-pentylnyl, 4-pentylnyl, 1-hexynyl, 2-hexynyl, 3-hexynyl, 4-hexynyl, 5-hexynyl and 4-methyl-2-pentylnyl.

In the present specification, examples of the "C₃₋₁₀ cycloalkyl group" include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, bicyclo[2.2.1]heptyl, bicyclo[2.2.2]octyl, bicyclo[3.2.1]octyl and adamantyl.

In the present specification, examples of the "optionally halogenated C₃₋₁₀ cycloalkyl group" include a C₃₋₁₀ cycloalkyl group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include cyclopropyl, 2,2-difluorocyclopropyl, 2,3-difluorocyclopropyl, cyclobutyl, difluorocyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl and cyclooctyl.

In the present specification, examples of the "C₃₋₁₀ cycloalkenyl group" include cyclopropenyl, cyclobutenyl, cyclopentenyl, cyclohexenyl, cycloheptenyl and cyclooctenyl.

In the present specification, examples of the "C₆₋₁₄ aryl group" include phenyl, 1-naphthyl, 2-naphthyl, 1-anthryl, 2-anthryl and 9-anthryl.

In the present specification, examples of the "C₇₋₁₆ aralkyl group" include benzyl, phenethyl, naphthylmethyl and phenylpropyl.

In the present specification, examples of the "C₁₋₆ alkoxy group" include methoxy, ethoxy, propoxy, isopropoxy, butoxy, isobutoxy, sec-butoxy, tert-butoxy, pentyloxy and hexyloxy.

In the present specification, examples of the "optionally halogenated C₁₋₆ alkoxy group" include a C₁₋₆ alkoxy group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include methoxy, difluoromethoxy, trifluoromethoxy, ethoxy, 2,2,2-trifluoroethoxy, propoxy, isopropoxy, butoxy, 4,4,4-trifluorobutoxy, isobutoxy, sec-butoxy, pentyloxy and hexyloxy.

In the present specification, examples of the "C₃₋₁₀ cycloalkyloxy group" include cyclopropyloxy, cyclobutyloxy, cyclopentyloxy, cyclohexyloxy, cycloheptyloxy and cyclooctyloxy.

In the present specification, examples of the “C₁₋₆ alkylthio group” include methylthio, ethylthio, propylthio, isopropylthio, butylthio, sec-butylthio, tert-butylthio, pentylthio and hexylthio.

In the present specification, examples of the “optionally halogenated C₁₋₆ alkylthio group” include a C₁₋₆ alkylthio group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include methylthio, difluoromethylthio, trifluoromethylthio, ethylthio, propylthio, isopropylthio, butylthio, 4,4,4-trifluorobutylthio, pentylthio and hexylthio.

In the present specification, examples of the “C₁₋₆ alkyl-carbonyl group” include acetyl, propanoyl, butanoyl, 2-methylpropanoyl, pentanoyl, 3-methylbutanoyl, 2-methylbutanoyl, 2,2-dimethylpropanoyl, hexanoyl and heptanoyl.

In the present specification, examples of the “optionally halogenated C₁₋₆ alkyl-carbonyl group” include a C₁₋₆ alkyl-carbonyl group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include acetyl, chloroacetyl, trifluoroacetyl, trichloroacetyl, propanoyl, butanoyl, pentanoyl and hexanoyl.

In the present specification, examples of the “C₁₋₆ alkoxy-carbonyl group” include methoxycarbonyl, ethoxycarbonyl, propoxycarbonyl, isopropoxycarbonyl, butoxycarbonyl, isobutoxycarbonyl, sec-butoxycarbonyl, tert-butoxycarbonyl, pentyloxycarbonyl and hexyloxycarbonyl.

In the present specification, examples of the “C₆₋₁₄ aryl-carbonyl group” include benzoyl, 1-naphthoyl and 2-naphthoyl.

In the present specification, examples of the “C₇₋₁₆ aralkyl-carbonyl group” include phenylacetyl and phenylpropionyl.

In the present specification, examples of the “5- to 14-membered aromatic heterocyclylcarbonyl group” include nicotinoyl, isonicotinoyl, thenoyl and furoyl.

In the present specification, examples of the “3- to 14-membered non-aromatic heterocyclylcarbonyl group” include morpholinylcarbonyl, piperidinylcarbonyl and pyrrolidinylcarbonyl.

In the present specification, examples of the “mono- or di-C₁₋₆ alkyl-carbamoyl group” include methylcarbamoyl, ethylcarbamoyl, dimethylcarbamoyl, diethylcarbamoyl and N-ethyl-N-methylcarbamoyl.

In the present specification, examples of the “mono- or di-C₇₋₁₆ aralkyl-carbamoyl group” include benzylcarbamoyl and phenethylcarbamoyl.

In the present specification, examples of the “C₁₋₆ alkylsulfonyl group” include methylsulfonyl, ethylsulfonyl, propylsulfonyl, isopropylsulfonyl, butylsulfonyl, sec-butylsulfonyl and tert-butylsulfonyl.

In the present specification, examples of the “optionally halogenated C₁₋₆ alkylsulfonyl group” include a C₁₋₆ alkylsulfonyl group optionally having 1 to 7, preferably 1 to 5, halogen atoms. Specific examples thereof include methylsulfonyl, difluoromethylsulfonyl, trifluoromethylsulfonyl, ethylsulfonyl, propylsulfonyl, isopropylsulfonyl, butylsulfonyl, 4,4,4-trifluorobutylsulfonyl, pentylsulfonyl and hexylsulfonyl.

In the present specification, examples of the “C₆₋₁₄ arylsulfonyl group” include phenylsulfonyl, 1-naphthylsulfonyl and 2-naphthylsulfonyl.

In the present specification, examples of the “substituent” include a halogen atom, a cyano group, a nitro group, an optionally substituted hydrocarbon group, an optionally substituted heterocyclic group, an acyl group, an optionally substituted amino group, an optionally substituted carbam-

oyl group, an optionally substituted thiocarbamoyl group, an optionally substituted sulfamoyl group, an optionally substituted hydroxy group, an optionally substituted sulfanyl (SH) group and an optionally substituted silyl group.

In the present specification, examples of the “hydrocarbon group” (including “hydrocarbon group” of “optionally substituted hydrocarbon group”) include a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₂₋₆ alkynyl group, a C₃₋₁₀ cycloalkyl group, a C₃₋₁₀ cycloalkenyl group, a C₆₋₁₄ aryl group and a C₇₋₁₆ aralkyl group.

In the present specification, examples of the “optionally substituted hydrocarbon group” include a hydrocarbon group optionally having substituent(s) selected from the following Substituent group A.

[Substituent Group A]

- (1) a halogen atom,
- (2) a nitro group,
- (3) a cyano group,
- (4) an oxo group,
- (5) a hydroxy group,
- (6) an optionally halogenated C₁₋₆ alkoxy group,
- (7) a C₆₋₁₄ aryloxy group (e.g., phenoxy, naphthoxy),
- (8) a C₇₋₁₆ aralkyloxy group (e.g., benzyloxy),
- (9) a 5- to 14-membered aromatic heterocyclyloxy group (e.g., pyridyloxy),
- (10) a 3- to 14-membered non-aromatic heterocyclyloxy group (e.g., morpholinyl, piperidinyl),
- (11) a C₁₋₆ alkyl-carbonyloxy group (e.g., acetoxy, propanoyloxy),
- (12) a C₆₋₁₄ aryl-carbonyloxy group (e.g., benzoyloxy, 1-naphthoyloxy, 2-naphthoyloxy),
- (13) a C₁₋₆ alkoxy-carbonyloxy group (e.g., methoxycarbonyloxy, ethoxycarbonyloxy, propoxycarbonyloxy, butoxycarbonyloxy),
- (14) a mono- or di-C₁₋₆ alkyl-carbamoyloxy group (e.g., methylcarbamoyloxy, ethylcarbamoyloxy, dimethylcarbamoyloxy, diethylcarbamoyloxy),
- (15) a C₆₋₁₄ aryl-carbamoyloxy group (e.g., phenylcarbamoyloxy, naphthylcarbamoyloxy),
- (16) a 5- to 14-membered aromatic heterocyclylcarbonyloxy group (e.g., nicotinoyloxy),
- (17) a 3- to 14-membered non-aromatic heterocyclylcarbonyloxy group (e.g., morpholinylcarbonyloxy, piperidinylcarbonyloxy),
- (18) an optionally halogenated C₁₋₆ alkylsulfonyloxy group (e.g., methylsulfonyloxy, trifluoromethylsulfonyloxy),
- (19) a C₆₋₁₄ arylsulfonyloxy group optionally substituted by a C₁₋₆ alkyl group (e.g., phenylsulfonyloxy, toluenesulfonyloxy),
- (20) an optionally halogenated C₁₋₆ alkylthio group,
- (21) a 5- to 14-membered aromatic heterocyclic group,
- (22) a 3- to 14-membered non-aromatic heterocyclic group,
- (23) a formyl group,
- (24) a carboxy group,
- (25) an optionally halogenated C₁₋₆ alkyl-carbonyl group,
- (26) a C₆₋₁₄ aryl-carbonyl group,
- (27) a 5- to 14-membered aromatic heterocyclylcarbonyl group,
- (28) a 3- to 14-membered non-aromatic heterocyclylcarbonyl group,
- (29) a C₁₋₆ alkoxy-carbonyl group,
- (30) a C₆₋₁₄ aryloxy-carbonyl group (e.g., phenyloxycarbonyl, 1-naphthyloxycarbonyl, 2-naphthyloxycarbonyl),

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- (31) a C_{7-16} aralkyloxy-carbonyl group (e.g., benzyloxy-carbonyl, phenethyloxy-carbonyl),
- (32) a carbamoyl group,
- (33) a thiocarbamoyl group,
- (34) a mono- or di- C_{1-6} alkyl-carbamoyl group,
- (35) a C_{6-14} aryl-carbamoyl group (e.g., phenylcarbamoyl),
- (36) a 5- to 14-membered aromatic heterocyclylcarbamoyl group (e.g., pyridylcarbamoyl, thienylcarbamoyl),
- (37) a 3- to 14-membered non-aromatic heterocyclylcarbamoyl group (e.g., morpholinylcarbamoyl, piperidinylcarbamoyl),
- (38) an optionally halogenated C_{1-6} alkylsulfonyl group,
- (39) a C_{6-14} arylsulfonyl group,
- (40) a 5- to 14-membered aromatic heterocyclylsulfonyl group (e.g., pyridylsulfonyl, thienylsulfonyl),
- (41) an optionally halogenated C_{1-6} alkylsulfinyl group,
- (42) a C_{6-14} arylsulfinyl group (e.g., phenylsulfinyl, 1-naphthylsulfinyl, 2-naphthylsulfinyl),
- (43) a 5- to 14-membered aromatic heterocyclylsulfinyl group (e.g., pyridylsulfinyl, thienylsulfinyl),
- (44) an amino group,
- (45) a mono- or di- C_{1-6} alkylamino group (e.g., methylamino, ethylamino, propylamino, isopropylamino, butylamino, dimethylamino, diethylamino, dipropylamino, dibutylamino, N-ethyl-N-methylamino),
- (46) a mono- or di- C_{6-14} arylamino group (e.g., phenylamino),
- (47) a 5- to 14-membered aromatic heterocyclylamino group (e.g., pyridylamino),
- (48) a C_{7-16} aralkylamino group (e.g., benzylamino),
- (49) a formylamino group,
- (50) a C_{1-6} alkyl-carbonylamino group (e.g., acetylamino, propanoylamino, butanoylamino),
- (51) a (C_{1-6} alkyl) (C_{1-6} alkyl-carbonyl) amino group (e.g., N-acetyl-N-methylamino),
- (52) a C_{6-14} aryl-carbonylamino group (e.g., phenylcarbonylamino, naphthylcarbonylamino),
- (53) a C_{1-6} alkoxy-carbonylamino group (e.g., methoxycarbonylamino, ethoxycarbonylamino, propoxycarbonylamino, butoxycarbonylamino, tert-butoxycarbonylamino),
- (54) a C_{7-16} aralkyloxy-carbonylamino group (e.g., benzyloxy-carbonylamino),
- (55) a C_{1-6} alkylsulfonylamino group (e.g., methylsulfonylamino, ethylsulfonylamino),
- (56) a C_{6-14} arylsulfonylamino group optionally substituted by a C_{1-6} alkyl group (e.g., phenylsulfonylamino, toluenesulfonylamino),
- (57) an optionally halogenated C_{1-6} alkyl group,
- (58) a C_{2-6} alkenyl group,
- (59) a C_{2-6} alkynyl group,
- (60) a C_{3-10} cycloalkyl group,
- (61) a C_{3-10} cycloalkenyl group, and
- (62) a C_{6-14} aryl group.

The number of the above-mentioned substituents in the “optionally substituted hydrocarbon group” is, for example, 1 to 5, preferably 1 to 3. When the number of the substituents is two or more, the respective substituents may be the same or different.

In the present specification, examples of the “heterocyclic group” (including “heterocyclic group” of “optionally substituted heterocyclic group”) include (i) an aromatic heterocyclic group, (ii) a non-aromatic heterocyclic group and (iii) a 7- to 10-membered bridged heterocyclic group, each

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containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

In the present specification, examples of the “aromatic heterocyclic group” (including “5- to 14-membered aromatic heterocyclic group”) include a 5- to 14-membered (preferably 5- to 10-membered) aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

Preferable examples of the “aromatic heterocyclic group” include 5- or 6-membered monocyclic aromatic heterocyclic groups such as thienyl, furyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, triazolyl, tetrazolyl, triazinyl and the like; and 8- to 14-membered fused polycyclic (preferably bi- or tricyclic) aromatic heterocyclic groups such as benzothiophenyl, benzofuranlyl, benzimidazolyl, benzoxazolyl, benzisoxazolyl, benzothiazolyl, benzisothiazolyl, benzotriazolyl, imidazopyridinyl, thienopyridinyl, furopyridinyl, pyrrolopyridinyl, pyrazolopyridinyl, oxazolopyridinyl, thiazolopyridinyl, imidazopyrazinyl, imidazopyrimidinyl, thienopyrimidinyl, furopyrimidinyl, pyrrolopyrimidinyl, pyrazolopyrimidinyl, oxazolopyrimidinyl, thiazolopyrimidinyl, pyrazolotriazinyl, naphtho[2,3-b]thienyl, phenoxathiinyl, indolyl, isoindolyl, 1H-indazolyl, purinyl, isoquinolyl, quinolyl, phthalazinyl, naphthyridinyl, quinoxalinyl, quinazolinyl, cinnolinyl, carbazolyl, β -carbolinyl, phenanthridinyl, acridinyl, phenaziny, phenothiazinyl, phenoxazinyl and the like.

In the present specification, examples of the “non-aromatic heterocyclic group” (including “3- to 14-membered non-aromatic heterocyclic group”) include a 3- to 14-membered (preferably 4- to 10-membered) non-aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

Preferable examples of the “non-aromatic heterocyclic group” include 3- to 8-membered monocyclic non-aromatic heterocyclic groups such as aziridinyl, oxiranyl, thiiranyl, azetidiny, oxetanyl, thietanyl, tetrahydrothienyl, tetrahydrofuranlyl, pyrrolinyl, pyrrolidinyl, imidazolinyl, imidazolidinyl, oxazolinyl, oxazolidinyl, pyrazolinyl, pyrazolidinyl, thiazolinyl, thiazolidinyl, tetrahydroisothiazolyl, tetrahydrooxazolyl, tetrahydroisoxazolyl, piperidinyl, piperazinyl, tetrahydropyridinyl, dihydropyridinyl, dihydrothiopyranyl, tetrahydropyrimidinyl, tetrahydropyridazinyl, dihydropyranlyl, tetrahydropyranyl, tetrahydrothiopyranyl, morpholinyl, thiomorpholinyl, azepanyl, diazepanyl, azepinyl, oxepanyl, azocanyl, diazocanyl and the like; and 9- to 14-membered fused polycyclic (preferably bi- or tricyclic) non-aromatic heterocyclic groups such as dihydrobenzofuranlyl, dihydrobenzimidazolyl, dihydrobenzoxazolyl, dihydrobenzothiazolyl, dihydrobenzisothiazolyl, dihydronaphtho[2,3-b]thienyl, tetrahydroisoquinolyl, tetrahydroquinolyl, 4H-quinoliziny, indoliny, isoindoliny, tetrahydrothieno[2,3-c]pyridinyl, tetrahydrobenzazepinyl, tetrahydroquinoxaliny, tetrahydrophenanthridinyl, hexahydrophenothiazinyl, hexahydrophenoxazinyl, tetrahydrophthalazinyl, tetrahydronaphthyridinyl, tetrahydroquinazolinyl, tetrahydrocinnolinyl, tetrahydrocarbazolyl, tetrahydro- β -carbolinyl, tetrahydroacrydinyl, tetrahydrophenazinyl, tetrahydrothioxantheny, octahydroisoquinolyl and the like.

In the present specification, preferable examples of the “7- to 10-membered bridged heterocyclic group” include quinuclidinyl and 7-azabicyclo[2.2.1]heptanyl.

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In the present specification, examples of the “nitrogen-containing heterocyclic group” include a “heterocyclic group” containing at least one nitrogen atom as a ring-constituting atom.

In the present specification, examples of the “optionally substituted heterocyclic group” include a heterocyclic group optionally having substituent(s) selected from the above-mentioned Substituent group A.

The number of the substituents in the “optionally substituted heterocyclic group” is, for example, 1 to 3. When the number of the substituents is two or more, the respective substituents may be the same or different.

In the present specification, examples of the “acyl group” include a formyl group, a carboxy group, a carbamoyl group, a thiocarbamoyl group, a sulfino group, a sulfo group, a sulfamoyl group and a phosphono group, each optionally having “1 or 2 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₃₋₁₀ cycloalkenyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a 5- to 14-membered aromatic heterocyclic group, a 3- to 14-membered non-aromatic heterocyclic group, an amino group and a mono- or di-C₁₋₆ alkyl-amino group, each of which optionally has 1 to 3 substituents selected from a halogen atom, an optionally halogenated C₁₋₆ alkoxy group, a hydroxy group, a nitro group, a cyano group, an amino group and a carbamoyl group”.

Examples of the “acyl group” also include a hydrocarbon-sulfonyl group, a heterocyclylsulfonyl group, a hydrocarbon-sulfinyl group and a heterocyclylsulfinyl group.

Here, the hydrocarbon-sulfonyl group means a hydrocarbon group-bonded sulfonyl group, the heterocyclylsulfonyl group means a heterocyclic group-bonded sulfonyl group, the hydrocarbon-sulfinyl group means a hydrocarbon group-bonded sulfinyl group and the heterocyclylsulfinyl group means a heterocyclic group-bonded sulfinyl group.

Preferable examples of the “acyl group” include a formyl group, a carboxy group, a C₁₋₆ alkyl-carbonyl group, a C₂₋₆ alkenyl-carbonyl group (e.g., crotonoyl), a C₃₋₁₀ cycloalkyl-carbonyl group (e.g., cyclobutanecarbonyl, cyclopentanecarbonyl, cyclohexanecarbonyl, cycloheptanecarbonyl), a C₃₋₁₀ cycloalkenyl-carbonyl group (e.g., 2-cyclohexenecarbonyl), a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a C₆₋₁₄ aryloxy-carbonyl group (e.g., phenyloxy-carbonyl, naphthyloxy-carbonyl), a C₇₋₁₆ aralkyloxy-carbonyl group (e.g., benzyloxy-carbonyl, phenethyloxy-carbonyl), a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group, a mono- or di-C₂₋₆ alkenyl-carbamoyl group (e.g., diallylcarbamoyl), a mono- or di-C₃₋₁₀ cycloalkyl-carbamoyl group (e.g., cyclopropylcarbamoyl), a mono- or di-C₆₋₁₄ aryl-carbamoyl group (e.g., phenylcarbamoyl), a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, a 5- to 14-membered aromatic heterocyclylcarbamoyl group (e.g., pyridylcarbamoyl), N—C₁₋₆ alkyl-N',N'-di-C₁₋₆ alkylhydrazine-carbonyl group, a thiocarbamoyl group, a mono- or di-C₁₋₆ alkyl-thiocarbamoyl group (e.g., methylthiocarbamoyl, N-ethyl-N-methylthiocarbamoyl), a mono- or di-C₂₋₆ alkenyl-thiocarbamoyl group (e.g., diallylthiocarbamoyl), a mono- or di-C₃₋₁₀ cycloalkyl-thiocarbamoyl group (e.g., cyclopropylthiocarbamoyl, cyclohexylthiocarbamoyl), a mono- or di-C₆₋₁₄ aryl-thiocarbamoyl group (e.g., phenylthiocarbamoyl), a mono- or di-C₇₋₁₆ aralkyl-thiocarbamoyl group (e.g., benzylthiocarbamoyl, phenethylthiocarbamoyl), a 5- to 14-membered aromatic heterocyclylthiocarbamoyl group (e.g., pyridylthiocarbamoyl), a sulfino group, a C₁₋₆

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alkylsulfinyl group (e.g., methylsulfinyl, ethylsulfinyl), a sulfo group, a C₁₋₆ alkylsulfonyl group, a C₆₋₁₄ arylsulfonyl group, a phosphono group and a mono- or di-C₁₋₆ alkylphosphono group (e.g., dimethylphosphono, diethylphosphono, diisopropylphosphono, dibutylphosphono).

In the present specification, examples of the “optionally substituted amino group” include an amino group optionally having “1 or 2 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a 5- to 14-membered aromatic heterocyclic group, a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group, a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, a C₁₋₆ alkylsulfonyl group and a C₆₋₁₄ arylsulfonyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted amino group include an amino group, a mono- or di-(optionally halogenated C₁₋₆ alkyl) amino group (e.g., methylamino, trifluoromethylamino, dimethylamino, ethylamino, diethylamino, propylamino, dibutylamino), a mono- or di-C₂₋₆ alkenylamino group (e.g., diallylamino), a mono- or di-C₃₋₁₀ cycloalkylamino group (e.g., cyclopropylamino, cyclohexylamino), a mono- or di-C₆₋₁₄ arylamino group (e.g., phenylamino), a mono- or di-C₇₋₁₆ aralkylamino group (e.g., benzylamino, dibenzylamino), a mono- or di-(optionally halogenated C₁₋₆ alkyl)-carbonylamino group (e.g., acetylamino, propionylamino), a mono- or di-C₆₋₁₄ aryl-carbonylamino group (e.g., benzoylamino), a mono- or di-C₇₋₁₆ aralkyl-carbonylamino group (e.g., benzylcarbonylamino), a mono- or di-5- to 14-membered aromatic heterocyclylcarbonylamino group (e.g., nicotinoylamino, isonicotinoylamino), a mono- or di-3- to 14-membered non-aromatic heterocyclylcarbonylamino group (e.g., piperidinylcarbonylamino), a mono- or di-C₁₋₆ alkoxy-carbonylamino group (e.g., tert-butoxycarbonylamino), a 5- to 14-membered aromatic heterocyclylamino group (e.g., pyridylamino), a carbamoylamino group, a (mono- or di-C₁₋₆ alkyl-carbamoyl) amino group (e.g., methylcarbamoylamino), a (mono- or di-C₇₋₁₆ aralkyl-carbamoyl) amino group (e.g., benzylcarbamoylamino), a C₁₋₆ alkylsulfonylamino group (e.g., methylsulfonylamino, ethylsulfonylamino), a C₆₋₁₄ arylsulfonylamino group (e.g., phenylsulfonylamino), a (C₁₋₆ alkyl) (C₁₋₆ alkyl-carbonyl) amino group (e.g., N-acetyl-N-methylamino) and a (C₁₋₆ alkyl) (C₆₋₁₄ aryl-carbonyl) amino group (e.g., N-benzoyl-N-methylamino).

In the present specification, examples of the “optionally substituted carbamoyl group” include a carbamoyl group optionally having “1 or 2 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a 5- to 14-membered aromatic heterocyclic group, a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group and a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted carbamoyl group include a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group, a mono- or di-C₂₋₆ alkenyl-carbamoyl group (e.g., diallylcarbamoyl), a mono- or di-C₃₋₁₀

cycloalkyl-carbamoyl group (e.g., cyclopropylcarbamoyl, cyclohexylcarbamoyl), a mono- or di-C₆₋₁₄ aryl-carbamoyl group (e.g., phenylcarbamoyl), a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbonyl-carbamoyl group (e.g., acetylcarbamoyl, propionylcarbamoyl), a mono- or di-C₆₋₁₄ aryl-carbonyl-carbamoyl group (e.g., benzoylcarbamoyl) and a 5- to 14-membered aromatic heterocyclylcarbamoyl group (e.g., pyridylcarbamoyl).

In the present specification, examples of the “optionally substituted thiocarbamoyl group” include a thiocarbamoyl group optionally having “1 or 2 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a 5- to 14-membered aromatic heterocyclic group, a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group and a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted thiocarbamoyl group include a thiocarbamoyl group, a mono- or di-C₁₋₆ alkyl-thiocarbamoyl group (e.g., methylthiocarbamoyl, ethylthiocarbamoyl, dimethylthiocarbamoyl, diethylthiocarbamoyl, N-ethyl-N-methylthiocarbamoyl), a mono- or di-C₂₋₆ alkenyl-thiocarbamoyl group (e.g., diallylthiocarbamoyl), a mono- or di-C₃₋₁₀ cycloalkyl-thiocarbamoyl group (e.g., cyclopropylthiocarbamoyl, cyclohexylthiocarbamoyl), a mono- or di-C₆₋₁₄ aryl-thiocarbamoyl group (e.g., phenylthiocarbamoyl), a mono- or di-C₇₋₁₆ aralkyl-thiocarbamoyl group (e.g., benzylthiocarbamoyl, phenethylthiocarbamoyl), a mono- or di-C₁₋₆ alkyl-carbonyl-thiocarbamoyl group (e.g., acetylthiocarbamoyl, propionylthiocarbamoyl), a mono- or di-C₆₋₁₄ aryl-carbonyl-thiocarbamoyl group (e.g., benzoylthiocarbamoyl) and a 5- to 14-membered aromatic heterocyclylthiocarbamoyl group (e.g., pyridylthiocarbamoyl).

In the present specification, examples of the “optionally substituted sulfamoyl group” include a sulfamoyl group optionally having “1 or 2 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a 5- to 14-membered aromatic heterocyclic group, a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group and a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted sulfamoyl group include a sulfamoyl group, a mono- or di-C₁₋₆ alkyl-sulfamoyl group (e.g., methylsulfamoyl, ethylsulfamoyl, dimethylsulfamoyl, diethylsulfamoyl, N-ethyl-N-methylsulfamoyl), a mono- or di-C₂₋₆ alkenyl-sulfamoyl group (e.g., diallylsulfamoyl), a mono- or di-C₃₋₁₀ cycloalkyl-sulfamoyl group (e.g., cyclopropylsulfamoyl, cyclohexylsulfamoyl), a mono- or di-C₆₋₁₄ aryl-sulfamoyl group (e.g., phenylsulfamoyl), a mono- or di-C₇₋₁₆ aralkyl-sulfamoyl group (e.g., benzylsulfamoyl, phenethylsulfamoyl), a mono- or di-C₁₋₆ alkyl-carbonyl-sulfamoyl group (e.g., acetylsulfamoyl, propionylsulfamoyl), a mono- or di-C₆₋₁₄ aryl-

carbonyl-sulfamoyl group (e.g., benzoylsulfamoyl) and a 5- to 14-membered aromatic heterocyclylsulfamoyl group (e.g., pyridylsulfamoyl).

In the present specification, examples of the “optionally substituted hydroxy group” include a hydroxy group optionally having “a substituent selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group, a C₇₋₁₆ aralkyl-carbonyl group, a 5- to 14-membered aromatic heterocyclylcarbonyl group, a 3- to 14-membered non-aromatic heterocyclylcarbonyl group, a C₁₋₆ alkoxy-carbonyl group, a 5- to 14-membered aromatic heterocyclic group, a carbamoyl group, a mono- or di-C₁₋₆ alkyl-carbamoyl group, a mono- or di-C₇₋₁₆ aralkyl-carbamoyl group, a C₁₋₆ alkylsulfonyl group and a C₆₋₁₄ arylsulfonyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted hydroxy group include a hydroxy group, a C₁₋₆ alkoxy group, a C₂₋₆ alkenyloxy group (e.g., allyloxy, 2-butenyloxy, 2-pentenylloxy, 3-hexenyloxy), a C₃₋₁₀ cycloalkyloxy group (e.g., cyclohexyloxy), a C₆₋₁₄ aryloxy group (e.g., phenoxy, naphthyloxy), a C₇₋₁₆ aralkyloxy group (e.g., benzyloxy, phenethylloxy), a C₁₋₆ alkyl-carbonyloxy group (e.g., acetyloxy, propionyloxy, butyryloxy, isobutyryloxy, pivaloyloxy), a C₆₋₁₄ aryl-carbonyloxy group (e.g., benzoyloxy), a C₇₋₁₆ aralkyl-carbonyloxy group (e.g., benzylcarbonyloxy), a 5- to 14-membered aromatic heterocyclylcarbonyloxy group (e.g., nicotinoyloxy), a 3- to 14-membered non-aromatic heterocyclylcarbonyloxy group (e.g., piperidinylcarbonyloxy), a C₁₋₆ alkoxy-carbonyloxy group (e.g., tert-butoxy-carbonyloxy), a 5- to 14-membered aromatic heterocyclyloxy group (e.g., pyridyloxy), a carbamoyloxy group, a C₁₋₆ alkyl-carbamoyloxy group (e.g., methylcarbamoyloxy), a C₇₋₁₆ aralkyl-carbamoyloxy group (e.g., benzylcarbamoyloxy), a C₁₋₆ alkylsulfonyloxy group (e.g., methylsulfonyloxy, ethylsulfonyloxy) and a C₆₋₁₄ arylsulfonyloxy group (e.g., phenylsulfonyloxy).

In the present specification, examples of the “optionally substituted sulfanyl group” include a sulfanyl group optionally having “a substituent selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group, a C₇₋₁₆ aralkyl group, a C₁₋₆ alkyl-carbonyl group, a C₆₋₁₄ aryl-carbonyl group and a 5- to 14-membered aromatic heterocyclic group, each of which optionally has 1 to 3 substituents selected from Substituent group A” and a halogenated sulfanyl group.

Preferable examples of the optionally substituted sulfanyl group include a sulfanyl (—SH) group, a C₁₋₆ alkylthio group, a C₂₋₆ alkenylthio group (e.g., allylthio, 2-butenylthio, 2-pentenylthio, 3-hexenylthio), a C₃₋₁₀ cycloalkylthio group (e.g., cyclohexylthio), a C₆₋₁₄ arylthio group (e.g., phenylthio, naphthylthio), a C₇₋₁₆ aralkylthio group (e.g., benzylthio, phenethylthio), a C₁₋₆ alkyl-carbonylthio group (e.g., acetylthio, propionylthio, butyrylthio, isobutyrylthio, pivaloylthio), a C₆₋₁₄ aryl-carbonylthio group (e.g., benzoylthio), a 5- to 14-membered aromatic heterocyclylthio group (e.g., pyridylthio) and a halogenated thio group (e.g., pentafluorothio).

In the present specification, examples of the “optionally substituted silyl group” include a silyl group optionally having “1 to 3 substituents selected from a C₁₋₆ alkyl group, a C₂₋₆ alkenyl group, a C₃₋₁₀ cycloalkyl group, a C₆₋₁₄ aryl group and a C₇₋₁₆ aralkyl group, each of which optionally has 1 to 3 substituents selected from Substituent group A”.

Preferable examples of the optionally substituted silyl group include a tri- C_{1-6} alkylsilyl group (e.g., trimethylsilyl, tert-butyl(dimethyl) silyl).

In the present specification, examples of the “hydrocarbon ring” include a C_{6-14} aromatic hydrocarbon ring, C_{3-10} cycloalkane and C_{3-10} cycloalkene.

In the present specification, examples of the “ C_{6-14} aromatic hydrocarbon ring” include benzene and naphthalene.

In the present specification, examples of the “ C_{3-10} cycloalkane” include cyclopropane, cyclobutane, cyclopentane, cyclohexane, cycloheptane and cyclooctane.

In the present specification, examples of the “ C_{3-10} cycloalkene” include cyclopropene, cyclobutene, cyclopentene, cyclohexene, cycloheptene and cyclooctene.

In the present specification, examples of the “heterocycle” include an aromatic heterocycle and a non-aromatic heterocycle, each containing, as a ring-constituting atom besides carbon atom, 1 to 4 hetero atoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

In the present specification, examples of the “aromatic heterocycle” include a 5- to 14-membered (preferably 5- to 10-membered) aromatic heterocycle containing, as a ring-constituting atom besides carbon atom, 1 to 4 hetero atoms selected from a nitrogen atom, a sulfur atom and an oxygen atom. Preferable examples of the “aromatic heterocycle” include 5- or 6-membered monocyclic aromatic heterocycles such as thiophene, furan, pyrrole, imidazole, pyrazole, thiazole, isothiazole, oxazole, isoxazole, pyridine, pyrazine, pyrimidine, pyridazine, 1,2,4-oxadiazole, 1,3,4-oxadiazole, 1,2,4-thiadiazole, 1,3,4-thiadiazole, triazole, tetrazole, triazine and the like; and 8- to 14-membered fused polycyclic (preferably bi- or tricyclic) aromatic heterocycles such as benzothiophene, benzofuran, benzimidazole, benzoxazole, benzisoxazole, benzothiazole, benzisothiazole, benzotriazole, imidazopyridine, thienopyridine, furopyridine, pyrrolopyridine, pyrazolopyridine, oxazolopyridine, thiazolopyridine, imidazopyrazine, imidazopyrimidine, thienopyrimidine, furopyrimidine, pyrrolopyrimidine, pyrazolopyrimidine, oxazolopyrimidine, thiazolopyrimidine, pyrazolopyrimidine, pyrazolotriazine, naphtho[2,3-b]thiophene, phenoxathiin, indole, isoindole, 1H-indazole, purine, isoquinoline, quinoline, phthalazine, naphthyridine, quinoxaline, quinazoline, cinnoline, carbazole, β -carboline, phenanthridine, acridine, phenazine, phenothiazine, phenoxazine and the like.

In the present specification, examples of the “non-aromatic heterocycle” include a 3- to 14-membered (preferably 4- to 10-membered) non-aromatic heterocycle containing, as a ring-constituting atom besides carbon atom, 1 to 4 hetero atoms selected from a nitrogen atom, a sulfur atom and an oxygen atom. Preferable examples of the “non-aromatic heterocycle” include 3- to 8-membered monocyclic non-aromatic heterocycles such as aziridine, oxirane, thiirane, azetidine, oxetane, thietane, tetrahydrothiophene, tetrahydrofuran, pyrroline, pyrrolidine, imidazoline, imidazolidine, oxazoline, oxazolidine, pyrazoline, pyrazolidine, thiazoline, thiazolidine, tetrahydroisothiazole, tetrahydrooxazole, tetrahydroisoxazole, piperidine, piperazine, tetrahydropyridine, dihydropyridine, dihydrothiopyran, tetrahydropyrimidine, tetrahydropyridazine, dihydropyran, tetrahydropyran, tetrahydrothiopyran, morpholine, thiomorpholine, azepane, diazepane, azepine, azocane, diazocane, oxepane and the like; and 9- to 14-membered fused polycyclic (preferably bi- or tricyclic) non-aromatic heterocycles such as dihydrobenzofuran, dihydrobenzimidazole, dihydrobenzoxazole, dihydrobenzothiazole, dihydrobenzisothiazole, dihydronaphtho[2,3-b]thiophene, tetrahydroisoquinoline,

tetrahydroquinoline, 4H-quinolizine, indoline, isoindoline, tetrahydrothieno[2,3-c]pyridine, tetrahydrobenzazepine, tetrahydroquinoxaline, tetrahydrophenanthridine, hexahydrophenothiazine, hexahydrophenoxazine, tetrahydrophtalazine, tetrahydronaphthyridine, tetrahydroquinazoline, tetrahydrocinnoline, tetrahydrocarbazole, tetrahydro- β -carboline, tetrahydroacridine, tetrahydrophenazine, tetrahydrothioxanthene, octahydroisoquinoline and the like.

In the present specification, examples of the “nitrogen-containing heterocycle” include a heterocycle containing at least one nitrogen atom as a ring-constituting atom, from among the “heterocycle”.

In the present specification, examples of the “ C_{3-6} cycloalkyl group” include cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl.

In the present specification, examples of the “mono- or di- C_{1-6} alkylamino group” include methylamino, dimethylamino, ethylamino, diethylamino, propylamino, isopropylamino, butylamino, isobutylamino, sec-butylamino, tert-butylamino, pentylamino, isopentylamino, neopentylamino, 1-ethylpropylamino, hexylamino, isohexylamino, 1,1-dimethylbutylamino, 2,2-dimethylbutylamino, 3,3-dimethylbutylamino, 2-ethylbutylamino and the like.

The definition of each symbol in the formula (I) is explained in detail in the following.

R^1 is an optionally substituted C_{1-6} alkyl group, an optionally substituted mono- or di- C_{1-6} alkylamino group, an optionally substituted C_{3-6} cycloalkyl group, or an optionally substituted 3- to 14-membered non-aromatic heterocyclic group.

Examples of the substituent of the above-mentioned “optionally substituted C_{1-6} alkyl group”, “optionally substituted mono- or di- C_{1-6} alkylamino group”, “optionally substituted C_{3-6} cycloalkyl group” and “optionally substituted 3- to 14-membered non-aromatic heterocyclic group” include substituents selected from Substituent group A. The number of the substituents is preferably 1 to 3. When the number of the substituents is 2 or more, the respective substituents may be the same or different.

R^1 is preferably

- (1) an optionally substituted C_{1-6} alkyl group (e.g., methyl, ethyl),
- (2) an optionally substituted mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino), or
- (3) an optionally substituted C_{3-6} cycloalkyl group (e.g., cyclopropyl).

R^1 is more preferably

- (1) a C_{1-6} alkyl group (e.g., methyl, ethyl),
- (2) a mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino), or
- (3) a C_{3-6} cycloalkyl group (e.g., cyclopropyl).

R^1 is further more preferably

- (1) a C_{1-6} alkyl group (e.g., methyl, ethyl), or
- (2) a mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino).

R^2 is a hydrogen atom, a fluorine atom, an optionally substituted C_{1-6} alkyl group, or an optionally substituted C_{3-6} cycloalkyl group.

Examples of the substituent of the above-mentioned “optionally substituted C_{1-6} alkyl group” and “optionally substituted C_{3-6} cycloalkyl group” include substituents selected from Substituent group A. The number of the substituents is preferably 1 to 3. When the number of the substituents is 2 or more, the respective substituents may be the same or different.

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R² is preferably

- (1) a hydrogen atom,
- (2) a fluorine atom, or
- (3) an optionally substituted C₁₋₆ alkyl group (e.g., methyl).

R² is more preferably

- (1) a hydrogen atom,
- (2) a fluorine atom, or
- (3) a C₁₋₆ alkyl group (e.g., methyl).

R² is further more preferably

- (1) a hydrogen atom, or
- (2) a fluorine atom.

R³ is an acyl group, an optionally substituted C₁₋₆ alkyl group, an optionally substituted C₃₋₆ cycloalkyl group, an optionally substituted C₆₋₁₄ aryl group, an optionally substituted 3- to 14-membered non-aromatic heterocyclic group, or an optionally substituted 5- to 14-membered aromatic heterocyclic group.

Examples of the substituent of the above-mentioned “optionally substituted C₁₋₆ alkyl group”, “optionally substituted C₃₋₆ cycloalkyl group”, “optionally substituted C₆₋₁₄ aryl group”, “optionally substituted 3- to 14-membered non-aromatic heterocyclic group” and “optionally substituted 5- to 14-membered aromatic heterocyclic group” include substituents selected from Substituent group A. The number of the substituents is preferably 1 to 3. When the number of the substituents is 2 or more, the respective substituents may be the same or different.

R³ is preferably an acyl group.

R³ is more preferably

- (1) an optionally substituted C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl, 2,2-dimethylpropanoyl, butanoyl, 2-methylbutanoyl),
- (2) an optionally substituted C₁₋₆ alkoxy-carbonyl group (e.g., ethoxycarbonyl, tert-butoxycarbonyl),
- (3) an optionally substituted C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)),
- (4) an optionally substituted 3- to 14-membered non-aromatic heterocyclylcarbonyl group (preferably a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, tetrahydrofurylcabonyl, azetidinylcabonyl)),
- (5) an optionally substituted mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl),
- (6) an optionally substituted N—C₁₋₆ alkyl-N—C₁₋₆ alkoxy-carbamoyl group (e.g., N-methyl-N-methoxycarbamoyl), or
- (7) a N—C₁₋₆ alkyl-N',N'-di-C₁₋₆ alkylhydrazine-carbonyl group (e.g., N-methyl-N',N'-dimethylhydrazinecarbonyl).

R³ is further more preferably

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl, 2,2-dimethylpropanoyl, butanoyl, 2-methylbutanoyl) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom (e.g., a fluorine atom),
 - (b) a hydroxy group, and
 - (c) a cyano group,
- (2) a C₁₋₆ alkoxy-carbonyl group (e.g., ethoxycarbonyl, tert-butoxycarbonyl),
- (3) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobu-

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tylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from

- (a) a halogen atom (e.g., a fluorine atom),
- (b) a hydroxy group,
- (c) a cyano group, and
- (d) a C₁₋₆ alkyl group (e.g., methyl),
- (4) a 3- to 14-membered non-aromatic heterocyclylcarbonyl group (preferably a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, tetrahydrofurylcabonyl, azetidinylcabonyl)) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom (e.g., a fluorine atom),
 - (b) a hydroxy group, and
 - (c) a C₁₋₆ alkyl group (e.g., methyl),
- (5) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl),
- (6) a N—C₁₋₆ alkyl-N—C₁₋₆ alkoxy-carbamoyl group (e.g., N-methyl-N-methoxycarbamoyl), or
- (7) a N—C₁₋₆ alkyl-N',N'-di-C₁₋₆ alkylhydrazine-carbonyl group (e.g., N-methyl-N',N'-dimethylhydrazinecarbonyl).

R³ is still more preferably

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl, butanoyl) optionally substituted by 1 to 3 hydroxy groups,
- (2) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom (e.g., a fluorine atom), and
 - (b) a hydroxy group,
- (3) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, tetrahydrofurylcabonyl, azetidinylcabonyl), or
- (4) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl).

R³ is even more preferably

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl) optionally substituted by 1 to 3 hydroxy groups,
- (2) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from
 - (a) a halogen atom (e.g., a fluorine atom), and
 - (b) a hydroxy group,
- (3) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, azetidinylcabonyl), or
- (4) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl).

R³ is particularly preferably

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl) optionally substituted by 1 to 3 hydroxy groups,
- (2) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, azetidinylcabonyl), or
- (3) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl).

Ring A is an optionally further substituted C₆₋₁₄ aromatic hydrocarbon ring, or an optionally further substituted 5- to 14-membered aromatic heterocycle.

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Ring A optionally has substituent(s) in addition to $-\text{CH}_2-$ pyrrolidine ring in the formula (I). Examples of the substituent include the above-mentioned "substituent". The number of the substituents is preferably 1 to 3. When the number of the substituents is 2 or more, the respective substituents may be the same or different.

Ring A is preferably

- (1) an optionally further substituted benzene ring, or
- (2) an optionally further substituted 5- or 6-membered aromatic heterocycle (e.g., thiazole, pyridine).

Ring A is more preferably

- (1) a benzene ring further substituted by one substituent selected from
 - (a) a C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom, a bromine atom),
 - (ii) an optionally halogenated C_{1-6} alkyl group (e.g., methyl, difluoromethyl), and
 - (iii) an optionally halogenated C_{1-6} alkoxy group (e.g., methoxy, difluoromethoxy), and
 - (b) a 5- to 14-membered aromatic heterocyclic group (preferably a 5- or 6-membered aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyridazinyl, thiazolyl, oxazolyl)) optionally substituted by 1 to 3 substituents selected from
 - (i) a C_{1-6} alkyl group (e.g., methyl, ethyl) optionally substituted by 1 to 3 substituents selected from a halogen atom (e.g., a fluorine atom) and a hydroxy group,
 - (ii) a C_{1-6} alkoxy group (e.g., methoxy),
 - (iii) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
 - (iv) a C_{1-6} alkoxy-carbonyl group (e.g., ethoxycarbonyl), and

optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or

- (2) a 5- or 6-membered aromatic heterocycle (e.g., thiazole, pyridine) further substituted by one C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Ring A is further more preferably a benzene ring further substituted by one C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

- (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
- (ii) an optionally halogenated C_{1-6} alkyl group (e.g., methyl, difluoromethyl), and

optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

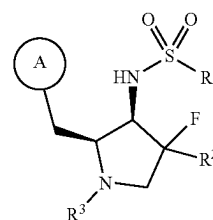
Ring A is particularly preferably a benzene ring further substituted by one C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

- (i) a halogen atom (e.g., a fluorine atom), and
- (ii) a C_{1-6} alkyl group (e.g., methyl), and

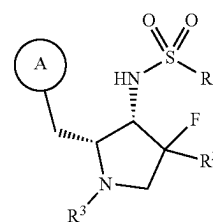
optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Regarding the pyrrolidine ring of compound (I), the configuration based on the carbon atom that $-\text{NHSO}_2\text{R}^1$ is bonded to and the carbon atom that $-\text{CH}_2-\text{Ring A}$ is bonded to is preferably cis-form. That is, compound (I) is preferably represented by the formula (IA) or (IB):

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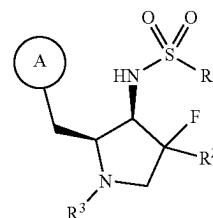


(IA)



(IB)

wherein each symbol is as defined above, more preferably represented by the formula (IA):



(IA)

wherein each symbol is as defined above.

Preferable examples of compound (I) include the following compounds. These compounds are preferably represented by the above formula (IA) or (IB), more preferably represented by the formula (IA).

[Compound A]

Compound (I) wherein

R^1 is

- (1) an optionally substituted C_{1-6} alkyl group (e.g., methyl, ethyl),
- (2) an optionally substituted mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino), or
- (3) an optionally substituted C_{3-6} cycloalkyl group (e.g., cyclopropyl);

R^2 is

- (1) a hydrogen atom,
- (2) a fluorine atom, or
- (3) an optionally substituted C_{1-6} alkyl group (e.g., methyl);

R^3 is

- (1) an optionally substituted C_{1-6} alkyl-carbonyl group (e.g., 2-methylpropanoyl, 2,2-dimethylpropanoyl, butanoyl, 2-methylbutanoyl),
- (2) an optionally substituted C_{1-6} alkoxy-carbonyl group (e.g., ethoxycarbonyl, tert-butoxycarbonyl),
- (3) an optionally substituted C_{3-10} cycloalkyl-carbonyl group (the C_{3-10} cycloalkyl moiety of the C_{3-10} cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)),
- (4) an optionally substituted 3- to 14-membered non-aromatic heterocyclylcarbonyl group (preferably a 3- to

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- 8-membered monocyclic non-aromatic heterocyclyl-carbonyl group (e.g., oxetanylcarbonyl, tetrahydrofurylcarbonyl, azetidiny carbonyl)),
- (5) an optionally substituted mono- or di- C_{1-6} alkyl-carbamoyl group (e.g., dimethylcarbamoyl), 5
- (6) an optionally substituted $N-C_{1-6}$ alkyl- $N-C_{1-6}$ alkoxy-carbamoyl group (e.g., N-methyl-N-methoxy-carbamoyl), or
- (7) a $N-C_{1-6}$ alkyl- N' , N' -di- C_{1-6} alkylhydrazine-carbonyl group (e.g., N-methyl- N' , N' -dimethylhydrazinecarbonyl); and Ring A is
- (1) an optionally further substituted benzene ring, or
- (2) an optionally further substituted 5- or 6-membered aromatic heterocycle (e.g., thiazole, pyridine).
- [Compound B] 15
- Compound (I) wherein
- R^1 is
- (1) a C_{1-6} alkyl group (e.g., methyl, ethyl),
- (2) a mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino), or 20
- (3) a C_{3-6} cycloalkyl group (e.g., cyclopropyl);
- R^2 is
- (1) a hydrogen atom,
- (2) a fluorine atom, or
- (3) a C_{1-6} alkyl group (e.g., methyl); 25
- R^3 is
- (1) a C_{1-6} alkyl-carbonyl group (e.g., 2-methylpropanoyl, 2,2-dimethylpropanoyl, butanoyl, 2-methylbutanoyl) optionally substituted by 1 to 3 substituents selected from 30
- (a) a halogen atom (e.g., a fluorine atom),
- (b) a hydroxy group, and
- (c) a cyano group,
- (2) a C_{1-6} alkoxy-carbonyl group (e.g., ethoxycarbonyl, tert-butoxycarbonyl), 35
- (3) a C_{3-10} cycloalkyl-carbonyl group (the C_{3-10} cycloalkyl moiety of the C_{3-10} cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from 40
- (a) a halogen atom (e.g., a fluorine atom),
- (b) a hydroxy group,
- (c) a cyano group, and
- (d) a C_{1-6} alkyl group (e.g., methyl), 45
- (4) a 3- to 14-membered non-aromatic heterocyclylcarbonyl group (preferably a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcarbonyl, tetrahydrofurylcarbonyl, azetidiny carbonyl)) optionally substituted by 1 to 3 substituents 50
- (a) a halogen atom (e.g., a fluorine atom),
- (b) a hydroxy group, and
- (c) a C_{1-6} alkyl group (e.g., methyl),
- (5) a mono- or di- C_{1-6} alkyl-carbamoyl group (e.g., dimethylcarbamoyl), 55
- (6) a $N-C_{1-6}$ alkyl- $N-C_{1-6}$ alkoxy-carbamoyl group (e.g., N-methyl-N-methoxycarbamoyl), or
- (7) a $N-C_{1-6}$ alkyl- N' , N' -di- C_{1-6} alkylhydrazine-carbonyl group (e.g., N-methyl- N' , N' -dimethylhydrazinecarbonyl); and Ring A is 60
- (1) a benzene ring
- further substituted by one substituent selected from
- (a) a C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from 65
- (i) a halogen atom (e.g., a fluorine atom, a chlorine atom, a bromine atom),

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- (ii) an optionally halogenated C_{1-6} alkyl group (e.g., methyl, difluoromethyl), and
- (iii) an optionally halogenated C_{1-6} alkoxy group (e.g., methoxy, difluoromethoxy), and
- (b) a 5- to 14-membered aromatic heterocyclic group (preferably a 5- or 6-membered aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyridazinyl, thiazolyl, oxazolyl)) optionally substituted by 1 to 3 substituents selected from
- (i) a C_{1-6} alkyl group (e.g., methyl, ethyl) optionally substituted by 1 to 3 substituents selected from a halogen atom (e.g., a fluorine atom) and a hydroxy group,
- (ii) a C_{1-6} alkoxy group (e.g., methoxy),
- (iii) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
- (iv) a C_{1-6} alkoxy-carbonyl group (e.g., ethoxycarbonyl), and
- optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a 5- or 6-membered aromatic heterocycle (e.g., thiazole, pyridine) further substituted by one C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).
- [Compound C] 25
- Compound (I) wherein
- R^1 is
- (1) a C_{1-6} alkyl group (e.g., methyl, ethyl),
- (2) a mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino), or
- (3) a C_{3-6} cycloalkyl group (e.g., cyclopropyl);
- R^2 is
- (1) a hydrogen atom, or
- (2) a fluorine atom;
- R^3 is
- (1) a C_{1-6} alkyl-carbonyl group (e.g., 2-methylpropanoyl, butanoyl) optionally substituted by 1 to 3 hydroxy groups,
- (2) a C_{3-10} cycloalkyl-carbonyl group (the C_{3-10} cycloalkyl moiety of the C_{3-10} cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from
- (a) a halogen atom (e.g., a fluorine atom), and
- (b) a hydroxy group,
- (3) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcarbonyl, tetrahydrofurylcarbonyl, azetidiny carbonyl), or
- (4) a mono- or di- C_{1-6} alkyl-carbamoyl group (e.g., dimethylcarbamoyl); and
- Ring A is a benzene ring
- further substituted by one C_{6-14} aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
- (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
- (ii) an optionally halogenated C_{1-6} alkyl group (e.g., methyl, difluoromethyl), and
- optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).
- [Compound D]
- Compound (I) wherein
- R^1 is
- (1) a C_{1-6} alkyl group (e.g., methyl, ethyl), or
- (2) a mono- or di- C_{1-6} alkylamino group (e.g., dimethylamino);

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R² is

- (1) a hydrogen atom, or
(2) a fluorine atom;

R³ is

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl) optionally substituted by 1 to 3 hydroxy groups,
(2) a C₃₋₁₀ cycloalkyl-carbonyl group (the C₃₋₁₀ cycloalkyl moiety of the C₃₋₁₀ cycloalkyl-carbonyl group is optionally bridged (e.g., cyclopropylcarbonyl, cyclobutylcarbonyl, bicyclo[1.1.1]pentan-1-ylcarbonyl)) optionally substituted by 1 to 3 substituents selected from

- (a) a halogen atom (e.g., a fluorine atom), and
(b) a hydroxy group,

- (3) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, azetidinylcabonyl), or

- (4) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl); and

Ring A is a benzene ring

further substituted by one C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

- (i) a halogen atom (e.g., a fluorine atom), and
(ii) a C₁₋₆ alkyl group (e.g., methyl), and

optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

[Compound E]

Compound (I) wherein

R¹ is

- (1) a C₁₋₆ alkyl group (e.g., ethyl), or
(2) a mono- or di-C₁₋₆ alkylamino group (e.g., dimethylamino);

R² is

- (1) a hydrogen atom, or
(2) a fluorine atom;

R³ is

- (1) a C₁₋₆ alkyl-carbonyl group (e.g., 2-methylpropanoyl) optionally substituted by 1 to 3 hydroxy groups,
(2) a 3- to 8-membered monocyclic non-aromatic heterocyclylcarbonyl group (e.g., oxetanylcabonyl, azetidinylcabonyl), or

- (3) a mono- or di-C₁₋₆ alkyl-carbamoyl group (e.g., dimethylcarbamoyl); and

Ring A is a benzene ring

further substituted by one C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

- (i) a halogen atom (e.g., a fluorine atom), and
(ii) a C₁₋₆ alkyl group (e.g., methyl), and

optionally further substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Specific examples of compound (I) include the compounds of the below-mentioned Examples 1 to 616.

Specifically, compound (I) is preferably

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 1);

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof (Example 2);

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 3);

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N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 5);

N-{(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 56);

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 66);

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 67);

N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof (Example 87);

N-{(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof (Example 91);

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 94);

N-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 144);

N-{(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 146);

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((1R,3S)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 225);

N-{(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 236);

N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 302);

N-{(2S,3R)-4,4-difluoro-1-((1R,3S)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 375);

N-{(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 380);

N-{(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof (Example 433);

N'-(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide or a salt thereof (Example 443);

N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 450);

N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof (Example 451);

(2S,3R,4S)-3-[(dimethylsulfamoyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide or a salt thereof (Example 459);

N'-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide or a salt thereof (Example 462);

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof (Example 463); or

N'-{(2S,3R)-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof (Example 542).

Compound (I) is particularly preferably

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 1);

N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 3);

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 94);

N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof (Example 443);

N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide or a salt thereof (Example 451);

(2S,3R,4S)-3-[(dimethylsulfamoyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide or a salt thereof (Example 459);

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof (Example 463); or

N'-{(2S,3R)-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof (Example 542).

As a salt of a compound represented by the formula (I), a pharmacologically acceptable salt is preferable, and examples of such salt include a salt with inorganic base, a salt with organic base, a salt with inorganic acid, a salt with organic acid, a salt with basic or acidic amino acid and the like.

Preferable examples of the salt with inorganic base include alkali metal salts such as sodium salt, potassium salt and the like, alkaline earth metal salts such as calcium salt, magnesium salt and the like, aluminum salt, ammonium salt and the like.

Preferable examples of the salt with organic base include salts with trimethylamine, triethylamine, pyridine, picoline, ethanolamine, diethanolamine, triethanolamine, tromethamine[tris(hydroxymethyl)methylamine], tert-butylamine, cyclohexylamine, benzylamine, dicyclohexylamine, N,N-dibenzylethylenediamine and the like.

Preferable examples of the salt with inorganic acid include salts with hydrochloric acid, hydrobromic acid, nitric acid, sulfuric acid, phosphoric acid and the like.

Preferable examples of the salt with organic acid include salts with formic acid, acetic acid, trifluoroacetic acid, phthalic acid, fumaric acid, oxalic acid, tartaric acid, maleic

acid, citric acid, succinic acid, malic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid and the like.

Preferable examples of the salt with basic amino acid include salts with arginine, lysine, ornithine and the like. Preferable examples of the salt with acidic amino acid include salts with aspartic acid, glutamic acid and the like.

The production method of the compound of the present invention is explained below.

The raw material compound and reagent used and the compound obtained in each step in the following production method may be each in a form of a salt, and examples of such salt include those similar to the salts of the compound represented by the formula (I), and the like.

When the compound obtained in each step is a free form, it can be converted to the objective salt according to a method known per se. When the compound obtained in each step is a salt, it can be converted to the objective free form or the other salt according to a method known per se.

The compound obtained in each step can be used directly as the reaction mixture or as a crude product for the next reaction. Alternatively, the compound obtained in each step can be isolated and purified from a reaction mixture according to a method known per se, for example, a separation means such as concentration, crystallization, recrystallization, distillation, solvent extraction, fractional distillation, column chromatography and the like.

When the raw material compound and reagent used in each step are commercially available, the commercially available product can also be used directly.

In the reaction in each step, while the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 10 min-8 hr, unless otherwise specified.

In the reaction in each step, while the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally -78° C.-300° C., preferably -78° C.-150° C., unless otherwise specified.

In the reaction in each step, while the pressure varies depending on the kind of the reagent and solvent to be used, it is generally 1 atm-20 atm, preferably 1 atm-3 atm, unless otherwise specified.

Microwave synthesizer such as Initiator manufactured by Biotage and the like may be used for the reaction in each step. While the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally room temperature -300° C., preferably 50° C.-250° C., unless otherwise specified. While the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 1 min-8 hr, unless otherwise specified.

In the reaction in each step, the reagent is used in an amount of 0.5 equivalents-20 equivalents, preferably 0.8 equivalents-5 equivalents, relative to the substrate, unless otherwise specified. When the reagent is used as a catalyst, the reagent is used in an amount of 0.001 equivalent-1 equivalent, preferably 0.01 equivalent-0.2 equivalent, relative to the substrate. When the reagent is used as a reaction solvent, the reagent is used in a solvent amount.

Unless otherwise specified, the reaction in each step is carried out without solvent, or by dissolving or suspending the raw material compound in a suitable solvent. Examples of the solvent include those described in Examples and the following solvents.

alcohols: methanol, ethanol, tert-butyl alcohol, 2-methoxyethanol and the like;

ethers: diethyl ether, diphenyl ether, tetrahydrofuran, 1,2-dimethoxyethane and the like;
 aromatic hydrocarbons: chlorobenzene, toluene, xylene and the like;
 saturated hydrocarbons: cyclohexane, hexane and the like;
 amides: N,N-dimethylformamide, N-methylpyrrolidone and the like;
 halogenated hydrocarbons: dichloromethane, carbon tetrachloride and the like;
 nitriles: acetonitrile and the like;
 sulfoxides: dimethyl sulfoxide and the like;
 aromatic organic bases: pyridine and the like;
 anhydrides: acetic anhydride and the like;
 organic acids: formic acid, acetic acid, trifluoroacetic acid and the like;
 inorganic acids: hydrochloric acid, sulfuric acid and the like;
 esters: ethyl acetate and the like;
 ketones: acetone, methyl ethyl ketone and the like;
 water.

The above-mentioned solvent can be used in a mixture of two or more kinds thereof in an appropriate ratio.

When a base is used for the reaction in each step, examples thereof include those described in Examples and the following bases.

inorganic bases: sodium hydroxide, magnesium hydroxide, sodium carbonate, calcium carbonate, sodium hydrogen carbonate and the like;
 organic bases: triethylamine, diethylamine, pyridine, 4-dimethylaminopyridine, N,N-dimethylaniline, 1,4-diazabicyclo[2.2.2]octane, 1,8-diazabicyclo[5.4.0]-7-undecene, imidazole, piperidine and the like;
 metal alkoxides: sodium ethoxide, potassium tert-butoxide and the like;
 alkali metal hydrides: sodium hydride and the like;
 metal amides: sodium amide, lithium diisopropylamide, lithium hexamethyldisilazide and the like;
 organic lithiums: n-butyllithium and the like.

When an acid or an acid catalyst is used for the reaction in each step, examples thereof include those described in Examples and the following acids and acid catalysts.

inorganic acids: hydrochloric acid, sulfuric acid, nitric acid, hydrobromic acid, phosphoric acid and the like;
 organic acids: acetic acid, trifluoroacetic acid, citric acid, p-toluenesulfonic acid, 10-camphorsulfonic acid and the like;
 Lewis acid: boron trifluoride diethyl ether complex, zinc iodide, anhydrous aluminum chloride, anhydrous zinc chloride, anhydrous iron chloride and the like.

Unless otherwise specified, the reaction in each step is carried out according to a method known per se, for example, the method described in Jikken Kagaku Kouza, 5th Edition, vol. 13-19 (the Chemical Society of Japan ed.); Shin Jikken Kagaku Kouza, vol. 14-15 (the Chemical Society of Japan ed.); Fine Organic Chemistry, Revised 2nd Edition (L. F. Fieser, Th. Eicher, Nankodo); Organic Name Reactions, the Reaction Mechanism and Essence, Revised Edition (Hideo Togo, Kodansha); ORGANIC SYNTHESIS Collective Volume I-VII (John Wiley & Sons Inc.); Modern Organic Synthesis in the Laboratory A Collection of Standard Experimental Procedures (Jie Jack Li, OXFORD UNIVERSITY); Comprehensive Heterocyclic Chemistry III, Vol. 1-Vol. 14 (Elsevier Japan); Strategic Applications of Named Reactions in Organic Synthesis (translated by Kiyoshi Tomioka, Kagakudojin); Comprehensive Organic

Transformations (VCH Publishers Inc.), 1989, or the like, or the method described in Examples.

In each step, the protection or deprotection reaction of a functional group is carried out according to a method known per se, for example, the method described in "Protective Groups in Organic Synthesis, 4th Ed", Wiley-Interscience, Inc., 2007 (Theodora W. Greene, Peter G. M. Wuts); "Protecting Groups 3rd Ed." Thieme, 2004 (P. J. Kocienski), or the like, or the method described in Examples.

Examples of the protecting group for a hydroxy group of an alcohol and the like and a phenolic hydroxy group include ether-type protecting groups such as methoxymethyl ether, benzyl ether, tert-butyldimethylsilyl ether, tetrahydropyranyl ether and the like; carboxylate ester-type protecting groups such as acetate ester and the like; sulfonate ester-type protecting groups such as methanesulfonate ester and the like; carbonate ester-type protecting groups such as tert-butylcarbonate and the like, and the like.

Examples of the protecting group for a carbonyl group of an aldehyde include acetal-type protecting groups such as dimethylacetal and the like; cyclic acetal-type protecting groups such as 1,3-dioxane and the like, and the like.

Examples of the protecting group for a carbonyl group of a ketone include ketal-type protecting groups such as dimethylketal and the like; cyclic ketal-type protecting groups such as 1,3-dioxane and the like; oxime-type protecting groups such as O-methyloxime and the like; hydrazone-type protecting groups such as N,N-dimethylhydrazone and the like, and the like.

Examples of the protecting group for a carboxyl group include ester-type protecting groups such as methyl ester and the like; amide-type protecting groups such as N,N-dimethylamide and the like, and the like.

Examples of the protecting group for a thiol include ether-type protecting groups such as benzyl thioether and the like; ester-type protecting groups such as thioacetate ester, thiocarbonate, thiocarbamate and the like, and the like.

Examples of the protecting group for an amino group and an aromatic heterocycle such as imidazole, pyrrole, indole and the like include carbamate-type protecting groups such as benzyl carbamate and the like; amide-type protecting groups such as acetamide and the like; alkyl amine-type protecting groups such as N-triphenylmethylamine and the like; sulfonamide-type protecting groups such as methanesulfonamide and the like, and the like.

The protecting groups can be removed according to a method known per se, for example, by employing a method using acid, base, ultraviolet rays, hydrazine, phenylhydrazine, sodium N-methyldithiocarbamate, tetrabutylammonium fluoride, palladium acetate, trialkylsilyl halide (e.g., trimethylsilyl iodide, trimethylsilyl bromide) and the like, a reduction method, and the like.

When reduction reaction is carried out in each step, examples of the reducing agent to be used include metal hydrides such as lithium aluminum hydride, sodium triacetoxyborohydride, sodium cyanoborohydride, diisobutylaluminum hydride (DIBAL-H), sodium borohydride, tetramethylammonium triacetoxyborohydride and the like; boranes such as borane tetrahydrofuran complex and the like; Raney nickel; Raney cobalt; hydrogen; formic acid; triethylsilane and the like. When carbon-carbon double bond or triple bond is reduced, a method using a catalyst such as palladium-carbon, Lindlar's catalyst and the like may be employed.

When oxidation reaction is carried out in each step, examples of the oxidizing agent to be used include peroxides such as m-chloroperbenzoic acid (mCPBA), hydrogen per-

oxide, tert-butylhydroperoxide and the like; perchlorates such as tetrabutylammonium perchlorate and the like; chlorates such as sodium chlorate and the like; chlorites such as sodium chlorite and the like; periodates such as sodium periodate and the like; hypervalent iodine reagents such as iodosylbenzene and the like; reagents containing manganese such as manganese dioxide, potassium permanganate and the like; leads such as lead tetraacetate and the like; reagents containing chromium such as pyridinium chlorochromate (PCC), pyridinium dichromate (PDC), Jones reagent and the like; halogen compounds such as N-bromosuccinimide (NBS) and the like; oxygen; ozone; sulfur trioxide-pyridine complex; osmium tetroxide; selenium dioxide; 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) and the like.

When radical cyclization reaction is carried out in each step, examples of the radical initiator to be used include azo compounds such as azobisisobutyronitrile (AIBN) and the like; water-soluble radical initiators such as 4-4'-azobis-4-cyanopentanoic acid (ACPA) and the like; triethylboron in the presence of air or oxygen; benzoyl peroxide and the like. Examples of the radical reagent to be used include tributylstannane, tris(trimethylsilyl)silane, 1,1,2,2-tetraphenyldisilane, diphenylsilane, samarium iodide and the like.

When Wittig reaction is carried out in each step, examples of the Wittig reagent to be used include alkylidene phosphoranes and the like. The alkylidene phosphoranes can be prepared according to a method known per se, for example, by reacting a phosphonium salt with a strong base.

When Horner-Emmons reaction is carried out in each step, examples of the reagent to be used include phosphonoacetates such as methyl dimethylphosphonoacetate, ethyl diethylphosphonoacetate and the like; and bases such as alkali metal hydrides, organic lithiums and the like.

When Friedel-Crafts reaction is carried out in each step, a combination of a Lewis acid and an acid chloride or a combination of a Lewis acid and an alkylating agent (e.g., an alkyl halide, an alcohol, an olefin etc.) is used as a reagent. Alternatively, an organic acid or an inorganic acid can also be used instead of a Lewis acid, and an anhydride such as acetic anhydride and the like can also be used instead of an acid chloride.

When aromatic nucleophilic substitution reaction is carried out in each step, a nucleophile (e.g., an amine, imidazole etc.) and a base (e.g., an organic base etc.) are used as a reagent.

When nucleophilic addition reaction by a carbo anion, nucleophilic 1,4-addition reaction (Michael addition reaction) by a carbo anion or nucleophilic substitution reaction by a carbo anion is carried out in each step, and examples of the base to be used for generation of the carbo anion include organic lithiums, metal alkoxides, inorganic bases, organic bases and the like.

When Grignard reaction is carried out in each step, examples of the Grignard reagent to be used include arylmagnesium halides such as phenylmagnesium bromide and the like; and alkylmagnesium halides such as methylmagnesium bromide and the like. The Grignard reagent can be prepared according to a method known per se, for example, by reacting an alkyl halide or an aryl halide with a metal magnesium in an ether or tetrahydrofuran as a solvent.

When Knoevenagel condensation reaction is carried out in each step, a compound having an activated methylene group with two electron withdrawing groups (e.g., malonic acid, diethyl malonate, malononitrile etc.) and a base (e.g., an organic base, a metal alkoxide, an inorganic base) are used as a reagent.

When Vilsmeier-Haack reaction is carried out in each step, phosphoryl chloride and an amide derivative (e.g., N,N-dimethylformamide etc.) are used as a reagent.

When azidation reaction of an alcohol, an alkyl halide or a sulfonate is carried out in each step, examples of the azidating agent to be used include diphenylphosphorylazide (DPPA), trimethylsilylazide, sodium azide and the like. For example, for the azidation reaction of an alcohol, a method using diphenylphosphorylazide and 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), a method using trimethylsilylazide and a Lewis acid, and the like are employed.

When reductive amination reaction is carried out in each step, examples of the reducing agent to be used include sodium triacetoxymethylborohydride, sodium cyanoborohydride, hydrogen, formic acid and the like. When the substrate is an amine compound, examples of the carbonyl compound to be used include paraformaldehyde, aldehydes such as acetaldehyde and the like, and ketones such as cyclohexanone and the like. When the substrate is a carbonyl compound, examples of the amine to be used include ammonia, primary amines such as methylamine and the like; secondary amines such as dimethylamine and the like, and the like.

When Mitsunobu reaction is carried out in each step, an azodicarboxylate (e.g., diethyl azodicarboxylate (DEAD), diisopropyl azodicarboxylate (DIAD) etc.) and triphenylphosphine are used as a reagent.

When esterification reaction, amidation reaction or urea formation reaction is carried out in each step, examples of the reagent to be used include acyl halides such as acid chlorides, acid bromides and the like; activated carboxylic acids such as acid anhydrides, activated esters, sulfates and the like. Examples of the activating agent of the carboxylic acid include carbodiimide condensing agents such as 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (WSCD) and the like; triazine condensing agents such as 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride n-hydrate (DMT-MM) and the like; carbonate condensing agents such as 1,1-carbonyldiimidazole (CDI) and the like; diphenylphosphoryl azide (DPPA); benzotriazol-1-yloxy-tris(dimethylaminophosphonium) salt (BOP reagent); 2-chloro-1-methyl-pyridinium iodide (Mukaiyama reagent); thionyl chloride; lower alkyl haloformates such as ethyl chloroformate and the like; O-(7-azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium

hexafluorophosphate (HATU); sulfuric acid; combinations thereof and the like. When carbodiimide condensing agent is used, an additive such as 1-hydroxybenzotriazole (HOBt), N-hydroxysuccinimide (HOSu), dimethylaminopyridine (DMAP) and the like may be added to the reaction system.

When coupling reaction is carried out in each step, examples of the metal catalyst to be used include palladium compounds such as palladium(II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), dichlorobis(triethylphosphine)palladium(II), tris(dibenzylideneacetone)dipalladium(0), 1,1'-bis(diphenylphosphino)ferrocene palladium(II) chloride and the like; nickel compounds such as tetrakis(triphenylphosphine)nickel(0) and the like; rhodium compounds such as tris(triphenylphosphine)rhodium(III) chloride and the like; cobalt compounds; copper compounds such as copper oxide, copper(I) iodide and the like; platinum compounds and the like. In addition, a base can be added to the reaction system, and examples thereof include inorganic bases and the like.

When thiocarbonylation reaction is carried out in each step, phosphorus pentasulfide is typically used as the thiocarbonylating agent. Alternatively, a reagent having a 1,3,2,4-dithiadiphosphetane-2,4-disulfide structure (e.g., 2,4-bis

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(4-methoxyphenyl)-1,3,2,4-dithiadiphosphetane-2,4-disulfide (Lawesson reagent) etc.) can also be used instead of phosphorus pentasulfide.

When Wohl-Ziegler reaction is carried out in each step, examples of the halogenating agent to be used include N-iodosuccinimide, N-bromosuccinimide (NBS), N-chlorosuccinimide (NCS), bromine, sulfuryl chloride and the like. In addition, the reaction can be accelerated by subjecting a radical initiator such as heat, light, benzoyl peroxide, azobisisobutyronitrile and the like to the reaction system reaction.

When halogenation reaction of a hydroxy group is carried out in each step, examples of the halogenating agent to be used include hydrohalic acids and acid halides of inorganic acids, specifically, hydrochloric acid, thionyl chloride, phosphorus oxychloride and the like for chlorination, 48% hydrobromic acid and the like for bromination. In addition, a method of producing an alkyl halide by reacting an alcohol with triphenylphosphine and carbon tetrachloride or carbon tetrabromide or the like can be employed. Alternatively, a method of producing an alkyl halide via two steps comprising converting an alcohol to the corresponding sulfonate, and then reacting the sulfonate with lithium bromide, lithium chloride or sodium iodide can also be employed.

When Arbuzov reaction is carried out in each step, examples of the reagent to be used include alkyl halides such as ethyl bromoacetate and the like; and phosphites such as triethyl phosphite, tri (isopropyl) phosphite and the like.

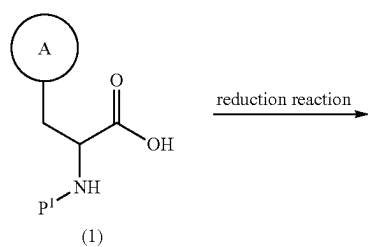
When sulfonate esterification reaction is carried out in each step, examples of the sulfonating agent to be used include methanesulfonyl chloride, p-toluenesulfonyl chloride, methanesulfonic anhydride, p-toluenesulfonic anhydride and the like.

When hydrolysis reaction is carried out in each step, an acid or a base is used as a reagent. For acid hydrolysis reaction of tert-butyl ester, formic acid, triethylsilane and the like may be added to reductively-trap tert-butyl cation which is by-produced.

When dehydration reaction is carried out in each step, examples of the dehydrating agent to be used include sulfuric acid, diphosphorus pentaoxide, phosphorus oxychloride, N,N'-dicyclohexylcarbodiimide, alumina, polyphosphoric acid and the like.

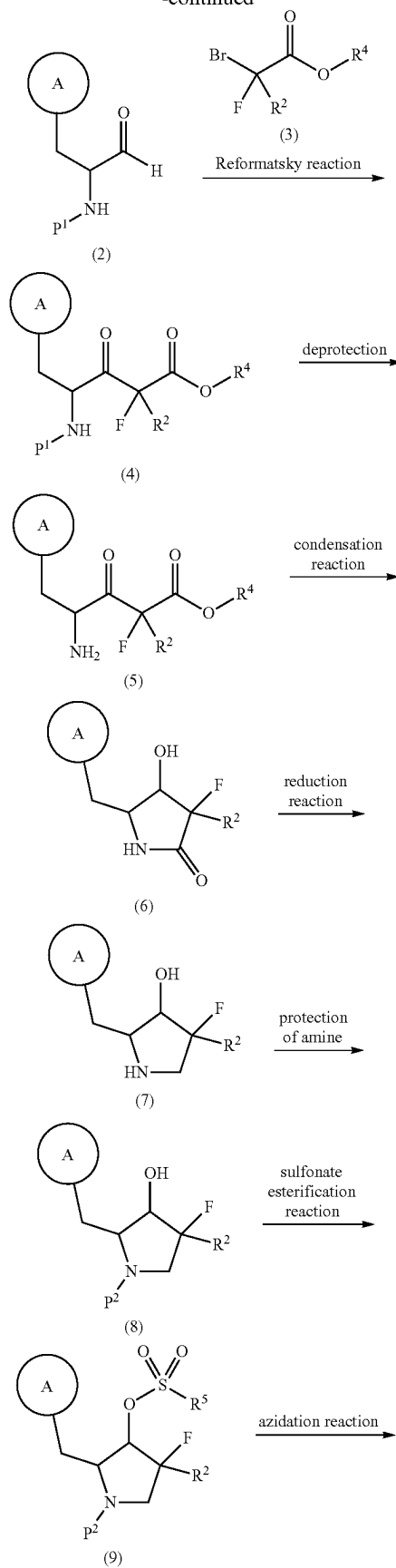
Compound (1) can be produced from compound (1) according to the method shown in the following Reaction Scheme 1. In the reaction scheme, LG¹ and LG² are each independently a leaving group, R⁴ and R⁵ are each independently an optionally substituted C₁₋₆ alkyl group or an optionally substituted C₆₋₁₄ aryl group, P¹ and P² are each independently a protecting group, and the other symbols are as defined above.

Reaction Scheme 1



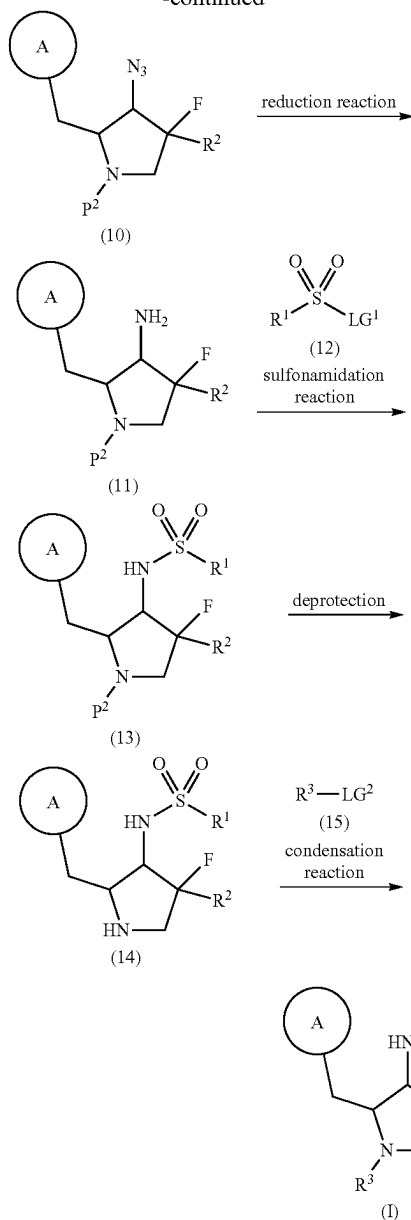
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Examples of the “leaving group” represented by LG^1 or LG^2 include halogen atoms, optionally halogenated C_{1-6} alkylsulfonyloxy groups (e.g., methanesulfonyloxy, ethanesulfonyloxy, trifluoromethanesulfonyloxy), and C_{6-14} arylsulfonyloxy groups optionally substituted by C_{1-6} alkyl group(s) (e.g., benzenesulfonyloxy, toluenesulfonyloxy) and the like.

Examples of the substituent of the “optionally substituted C_{1-6} alkyl group” and “optionally substituted C_{6-14} aryl group” represented by R^4 or R^5 include substituents selected from Substituent group A. The number of the substituents is preferably 1 to 3. When the number of the substituents is 2 or more, the respective substituents may be the same or different.

Examples of the “protecting group” represented by P^1 or P^2 include those exemplified as the above-mentioned “protecting group for an amino group and an aromatic heterocycle such as imidazole, pyrrole, indole and the like”.

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Compound (1) may be commercially available, or can be produced according to a method known per se or a method analogous thereto.

Compound (4) can be produced by subjecting compound (2) to the Reformatsky reaction with compound (3) in the presence of a metal. Examples of the metal to be used include zinc, samarium iodide, indium and the like. Compound (3) may be commercially available, or can be produced according to a method known per se or a method analogous thereto.

Compound (6) can be produced by subjecting compound (5) to a condensation reaction in the presence of a base. Examples of the base to be used include inorganic bases, organic bases, alkali metal hydrides and the like.

Compound (9) can be produced by subjecting compound (8) to a sulfonate esterification reaction. Examples of the sulfonating agent to be used include methanesulfonyl chloride, p-toluenesulfonyl chloride, methanesulfonic anhydride, p-toluenesulfonic anhydride, trifluoromethanesulfonic anhydride and the like.

Compound (10) can be produced by subjecting compound (9) to an azidation reaction. Examples of the azidating agent to be used include tetra-n-butylammonium azide, trimethylsilylazide, sodium azide and the like.

Compound (13) can be produced by subjecting compound (11) to a sulfonamidation reaction with compound (12). Examples of compound (12) to be used include sulfonyl chlorides, sulfamoyl chlorides and the like. Compound (12) may be commercially available, or can be produced according to a method known per se or a method analogous thereto.

Compound (I) can be produced by subjecting compound (14) to a condensation reaction with compound (15). Examples of compound (15) to be used include acyl halides such as acid chlorides, acid bromides, alkyl chloroformates, carbamoyl chlorides and the like; and activated carboxylic acids such as acid anhydrides, activated esters, sulfates and the like. Examples of the activating agent of the carboxylic acid include carbodiimide condensing agents such as 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (WSCD) and the like; triazine condensing agents such as 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride-n-hydrate (DMT-MM) and the like; carbonate ester condensing agents such as 1,1-carbonyldiimidazole (CDI) and the like; diphenylphosphoryl azide (DPPA); benzotriazol-1-yloxy-trisdimethylaminophosphonium salt (BOP reagent); 2-chloro-1-methyl-pyridinium iodide (Mukaiyama reagent); thionyl chloride; lower alkyl haloformates such as chloroethyl formate and the like; O-(7-azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium

hexafluorophosphate (HATU); sulfuric acid; combinations thereof, and the like. In addition, a base may be added to the reaction system. Examples of the base include inorganic bases, organic bases and the like. When a carbodiimide condensing agent is used, an additive such as 1-hydroxybenzotriazole (HOBt), N-hydroxysuccinimide (HOSu), dimethylaminopyridine (DMAP) and the like may be further added to the reaction system.

In the thus-obtained compound (I), an intramolecular functional group can also be converted to an object functional group by a combination of chemical reactions known per se. Examples of the chemical reaction include oxidation reaction, reduction reaction, alkylation reaction, acylation reaction, ureation reaction, hydrolysis reaction, amination reaction, esterification reaction, aryl coupling reaction, deprotection reaction and the like.

In the above-mentioned production method, when a starting compound has an amino group, a carboxyl group, a

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hydroxy group, a carbonyl group or a mercapto group as a substituent, a protecting group generally used in the peptide chemistry may be introduced into these groups, and the object compound can be obtained by removing the protecting group as necessary after the reaction.

Compound (I) obtained by the above-mentioned production method can be isolated and purified by a known means, such as solvent extraction, liquid conversion, phase transfer, crystallization, recrystallization, chromatography and the like.

When compound (I) contains optical isomer, stereoisomer, regio isomer and rotamer, these compounds are also included in compound (I), and each can be obtained as a single product by a synthesis method or a separation method known per se. For example, when an optical isomer exists in compound (I), an optical isomer resolved from the compound is also encompassed in compound (I).

Here, an optical isomer can be produced by a method known per se.

Compound (I) may be a crystal.

A crystal of compound (I) (hereinafter sometimes to be abbreviated as the crystal of the present invention) can be produced by crystallizing compound (I), by applying a crystallization method known per se.

In the present specification, the melting point means a melting point measured, for example, by micro melting point apparatus (Yanako, MP-500D or Buchi, B-545), DSC (differential scanning calorimetry analysis) apparatus (METTLER TOLEDO, DSC1) and the like.

Generally, the melting point sometimes varies depending on the measurement device, measurement condition and the like. The crystal in the present specification may be a crystal showing a melting point different from the values described in the present specification as long as the difference is within a general error range.

The crystal of the present invention is superior in the physicochemical properties (e.g., melting point, solubility, stability) and biological properties (e.g., pharmacokinetics (absorbability, distribution, metabolism, excretion), efficacy expression), and is extremely useful as a medicament.

Compound (I) may be used as a prodrug. A prodrug of the compound (I) means a compound which is converted to the compound (I) of the present invention with a reaction due to an enzyme, an gastric acid, etc. under the physiological condition in the living body, that is, a compound which is converted to the compound (I) of the present invention with oxidation, reduction, hydrolysis, etc. according to an enzyme; a compound which is converted to the compound (I) of the present invention by hydrolysis etc. due to gastric acid, etc.

A prodrug of compound (I) may be

a compound obtained by subjecting an amino group in compound (I) to an acylation, alkylation or phosphorylation (e.g., a compound obtained by subjecting an amino group in compound (I) to an eicosanoylation, alanylation, pentylaminocarbonylation, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methoxycarbonylation, tetrahydrofuranylation, pyrrolidylmethylation, pivaloyloxymethylation and tert-butylation, etc.);

a compound obtained by subjecting a hydroxy group in compound (I) to an acylation, alkylation, phosphorylation or boration (e.g., a compound obtained by subjecting an hydroxy group in compound (I) to an acetylation, palmitoylation, propanoylation, pivaloylation, succinylation, fumarylation, alanylation, dimethylaminomethylcarbonylation, etc.);

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a compound obtained by subjecting a carboxyl group in compound (I) to an esterification or amidation (e.g., a compound obtained by subjecting a carboxyl group in compound (I) to an ethyl esterification, phenyl esterification, carboxymethyl esterification, dimethylaminomethyl esterification, pivaloyloxymethyl esterification, ethoxycarbonyloxyethyl esterification, phthalidyl esterification, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methyl esterification, cyclohexyloxycarbonylethyl esterification and methylamidation, etc.)

and the like. Any of these compounds can be produced from compound (I) by a method known per se.

A prodrug for compound (I) may also be one which is converted into compound (I) under a physiological condition, such as those described in *IYAKUHIN no KAIHATSU* (Development of Pharmaceuticals), Vol. 7, Design of Molecules, p. 163-198, Published by HIROKAWA SHOTEN (1990).

In the present specification, a prodrug may form a salt, and as such salt, those exemplified as a salt of the compound represented by the above-mentioned formula (I) can be mentioned.

Compound (I) may be labeled with an isotope (e.g., ^3H , ^{13}C , ^{14}C , ^{18}F , ^{35}S , ^{125}I) and the like.

Compound (I) labeled with or substituted by an isotope can be used, for example, as a tracer used for Positron Emission Tomography (PET) (PET tracer), and is useful in the field of medical diagnosis and the like.

Furthermore, compound (I) may be a hydrate or a non-hydrate, or a non-solvate (e.g., anhydride), or a solvate (e.g., hydrate).

Compound (I) also encompasses a deuterium conversion form wherein ^1H is converted to $^2\text{H}(\text{D})$.

Furthermore, compound (I) may be a pharmaceutically acceptable cocrystal or cocrystal salt. The cocrystal or cocrystal salt means a crystalline substance constituted with two or more special solids at room temperature, each having different physical properties (e.g., structure, melting point, melting heat, hygroscopicity, solubility and stability). The cocrystal or cocrystal salt can be produced by a cocrystallization method known per se.

Compound (I) or a prodrug thereof (hereinafter sometimes to be simply abbreviated as the compound of the present invention) can be used as it is or in the form of a pharmaceutical composition (also referred to as a medicament) by mixing with a pharmacologically acceptable carrier etc. to mammals (e.g., human, mouse, rat, rabbit, dog, cat, bovine, horse, swine, monkey) as an agent for the prophylaxis or treatment of various diseases mentioned below.

As pharmacologically acceptable carriers, various organic or inorganic carrier substances conventionally used as preparation materials can be used. These are incorporated as excipient, lubricant, binder and disintegrant for solid preparations; or solvent, solubilizing agent, suspending agent, isotonicity agent, buffer and soothing agent for liquid preparations; and the like; and preparation additives such as preservative, antioxidant, colorant, sweetening agent and the like can be added as necessary.

Preferable examples of the excipient include lactose, sucrose, D-mannitol, D-sorbitol, starch, gelatinated starch, dextrin, crystalline cellulose, low-substituted hydroxypropylcellulose, sodium carboxymethylcellulose, gum arabic, pullulan, light anhydrous silicic acid, synthetic aluminum silicate and magnesium alumino metasilicate.

Preferable examples of the lubricant include magnesium stearate, calcium stearate, talc and colloidal silica.

Preferable examples of the binder include gelatinated starch, sucrose, gelatin, gum arabic, methylcellulose, carboxymethylcellulose, sodium carboxymethylcellulose, crystalline cellulose, sucrose, D-mannitol, trehalose, dextrin, pullulan, hydroxypropylcellulose, hydroxypropylmethylcellulose and polyvinylpyrrolidone.

Preferable examples of the disintegrant include lactose, sucrose, starch, carboxymethylcellulose, calcium carboxymethylcellulose, croscarmellose sodium, sodium carboxymethyl starch, light anhydrous silicic acid and low-substituted hydroxypropylcellulose.

Preferable examples of the solvent include water for injection, physiological brine, Ringer's solution, alcohol, propylene glycol, polyethylene glycol, sesame oil, corn oil, olive oil and cottonseed oil.

Preferable examples of the solubilizing agent include polyethylene glycol, propylene glycol, D-mannitol, trehalose, benzyl benzoate, ethanol, trisaminomethane, cholesterol, triethanolamine, sodium carbonate, sodium citrate, sodium salicylate and sodium acetate.

Preferable examples of the suspending agent include surfactants such as stearyltriethanolamine, sodium lauryl sulfate, lauryl aminopropionate, lecithin, benzalkonium chloride, benzethonium chloride, glycerol monostearate and the like; hydrophilic polymers such as poly(vinyl alcohol), polyvinylpyrrolidone, carboxymethylcellulose sodium, methylcellulose, hydroxymethylcellulose, hydroxyethylcellulose, hydroxypropylcellulose and the like, polysorbates; and polyoxyethylene hydrogenated castor oil.

Preferable examples of the isotonicity agent include sodium chloride, glycerol, D-mannitol, D-sorbitol and glucose.

Preferable examples of the buffer include buffers of phosphate, acetate, carbonate, citrate etc.

Preferable examples of the soothing agent include benzyl alcohol.

Preferable examples of the preservative include p-oxybenzoate esters, chlorobutanol, benzyl alcohol, phenethyl alcohol, dehydroacetic acid and sorbic acid.

Preferable examples of the antioxidant include sulfite salts and ascorbate salts.

Preferable examples of the colorant include aqueous food tar colors (e.g., food colors such as Food Color Red Nos. 2 and 3, Food Color Yellow Nos. 4 and 5, Food Color Blue Nos. 1 and 2 and the like food colors), water insoluble lake dyes (e.g., aluminum salt of the above-mentioned aqueous food tar color), natural dyes (e.g., β -carotene, chlorophyll, red iron oxide) and the like.

Preferable examples of the sweetening agent include saccharin sodium, dipotassium glycyrrhizinate, aspartame and stevia.

Examples of the dosage form of the above-mentioned pharmaceutical composition include oral preparations such as tablet (including sugar-coated tablet, film-coated tablet, sublingual tablet, orally disintegrating tablet, buccal tablet), capsule (including soft capsule, microcapsule), pill, granule, powder, troche, syrup, liquid, emulsion, suspension, aerosol, films (e.g., orally disintegrable films, oral mucosa-adhesive film) and the like; and parenteral agents such as injection (e.g., subcutaneous injection, intravenous injection, intramuscular injection, intraperitoneal injection, drip infusion), external preparation (e.g., transdermal absorption type preparation, ointment, lotion, adhesive preparation), suppository (e.g., rectal suppository, vaginal suppository), pellet, nasal preparation, pulmonary preparation (inhalant), eye drop and the like. The compound and medicament of the present invention can be respectively safely administered

orally or parenterally (e.g., intrarectal, intravenous, intraarterial, intramuscular, subcutaneous, intraorgan, intranasal, intradermal, instillation, intracerebral, intravaginal, intraperitoneal, intratumoral, proximal tumor administrations, and administration to the lesion).

These preparations may be a release control preparation (e.g., sustained-release microcapsule) such as an immediate-release preparation, a sustained-release preparation and the like.

The pharmaceutical composition can be produced according to a method conventionally used in the field of pharmaceutical formulation, for example, the method described in the Japanese Pharmacopoeia, and the like.

While the content of the compound of the present invention in the pharmaceutical composition of the present invention varies depending on the dosage form, dose of the compound of the present invention and the like, it is, for example, about 0.1 to 100 wt %.

When an oral preparation is produced, coating may be applied where necessary for the purpose of taste masking, enteric solubility or sustainability.

Examples of the coating base used for coating include sugar coating base, water-soluble film coating base, enteric film coating base, and sustained-release film coating base.

As the sugar coating base, sucrose is used, and one or more kinds selected from talc, and the precipitated calcium carbonate, gelatin, gum arabic, pullulan, carnauba wax and the like may be further used in combination.

Examples of the water-soluble film coating base include cellulose polymers such as hydroxypropylcellulose, hydroxypropylmethylcellulose, hydroxyethylcellulose, methylhydroxyethylcellulose and the like; synthetic polymers such as polyvinyl acetal diethylaminoacetate, aminoalkylmethacrylate copolymer E [Eudragit E (trade name)], polyvinylpyrrolidone and the like; and polysaccharides such as pullulan and the like.

Examples of the enteric film coating base include cellulose polymers such as hydroxypropylmethylcellulose phthalate, hydroxypropylmethylcellulose acetate succinate, carboxymethylcellulose, cellulose acetate phthalate and the like; acrylic acid polymers such as methacrylic acid copolymer L [Eudragit L (trade name)], methacrylic acid copolymer LD [Eudragit L-30D-55 (trade name)], methacrylic acid copolymer S [Eudragit S (trade name)] and the like; and naturally-occurring substances such as shellac and the like.

Examples of the sustained-release film coating base include cellulose polymers such as ethylcellulose and the like; and acrylic acid polymers such as aminoalkylmethacrylate copolymer RS [Eudragit RS (trade name)], ethyl acrylate-methyl methacrylate copolymer suspension [Eudragit NE (trade name)] and the like.

Two or more kinds of the above-mentioned coating bases may be used in a mixture at an appropriate ratio. In addition, for example, light shielding agents such as titanium oxide, red ferric oxide and the like may also be used during coating.

Since the compound of the present invention shows low toxicity (e.g., acute toxicity, chronic toxicity, genetic toxicity, reproductive toxicity, cardiotoxicity, carcinogenicity) and less side effects, it can be used as a prophylactic or therapeutic agent, or diagnostic agent for various diseases in mammals (e.g., human, bovine, horse, dog, cat, monkey, mouse, rat).

Moreover, the compound of the present invention is expected to be superior in central migration.

The compound of the present invention has an excellent an orexin type 2 receptor agonist activity, and may treat,

prevent or ameliorate the risk of various neurological and psychiatric diseases associated with an orexin type 2 receptor. The compound of the present invention is useful as an agent for the prophylaxis or treatment of various diseases such as narcolepsy, idiopathic hypersomnia, hypersomnia, sleep apnea syndrome, narcolepsy syndrome accompanied by narcolepsy-like symptoms, hypersomnia syndrome accompanied by daytime hypersomnia (e.g., Kleine Levin syndrome, major depression with hypersomnia, Lewy body dementia, Parkinson's disease, progressive supranuclear paralysis, Prader-Willi syndrome, Moebius syndrome, hypoventilation syndrome, Niemann-Pick disease type C, brain contusion, cerebral infarction, brain tumor, muscular dystrophy, multiple sclerosis, acute disseminated encephalomyelitis, Guillain-Barre syndrome, Rasmussen's encephalitis, Wernicke's encephalitis, limbic encephalitis, Hashimoto's encephalopathy), coma, loss of consciousness, obesity (e.g., malignant mastocytosis, exogenous obesity, hyperinsulinar obesity, hyperplasmic obesity, hypop hyseal adiposity, hypoplasmic obesity, hypothyroid obesity, hypothalamic obesity, symptomatic obesity, infantile obesity, upper body obesity, alimentary obesity, hypogonadal obesity, systemic mastocytosis, simple obesity, central obesity), insulin resistance syndrome, Alzheimer's disease, disturbance of consciousness such as coma and the like, side effects and complications due to anesthesia, sleep disturbance, sleep problem, insomnia, Intermittent sleep, nocturnal myoclonus, REM sleep interruption, jet lag, jet lag syndrome, sleep disorder of alternating worker, sleep disorder, night terror, depression, major depression, sleepwalking disease, enuresis, sleep disorder, Alzheimer's dusk, diseases associated with circadian rhythm, fibromyalgia, condition arising from decline in the quality of sleep, overeating, obsessive compulsive eating disorder, obesity-related disease, hypertension, diabetes, elevated plasma insulin concentration and insulin resistance, hyperlipidemia, hyperlipemia, endometrial cancer, breast cancer, prostate cancer, colorectal cancer, cancer, osteoarthritis, obstructive sleep apnea, cholelithiasis, gallstones, cardiac disease, abnormal heartbeat, arrhythmia, myocardial infarction, congestive cardiac failure, cardiac failure, coronary heart disease, cardiovascular disorder, sudden death, polycystic ovarian disease, craniopharyngioma, Froelich's syndrome, growth hormone deficient, normal mutant short stature, Turner's syndrome, children suffering from acute lymphoblastic leukemia, syndrome X, reproductive hormone abnormality, declining fertility, infertility, male gonadal function decline, sexual and reproductive dysfunction such as female male hirsutism, fetal defects associated with pregnant women obesity, gastrointestinal motility disorders such as obesity-related gastroesophageal reflux, obesity hypoventilation syndrome (Pickwick syndrome), respiratory diseases such as dyspnea, inflammation such as systemic inflammation of the vascular system, arteriosclerosis, hypercholesterolemia, hyperuricemia, lower back pain, gall bladder disease, gout, kidney cancer, risk of secondary outcomes of obesity such as lowering the risk of left ventricular hypertrophy, migraine pain, headache, neuropathic pain, Parkinson's disease, psychosis, schizophrenia, facial flushing, night sweats, diseases of the genital/urinary system, diseases related to sexual function or fertility, dysthymic disorder, bipolar disorder, bipolar I disorder, bipolar II disorder, cyclothymic disorder, acute stress disorder, agoraphobia, generalized anxiety disorder, obsessive disorder, panic attack, panic disorder, posttraumatic stress disorder, separation anxiety disorder, social phobia, anxiety disorder, acute neurological and psychiatric disorders such as cardiac bypass surgery and post-transplant cerebral defi-

cit, stroke, ischemic stroke, cerebral ischemia, spinal cord trauma, head trauma, perinatal hypoxia, cardiac arrest, hypoglycemic nerve injury, Huntington's disease, amyotrophic lateral sclerosis, multiple sclerosis, eye damage, retinopathy, cognitive impairment, muscle spasm, tremor, epilepsy, disorders associated with muscle spasticity, delirium, amnesic disorder, age-related cognitive decline, schizoaffective disorder, delusional disorder, drug addiction, dyskinesia, chronic fatigue syndrome, fatigue, medication-induced Parkinsonism syndrome, Jilt-do La Tourette's syndrome, chorea, myoclonus, tic, restless legs syndrome, dystonia, dyskinesia, attention deficit hyperactivity disorder (ADHD), behavior disorder, urinary incontinence, withdrawal symptoms, trigeminal neuralgia, hearing loss, tinnitus, nerve damage, retinopathy, macular degeneration, vomiting, cerebral edema, pain, bone pain, arthralgia, toothache, cataplexy, and traumatic brain injury (TBI).

Particularly, the compound of the present invention is useful as an agent for the prophylaxis or treatment of narcolepsy, idiopathic hypersomnia, hypersomnia, sleep apnea syndrome, narcolepsy syndrome accompanied by narcolepsy-like symptoms, hypersomnia syndrome accompanied by daytime hypersomnia (e.g., Parkinson's disease, Guillain-Barre syndrome and Kleine Levin syndrome), Alzheimer's disease, obesity, insulin resistance syndrome, cardiac failure, diseases related to bone loss, sepsis, disturbance of consciousness such as coma and the like, side effects and complications due to anesthesia, and the like, or anesthetic antagonist.

While the dose of the compound of the present invention varies depending on the subject of administration, administration route, target disease, symptom and the like, for example, when the compound of the present invention is administered orally or parenterally to an adult patient, its dose is for example, about 0.01 to 100 mg/kg body weight per dose, preferably 0.1 to 50 mg/kg body weight per dose and more preferably 0.5 to 20 mg/kg body weight per dose. This amount is desirably administered in one to 3 portions daily.

The compound of the present invention can be used in combination with other drugs (hereinafter to be abbreviated as concomitant drug).

By combining the compound of the present invention and a concomitant drug, a superior effect, for example,

- (1) the dose can be reduced as compared to single administration of the compound of the present invention or a concomitant drug,
- (2) the drug to be combined with the compound of the present invention can be selected according to the condition of patients (mild case, severe case and the like),
- (3) the period of treatment can be set longer by selecting a concomitant drug having different action and mechanism from the compound of the present invention,
- (4) a sustained treatment effect can be designed by selecting a concomitant drug having different action and mechanism from the compound of the present invention,
- (5) a synergistic effect can be afforded by a combined use of the compound of the present invention and a concomitant drug, and the like, can be achieved.

In the present specification, the compound of the present invention and a concomitant drug used in combination are referred to as the "combination agent of the present invention".

When using the combination agent of the present invention, the administration time of the compound of the present

invention and the concomitant drug is not restricted, and the compound of the present invention or a pharmaceutical composition thereof, or the concomitant drug or a pharmaceutical composition thereof can be administered to an administration subject simultaneously, or may be administered at different times. The dosage of the concomitant drug may be determined according to the dose clinically used, and can be appropriately selected depending on an administration subject, administration route, disease, combination and the like.

The administration mode of the combination agent of the present invention and the concomitant drug is not particularly limited, and the compound of the present invention and the concomitant drug only need to be combined on administration. Examples of such administration mode include the following: (1) administration of a single preparation obtained by simultaneously processing the compound of the present invention and the concomitant drug, (2) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route, (3) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route in a staggered manner, (4) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes, (5) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes in a staggered manner (e.g., administration in the order of the compound of the present invention and the concomitant drug, or in the reverse order) and the like.

The dose of the concomitant drug can be appropriately determined based on the dose employed in clinical situations. The mixing ratio of the compound of the present invention and a concomitant drug can be appropriately determined depending on the administration subject, administration route, target disease, symptom, combination and the like.

For example, the content of the compound of the present invention in the combination agent of the present invention differs depending on the form of a preparation, and usually from about 0.01 to about 100 wt %, preferably from about 0.1 to about 50 wt %, further preferably from about 0.5 to about 20 wt %, based on the whole preparation.

The content of the concomitant drug in the combination agent of the present invention differs depending on the form of a preparation, and usually from about 0.01 to about 100 wt %, preferably from about 0.1 to about 50 wt %, further preferably from about 0.5 to about 20 wt %, based on the whole preparation.

The content of additives such as a carrier and the like in the combination agent of the present invention differs depending on the form of a preparation, and usually from about 1 to about 99.99 wt %, preferably from about 10 to about 90 wt %, based on the preparation.

Similar contents may be employed even when the compound of the present invention and a concomitant drug are separately formulated into preparations.

Examples of the concomitant drug include the followings. A therapeutic drug for narcolepsy (e.g., methylphenidate, amphetamine, pemoline, phenelzine, protriptyline, sodium oxybate, modafinil, caffeine), antiobesity drug (amphetamine, benzphetamine, bromocriptine, bupropion, diethylpro-

pion, exenatide, fenfluramine, liothyronine, liraglutide, mazindol, methamphetamine, octreotide, orlistat, phendimetrazine, phendimetrazine, phenmetrazine, phentermine, Qnexa (registered trade mark), phenylpropanolamine, pramlintide, propylhexedrine, recombinant leptin, sibutramine, topiramate, zimeclidine, zonisamide, Lorca-serin, metformin), acetylcholine esterase inhibitor (e.g., donepezil, rivastigmine, galanthamine, zanzepil, idebenone, tacrine), antidementia agent (e.g., memantine), inhibitor of β amyloid protein production, secretion, accumulation, aggregation and/or deposition, β secretase inhibitor (e.g., 6-(4-biphenyl)methoxy-2-[2-(N,N-dimethylamino)ethyl]tetralin, 6-(4-biphenyl)methoxy-2-(N,N-dimethylamino)methyltetralin, 6-(4-biphenyl)methoxy-2-(N,N-dipropylamino)methyltetralin, 2-(N,N-dimethylamino)methyl-6-(4'-methoxybiphenyl-4-yl)methoxytetralin, 6-(4-biphenyl)methoxy-2-[2-(N,N-diethylamino)ethyl]tetralin, 2-[2-(N,N-dimethylamino)ethyl]-6-(4'-methylbiphenyl-4-yl)methoxytetralin, 2-[2-(N,N-dimethylamino)ethyl]-6-(4'-methoxybiphenyl-4-yl)methoxytetralin, 6-(2',4'-dimethoxybiphenyl-4-yl)methoxy-2-[2-(N,N-dimethylamino)ethyl]tetralin, 6-[4-(1,3-benzodioxol-5-yl)phenyl]methoxy-2-[2-(N,N-dimethylamino)ethyl]tetralin, 6-(3',4'-dimethoxybiphenyl-4-yl)methoxy-2-[2-(N,N-dimethylamino)ethyl]tetralin, an optically active form thereof, a salt thereof and a hydrate thereof, OM99-2 (WO01/00663)), Y secretase inhibitor, β amyloid protein aggregation inhibitor (e.g., PTI-00703, ALZHEMED (NC-531), PPI-368 (National Publication of International Patent Application No. 11-514333), PPI-558 (National Publication of International Patent Application No. 2001-500852), SKF-74652 (Biochem. J. (1999), 340(1), 283-289)), β amyloid vaccine, β amyloid-degrading enzyme and the like, brain function enhancer (e.g., aniracetam, nicergoline), therapeutic drug for Parkinson's disease [(e.g., dopamine receptor agonist (e.g., L-DOPA, bromocriptine, pergolide, talipexole, pramipexole, cabergoline, amantadine), monoamine oxidase enzyme (MAO) inhibitor (e.g., deprenyl, selegiline, remacemide, riluzole), anticholinergic agent (e.g., trihexyphenidyl, biperiden), COMT inhibitor (e.g., entacapone)], therapeutic drug for amyotrophic lateral sclerosis (e.g., riluzole etc., neurotrophic factor), therapeutic drug for abnormal behavior accompanying progress of dementia, wandering and the like (e.g., sedative, anti-anxiety drug), apoptosis inhibitor (e.g., CPI-1189, IDN-6556, CEP-1347), neuronal differentiation-regenerate promoter (e.g., leteprinin, xaliproden; SR-57746-A), SB-216763, Y-128, VX-853, prosaptide, 5,6-dimethoxy-2-[2,2,4,6,7-pentamethyl-3-(4-methylphenyl)-2,3-dihydro-1-benzofuran-5-yl]isoindoline, 5,6-dimethoxy-2-[3-(4-isopropylphenyl)-2,2,4,6,7-pentamethyl-2,3-dihydro-1-benzofuran-5-yl]isoindoline, 6-[3-(4-isopropylphenyl)-2,2,4,6,7-pentamethyl-2,3-dihydro-1-benzofuran-5-yl]-6,7-dihydro-5H-[1,3]dioxolo[4,5-f]isoindole and an optically active form, salt or hydrate thereof), non-steroidal antiinflammatory agents (meloxicam, tenoxicam, indomethacin, ibuprofen, celecoxib, rofecoxib, aspirin, indomethacin etc.), steroid drug (dexamethasone, hexestrol, cortisone acetate etc.), disease-modifying anti-rheumatic drug (DMARDs), anti-cytokine drug (e.g., TNF inhibitor, MAP kinase inhibitor), therapeutic agent for incontinence, frequent urination (e.g., flavoxate hydrochloride, oxybutynin hydrochloride, propiverine hydrochloride), phosphodiesterase inhibitor (e.g., sildenafil (citrate)), dopamine agonist (e.g., apomorphine), antiarrhythmic drugs (e.g., mexiletine), sex hormone or a derivative thereof (e.g., progesterone, estradiol, estradiol benzoate), therapeutic agent for osteoporosis (e.g., alfacal-

cidol, calcitriol, elcatonin, calcitonin salmon, estriol, ipriflavone, pamidronate disodium, alendronate sodium hydrate, incadronate disodium), parathyroid hormone (PTH), calcium receptor antagonists, therapeutic drug for insomnia (e.g., benzodiazepines medicament, non-benzodiazepines medicament, melatonin agonist, orexin receptor antagonists), therapeutic drug for schizophrenia (e.g., typical antipsychotic agents such as haloperidol and the like; atypical antipsychotic agents such as clozapine, olanzapine, risperidone, aripiprazole and the like; medicament acting on metabotropic glutamate receptor or ion channel conjugated-type glutamate receptor; phosphodiesterase inhibitor), benzodiazepines medicament (chlordiazepoxide, diazepam, potassium clorazepate, lorazepam, clonazepam, alprazolam etc.), L-type calcium channel inhibitor (pregabalin etc.), tricyclic or tetracyclic antidepressant (imipramine hydrochloride, amitriptyline hydrochloride, desipramine hydrochloride, clomipramine hydrochloride etc.), selective serotonin reuptake inhibitor (fluvoxamine maleate, fluoxetine hydrochloride, citalopram hydrobromide, sertraline hydrochloride, paroxetine hydrochloride, escitalopram oxalate etc.), serotonin-noradrenaline reuptake inhibitor (venlafaxine hydrochloride, duloxetine hydrochloride, desvenlafaxine hydrochloride etc.), noradrenaline reuptake inhibitor (reboxetine mesylate etc.), mirtazapine, trazodone hydrochloride, nefazodone hydrochloride, bupropion hydrochloride, setiptiline maleate, 5-HT_{1A} agonist, (buspirone hydrochloride, tandospirone citrate, osetmozotan hydrochloride etc.), 5-HT_{2A} antagonist, 5-HT_{2A} inverse agonist, 5-HT₃ antagonist (cyamemazine etc.), heart non-selective β inhibitor (propranolol hydrochloride, oxprenolol hydrochloride etc.), histamine H₁ antagonist (hydroxyzine hydrochloride etc.), CRF antagonist, other antianxiety drug (meprobamate etc.), tachykinin antagonist (MK-869, saredutant etc.), medicament that acts on metabotropic glutamate receptor, CCK antagonist, β_3 adrenaline antagonist (amibegron hydrochloride etc.), GAT-1 inhibitor (tiagabine hydrochloride etc.), N-type calcium channel inhibitor, carbonic anhydrase II inhibitor, NMDA glycine moiety agonist, NMDA antagonist (memantine etc.), peripheral benzodiazepine receptor agonist, vasopressin antagonist, vasopressin V1b antagonist, vasopressin V1a antagonist, phosphodiesterase inhibitor, opioid antagonist, opioid agonist, uridine, nicotinic acid receptor agonist, thyroid hormone (T3, T4), TSH, TRH, MAO inhibitor (phenelzine sulfate, tranylcypromine sulfate, moclobemide etc.), therapeutic drug for bipolar disorder (lithium carbonate, sodium valproate, lamotrigine, riluzole, felbamate etc.), cannabinoid CB1 antagonist (rimonabant etc.), FAAH inhibitor, sodium channel inhibitor, anti-ADHD drug (methylphenidate hydrochloride, methamphetamine hydrochloride etc.), therapeutic drug for alcoholism, therapeutic drug for autism, therapeutic drug for chronic fatigue syndrome, therapeutic drug for spasm, therapeutic drug for fibromyalgia syndrome, therapeutic drug for headache, therapeutic drug for quitting smoking, therapeutic drug for myasthenia gravis, therapeutic drug for cerebral infarction, therapeutic drug for mania, therapeutic drug for hypersomnia, therapeutic drug for pain, therapeutic drug for dysthymia, therapeutic drug for autonomic ataxia, therapeutic drug for male and female sexual dysfunction, therapeutic drug for migraine, therapeutic drug for pathological gambler, therapeutic drug for restless legs syndrome, therapeutic drug for substance addiction, therapeutic drug for alcohol-related syndrome, therapeutic drug for irritable bowel syndrome, therapeutic drug for lipid abnormality such as cholesterol-lowering drug (statin series (pravastatin sodium, atorvastatin, simvastatin, rosuvastatin etc.), fibrate (clofibrate etc.),

squalene synthetase inhibitor), therapeutic drug for abnormal behavior or suppressant of dromomania due to dementia (sedatives, antianxiety drug etc.), therapeutic drug for diabetes, therapeutic agent for diabetic complications, therapeutic drug for hypertension, therapeutic drug for hypotension, diuretic, chemotherapeutic agent, immunotherapeutic agent, antithrombotic agent, anti-cancer agent and the like.

Two or more kinds of the above-mentioned concomitant drug may be used in a mixture at an appropriate ratio.

When the compound of the present invention is applied to each of the above-mentioned diseases, it can also be used in combination with biologics (e.g., antibody drug, nucleic acid or nucleic acid derivative, aptamer drug, vaccine preparation), or can be used in combination with a gene therapy method and the like, or can also be used in combination with a treatment in psychiatric field without using drugs.

Examples of the antibody drug and vaccine preparation include vaccine preparation against angiotensin II, vaccine preparation against CETP, CETP antibody, antibody against TNF α antibody and other cytokines, amyloid β vaccine preparation, vaccine for type 1 diabetes (e.g., DIAPEP-277 of Peptor), anti-HIV antibody and HIV vaccine preparation, as well as antibodies or vaccine preparations against cytokines, renin-angiotensin type enzymes and products thereof, antibodies or vaccine preparations against enzymes or proteins involved in blood lipid metabolism, antibodies or vaccines relating to enzymes and proteins involved in blood coagulation or fibrinolysis system, antibodies or vaccine preparations against proteins involved in sugar metabolism and insulin resistance, and the like. In addition, it can be used in combination with biologics relating to growth factors such as GH, IGF and the like.

Examples of the gene therapy method include a treatment method using gene relating to cytokine, renin-angiotensin type enzyme and product thereof, G protein, G protein conjugated receptor and phosphorylating enzyme thereof, a treatment method using a DNA decoy such as NF κ B decoy and the like, a treatment method using antisense, a treatment method using a gene relating to an enzyme or protein involved in blood lipid metabolism (e.g., a gene relating to metabolism, excretion and absorption of cholesterol or triglyceride or HDL-cholesterol or blood phospholipid), a treatment method using a gene relating to an enzyme or protein involved in angiogenesis therapy for peripheral vascular obstruction and the like (e.g., growth factors such as HGF, VEGF etc.), a treatment method using a gene relating to a protein involved in glucose metabolism and insulin resistance, antisense against cytokines such as TNF etc., and the like.

Examples of the treatment method in the psychiatric field without using drug include modified electroconvulsive therapy, deep brain stimulation therapy, repetitive transcranial magnetic stimulation therapy, psychotherapy including cognitive behavioral therapy and the like.

The compound of the present invention can also be used in combination with various organ regeneration methods such as cardiac regeneration, renal regeneration, pancreatic regeneration, revascularization and the like, cell transplantation therapy utilizing bone marrow cells (bone marrow-derived mononuclear cell, myelogenic stem cell), or artificial organ utilizing tissue engineering (e.g., artificial blood vessel, cardiomyocyte sheet).

EXAMPLES

The present invention is explained in detail in the following by referring to Examples, Experimental Examples and

Formulation Examples. However, the examples do not limit the present invention and the examples can be modified within the scope of the present invention.

The "room temperature" in the following Examples is generally about 10° C. to about 35° C. The ratio for mixed solvent is, unless otherwise specified, a volume mixing ratio and % means wt % unless otherwise specified.

The elution by column chromatography in the Examples was performed under the observation by TLC (Thin Layer Chromatography) unless otherwise specified. In the observation by TLC, 60 F₂₅₄ manufactured by Merck was used as a TLC plate, the solvent used as an elution solvent in column chromatography was used as an eluent, and UV detector was used for the detection. In silica gel column chromatography, the indication of NH means use of aminopropylsilane-bonded silica gel and the indication of Diol means use of 3-(2,3-dihydroxypropoxy)propylsilane-bonded silica gel. In preparative HPLC (high performance liquid chromatography), the indication of C18 means use of octadecyl-bonded silica gel. The ratio for elution solvent is, unless otherwise specified, a volume mixing ratio.

For the analysis of ¹H NMR, ACD/SpecManager (trade name) software and the like were used. Peaks of a hydroxyl group, an amino group and the like, having very mild proton peak, are not sometimes described.

MS was measured by LC/MS. As the ionization method, ESI method, or APCI method was used. The data indicates actual measured value (found). While molecular ion peak is generally observed, a fragment ion is sometimes observed. In the case of a salt, a molecular ion peak or fragment ion peak of free form is generally observed.

The unit of sample concentration (c) for optical rotation ([α]_D) is g/100 mL.

Elemental analysis value (Anal.) is described as calculated value (Calcd) and actual measured value (Found).

The retention time in the Examples was measured by liquid chromatograph method. The measurement conditions are as follows, unless otherwise specified.

column: YMC PackPro C18 2.0 mm i.d.×75 mm (3 μm) mobile phase: the solution prepared by adding 0.04 M Britton-Robinson buffer solution (pH 6.5) to methanol (5:2), and mixing them well, and adjusting the pH to 7.4 with 0.2 M sodium hydroxide solution.

Peaks by powder X-ray diffraction in the Examples mean peaks measured at room temperature by Ultima IV (Rigaku Corporation, Japan) using Cu Kα radiation as a radiation source. The measurement conditions are as follows.

Electric pressure/Electric current: 40 kV/50 mA

Scan speed: 6 degrees/min

Scan range of 2 Theta: 2-35 degrees

The crystallinity by powder X-ray diffraction in the Examples was calculated by the Hermans method.

In the following Examples, the following abbreviations are used.

mp: melting point

MS: mass spectrum

M: mol concentration

N: normality

CDCl₃: deuteriochloroform

DMSO-d₆: deuterodimethyl sulfoxide

¹H NMR: proton nuclear magnetic resonance

LC/MS: liquid chromatograph mass spectrometer

ESI: Electrospray Ionization

APCI: Atmospheric Pressure Chemical Ionization

HATU: (dimethylamino)-N,N-dimethyl 3H-[1,2,3]-triazolo[4,5-b]pyridin-3-yloxy)methaneiminium hexafluorophosphate

PPh₃: triphenylphosphine

TFA: trifluoroacetic acid

DMAP: N,N-dimethyl-4-aminopyridine

CPME: cyclopentyl methyl ether

WSC: N-(3-(dimethylamino)propyl)-N'-ethylcarbodiimide

IPE: 2-isopropoxypropane

DIPEA: N-ethyl-N-isopropylpropan-2-amine

DMF: N,N-dimethylformamide

HOBt: 1H-benzotriazol-1-ol

THF: tetrahydrofuran

DME: 1,2-dimethoxyethane

MeOH: methanol

WSC·HCl: N-(3-(dimethylamino)propyl)-N'-ethylcarbodiimide hydrochloride (1:1)

EtOH: ethanol

Boc₂O: di-tert-butyl dicarbonate

TEA: triethylamine

Et₂O: ethoxyethane

EtOAc: ethyl acetate

CH₃CN: acetonitrile

TMSCl: trimethylsilyl chloride

XPhos Pd G3: methanesulfonato(2-dicyclohexylphosphino-2',4',6'-triisopropyl-1,1'-biphenyl)(2'-amino-1,1'-biphenyl-2-yl)palladium(II)

PdCl₂(dppf): [1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloride

Example 1

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4'-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

A) 3-chloro-2-fluorophenylalanine Hydrochloride

To a mixture of 1-(bromomethyl)-3-chloro-2-fluorobenzene (55.0 g), diethyl acetamidomalonate (56.1 g) and EtOH (400 mL) was added 20% sodium ethoxide ethanol solution (88 g) at 0° C. The mixture was refluxed for 2 hr 30 min, and cooled to room temperature. The impurity was removed by filtration, the filtrate was concentrated under reduced pressure, and a mixture of the residue and 6 M hydrochloric acid (500 mL) was refluxed for 15 hr. The reaction solution was concentrated under reduced pressure, and the obtained residue was washed with isopropanol/diisopropyl ether to give the title compound (67.1 g).

MS: [M+H]⁺ 217.8.

B) N-(tert-butoxycarbonyl)-3-chloro-2-fluorophenylalanine

To a mixture of 3-chloro-2-fluorophenylalanine hydrochloride (67.1 g), 1 M aqueous sodium hydroxide solution (528 mL) and DME (480 mL) was added Boc₂O (63.4 g) at room temperature. The mixture was stirred at room temperature for 2 hr 30 min, and poured into ice water. The mixture was basified with 1 M aqueous sodium hydroxide solution, and the aqueous layer was washed with diethyl ether. The aqueous layer was acidified with 1 M hydrochloric acid, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under

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reduced pressure. The obtained solid was washed with diisopropyl ether/hexane to give the title compound (63.8 g).
MS: $[M-H]^-$ 316.0.

C) tert-butyl {3-(3-chloro-2-fluorophenyl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl} carbamate

To a mixture of N-(tert-butoxycarbonyl)-3-chloro-2-fluorophenylalanine (63.8 g), N-methoxymethanamine hydrochloride (21.5 g), HOBt (29.8 g), TEA (44.7 g) and DMF (425 mL) was added WSC-HCl (46.2 g) at 0° C. The mixture was stirred at room temperature for 15 hr, and the reaction mixture was added to aqueous sodium hydrogen carbonate solution, and extracted with EtOAc/THF. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The obtained solid was washed with diisopropyl ether/hexane to give the title compound (70.0 g).
MS, found: 260.9.

D) tert-butyl {3-(3-chloro-2-fluorophenyl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl}(4-methoxybenzyl)carbamate

To a mixture of tert-butyl {3-(3-chloro-2-fluorophenyl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl} carbamate (70.0 g) and DMF (390 mL) was added 60% sodium hydride (10.1 g) at 0° C. The mixture was stirred at 0° C. for 5 min, and then at room temperature for 10 min, and to the reaction mixture were added 1-(chloromethyl)-4-methoxybenzene (60.7 g) and tetrabutylammonium iodide (7.16 g) at 0° C. The mixture was stirred at room temperature for 2 hr 30 min, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (90.7 g).

MS: $[M+H]^+$ 481.1.

E) tert-butyl [1-(3-chloro-2-fluorophenyl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate

To a mixture of tert-butyl {3-(3-chloro-2-fluorophenyl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl}(4-methoxybenzyl)carbamate (90.7 g) and Et₂O (500 mL) was added lithium aluminium hydride (9.30 g) by small and small at -78° C. The mixture was stirred at -15° C. for 1 hr 30 min, and to the reaction mixture were added dropwise sequentially EtOAc and 10% aqueous potassium hydrogensulfite solution at -78° C. The mixture was stirred at room temperature for 15 min, and to the mixture was added water. The insoluble substance was removed by filtration through Celite, and the filtrate was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (74.7 g).

MS: $[M-H]^-$ 420.0.

F) ethyl 4-{(tert-butoxycarbonyl)[(4-methoxyphenyl)methyl]amino}-5-(3-chloro-2-fluorophenyl)-2,4,5-trideoxy-2,2-difluoropentionate

To a mixture of zinc (37.0 g) and THF (500 mL) was added TMSCl (3.85 g) under argon atmosphere at room

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temperature. The mixture was stirred at room temperature for 15 min, and ethyl bromo (difluoro)acetate (71.9 g) was added dropwise to the mixture with vigorously stirring at room temperature while keeping the internal temperature of about 50° C. The mixture was stirred at room temperature for 15 min, and to the mixture was added dropwise a mixture of tert-butyl [1-(3-chloro-2-fluorophenyl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate (74.7 g) and THF (100 mL), while keeping the internal temperature of about 40° C. The mixture was stirred at room temperature for 2 hr, and added to 5% aqueous potassium hydrogensulfite solution under ice cooling, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (61.6 g).

MS, found: 490.0.

G) 5-[(3-chloro-2-fluorophenyl)methyl]-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one

To a mixture of ethyl 4-{(tert-butoxycarbonyl)[(4-methoxyphenyl)methyl]amino}-5-(3-chloro-2-fluorophenyl)-2,4,5-trideoxy-2,2-difluoropentionate (61.6 g) and EtOH (160 mL) was added 4 M hydrogen chloride CPME solution (282 mL) at room temperature. The mixture was stirred at room temperature for 2 hr, and the reaction solution was concentrated. To the obtained residue were added EtOH (360 mL) and DIPEA (43.8 g). The mixture was stirred at 70° C. for 1 hr 30 min, and the reaction mixture was poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane), and the obtained solid was washed with diisopropyl ether/hexane to give the title compound (36.1 g).

MS: $[M+H]^+$ 399.9.

H) 5-[(3-chloro-2-fluorophenyl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one

To a mixture of 5-[(3-chloro-2-fluorophenyl)methyl]-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one (36.1 g), CH₃CN (315 mL) and water (105 mL) was added ammonium hexanitratocerate (IV) (99 g) at room temperature. The mixture was stirred at room temperature for 4 hr, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane), and then silica gel column chromatography (NH₄ methanol/EtOAc) to give the title compound (22.7 g).

MS: $[M-H]^-$ 278.0.

I) tert-butyl 2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate

To a mixture of 5-[(3-chloro-2-fluorophenyl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one (22.7 g) and THF (350 mL) was added dropwise 1 M borane-THF complex THF solution (284 mL) at room temperature. The mixture was slowly warmed to 60° C., and stirred for 4 hr. Water was added dropwise to the mixture at 0° C., and the mixture was

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stirred at room temperature for 10 min, and concentrated under reduced pressure. To the residue was added 1 M hydrochloric acid (500 mL), and the mixture was vigorously stirred at 60° C. for 1 hr 30 min. The mixture was slowly added to 1 M aqueous sodium hydroxide solution under ice cooling, basified with potassium carbonate, saturated with salt, and extracted with EtOAc/THF. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. To a mixture of the obtained residue, sodium hydrogen carbonate (6.83 g) and THF (190 mL)/water (210 mL) was added a solution of Boc₂O (19.5 g) in THF (20 mL) at room temperature. The mixture was vigorously stirred at room temperature for 15 hr, poured into water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (22.2 g).

MS, found: 265.9.

J) rac-tert-butyl (2S,3S)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate

To a mixture of tert-butyl 2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate (22.1 g), pyridine (96 g) and Et₂O (355 mL) was added dropwise trifluoromethanesulfonic anhydride (51.2 g) under argon atmosphere at 0° C. The mixture was stirred at room temperature for 2 hr 30 min, poured into ice water, and extracted with EtOAc/hexane. The organic layer was separated, washed with 10% aqueous citric acid solution, aqueous sodium hydrogen carbonate solution, water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (22.4 g).

MS, found: 397.9.

K) rac-tert-butyl (2S,3R)-3-azido-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of rac-tert-butyl (2S,3S)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate (22.4 g) and CH₃CN (265 mL) was added tetra-n-butylammonium azide (38.3 g) at room temperature. The mixture was slowly warmed to 80° C., stirred for 1 hr, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (16.2 g).

MS, found: 290.9.

L) rac-tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of rac-tert-butyl (2S,3R)-3-azido-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (16.2 g) and THF (200 mL)/water (5 mL) was added PPh₃ (13.1 g) under argon atmosphere at room tem-

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perature. The mixture was stirred at 55° C. for 18 hr, added to aqueous sodium hydrogen carbonate solution under ice cooling, and extracted with EtOAc/THF. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound and by-product, respectively. To a mixture of the obtained by-product and THF (100 mL)/water (10 mL) was added 40% aqueous methanamine solution (3.22 g) at room temperature. The mixture was stirred at 70° C. for 15 hr, poured into water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (14.6 g), combined with the title compound obtained above.

MS, found: 308.9.

M) tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

rac-tert-Butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (14.6 g) was resolved by HPLC (column: CHIRALPAK IA, 50 mmIDx500 mmL, 20 μm, mobile phase: hexane/ethanol/diethylamine=800/200/1) to give the title compound (6.84 g) with shorter retention time (column: CHIRALPAK IA, 4.6 mmIDx250 mmL, 5 μm, mobile phase: hexane/ethanol/diethylamine=800/200/1).

MS, found: 309.1.

N) tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (1.04 g), TEA (0.577 g), DMAP (0.174 g) and THF (30 mL) was added ethanesulfonyl chloride (0.550 g) at 0° C. The mixture was stirred at room temperature for 5 hr, water was added to mixture at room temperature, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (1.25 g).

MS: [M-H]⁻ 455.1.

O) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-3-yl]ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (24.9 mg), (3-fluorophenyl)boronic acid (15.3 mg), XPhos Pd G3 (4.61 mg), 1 M aqueous potassium phosphate solution (0.163 mL) and DME (0.6 mL) was stirred at 80° C. for 2 hr. The reaction solution was concentrated, and the residue was purified by silica gel column chromatography (NH, EtOAc/hexane). A mixture of the obtained residue (27 mg) and 4 M hydrogen chloride CPME

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solution (2 mL) was stirred at room temperature for 4 hr. The insoluble substance was collected by filtration to give the title compound (17 mg).

MS: $[M+H]^+$ 417.1.

P) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (250 mg), oxetane-2-carboxylic acid (67.6 mg) and DMF (2 mL) were added HATU (315 mg) and DIPEA (357 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added saturated aqueous ammonium chloride solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH₄ EtOAc/hexane), and the obtained solid was crystallized from EtOAc/hexane to give the title compound (66 mg) with shorter retention time.

¹H NMR (400 MHz, CDCl₃) δ 1.27-1.47 (3H, m), 2.68-5.20 (14H, m), 7.03-7.13 (1H, m), 7.16-7.45 (6H, m).

Example 2

N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) tert-butyl (2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoro-3-(methylsulfonamido)pyrrolidine-1-carboxylate

Methanesulfonic anhydride (143 mg) was added to a mixture of tert-butyl (2S,3R)-3-amino-2-(3-chloro-2-fluorobenzyl)-4,4-difluoropyrrolidine-1-carboxylate (200 mg), TEA (0.229 mL) and THF (5 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/hexane) to give the title compound (244 mg).

MS: $[M-H]^-$ 441.1.

B) tert-butyl (2S,3R)-4,4-difluoro-3-(methylsulfonamido)-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoro-3-(methylsulfonamido)pyrrolidine-1-carboxylate (4.52 g), (3,5-difluorophenyl)boronic acid (3.22 g), XPhos Pd G3 (0.432 g) and 1 M aqueous potassium phosphate solution (30.6 mL) in DME (100 mL) was stirred at 80° C. for 3 h under nitrogen atmosphere. The mixture was purified by column chromatography (silica gel, EtOAc/hexane, then NH silica gel, EtOAc/hexane) to give the title compound (5.30 g).

MS: $[M-H]^-$ 519.1.

C) N-[(2S,3R)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-4,4-difluoro-3-(methylsulfonamido)-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)

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methyl)pyrrolidine-1-carboxylate (73 mg) and 4 M HCl/CPME solution (1 mL) was stirred overnight at room temperature. The mixture was diluted with EtOAc and precipitate was collected by filtration to give the title compound (50.0 mg).

MS: $[M+H]^+$ 421.0.

D) N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

To a mixture of N-[(2S,3R)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (4.3 g) and DIPEA (8.14 mL) in THF (60 mL) was added alpha-acetoxyisobutryl chloride (1.64 mL) at 0° C., and the mixture was stirred at same temperature for 10 min. To the mixture were added water (20 mL) and 4 M lithium hydroxide solution (23.5 mL), and the mixture was stirred overnight at room temperature. The mixture was diluted with saturated brine and extracted with EtOAc. The extract was dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/hexane) to give the title compound (4.40 g).

¹H NMR (400 MHz, CDCl₃) δ 1.31-1.47 (6H, m), 2.25-2.60 (1H, m), 2.88-3.02 (4H, m), 3.10 (1H, dd, J=14.2, 7.5 Hz), 4.01-4.49 (3H, m), 4.92-5.17 (2H, m), 6.75-6.87 (1H, m), 7.01-7.11 (2H, m), 7.16-7.23 (1H, m), 7.26-7.32 (1H, m), 7.36-7.46 (1H, m).

Example 3

N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A) tert-butyl (2S,3R)-3-(ethylsulfonamido)-4,4-difluoro-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3R)-2-(3-chloro-2-fluorobenzyl)-3-(ethylsulfonamido)-4,4-difluoropyrrolidine-1-carboxylate (3.70 g), (3,5-difluorophenyl)boronic acid (2.56 g) and 1 M aqueous potassium phosphate solution (24.3 mL) in DME (50 mL) was added XPhos Pd G3 (0.343 g) at room temperature. The mixture was stirred at 90° C. under nitrogen atmosphere for 15 h. The reaction mixture was poured into water and extracted with EtOAc. The organic layer was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/hexane) to give the title compound (3.30 g).

MS: $[M-H]^-$ 533.2.

B) N-[(2S,3R)-4,4-difluoro-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl]ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-3-(ethylsulfonamido)-4,4-difluoro-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (3.30 g) and 4 M HCl/CPME solution (30 mL) was stirred overnight at room temperature. By filtration, the title compound (2.86 g) was obtained.

MS: $[M+H]^+$ 435.1.

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C) N-((2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)ethanesulfonamide

To a mixture of N-((2S,3R)-4,4-difluoro-2-((2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)ethanesulfonamide hydrochloride (200 mg) and DIPEA (0.367 mL) in THF (3 mL) was added alpha-acetoxy-isobutyryl chloride (0.074 mL) at 0° C., and the mixture was stirred at same temperature for 10 min. To the mixture were added water (1 mL) and 4 M lithium hydroxide solution (1.06 mL), and the mixture was stirred overnight at room temperature. The mixture was diluted with saturated brine and extracted with EtOAc. The extract was dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/hexane) and recrystallized from EtOAc/hexane to give the title compound (154 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.32-1.40 (9H, m), 2.27-2.54 (1H, m), 2.88-3.16 (4H, m), 4.02-4.49 (3H, m), 4.86-5.20 (2H, m), 6.78-6.86 (1H, m), 7.02-7.10 (2H, m), 7.16-7.22 (1H, m), 7.27-7.31 (1H, m), 7.35-7.43 (1H, m).

Example 4

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A) 2-amino-3-(3-bromo-2-fluorophenyl)propanoic Acid Hydrochloride

Sodium ethoxide (121 g) was added to a solution of 1-bromo-3-(bromomethyl)-2-fluorobenzene (91 g) and diethyl 2-acetamidomalonate (77 g) in EtOH (566 ml) at room temperature. After being refluxed for 2 h, the reaction mixture was cooled to room temperature, and the insoluble substance was filtered off. The filtrate was concentrated under reduced pressure, and the residue was diluted with 8 M hydrochloric acid (849 ml), and the mixture was refluxed for 17 h. The reaction mixture was concentrated under reduced pressure to give crystals, which were collected by filtration and washed with 2-propanol/IPE to give the title compound (100 g).

MS: [M+H]⁺ 261.9.

B) 3-(3-bromo-2-fluorophenyl)-2-((tert-butoxycarbonyl)amino)propanoic Acid

Boc₂O (92 ml) was added to a solution of 2-amino-3-(3-bromo-2-fluorophenyl)propanoic acid hydrochloride (100 g) and 0.75 M sodium hydroxide solution (893 ml) in DME (918 ml) at 0° C. After being vigorously stirred at room temperature for 16 h, the reaction mixture was poured into ice water, basified with 1 M NaOH aqueous solution and washed with Et₂O. The aqueous layer was acidified to pH 3 with 1 M hydrochloric acid and extracted with EtOAc. The organic layer was dried over magnesium sulfate and concentrated under reduced pressure to give the title compound (105 g).

MS, found: 261.8.

C) tert-butyl (3-(2-fluoro-[1,1'-biphenyl]-3-yl)-1-(methoxy(methyl)amino)-1-oxopropan-2-yl)carbamate

To a mixture of 3-(3-bromo-2-fluorophenyl)-2-((tert-butoxycarbonyl)amino)propanoic acid (250 g) in 1,4-dioxane

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(1.2 L) were added phenylboronic acid (101 g) and Pd(dppf)Cl₂·CH₂Cl₂ (28.2 g) at 15° C. under nitrogen atmosphere. After stirring for 15 min, a solution of K₂CO₃ (191 g) in H₂O (600 mL) was added thereto. The reaction mixture was stirred at 100° C. for 16 h. The reaction mixture was poured into water and it was adjusted to pH 2 with 2 M hydrochloric acid and then it was extracted with EtOAc. The combined organic layer was washed with saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure. To a stirred solution of the obtained residue (250 g) and N, O-dimethylhydroxylamine (71.3 g) in DMF (2 L) was added HATU (250 g) and then DIPEA (331 g) was added thereto at 0° C. The reaction mixture was stirred at 15° C. under nitrogen atmosphere for 16 h. The reaction mixture was poured into water and the organic layer was separated. The water layer was extracted with EtOAc. The combined organic layer was washed with saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was collected by filtration and washed with petroleum ether/EtOAc to give the title compound (180 g).

¹H NMR (400 MHz, CDCl₃) δ 1.38 (9H, s), 3.00-3.18 (2H, m), 3.20 (3H, s), 3.77 (3H, s), 4.91-5.09 (1H, m), 5.17-5.30 (1H, m), 7.09-7.19 (2H, m), 7.28-7.48 (4H, m), 7.50-7.57 (2H, m).

D) ethyl 4-((tert-butoxycarbonyl)amino)-2,2-difluoro-5-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-hydroxypentanoate

To a stirred solution of tert-butyl (3-(2-fluoro-[1,1'-biphenyl]-3-yl)-1-(methoxy(methyl)amino)-1-oxopropan-2-yl)carbamate (40.0 g) in THF (1.2 L) was added lithium aluminium hydride (4.53 g) in portions at -78° C., and the reaction mixture was stirred at -10° C. under nitrogen atmosphere for 3 h. The reaction mixture was poured into cooled aqueous saturated ammonium chloride solution, and the mixture was filtered. The filter cake was washed with EtOAc and the filtrate was extracted with EtOAc. The combined organic layer was washed with 1 M hydrochloric acid, saturated aqueous sodium hydrogen carbonate solution and saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure to give the tert-butyl (1-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-oxopropan-2-yl)carbamate (40.0 g). To the mixture of activated zinc (56.8 g) in THF (300 mL) was added TMSCl (4.43 g) at 20° C., and the resulting mixture was stirred at 20° C. for min under nitrogen atmosphere. Then a solution of ethyl 2-bromo-2,2-difluoroacetate (82.8 g) in THF (500 mL) was added dropwise to the above mixture at 40° C., and it was stirred at 40° C. for another 1.5 h under nitrogen atmosphere to give 0.51 M (2-ethoxy-1,1-difluoro-2-oxoethyl)zinc(II) bromide/THF solution (800 mL). To a stirred solution of tert-butyl (1-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-oxopropan-2-yl)carbamate (27.0 g) in THF (150 mL) was added 0.51 M (2-ethoxy-1,1-difluoro-2-oxoethyl)zinc(II) bromide/THF solution (385 mL) at 15° C., and the reaction mixture was stirred at 15° C. under nitrogen atmosphere for 20 min. Then it was stirred at 55° C. for another 20 min. The reaction mixture was poured into aqueous ammonium chloride solution and it was extracted with EtOAc. The combined organic layer was washed with saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/petroleum ether) to give the title compound (12.0 g).

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¹H NMR (400 MHz, CDCl₃) δ 1.30-1.54 (12H, m), 3.00-3.35 (2H, m), 3.70-4.55 (3H, m), 4.74-5.19 (1H, m), 7.08-7.25 (2H, m), 7.35-7.57 (7H, m).

E) ethyl 4-amino-2,2-difluoro-5-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-hydroxypentanoate Hydrochloride

To a stirred solution of ethyl 4-(tert-butoxycarbonyl)amino-2,2-difluoro-5-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-hydroxypentanoate (39.0 g) in 1,4-dioxane (80 mL) was added 4 M HCl/1,4-dioxane solution (250 mL), and the reaction mixture was stirred at 15° C. for 3 h. The mixture was concentrated under reduced pressure to give the title compound (34.0 g).

MS: [M+H]⁺ 368.1.

F) 3,3-difluoro-5-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-4-hydroxypyrrolidin-2-one

To a stirred solution of ethyl 4-amino-2,2-difluoro-5-(2-fluoro-[1,1'-biphenyl]-3-yl)-3-hydroxypentanoate hydrochloride (66.0 g) in EtOH (850 mL) was added DIPEA (66.0 g), and the reaction mixture was stirred at 80° C. under nitrogen atmosphere for 0.5 h. The reaction mixture was concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/petroleum ether) to give the title compound (35.0 g).

MS: [M+H]⁺ 322.0.

G) 4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-ol

To a stirred solution of 3,3-difluoro-5-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-4-hydroxypyrrolidin-2-one (36.0 g) in THF (400 mL) was added 1 M BH₃-THF complex/THF solution (336 mL) at 15° C., and the reaction mixture was stirred at 70° C. for 16 h under nitrogen atmosphere. Then the reaction mixture was quenched with water dropwise at 0° C., and then 1 M hydrochloric acid was added thereto, and it was stirred at 66° C. for 2 h. The reaction mixture was adjusted to pH 8 with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The combined organic layer was washed with water, dried over anhydrous sodium sulfate and concentrated under reduced pressure to give the title compound (35.0 g).

MS: [M+H]⁺ 308.1.

H) rac-tert-butyl (2S,3S)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-hydroxypyrrolidine-1-carboxylate

To 4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-ol (37.0 g) was added tert-butoxycarbonyl tert-butyl carbonate (26.3 g) in THF (300 mL), and the mixture was basified to pH 8 with saturated aqueous sodium hydrogen carbonate solution, and the reaction mixture was stirred at 15° C. for 2 h. The reaction mixture was poured into water and extracted with EtOAc. The combined organic layer was washed with saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/petroleum ether), and then purified by preparative HPLC (column: C18, mobile phase: CH₃CN/water (containing 10 mM ammonium bicarbonate)) to give the title compound (9 g).

¹H NMR (400 MHz, CDCl₃) δ 1.48 (9H, s), 2.14-2.28 (1H, m), 2.78-3.00 (1H, m), 3.24-3.40 (1H, m), 3.69-4.00

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(2H, m), 4.00-4.08 (1H, m), 4.14-4.20 (1H, m), 7.13-7.24 (2H, m), 7.32-7.50 (4H, m), 7.52-7.59 (2H, m).

I) rac-tert-butyl (2S,3S)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-(((trifluoromethyl)sulfonyl)oxy)pyrrolidine-1-carboxylate

To a stirred solution of rac-tert-butyl (2S,3S)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-hydroxypyrrolidine-1-carboxylate (9.00 g) in dichloromethane (150 mL) was added pyridine (8.74 g), and then a solution of trifluoromethanesulfonic anhydride (15.6 g) in dichloromethane (50 mL) was added dropwise thereto at -10° C., and the reaction mixture was stirred at -10° C. under nitrogen atmosphere for 1 h. The reaction mixture was poured into water and it was extracted with dichloromethane. The combined organic layer was washed with 5% aqueous citric acid solution and saturated brine, dried over anhydrous sodium sulfate and concentrated under reduced pressure to give the title compound (11.0 g).

¹H NMR (400 MHz, CDCl₃) δ 1.51 (9H, s), 2.76-2.97 (1H, m), 3.34-3.52 (1H, m), 3.84-3.99 (2H, m), 4.32-4.51 (1H, m), 4.89-5.05 (1H, m), 7.10-7.25 (2H, m), 7.34-7.56 (6H, m).

J) rac-tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate

To a stirred solution of rac-tert-butyl (2S,3S)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-(((trifluoromethyl)sulfonyl)oxy)pyrrolidine-1-carboxylate (11.0 g) in N,N-Dimethylacetamide (450 mL) was added sodium azide (5.30 g), and the reaction mixture was stirred at 130° C. for 3 h. The mixture was diluted with water and it was adjusted to pH 9-10 with aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The combined organic layers were washed with water, dried over anhydrous sodium sulfate and concentrated reduced pressure to give rac-tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (9.00 g). To a stirred solution of rac-tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (1.00 g) in MeOH (30 mL) was added 10% Pd on carbon (100 mg), and the mixture was degassed under reduced pressure and purged with hydrogen several times and then stirred at 25° C. under hydrogen atmosphere (15 psi) for 16 h. The reaction mixture was filtered through a pad of celite and the filter cake was washed with MeOH and the combined filtrate was concentrated under reduced pressure to give crude product as 1st batch. To a stirred solution of rac-tert-butyl (2S,3S)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-(((trifluoromethyl)sulfonyl)oxy)pyrrolidine-1-carboxylate (8.00 g) in MeOH (200 mL) was added 10% Pd on carbon (800 mg), and the mixture was degassed under reduced pressure and purged with hydrogen several times and then stirred at 25° C. under hydrogen atmosphere (15 psi) for 3 h. The reaction mixture was filtered through a pad of celite and the filter cake was washed with MeOH. The combined filtrate was concentrated under reduced pressure. The residue and crude product of 1st batch were purified by column chromatography (silica gel, EtOAc/petroleum ether) to give the title compound (6.04 g).

¹H NMR (400 MHz, DMSO-d₆) δ 0.89-1.32 (9H, m), 1.86-2.01 (2H, m), 2.53-2.78 (1H, m), 2.94-3.13 (1H, m), 3.63-3.95 (3H, m), 4.11-4.36 (1H, m), 7.13-7.55 (8H, m).

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K) tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate

Optical resolution of rac-tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (6.03 g) was performed using preparative HPLC (Column: CHIRALPAK AD, mobile phase: hexane/EtOH=900/100 (v/v)) to afford the title compound (3.09 g) with shorter retention time (Column: CHIRALPAK AD-H, eluted with hexane/EtOH=900/100 (v/v)).

MS, found: 351.1.

L) tert-butyl (2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-(methylsulfonamido)pyrrolidine-1-carboxylate

Methanesulfonic anhydride (292 mg) was added to a stirred solution of tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (355 mg) and TEA (0.365 mL) in THF (8 mL) at room temperature. After 0.5 h, the reaction mixture was concentrated under reduced pressure, and the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (427 mg).

MS, found: 385.0.

M) N-((2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)-3-(methylsulfonamido)pyrrolidine-1-carboxylate (427 mg) in 4 M HCl/CPME solution (12 mL) was stirred at room temperature for 2 h. The solid was collected by filtration to give the title compound (340 mg).

MS: [M+H]⁺ 385.0.

N) N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A mixture of N-((2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)methanesulfonamide hydrochloride (45 mg) and DIPEA (0.092 mL) in THF (1.28 mL) was stirred at room temperature for 30 min. To the suspension was added dropwise alpha-acetoxy-isobutyryl chloride (0.0186 mL) at 0° C., and the mixture was stirred overnight at same temperature. To the mixture were added water (0.855 mL) and 4M lithium hydroxide solution (0.267 mL), and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, EtOAc/hexane) to give the title compound (45 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.91-1.26 (6H, m), 2.62-2.72 (1H, m), 2.97-3.07 (4H, m), 4.09-4.44 (2H, m), 4.59-4.81 (1H, m), 4.82-5.02 (1H, m), 5.27 (1H, s), 7.07-

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7.17 (1H, m), 7.25-7.34 (2H, m), 7.34-7.41 (1H, m), 7.42-7.49 (2H, m), 7.49-7.56 (2H, m), 8.09-8.19 (1H, m).

Example 5

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A) tert-butyl (2S,3R)-3-(ethylsulfonamido)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate

Ethanesulfonyl chloride (0.233 mL) was added to a solution of tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (500 mg), TEA (0.514 mL) and DMAP (75 mg) in THF (5 mL) at room temperature. After being stirred at room temperature for 2 h, the reaction mixture was poured into water and extracted with EtOAc. The organic layer was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residual oil was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (441 mg).

MS, found: 399.1.

B) N-((2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-3-(ethylsulfonamido)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidine-1-carboxylate (441 mg) and 4 M HCl/CPME (5 mL) was stirred at room temperature for 2 h. The mixture was concentrated under reduced pressure to give the title compound (360 mg).

MS: [M+H]⁺ 399.1.

C) N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-((2S,3R)-4,4-difluoro-2-((2-fluoro-[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)ethanesulfonamide hydrochloride (60 mg) and DIPEA (0.119 mL) in THF (1.66 mL) was stirred at room temperature for 30 min. To the suspension was added dropwise alpha-acetoxy-isobutyryl chloride (0.024 mL) at 0° C., and the mixture was stirred overnight at same temperature. To the mixture were added water (1.10 mL) and 4 M lithium hydroxide solution (0.345 mL), and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (42.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.95-1.36 (9H, m), 2.60-2.73 (1H, m), 2.99-3.15 (3H, m), 4.09-4.51 (2H, m), 4.58-4.98 (2H, m), 5.26 (1H, s), 7.07-7.18 (1H, m), 7.24-7.33 (2H, m), 7.34-7.42 (1H, m), 7.42-7.56 (4H, m), 8.13 (1H, brs).

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Example 7

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A) tert-butyl {3-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-1-(methoxy(methyl)amino)-1-oxopropan-2-yl}carbamate

WSC (63.7 mL) was added to a mixture of 3-(3-bromo-2-fluorophenyl)-2-((tert-butoxycarbonyl)amino)propanoic acid (105 g), N, O-dimethylhydroxylamine hydrochloride (31.1 g), HOBT (43.1 g), TEA (48.5 mL) and DMF (580 mL) at 0° C. After being stirred at room temperature for 15 h, the reaction mixture was poured into half-saturated aqueous sodium hydrogen carbonate solution and stirred at room temperature for 20 min to give crystals, which were collected by filtration and washed successively with water, 2-propanol and IPE. A mixture of the obtained solid, (3-fluorophenyl)boronic acid (48.7 g), XPhos Pd G3 (2.45 g), 2 M aqueous potassium phosphate solution (435 mL) and THF (537 mL) was stirred at 70° C. for 1 h under argon atmosphere. The reaction mixture was poured into half-saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was washed with water and saturated brine, dried over magnesium sulfate, passed through NH silica gel pad (eluted with EtOAc) and concentrated under reduced pressure to give crystals, which were collected by filtration and washed with IPE to give the title compound (92.8 g).

MS, found: 321.0.

B) tert-butyl {3-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl}[(4-methoxyphenyl)methyl]carbamate

To a stirred suspension of tert-butyl {3-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-1-(methoxy(methyl)amino)-1-oxopropan-2-yl}carbamate (77.0 g) in DMF (366 mL) was added NaH (60% in oil) (9.52 g) at 0° C. After being stirred for 10 min at room temperature, alpha-chloro-4-methoxytoluene (49.7 mL) and tetrabutylammonium iodide (6.76 g) were added thereto at 0° C. After 1.5 h at room temperature, the reaction mixture was poured into a stirred mixture of EtOAc and 5% KHSO₄ aqueous solution at 0° C., and the organic layer was separated. The organic layer was washed with saturated brine, dried over sodium sulfate, filtrated and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (83 g).

MS, found: 441.2.

C) tert-butyl [1-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate

Lithium aluminium hydride (0.983 g) was added to a stirred solution of tert-butyl {3-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-1-[methoxy(methyl)amino]-1-oxopropan-2-yl}[(4-methoxyphenyl)methyl]carbamate (10 g) in Et₂O (83 mL) at -78° C. The mixture was warmed up to 0° C. After 0.5 h, the reaction mixture was quenched with EtOAc (3.61 mL) with keeping temperature of the reaction mixture under 10° C., and a solution of potassium bisulfate (6.30 g) in water (83 mL) was added to the mixture. The mixture was extracted with Et₂O. The organic layer was separated,

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washed with water and a mixture of saturated brine and aqueous sodium hydrogen carbonate solution, and through silica gel pad (eluted with EtOAc/hexane) to give the title compound (8.43 g).

MS, found: 426.0.

D) ethyl 4-[(tert-butoxycarbonyl)[(4-methoxyphenyl)methyl]amino]-2,4,5-trideoxy-5-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-2,2-difluoropentionate

Chlorotrimethylsilane (0.305 mL) was added to a stirred suspension of zinc (2.75 g) in THF (24 mL) at room temperature. After 10 min, ethyl 2-bromo-2,2-difluoroacetate (3.08 mL) was added to the mixture under water bath cooling with keeping temperature of the reaction mixture under 40° C. After 10 min, a solution of tert-butyl [1-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate (5.78 g) in THF (6 mL) was added to the mixture at room temperature. After 1.5 h, the reaction mixture was poured into a stirred mixture of 10% KHSO₄ aqueous solution (60 mL) and EtOAc (120 mL). The organic layer was separated, washed with saturated brine, dried over sodium sulfate, filtrated, and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (6.71 g).

MS, found: 550.1

E) 5-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one

4 M HCl/CPME (30.1 mL) was added to a solution of ethyl 4-[(tert-butoxycarbonyl)[(4-methoxyphenyl)methyl]amino]-2,4,5-trideoxy-5-(2,3'-difluoro[1,1'-biphenyl]-3-yl)-2,2-difluoropentionate (7.28 g) in EtOH (3.8 mL) at room temperature. After being stirred at room temperature for 1 h, the reaction mixture was concentrated under reduced pressure. The residual oil was dissolved in EtOH (38 mL), and DIPEA (6.28 mL) was added thereto at room temperature. After being stirred at 80° C. for 0.5 h, the reaction mixture was concentrated under reduced pressure. The residue was poured into water, and the mixture was extracted with EtOAc. The organic layer was concentrated under reduced pressure, and the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (5.00 g).

MS: [M+H]⁺ 460.2.

F) 5-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one

Ceric ammonium nitrate (14.0 g) in water (15.9 mL) was added to a solution of 5-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one (5.85 g) in CH₃CN (47.7 mL) at room temperature. The mixture was stirred at room temperature for 2 h. The mixture was neutralized with saturated aqueous sodium hydrogen carbonate solution and diluted with EtOAc. The insoluble material was removed by filtration, and the filtrate was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography

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(silica gel, eluted with EtOAc/hexane) to give the title compound (3.80 g).

MS: [M-H]⁻ 338.2.

G) tert-butyl 2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate

0.9 M Borane-THF complex (37.3 mL) was added dropwise to a solution of 5-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one (3.80 g) in THF (37.3 mL) at room temperature. After being stirred at 60° C. for 3 h, the reaction mixture was quenched by dropwise addition of water at 0° C. The mixture was stirred at room temperature for 10 min and concentrated under reduced pressure. The residue was diluted with EtOH (25 mL) and 1 M hydrochloric acid (125 mL), and the mixture was stirred at 60° C. for 1 h. The reaction mixture was poured carefully into iced saturated aqueous sodium hydrogen carbonate solution (250 mL) and extracted with EtOAc. The organic layer was washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was mixed with sodium hydrogen carbonate (0.941 g), THF (28.0 mL) and water (28.0 mL), and Boc₂O (2.83 mL) was added to the mixture at room temperature. The mixture was stirred at room temperature under nitrogen atmosphere for 3 h. The mixture was poured into water at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residual oil was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (4.00 g).

MS, found: 326.2.

H) rac-tert-butyl (2S,3S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate

Trifluoromethanesulfonic anhydride (3.16 mL) was added to a solution of tert-butyl 2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate (4.0 g) and pyridine (3.80 mL) in Et₂O (47.0 mL) at 0° C. After being stirred at room temperature for 20 h, the mixture was quenched with water at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The organic layer was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (3.22 g).

¹H NMR (400 MHz, CDCl₃) δ 1.50 (9H, s), 2.74-2.99 (1H, m), 3.33-3.51 (1H, m), 3.72-4.02 (2H, m), 4.33-4.49 (1H, m), 4.86-5.02 (1H, m), 7.03-7.13 (1H, m), 7.17-7.32 (4H, m), 7.33-7.47 (2H, m).

I) rac-tert-butyl (2S,3R)-3-azido-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

Tetra-n-butylammonium azide (24.9 g) was added to a solution of rac-tert-butyl (2S,3S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate (24.4 g) in CH₃CN (292 mL) at room temperature. After being stirred at 80° C. for 1 h, the reaction mixture was poured into iced water and extracted with EtOAc. The organic layer was washed with water and saturated brine, dried over magnesium sulfate and

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concentrated under reduced pressure. The residual oil was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (19.5 g).

MS, found: 351.0.

J) rac-tert-butyl (2S,3R)-3-amino-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

A solution of rac-tert-butyl (2S,3R)-3-azido-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (19.5 g) in MeOH (433 mL) was hydrogenated in the presence of 20% Pd(OH)₂ on carbon (50% wet) (1.95 g) at room temperature under ordinary pressure for 1 h. After removal of the catalyst by filtration, the filtrate was concentrated under reduced pressure. The residual oil was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (16.5 g).

MS, found: 369.0.

K) tert-butyl (2S,3R)-3-amino-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

Optical resolution of rac-tert-butyl (2S,3R)-3-amino-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (16.5 g) was performed using preparative HPLC (Column: CHIRALPAK AD, mobile phase: hexane/EtOH/diethylamine=900/100/1 (v/v/v)) to afford the title compound (7.32 g) with shorter retention time (Column: CHIRALPAK AD-H, mobile phase: hexane/EtOH/diethylamine=900/100/1 (v/v/v)).

MS, found: 369.1.

L) tert-butyl (2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

Methanesulfonic anhydride (185 mg) was added to a stirred mixture of tert-butyl (2S,3R)-3-amino-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (300 mg) and TEA (0.296 mL) in THF (5 mL) at room temperature and the mixture was stirred overnight. The reaction mixture was directly purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (309 mg).

MS: [M-H]⁻ 501.1.

M) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (309 mg) and 4 M HCl/EtOAc (3 mL) was stirred at room temperature for 1 h. The mixture was diluted with EtOAc. The resulting solid was collected by filtration to give the title compound (230 mg).

MS: [M-H]⁻ 401.0.

N) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (40 mg) and DIPEA

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(0.079 mL) in THF (1.09 mL) was stirred at room temperature for 30 min. To the suspension was added dropwise alpha-acetoxy-isobutyl chloride (0.159 mL) at 0° C., and the mixture was stirred overnight at same temperature. To the mixture were added water (0.729 mL) and 4 M aqueous lithium hydroxide solution (0.228 mL), and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (41 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.95-1.34 (6H, m), 2.61-2.75 (1H, m), 2.96-3.10 (4H, m), 4.10-4.47 (2H, m), 4.60-4.81 (1H, m), 4.84-5.01 (1H, m), 5.28 (1H, s), 7.08-7.18 (1H, m), 7.20-7.28 (1H, m), 7.31-7.41 (4H, m), 7.47-7.56 (1H, m), 8.14 (1H, brs).

Example 8

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-((2S,3R)-2-((2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl)-4,4-difluoropyrrolidin-3-yl)ethanesulfonamide hydrochloride (57 mg) and DIPEA (81 mg) in THF (2 mL) was added 1-chloro-2-methyl-1-oxopropan-2-yl acetate (31.1 mg) at room temperature. The mixture was stirred at room temperature for 1 h. To the mixture were added water (1 mL) and 4 M aqueous lithium hydroxide solution (0.314 mL) at room temperature. The mixture was stirred at room temperature for 1 h. After azeotropic evaporation with toluene, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (17.0 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 0.87-1.43 (9H, m), 2.60-2.77 (1H, m), 2.97-3.17 (3H, m), 4.07-5.01 (3H, m), 5.27 (1H, s), 7.03-7.62 (8H, m), 8.14 (1H, brs).

Example 13

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (29 mg), DIPEA (42.7 mg) and 1-hydroxycyclobutane-1-carboxylic acid (9.21 mg) in THF (4 mL) was added HATU (37.7 mg) at room temperature. The mixture was stirred at room temperature for 15 h. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with saturated brine (5 mL), dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (29.0 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.02-1.17 (1H, m), 1.44-1.66 (1H, m), 1.79-2.04 (2H, m), 2.16-2.42 (2H, m), 2.65-2.84 (1H, m), 2.95-3.10 (4H, m), 3.91-4.11 (1H, m), 4.25-4.53 (2H, m), 4.82-4.96 (1H, m), 5.95 (1H, s), 7.09-7.42 (6H, m), 7.45-7.59 (1H, m), 8.19 (1H, brs).

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Example 21

N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) tert-butyl(S)-(3-(3-chlorophenyl)-1-(methoxymethyl)amino)-1-oxopropan-2-yl)carbamate

To a mixture of (S)-2-((tert-butoxycarbonyl)amino)-3-(3-chlorophenyl)propanoic acid (25.3 g), DIPEA (27.3 g) and HATU (35.3 g) in THF (250 mL) was added N, O-dimethylhydroxylamine hydrochloride (12.4 g) at room temperature. The mixture was stirred at room temperature for 15 h. The mixture was diluted with EtOAc (250 mL). The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution (200 mL) at room temperature. The organic layer was separated, washed with saturated brine (100 mL), dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (28.5 g).

MS, found: 243.0.

B) tert-butyl(S)-(3-(3-chlorophenyl)-1-(methoxymethyl)amino)-1-oxopropan-2-yl)(4-methoxybenzyl)carbamate

To a mixture of tert-butyl(S)-(3-(3-chlorophenyl)-1-(methoxymethyl)amino)-1-oxopropan-2-yl)carbamate (13.7 g) in DMF (80 mL) was added sodium hydride (2.08 g) at 0° C. After being stirred for 10 min at room temperature, 1-(chloromethyl)-4-methoxybenzene (12.5 g) and tetrabutylammonium iodide (1.48 g) were added thereto at 0° C. After being stirred 1.5 h at room temperature, the reaction mixture was poured into a stirred mixture of EtOAc (200 mL) and 5% KHSO₄ aqueous solution (100 mL) at 0° C., and the organic layer was separated. The organic layer was washed with water and saturated brine, dried over magnesium sulfate, filtrated and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (18.3 g).

MS: [M+H]⁺ 463.1.

C) tert-butyl [(2)-1-(3-chlorophenyl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate

Aluminum(III) lithium hydride (2.10 g) was added to a stirred solution of tert-butyl(S)-(3-(3-chlorophenyl)-1-(methoxymethyl)amino)-1-oxopropan-2-yl)(4-methoxybenzyl)carbamate (18.3 g) in Et₂O (150 mL) at -78° C. The mixture was warmed up to 0° C. After 0.5 h, the reaction mixture was quenched with EtOAc (6.97 g) and a solution of potassium hydrogen sulfate (13.5 g) in water (150 mL). And then, EtOAc and water were added thereto. The organic layer was separated, washed with water and a mixture of saturated brine and aqueous sodium hydrogen carbonate solution, dried over magnesium sulfate, filtrated and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (12.8 g).

MS: [M-H]⁻ 402.1.

D) ethyl (4S)-4-((tert-butoxycarbonyl)(4-methoxybenzyl)amino)-5-(3-chlorophenyl)-2,2-difluoro-3-hydroxypentanoate

To a stirred mixture of zinc (0.456 g) in THF (6 mL) was added chlorotrimethylsilane (0.051 g) at room temperature.

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After 5 min, ethyl 2-bromo-2,2-difluoroacetate (0.945 g) was added to the mixture under water bath cooling. After 10 min, a solution of tert-butyl [(2S)-1-(3-chlorophenyl)-3-oxopropan-2-yl][(4-methoxyphenyl)methyl]carbamate (0.94 g) in THF (2 mL) was added to the mixture at room temperature. The mixture was stirred at room temperature under nitrogen atmosphere for 2 h. The reaction mixture was poured into a stirred mixture of 10% aqueous citric acid solution and EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate, filtrated, and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (1.11 g).

MS, found: 472.1.

E) ethyl (4S)-5-(3-chlorophenyl)-2,2-difluoro-3-hydroxy-4-((4-methoxybenzyl)amino)pentanoate
Hydrochloride

To a mixture of ethyl (4S)-4-((tert-butoxycarbonyl)(4-methoxybenzyl)amino)-5-(3-chlorophenyl)-2,2-difluoro-3-hydroxypentanoate (1.11 g) in EtOH (3 mL) was added 4 M HCl/CPME solution (5.26 mL) at room temperature. The mixture was stirred at room temperature for 2 h. Evaporation gave the title compound (0.976 g).

MS: [M+H]⁺ 428.0.

F) (5S)-5-[(3-chlorophenyl methyl)-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one

To a mixture of ethyl (4S)-5-(3-chlorophenyl)-2,2-difluoro-3-hydroxy-4-((4-methoxybenzyl)amino)pentanoate hydrochloride (0.976 g) in EtOH (5 mL) was added DIPEA (2.72 g) at room temperature. The mixture was stirred at room temperature for 2 h. After evaporation of the solvent, the residue was poured into water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (0.770 g).

MS: [M-H]⁻ 380.1.

G) (5S)-5-[(3-chlorophenyl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one

To a mixture of (5S)-5-[(3-chlorophenyl)methyl]-3,3-difluoro-4-hydroxy-1-[(4-methoxyphenyl)methyl]pyrrolidin-2-one (0.77 g) in CH₃CN (7 mL) and water (2.33 mL) was added ammonium cerium (IV) nitrate (2.21 g) at room temperature. The mixture was stirred at room temperature for 2 h. The mixture was quenched with saturated aqueous sodium carbonate solution and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (0.460 g).

MS: [M-H]⁻ 260.0.

H) (2S)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-ol

To a mixture of (5S)-5-[(3-chlorophenyl)methyl]-3,3-difluoro-4-hydroxypyrrolidin-2-one (3.45 g) in THF (40 mL)

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was added 0.9 M borane-THF complex/THF solution (44.0 mL) at room temperature. The mixture was stirred at 60° C. under nitrogen atmosphere for 3 h. The mixture was quenched with water at 0° C. The mixture was stirred at room temperature for 10 min and concentrated under reduced pressure. The residue was diluted with EtOH (10 mL) and 1 M hydrochloric acid (50 mL), and the mixture was stirred at 60° C. for 1 h. Sodium hydrogen carbonate was carefully added to the reaction mixture in the iced bath to bring the pH of the solution to 8. Evaporation gave the title compound (3.27 g).

MS, found: 247.9.

I) tert-butyl (2S)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate

To a crude mixture of (2S)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-ol (2.27 g) and TEA (1.86 g) in THF (30 mL) was added Boc₂O (3.00 g) at room temperature. The mixture was stirred at room temperature under nitrogen atmosphere for 3 h. After evaporation, the mixture was diluted with water and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (2.84 g).

MS, found: 247.9.

J) tert-butyl (2S,3S)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate

Trifluoromethanesulfonic anhydride (406 μL) was added to a solution of tert-butyl (2S)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-hydroxypyrrolidine-1-carboxylate (420 mg) and pyridine (488 μL) in Et₂O (6.04 mL) at 0° C. After being stirred at room temperature for 2 h and then at 40° C. for 2 h, the mixture was quenched with water at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The organic layer was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (410 mg).

MS, found: 379.9.

K) tert-butyl (2S,3R)-3-azido-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3S)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate (2.43 g) in CH₃CN (20 mL) was added tetrabutylammonium azide (2.16 g) at room temperature. The mixture was stirred at 80° C. for 1 h. The mixture was quenched with water at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (1.85 g).

MS, found: 272.9.

L) tert-butyl (2S,3R)-3-amino-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3R)-3-azido-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (1.57

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g) in THF (20 mL) was added PPh₃ (1.33 g) at room temperature. The mixture was stirred at 50° C. for 15 h. After evaporation, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane). The residue was further purified by preparative HPLC (Column: CHIRALPAK IA, mobile phase: hexane/EtOH/diethylamine=650/350/1 (v/v/v)) to afford the title compound (1.13 g) with shorter retention time (Column: CHIRALPAK IA, mobile phase: hexane/EtOH/diethylamine=650/350/1 (v/v/v)).

MS, found: 291.1.

M) tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

Methanesulfonic anhydride (166 mg) was added to a stirred solution of tert-butyl (2S,3R)-3-amino-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (165 mg) and TEA (0.199 mL) in THF (3 mL) at room temperature. After 0.5 h, the reaction mixture was concentrated under reduced pressure, and the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (186 mg).

MS: [M-H]⁻ 423.1.

N) tert-butyl (2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (186 mg), (3-fluorophenyl)boronic acid (92 mg), XPhos Pd G3 (37.1 mg) and 1 M aqueous potassium phosphate solution (1.31 mL) in DME (1.46 mL) was stirred overnight at 90° C. The mixture was purified by column chromatography (NH silica gel, EtOAc/hexane) to give the title compound (205 mg).

MS: [M-H]⁻ 483.2.

O) N-{(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (205 mg) and 4 M HCl/CPME solution (5 mL) was stirred at room temperature for 2 h. The solid was collected by filtration to give the title compound (175 mg).

MS: [M+H]⁺ 385.0.

P) N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carboxyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide

HATU (43.4 mg) was added to a solution of N-{(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide hydrochloride (40 mg), bicyclo[1.1.1]pentane-1-carboxylic acid (12.8 mg) and TEA (0.066 mL) in DMF (1 mL) at room temperature. The mixture was stirred at room temperature under a dry atmosphere for 30 min. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The

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residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (32.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.21-1.76 (3H, m), 1.98-3.18 (9H, m), 3.70-5.01 (5H, m), 6.97-7.53 (8H, m).

Example 25

N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl}methanesulfonamide

A mixture of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl}methanesulfonamide (35 mg), m-tolylboronic acid (16.6 mg), XPhos Pd G3 (6.91 mg) and 1 M aqueous potassium phosphate solution (245 μL) in THF (408 μL) was stirred at 90° C. for 1 h. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (33.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.97-1.23 (6H, m), 2.36 (3H, s), 2.61-2.71 (1H, m), 2.99 (3H, s), 3.00-3.08 (1H, m), 4.08-4.45 (2H, m), 4.58-5.02 (2H, m), 5.28 (1H, s), 7.05-7.15 (1H, m), 7.17-7.22 (1H, m), 7.24-7.39 (5H, m), 8.05-8.23 (1H, m).

Example 26

N-{(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl}methanesulfonamide

HATU (43.4 mg) was added to a solution of N-{(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide hydrochloride (40 mg), (R)-tetrahydrofuran-2-carboxylic acid (13.2 mg) and TEA (0.066 mL) in DMF (0.5 mL) at room temperature. The mixture was stirred at room temperature under a dry atmosphere for 30 min. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc then, NH silica gel, eluted with EtOAc/hexane) to give the title compound (39.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 1.54-1.81 (3H, m), 1.88-2.06 (1H, m), 2.62-3.14 (5H, m), 3.41-4.76 (7H, m), 7.12-7.23 (1H, m), 7.29-7.44 (2H, m), 7.45-7.57 (4H, m), 7.59-7.76 (1H, m), 8.14-8.30 (1H, m).

Example 35

N-{(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl}methanesulfonamide

A) tert-butyl (2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (90 mg), (3,5-difluorophenyl)boronic acid (50.2 mg), XPhos Pd G3 (17.9 mg) and 1 M aqueous potassium phosphate solution (635 μL) in DME (1.06 mL) was stirred overnight at 90° C. The mixture was quenched

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with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (104 mg).

MS, found: 403.0.

B) N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (104 mg) and 4 M HCl/CPME solution (2 mL) was stirred at room temperature for 2 h. The solid was collected by filtration to give the title compound (85 mg).

MS: [M+H]⁺ 403.0.

C) N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (20.0 mg) and isobutyric acid (5.09 μ L) in DMF (1.5 mL) were added HATU (26.0 mg) and DIPEA (0.032 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (19.7 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.11-0.98 (6H, m), 1.82-1.91 (1H, m), 2.54-3.15 (5H, m), 3.81-4.25 (2H, m), 4.37-4.70 (2H, m), 7.16-7.84 (7H, m), 8.23 (1H, brs).

Example 44

N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A mixture of N-((2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoropyrrolidin-3-yl)methanesulfonamide hydrochloride (200 mg) and DIPEA (456 μ L) in THF (3.16 mL) was stirred at room temperature for 30 min. To the suspension was added dropwise alpha-acetoxy-isobutyl chloride (92 μ L) at 0° C., and the mixture was stirred overnight at same temperature. To the mixture were added water (2.11 mL) and 4 M aqueous lithium hydroxide solution (1.32 mL), and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel,

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eluted with EtOAc/hexane) to give the title compound (184 mg).

MS: [M+H]⁺ 429.0.

B) N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

The mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide (24 mg), 4,4,4',4',5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (17.1 mg), XPhos Pd G3 (9.47 mg), potassium acetate (11.0 mg) and toluene (1 mL) was stirred at 80° C. under nitrogen atmosphere for 4 h. The mixture was quenched with water at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The mixture of the obtained residue, cesium carbonate (58.6 mg), 1-chloro-3-iodobenzene (0.011 mL), PdCl₂ (dppf) (8.78 mg) and DME (1 mL)/water (0.300 mL) was stirred at 80° C. under nitrogen atmosphere for 1 h. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane). The obtained solid was purified by preparative HPLC to give the title compound (7.50 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 0.93-1.28 (6H, m), 2.61-2.74 (1H, m), 2.93-3.12 (4H, m), 4.10-4.48 (2H, m), 4.57-4.81 (1H, m), 4.84-5.02 (1H, m), 5.29 (1H, s), 7.10-7.18 (1H, m), 7.29-7.39 (2H, m), 7.42-7.53 (3H, m), 7.57 (1H, s), 8.16 (1H, brs).

Example 45

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

To a solution of N-((2S,3R)-4,4-difluoro-2-((2-fluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)ethanesulfonamide hydrochloride (40.4 mg) and oxetane-2-carboxylic acid (9.33 μ L) in DMF (3 mL) were added HATU (53.0 mg) and DIPEA (0.065 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (12.1 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.19-1.33 (3H, m), 2.20-2.44 (1H, m), 2.60-3.15 (5H, m), 3.80-5.22 (7H, m), 7.11-7.57 (8H, m), 8.09-8.29 (1H, m).

Example 46

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Longer Retention Time

To a solution of N-((2S,3R)-4,4-difluoro-2-((2-fluoro[1,1'-biphenyl]-3-yl)methyl)pyrrolidin-3-yl)ethanesulfonamide hydrochloride (40.4 mg) and oxetane-2-carboxylic acid (9.33 μ L) in DMF (3 mL) were added HATU (53.0 mg) and DIPEA (0.065 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried

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over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (13.3 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.15-1.43 (3H, m), 1.95-2.41 (1H, m), 2.61-3.22 (5H, m), 3.76-5.13 (7H, m), 7.11-7.61 (8H, m), 8.13-8.38 (1H, m).

Example 50

N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A) tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

Ethanesulfonyl chloride (0.065 mL) was added to a mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (158 mg), TEA (0.127 mL), DMAP (11.2 mg) and THF (5 mL) at room temperature. After being stirred at room temperature for 2 h, the reaction mixture was quenched with saturated brine and concentrated under reduced pressure. The residual oil was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (198 mg).

MS, found: 338.9.

B) N-{(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (191 mg) and 4 M HCl/CPME solution (4 mL) was stirred at room temperature for 2 h. The solid was collected by filtration to give the title compound (149 mg).

MS: [M+H]⁺ 338.9.

C) N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

To a solution of N-{(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide hydrochloride (145 mg) and oxetane-2-carboxylic acid (0.049 mL) in DMF (5 mL) were added HATU (220 mg) and DIPEA (0.270 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (125 mg).

MS: [M+H]⁺ 423.0.

D) N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide (52.7 mg), phenylboronic acid (30.4 mg), XPhos Pd G3 (10.6 mg) and 1 M aqueous potassium

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phosphate solution (0.374 mL) in DME (1 mL) was stirred overnight at 90° C. After cooling back to room temperature, the solution was diluted with EtOAc and dried over sodium sulfate. After removal of the solvent under reduced pressure, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane), and then by preparative HPLC to give the title compound (8.7 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.01-1.29 (3H, m), 2.53-3.18 (6H, m), 3.75-5.29 (7H, m), 7.20-7.76 (9H, m), 8.18 (1H, d, J=9.8 Hz).

Example 51

N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Longer Retention Time

A mixture of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide (52.7 mg), phenylboronic acid (30.4 mg), XPhos Pd G3 (10.6 mg) and 1 M aqueous potassium phosphate solution (0.374 mL) in DME (1 mL) was stirred overnight at 90° C. After cooling back to room temperature, the solution was diluted with EtOAc and dried over sodium sulfate. After removal of the solvent under reduced pressure, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane), and then by preparative HPLC to give the title compound (11.4 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.01-1.32 (3H, m), 1.75-2.28 (1H, m), 2.52-3.23 (5H, m), 3.73-5.22 (7H, m), 7.18-7.71 (9H, m), 8.07-8.36 (1H, m).

Example 52

N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide (50.9 mg), (3-fluorophenyl)boronic acid (33.7 mg), XPhos Pd G3 (10.2 mg) and 1 M aqueous potassium phosphate solution (0.361 mL) in DME (1 mL) was stirred overnight at 90° C. After cooling back to room temperature, the solution was diluted with EtOAc and dried over sodium sulfate. After removal of the solvent under reduced pressure, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane), and then by preparative HPLC to give the title compound (9.6 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.03-1.25 (3H, m), 2.52-3.19 (6H, m), 3.84-5.25 (7H, m), 7.11-7.82 (8H, m), 8.07-8.30 (1H, m).

Example 56

N-{(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A) N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (580 mg) and 4 M hydrogen chloride CPME solution (10 mL) was stirred overnight at room

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temperature. The insoluble substance was collected by filtration to give the title compound (464 mg).

MS, found: 356.9.

B) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (200 mg), oxetane-2-carboxylic acid (78 mg) and DMF (3 mL) were added HATU (290 mg) and DIPEA (263 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added saturated aqueous ammonium chloride solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (147 mg).

MS: [M+H]⁺ 441.0.

C) N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide (147 mg), (3,5-difluorophenyl)boronic acid (105 mg), XPhos Pd G3 (28.2 mg), 1 M aqueous potassium phosphate solution (1.00 mL) and DME (2 mL) was stirred at 70° C. for 1 hr. The mixture was cooled to room temperature, and purified by silica gel column chromatography (EtOAc/hexane). The obtained solid was crystallized from EtOAc/hexane to give the title compound (101 mg) with shorter retention time.

¹H NMR (400 MHz, DMSO-d₆) δ 1.18-1.33 (3H, m), 2.26-2.45 (1H, m), 2.58-2.83 (2H, m), 2.95-3.22 (3H, m), 3.74-5.20 (7H, m), 7.14-7.52 (6H, m), 8.21 (1H, brs).

Example 66

N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (160 mg) and DIPEA (0.352 mL) in THF (2 mL) was added alpha-acetoxy-isobutyryl chloride (0.071 mL) at 0° C. and the mixture was stirred at same temperature for 10 min. To the mixture were added water (1 mL) and 4 M aqueous lithium hydroxide solution (1.02 mL) and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel,

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eluted with EtOAc/hexane) to give the title compound (172 mg).

MS: [M+H]⁺ 443.0.

B) N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (20 mg), (2,3-difluorophenyl)boronic acid (14.3 mg), XPhos Pd G3 (7.64 mg) and 1 M aqueous potassium phosphate solution (0.135 mL) in THF (0.5 mL) was stirred at 90° C. in sealed tube for 6 h. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (19.2 mg).

¹H NMR (300 MHz, CDCl₃) δ 1.30-1.43 (9H, m), 2.05-3.25 (5H, m), 3.99-4.53 (3H, m), 4.87 (1H, d, J=9.1 Hz), 4.99-5.33 (1H, m), 7.07-7.30 (5H, m), 7.35-7.48 (1H, m).

Example 67

N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (20 mg), (2,5-difluorophenyl)boronic acid (14.3 mg), XPhos Pd G3 (7.64 mg) and 1 M aqueous potassium phosphate solution (0.135 mL) in THF (0.5 mL) was stirred at 90° C. in sealed tube for 6 h. The mixture was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (17.4 mg).

¹H NMR (300 MHz, CDCl₃) δ 1.28-2.66 (10H, m), 2.87-3.22 (4H, m), 3.94-4.57 (3H, m), 4.86 (1H, d, J=10.6 Hz), 4.99-5.30 (1H, m), 6.99-7.24 (5H, m), 7.35-7.48 (1H, m).

Example 68

N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A) N-[(2S,3R)-4,4-difluoro-2-[[2-fluoro-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (75 mg), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (51.6 mg), XPhos Pd G3 (28.7 mg) and potassium acetate (33.2 mg) in toluene (1 mL) was stirred at 80° C. under nitrogen atmosphere for 5 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure to give the title compound (151 mg).

MS: [M+H]⁺ 535.2.

B) N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-4,4-difluoro-2-[[2-fluoro-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]methyl]-

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1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (0.022 g), cesium carbonate (0.040 g), 1-chloro-3-iodobenzene (7.65 μ L), PdCl₂(dppf) (6.02 mg) and DME (0.5 mL)/water (0.150 mL) was stirred overnight at 80° C. under nitrogen atmosphere. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (0.017 g).

¹H NMR (300 MHz, CDCl₃) δ 1.31-1.42 (9H, m), 2.21-2.58 (1H, m), 2.86-3.18 (4H, m), 3.99-4.53 (3H, m), 4.90 (1H, d, J=8.3 Hz), 5.01-5.29 (1H, m), 7.14-7.22 (1H, m), 7.27-7.44 (5H, m), 7.47-7.56 (1H, m).

Example 73

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide

A) tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (200 mg), TEA (166 mg) and THF (5 mL) was added methanesulfonic anhydride (143 mg) at room temperature. The reaction mixture was stirred overnight at room temperature, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (244 mg).

MS: [M-H]⁻ 441.1.

B) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (360 mg) and 4 M hydrogen chloride CPME solution (5 mL) was stirred at room temperature for 2 hr. The reaction solution was concentrated to give the title compound (300 mg).

MS, found: 342.9.

C) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(2R)-oxolan-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (300 mg), TEA (400 mg), (2R)-tetrahydrofuran-2-carboxylic acid (110 mg) and DMF (5 mL) was added HATU (361 mg) at room temperature. The mixture was stirred at room temperature for 30 min, to the mixture was added saturated aqueous sodium hydrogen carbonate solution at room temperature, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane). The obtained solid was washed with IPE to give the title compound (350 mg).

MS: [M+H]⁺ 441.0.

D) N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolan-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(2R)-oxolan-2-carbonyl]pyrrolidin-

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3-yl]methanesulfonamide (30 mg), phenylboronic acid (12.5 mg), 1 M aqueous potassium phosphate solution (0.204 mL), XPhos Pd G3 (5.76 mg) and THF (0.340 mL) was stirred at 90° C. for 1 hr. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (22 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 1.37-2.08 (4H, m), 2.60-3.12 (5H, m), 3.31-4.95 (7H, m), 7.08-7.55 (8H, m), 8.14-8.29 (1H, m).

Example 76

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (35.0 mg) and 1-hydroxycyclobutane-1-carboxylic acid (9.50 μ L) in DMF (1.5 mL) were added HATU (47.4 mg) and DIPEA (0.058 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (23.5 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.79-2.12 (4H, m), 2.51-2.62 (2H, m), 2.87 (3H, s), 2.92-3.14 (2H, m), 3.93-4.13 (1H, m), 4.19-4.41 (2H, m), 4.97-5.13 (2H, m), 7.16-7.21 (1H, m), 7.27-7.54 (8H, m).

Example 77

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (35.0 mg) and isobutyric acid (0.012 mL) in DMF (1.5 mL) were added HATU (47.4 mg) and DIPEA (0.058 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (33.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.45-1.15 (6H, m), 1.94-2.49 (1H, m), 2.74-3.17 (5H, m), 3.72-4.02 (1H, m), 4.28-4.60 (2H, m), 4.87-5.19 (1H, m), 7.09-7.24 (2H, m), 7.29-7.54 (6H, m).

Example 78

N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (35.0 mg) and bicyclo[1.1.1]pentane-1-

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carboxylic acid (9.17 μ L) in DMF (1.5 mL) were added HATU (47.4 mg) and DIPEA (0.058 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (33.2 mg).

^1H NMR (400 MHz, CDCl_3) δ 1.78-1.91 (2H, m), 2.10 (4H, s), 2.26-2.51 (1H, m), 2.77-3.12 (5H, m), 3.92-4.08 (1H, m), 4.17-4.44 (1H, m), 4.71-5.20 (2H, m), 7.11-7.24 (2H, m), 7.27-7.55 (7H, m).

Example 79

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Shorter Retention Time

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (110 mg) and oxetane-2-carboxylic acid (0.026 mL) in DMF (5 mL) were added HATU (149 mg) and DIPEA (0.183 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane), and then by preparative HPLC to give the title compound (33.2 mg).

^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 2.23-2.47 (1H, m), 2.61-3.09 (6H, m), 3.72-5.19 (7H, m), 7.11-7.56 (8H, m), 8.23 (1H, brs).

Example 81

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (100 mg) and oxetane-2-carboxylic acid (0.023 mL) in DMF (2 mL) were added HATU (130 mg) and DIPEA (0.159 mL) at room temperature. The mixture was stirred at room temperature for 1 h. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (41 mg).

^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 2.30-2.47 (1H, m), 2.64-2.78 (2H, m), 2.96-3.10 (4H, m), 3.75-5.19 (7H, m), 7.11-7.57 (7H, m), 8.14-8.32 (1H, m).

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Example 87

N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (157 mg), bicyclo[1.1.1]pentane-1-carboxylic acid (55.7 mg), TEA (209 mg) and DMF (4 mL) was added HATU (236 mg) at room temperature. The mixture was stirred at room temperature for 1 hr, to the mixture was added water, and the insoluble substance was collected by filtration to give the title compound (180 mg).

MS: $[\text{M}+\text{H}]^+$ 437.0.

B) N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide (180 mg), (3,5-difluorophenyl)boronic acid (130 mg), XPhos Pd G3 (69.8 mg), 1 M aqueous potassium phosphate solution (1.24 mL) and DME (3 mL) was stirred at 90° C. for 2 hr. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane), and then silica gel column chromatography (EtOAc/hexane). The obtained residue was crystallized from ethanol/water to give the title compound (121 mg).

^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 1.20-1.68 (2H, m), 1.90 (4H, s), 2.08-2.40 (1H, m), 2.61-2.77 (1H, m), 2.90-3.16 (4H, m), 3.70-4.31 (2H, m), 4.37-4.86 (2H, m), 7.08-7.57 (6H, m), 8.02-8.30 (1H, m).

Example 88

N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

A) tert-butyl (2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (97 mg), (3-fluorophenyl)boronic acid (46.4 mg), XPhos Pd G3 (18.7 mg) and 1 M aqueous potassium phosphate solution (0.663 mL) in THF (1 mL) was heated at 80° C. under nitrogen atmosphere for 3 h. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (100 mg).

MS: $[\text{M}-\text{H}]^-$ 497.2.

B) N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (100 mg) and 4 M HCl/CPME solution (5 mL) was stirred overnight at room

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temperature. The mixture was concentrated under reduced pressure to give the title compound (86 mg).

MS: [M+H]⁺ 399.0.

C) N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

HATU (22.7 mg) was added to a solution of N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide hydrochloride (20 mg), (R)-tetrahydrofuran-2-carboxylic acid (6.94 mg) and TEA (0.032 mL) in DMF (1 mL) at room temperature. The mixture was stirred at room temperature under a dry atmosphere for 30 min. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (18.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 1.03-2.06 (7H, m), 2.66-3.14 (4H, m), 3.41-4.74 (7H, m), 7.10-7.74 (8H, m), 8.07-8.31 (1H, m).

Example 91

N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (188 mg), oxetane-2-carboxylic acid (60.7 mg), TEA (251 mg) and DMF (4 mL) was added HATU (283 mg) at room temperature. The mixture was stirred at room temperature for 6 hr, to the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated aqueous sodium hydrogen carbonate solution, water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane), and then silica gel column chromatography (NH, EtOAc/hexane), and then HPLC (column: L-Column2 ODS (3.0 mmI.D.×50 mm, 3 μm), mobile phase: water/CH₃CN (containing 5 mM ammonium acetate)) to give the title compound (65.2 mg) with shorter retention time.

MS: [M+H]⁺ 426.9.

B) N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide (30 mg), (3,5-difluorophenyl)boronic acid (22.2 mg), XPhos Pd G3 (11.9 mg), 1 M aqueous potassium phosphate solution (0.211 mL) and THF (0.5 mL) was stirred at 90° C. for 4 hr. The mixture was

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purified by silica gel column chromatography (NH, EtOAc/hexane), and recrystallized from EtOAc/hexane to give the title compound (27.6 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.77-3.25 (7H, m), 3.68-3.92 (1H, m), 4.16-4.98 (5H, m), 5.01-5.21 (2H, m), 6.76-6.87 (1H, m), 7.01-7.10 (2H, m), 7.16-7.50 (3H, m).

Example 92

N-[(2S,3R)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

HATU (43.7 mg) was added to a solution of N-[(2S,3R)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide hydrochloride (35 mg), oxetane-2-carboxylic acid (9.39 mg) and TEA (0.053 mL) in DMF (0.5 mL) at room temperature. The mixture was stirred at room temperature for 30 min, and then additional oxetane-2-carboxylic acid (9.39 mg) was added to the mixture. The mixture was stirred at room temperature for 30 min. The mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated aqueous sodium hydrogen carbonate solution, water and saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (10.4 mg).

¹H NMR (300 MHz, CDCl₃) δ 2.31-3.21 (7H, m), 3.77-4.66 (6H, m), 4.96-5.11 (2H, m), 6.75-6.89 (1H, m), 6.99-7.15 (2H, m), 7.17-7.47 (3H, m).

Example 94

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (15.2 mg), DIPEA (21.7 mg) and THF (0.5 mL) was added 2-methylpropanoyl chloride (5.36 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added saturated aqueous sodium hydrogen carbonate solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (13.6 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.17-0.92 (6H, m), 1.22-1.35 (3H, m), 1.79-2.48 (1H, m), 2.60-2.77 (1H, m), 2.97-3.24 (3H, m), 3.70-4.22 (2H, m), 4.35-4.93 (2H, m), 7.03-7.65 (7H, m), 8.01-8.45 (1H, m).

Example 97

N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

HATU (20.46 mg) was added to a solution of N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide hydrochloride (18 mg), isobutyric acid (5.47 mg) and TEA (0.029 mL) in DMF (0.5 mL) at room temperature. The mixture was stirred at room

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temperature under a dry atmosphere for 30 min. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (12.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.11-1.38 (9H, m), 1.75-2.04 (1H, m), 2.52-2.92 (2H, m), 2.96-3.12 (1H, m), 3.15-3.24 (1H, m), 3.81-4.22 (2H, m), 4.35-4.65 (2H, m), 7.12-7.79 (8H, m), 8.10-8.30 (1H, m).

Example 98

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (20 mg) and isobutyric acid (3.95 μL) in DMF (0.5 mL) were added HATU (26.0 mg) and DIPEA (0.032 mL) at room temperature. The mixture was stirred at room temperature for 1 h. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (16 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.17-0.92 (6H, m), 1.11-1.94 (1H, m), 2.58-3.22 (5H, m), 3.65-4.26 (2H, m), 4.36-4.89 (2H, m), 7.06-7.57 (7H, m), 8.13-8.34 (1H, m).

Example 106

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide hydrochloride (30 mg) and isobutyric acid (7.29 mg) in DMF (0.5 mL) were added HATU (39.3 mg) and DIPEA (0.048 mL) at room temperature. The mixture was stirred at room temperature for 1 h. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (29.0 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.16-0.94 (6H, m), 1.19-1.95 (4H, m), 2.59-2.80 (1H, m), 2.97-3.25 (3H, m), 3.65-4.21 (2H, m), 4.33-4.89 (2H, m), 7.05-7.55 (8H, m), 8.03-8.43 (1H, m).

Example 116

N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A) N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a solution of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride

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(128 mg) and isobutyric acid (0.038 mL) in DMF (5 mL) were added HATU (195 mg) and DIPEA (0.238 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (108 mg).

MS: [M+H]⁺ 409.0.

B) N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (17 mg), (3,5-difluorophenyl)boronic acid (13.1 mg), XPhos Pd G3 (3.52 mg) and 1 M aqueous potassium phosphate solution (0.125 mL) in DME (1 mL) was stirred overnight at 80° C. After cooling back to room temperature, the solution was diluted with EtOAc and dried over sodium sulfate. After removal of the solvent under reduced pressure, the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (18.2 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.12-0.98 (6H, m), 1.03-1.88 (4H, m), 2.53-3.24 (4H, m), 3.81-4.23 (2H, m), 4.36-4.63 (2H, m), 7.16-7.78 (7H, m), 8.12-8.32 (1H, m).

Example 121

N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide (17 mg), (2,3-difluorophenyl)boronic acid (12.6 mg), XPhos Pd G3 (6.74 mg) and 1 M aqueous potassium phosphate solution (0.119 mL) in THF (0.5 mL) was stirred at 90° C. in sealed tube for 1 h. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (15.3 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.67-3.30 (7H, m), 3.67-3.93 (1H, m), 4.18-5.20 (7H, m), 7.06-7.49 (6H, m).

Example 122

N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide (17 mg), (2,5-difluorophenyl)boronic acid (12.6 mg), XPhos Pd G3 (6.74 mg) and 1 M aqueous potassium phosphate solution (0.119 mL) in THF (0.5 mL) was stirred at 90° C. in sealed tube for 1 h. The mixture was directly purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (10.9 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.66-3.32 (7H, m), 3.64-3.91 (1H, m), 4.17-5.21 (7H, m), 6.99-7.17 (3H, m), 7.17-7.30 (2H, m), 7.32-7.49 (1H, m).

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Example 124

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with Shorter Retention Time

A) tert-butyl (2S,3R)-3-[(cyclopropanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

Cyclopropanesulfonyl chloride (0.081 mL) was added to a stirred solution of tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (130 mg) and TEA (0.178 mL) in THF (2 mL) at room temperature. The mixture was stirred at the same temperature overnight, heated up to at 70° C. and stirred for 1 h. Additional cyclopropanesulfonyl chloride (0.081 mL) and TEA (0.178 mL) were added to the mixture, and the mixture was stirred overnight at 70° C. The reaction mixture was purified by column chromatography on silica gel (eluted with EtOAc/hexane) to give the title compound (130 mg).

MS, found: 411.1.

B) N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}cyclopropanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-3-[(cyclopropanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (186 mg) and 4 M HCl/CPME solution (4.0 mL) was stirred overnight at room temperature, and concentrated under reduced pressure to give the title compound (150 mg).

MS: [M+H]⁺ 411.1.

C) N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with Shorter Retention Time

To a mixture of N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}cyclopropanesulfonamide hydrochloride (80.0 mg), oxetane-2-carboxylic acid (29.3 mg), HATU (117 mg) and DMF (2.0 mL) was added DIPEA (0.200 mL) at room temperature. The mixture was stirred at room temperature for 1 h. To the mixture was added saturated aqueous sodium hydrogen carbonate solution, and the mixture was extracted with EtOAc/hexane. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane, then NH silica gel, eluted with EtOAc/hexane) to give the title compound (21.2 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.90-1.08 (3H, m), 1.21-1.33 (1H, m), 2.24-2.38 (1H, m), 2.54-2.81 (3H, m), 3.01-3.15 (1H, m), 3.70-5.22 (7H, m), 7.07-7.58 (8H, m), 8.05-8.42 (1H, m).

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Example 129

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with Shorter Retention Time

A) tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(cyclopropanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

To a mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (100 mg), TEA (83 mg) and THF (3 mL) was added cyclopropanesulfonyl chloride (96 mg) at room temperature. The mixture was stirred overnight at 70° C., and purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (75 mg).

MS: [M-H]⁻ 467.0.

B) N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}cyclopropanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(cyclopropanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (328 mg) and 4 M hydrogen chloride CPME solution (6 mL) was stirred at room temperature for 4 hr. The insoluble substance was collected by filtration to give the title compound (240 mg).

MS, found: 369.0.

C) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with Shorter Retention Time (Obtained by HPLC (Column: L-Column2 ODS (3.0 mm I.D.×50 mm, 3 μm), Mobile Phase: Water/CH₃CN (Containing 5 mM Ammonium Acetate)))

To a mixture of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}cyclopropanesulfonamide hydrochloride (35 mg), oxetane-2-carboxylic acid (13.2 mg) and DMF (1.5 mL) were added HATU (49.3 mg) and DIPEA (44.6 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added saturated aqueous ammonium chloride solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (11 mg).

MS: [M+H]⁺ 453.1.

D) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide (10 mg) with shorter retention time (obtained by HPLC (column: L-Column2 ODS (3.0 mm I.D.×50 mm, 3 μm), mobile phase: water/CH₃CN (containing 5 mM ammonium acetate))), (3-fluorophenyl)boronic acid (6.18 mg), Xphos Pd G3 (1.87 mg), 1 M aqueous

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potassium phosphate solution (0.066 mL) and DME (0.5 mL) was stirred overnight at 80° C. The mixture was cooled to room temperature, and EtOAc was added thereto. The mixture was dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (3.6 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.91-1.21 (4H, m), 2.15-2.43 (1H, m), 2.71-3.04 (3H, m), 3.08-3.26 (1H, m), 3.68-3.88 (1H, m), 4.16-5.18 (7H, m), 7.03-7.25 (3H, m), 7.27-7.44 (4H, m).

Example 131

N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (187 mg) and 4 M HCl/CPME solution (5 mL) was stirred at room temperature for 2 h. The solid was collected by filtration to give the title compound (145 mg).

MS: [M+H]⁺ 325.0.

B) N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (51.1 mg) and isobutyric acid (0.020 mL) in DMF (2 mL) were added HATU (81 mg) and DIPEA (0.099 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (53.0 mg).

MS: [M+H]⁺ 395.0.

C) N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chlorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide (14.5 mg), (2,3-difluorophenyl)boronic acid (11.6 mg), XPhos Pd G3 (3.11 mg) and 1 M aqueous potassium phosphate solution (0.110 mL) in DME (0.5 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (17.1 mg).

¹H NMR (400 MHz, DMSO-d₆) δ 0.16-0.98 (6H, m), 1.78-2.36 (1H, m), 2.54-3.13 (5H, m), 3.75-4.72 (4H, m), 7.26-7.64 (7H, m), 8.10-8.36 (1H, m).

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Example 133

N-[(2S,3R)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) N-((2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)pyrrolidin-3-yl) methanesulfonamide

HATU (283 mg) was added to a solution of N-((2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoropyrrolidin-3-yl) methanesulfonamide hydrochloride (188 mg), oxetane-2-carboxylic acid (60.7 mg) and TEA (0.345 mL) in DMF (4 mL) at room temperature. The mixture was stirred at room temperature for 6 h. The mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated aqueous sodium hydrogen carbonate solution, water and saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane, then NH silica gel, eluted with EtOAc/hexane) to give the title compound (84.5 mg).

MS: [M+H]⁺ 426.9

B) N-[(2S,3R)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-((2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)pyrrolidin-3-yl) methanesulfonamide (20 mg), (2,3-difluorophenyl)boronic acid (14.8 mg), XPhos Pd G3 (7.93 mg) and 1 M aqueous potassium phosphate solution (0.141 mL) in THF (0.5 mL) was stirred at 90° C. in sealed tube for 1 h. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (10.1 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.31-3.22 (7H, m), 3.77-3.94 (1H, m), 4.01-4.20 (1H, m), 4.24-5.17 (6H, m), 7.06-7.48 (6H, m).

Example 144

N-[(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (146 mg), cyclopropanecarboxylic acid (47.8 mg) and DMF (4 mL) were added HATU (211 mg) and DIPEA (191 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added saturated aqueous ammonium chloride solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (144 mg).

MS: [M-H]⁻ 423.0.

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B) N'-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

To a mixture of N'-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (16.0 mg), (3,5-difluorophenyl)boronic acid (12.0 mg), 1 M aqueous potassium phosphate solution (0.116 mL) and DME (0.500 mL) was added XPhos Pd G3 (3.2 mg) at room temperature. The mixture was stirred overnight under nitrogen atmosphere at 90° C., and purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (15.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.43-1.49 (8H, m), 2.85-5.03 (9H, m), 6.72-7.61 (6H, m).

Example 145

N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric Diamide

A) tert-butyl (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

A mixture of dimethylsulfamoyl chloride (3 mL), tert-butyl (2S,3R)-3-amino-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (400 mg) and DMAP (240 mg) was stirred overnight at 50° C. under nitrogen atmosphere. The reaction mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (500 mg).

MS: [M-H]⁻ 512.1.

B) N'-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide Hydrochloride

A mixture of 4 M HCl/CPME solution (12.2 mL) and tert-butyl (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (500 mg) was stirred overnight at 45° C. under nitrogen atmosphere. The reaction mixture was concentrated, and the solids were washed with EtOAc and filtered to afford the title compound (435 mg).

MS: [M+H]⁺ 414.0.

C) N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric Diamide

To a solution of N'-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide hydrochloride (20 mg), 1-hydroxycyclobutanecarboxylic acid (5.16 mg) and DIPEA (7.76 μL) in DMF (1 mL) was added HATU (16.9 mg) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (12.7 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.40-1.59 (1H, m), 1.74-1.91 (1H, m), 1.92-2.08 (2H, m), 2.47-2.60 (2H, m), 2.69-

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2.75 (6H, m), 2.89-3.02 (2H, m), 3.06-3.22 (1H, m), 3.90-4.07 (1H, m), 4.08-4.22 (1H, m), 4.22-4.43 (1H, m), 5.04 (2H, brs), 7.10-7.19 (1H, m), 7.24-7.38 (3H, m), 7.39-7.45 (2H, m), 7.48-7.53 (2H, m).

Example 146

N'-{(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A) N'-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N'-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide hydrochloride (47.9 mg), isobutyric acid (20.2 mg), DIPEA (0.080 mL) and DMF (0.80 mL) was added HATU (96.2 mg) at room temperature. The mixture was stirred overnight at room temperature. The mixture was diluted with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (44.6 mg).

MS: [M+H]⁺ 427.0.

B) N'-{(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

To a mixture of N'-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (15.9 mg), (3,5-difluorophenyl)boronic acid (12.0 mg), 1 M aqueous potassium phosphate solution (0.100 mL) and DME (0.300 mL) was added XPhos Pd G3 (3.0 mg) at room temperature. The mixture was stirred overnight at 90° C. under nitrogen atmosphere. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (13.4 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.52 (1H, d, J=6.4 Hz), 0.85-1.49 (8H, m), 1.82-5.09 (10H, m), 6.75-7.55 (6H, m).

Example 147

N'-{(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

To a mixture of N'-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (14.5 mg), (2,5-difluorophenyl)boronic acid (10.0 mg), 1 M aqueous potassium phosphate solution (0.100 mL) and DME (0.300 mL) was added XPhos Pd G3 (3.0 mg) at room temperature. The mixture was stirred at 120° C. for 20 min under microwave irradiation. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (9.70 mg).

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¹H NMR (400 MHz, CDCl₃) δ 0.53 (1H, d, J=6.5 Hz), 0.83-1.48 (8H, m), 1.90-5.01 (10H, m), 6.98-7.55 (6H, m).

Example 171

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]cyclopropanesulfonamide

To a mixture of N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}cyclopropanesulfonamide hydrochloride (44.3 mg), isobutyric acid (15 μL), DIPEA (50 μL) and DMF (0.80 mL) was added HATU (60.0 mg) at room temperature. The mixture was stirred at room temperature for 3 days. To the mixture was added saturated aqueous sodium hydrogen carbonate solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (30.6 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.52 (1H, d, J=6.5 Hz), 0.76-1.39 (9H, m), 1.84-5.16 (9H, m), 6.98-7.54 (8H, m).

Example 202

N-{(2S,3R)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (15 mg), (3-fluorophenyl)boronic acid (9.88 mg), XPhos Pd G3 (2.99 mg) and 1 M aqueous potassium phosphate solution (0.106 mL) in DME (1 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (17.1 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.05-1.49 (8H, m), 2.86-5.01 (9H, m), 7.04-7.44 (7H, m).

Example 205

N-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (13 mg), phenylboronic acid (7.46 mg), XPhos Pd G3 (2.59 mg) and 1 M aqueous potassium phosphate solution (0.092 mL) in DME (1 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (4.90 mg).

¹H NMR (400 MHz, CDCl₃) δ -0.13-0.18 (2H, m), 0.49-1.48 (6H, m), 2.83-4.97 (9H, m), 7.15-7.54 (8H, m).

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Example 210

(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide

A) tert-butyl 2-[(3-chloro-2-fluorophenyl)methyl]-1H-pyrrole-1-carboxylate

To a mixture of 1-(bromomethyl)-3-chloro-2-fluorobenzene (40.0 g), (1-(tert-butoxycarbonyl)-1H-pyrrol-2-yl)boronic acid (49.1 g), 1 M aqueous potassium phosphate solution (537 mL) and THF (900 mL) was added PdCl₂(dppf) (2.62 g) under argon atmosphere at room temperature. The mixture was refluxed for 16 hr, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (54.7 g).

MS: [M-H]⁻ 308.0.

B) 2-[(3-chloro-2-fluorophenyl)methyl]-1H-pyrrole

To a mixture of tert-butyl 2-[(3-chloro-2-fluorophenyl)methyl]-1H-pyrrole-1-carboxylate (54.7 g) and MeOH (250 mL) was added 28% sodium methoxide methanol solution (170 g) under argon atmosphere at room temperature. The mixture was stirred at room temperature for 5 hr, and the reaction mixture was added to aqueous ammonium chloride solution under ice cooling, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure to give the title compound (35.9 g).

MS: [M+H]⁺ 209.9.

C) benzyl 2-[(3-chloro-2-fluorophenyl)methyl]-2,5-dihydro-1H-pyrrole-1-carboxylate

To a mixture of 2-[(3-chloro-2-fluorophenyl)methyl]-1H-pyrrole (35.9 g), zinc (112 g) and EtOH (500 mL) was added dropwise conc. hydrochloric acid (171 mL) at 85° C. The mixture was stirred at 85° C. for 30 min, and the reaction mixture was poured into ice water, and basified with 8 M aqueous sodium hydroxide solution (pH=8). The insoluble substance was removed by filtration, and the filtrate was extracted with EtOAc/THF. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. To a mixture of the obtained residue, THF (400 mL) and water (400 mL) were added sodium hydrogen carbonate (21.6 g) and 1-(((benzyloxy)carbonyl)oxy)pyrrolidine-2,5-dione (44.8 g) at room temperature. The mixture was stirred at room temperature for 15 hr, and the reaction mixture was poured into water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane), and then silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (25.7 g).

MS: [M+H]⁺ 346.0.

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D) rac-benzyl (1S,2S,5R)-2-[(3-chloro-2-fluorophenyl)methyl]-6-oxa-3-azabicyclo[3.1.0]hexane-3-carboxylate

To a mixture of benzyl 2-[(3-chloro-2-fluorophenyl)methyl]-2,5-dihydro-1H-pyrrole-1-carboxylate (25.7 g), 1,1,1-trifluoroacetone (70.3 g), sodium 2,2'-((2-(bis(carboxymethyl)amino)ethyl)imino)diacetate dihydrate (0.055 g) and CH₃CN (180 mL)/water (120 mL) were added potassium peroxymonosulfate (137 g) and sodium hydrogen carbonate (94 g) over 35 min at 0° C. The mixture was vigorously stirred at 0° C. for 2 hr 30 min, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane), and then silica gel column chromatography (NH₃, EtOAc/hexane) to give the title compound (24.6 g).

MS: [M+H]⁺ 362.0.

E) rac-benzyl (2S,3S,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-hydroxypyrrolidine-1-carboxylate

A mixture of rac-benzyl (1S,2S,5R)-2-[(3-chloro-2-fluorophenyl)methyl]-6-oxa-3-azabicyclo[3.1.0]hexane-3-carboxylate (24.0 g) and N,N-diethylethanamine trihydrofluoride (64.1 g) was stirred at 120° C. for 17 hr. The mixture was poured into ice water, and extracted with EtOAc. The organic layer was separated, washed with aqueous sodium hydrogen carbonate solution and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (24.2 g).

MS: [M+H]⁺ 382.0.

F) rac-benzyl (2S,3S,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate

To a mixture of rac-benzyl (2S,3S,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-hydroxypyrrolidine-1-carboxylate (24.2 g), pyridine (100 g) and Et₂O (370 mL) was added dropwise trifluoromethanesulfonic anhydride (53.6 g) under argon atmosphere at 0° C. The mixture was stirred at room temperature for 2 hr 30 min, poured into ice water, and extracted with EtOAc. The organic layer was separated, washed successively with 10% aqueous citric acid solution, aqueous sodium hydrogen carbonate solution and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (29.0 g).

MS: [M+H]⁺ 514.0.

G) rac-benzyl (2S,3R,4S)-3-azido-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate

To a mixture of rac-benzyl (2S,3S,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-[(trifluoromethanesulfonyl)oxy]pyrrolidine-1-carboxylate (29.0 g) and CH₃CN (335 mL) was added tetra-n-butylammonium azide (48.2 g) at room temperature. The mixture was stirred at 80° C. for 1 hr, poured into ice water, and extracted with EtOAc. The

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organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (22.2 g).

MS: [M+H]⁺ 407.0.

H) rac-benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate

To a mixture of rac-benzyl (2S,3R,4S)-3-azido-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate (2.17 g) and THF (42.7 mL)/water (10.7 mL) was added triphenylphosphine (1.68 g) at room temperature. The mixture was stirred at 50° C. for 15 hr, to the mixture was added saturated brine at room temperature, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (methanol/EtOAc) to give the title compound (1.74 g).

MS: [M+H]⁺ 381.0.

I) benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate

rac-Benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate (668 mg) was resolved by HPLC (column: CHIRALPAK ID, 50 mmID×500 mmL, 20 μm, mobile phase: hexane/2-propanol/diethylamine=600/400/1) to give the title compound (251 mg) with longer retention time.

MS: [M+H]⁺ 381.1.

J) benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate

To a mixture of benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate (289 mg), TEA (189 mg) and THF (4.0 mL) was added ethanesulfonyl chloride (190 mg) at 0° C. The mixture was stirred at room temperature for 2 hr, to the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (295 mg).

MS: [M+H]⁺ 473.0.

K) benzyl (2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate

To a mixture of benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (340 mg), (3-fluorophenyl)boronic acid (150 mg), XPhos Pd G3 (30.9 mg) and DME (4.0 mL) was added 1 M aqueous potassium phosphate solution (1.2 mL) at room temperature, and the mixture was stirred under nitrogen atmosphere at 90° C. for 2 hr. To the reaction mixture were added 1 M aqueous potassium phosphate solution (0.4 mL), (3-fluorophenyl)boronic acid (50.5 mg) and XPhos Pd G3 (15.2 mg) at 90° C. The mixture was stirred under nitrogen atmosphere at 90° C. for 1 hr, to the

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reaction mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (250 mg).

MS: [M+H]⁺ 533.2.

L) N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide

A mixture of benzyl (2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (240 mg), 10% palladium on carbon (24.0 mg) and MeOH (5.0 mL)/THF (3.0 mL) was stirred overnight under normal pressure of hydrogen atmosphere at room temperature. The catalyst was removed by filtration, and the filtrate was concentrated under reduced pressure to give the title compound (178 mg).

MS: [M+H]⁺ 399.0.

M) (2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide

To a mixture of N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide (10.0 mg), DIPEA (22.2 mg) and THF (0.200 mL) was added dimethylcarbonyl chloride (11.7 mg) at 0° C. The mixture was stirred at room temperature for 3 hr, and the reaction solution was dried by blowing of nitrogen. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (6.7 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.81-1.44 (3H, m), 2.59-4.17 (13H, m), 4.73-4.90 (2H, m), 5.06-5.27 (1H, m), 6.98-7.53 (7H, m).

Example 211

N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide (8.9 mg), bis(trichloromethyl) carbonate (4.0 mg) and THF (0.200 mL) was added DIPEA (7.40 mg) at 0° C. The mixture was stirred at 0° C. for 30 min, and the reaction solution was concentrated under reduced pressure. The obtained residue was diluted with THF (0.200 mL), azetidine (3.83 mg) was added thereto at room temperature, and the mixture was stirred at room temperature for 1 hr. The mixture was dried by blowing of nitrogen, and the residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (4.8 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.81-1.35 (3H, m), 2.06-2.16 (2H, m), 2.85-4.23 (11H, m), 4.64-5.29 (3H, m), 6.98-7.54 (7H, m).

Example 212

N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-

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yl]ethanesulfonamide (11.6 mg), cyclopropanecarboxylic acid (4.6 mg), DIPEA (11.8 mg) and DMF (0.200 mL) was added HATU (20.5 mg) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was dried by blowing of nitrogen. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (8.4 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.05-1.47 (7H, m), 2.64-5.43 (11H, m), 6.98-7.54 (7H, m).

Example 220

N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A) N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide Hydrobromide

To benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (78 mg) was added 30% hydrogen bromide-acetic acid solution (0.8 mL) at room temperature. The mixture was stirred at room temperature for 1 hr, and the reaction solution was concentrated under reduced pressure. The obtained residue was subjected to azeotrope with toluene, and suspended in IPE, and the insoluble substance was collected by filtration to give the title compound (65 mg).

MS, found: 338.9.

B) N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide hydrobromide (65 mg), DIPEA (60.0 mg) and THF (0.8 mL) was added 1-chloro-2-methyl-1-oxopropan-2-yl acetate (30.6 mg) at 0° C. The mixture was stirred at the same temperature for 1 hr, and water (0.4 mL) and 4 M aqueous lithium hydroxide solution (0.387 mL) were added thereto at 0° C. The mixture was stirred at room temperature for 1 hr, to the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure to give the title compound (66 mg).

MS: [M+H]⁺ 425.0.

C) N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (22 mg), (3,5-difluorophenyl)boronic acid (16.4 mg), XPhos Pd G3 (4.38 mg), 1 M aqueous potassium phosphate solution (0.155 mL) and DME (0.8 mL) was stirred under nitrogen atmosphere at 80° C. for 1 hr. The mixture was purified by silica gel column chromatography (EtOAc/hexane), and then silica gel column chromatography (NH₄ EtOAc/hexane) to give the title compound (20.2 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.20 (3H, t, J=7.0 Hz), 1.40 (6H, s), 2.85 (2H, q, J=6.9 Hz), 3.02 (2H, dd, J=13.4, 6.1

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Hz), 3.22 (1H, dd, J=13.9, 7.8 Hz), 3.96-4.30 (3H, m), 4.75-5.02 (2H, m), 5.10-5.32 (1H, m), 6.82 (1H, t, J=8.9 Hz), 7.02-7.13 (2H, m), 7.16-7.23 (1H, m), 7.27-7.50 (2H, m).

Example 225

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

A) N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl}ethanesulfonamide

To a solution of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide hydrochloride (0.310 g) and 3-fluorocyclobutane-1-carboxylic acid (0.128 mL) in DMF (10 mL) were added HATU (0.450 g) and DIPEA (0.551 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane), and then purified by preparative HPLC (Column: CHIRALCEL OD-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=850/150 (v/v) to give the title compound (177 mg) with shorter retention time by HPLC (Column: CHIRALCEL OD-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=850/150 (v/v)).

MS: [M+H]⁺ 457.1.

B) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl}ethanesulfonamide (50 mg), (3-fluorophenyl)boronic acid (30.6 mg) and XPhos Pd G3 (9.26 mg) were added DME (2 mL) and 1 M aqueous potassium phosphate solution (0.328 mL) at room temperature. The mixture was stirred at 80° C. under nitrogen atmosphere for 1 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (54.1 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.29-1.48 (3H, m), 1.55-2.03 (2H, m), 2.25-3.26 (7H, m), 3.33-4.55 (4H, m), 4.88-5.24 (2H, m), 7.03-7.48 (7H, m).

Example 226

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(1s,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

A) N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1s,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl}ethanesulfonamide

To a solution of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide hydrochloride (0.310 g) and 3-fluorocyclobutane-1-carbox-

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ylic acid (0.128 mL) in DMF (10 mL) were added HATU (0.450 g) and DIPEA (0.551 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane), and then purified by preparative HPLC (Column: CHIRALCEL OD-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=850/150 (v/v) to give the title compound (134 mg) with longer retention time by HPLC (Column: CHIRALCEL OD-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=850/150 (v/v)).

MS: [M+H]⁺ 457.1.

B) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(1s,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1s,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl}ethanesulfonamide (45 mg), (3-fluorophenyl)boronic acid (17.9 mg), XPhos Pd G3 (8.34 mg) and 1 M aqueous potassium phosphate solution (0.295 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane). The residue was crystallized from EtOAc/hexane to give the title compound (19.5 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.29-1.47 (3H, m), 1.80-3.92 (11H, m), 4.19-5.01 (4H, m), 7.02-7.46 (7H, m).

Example 236

N-{(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A) N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide

A mixture of benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (179 mg) and 30% hydrogen bromide/acetic acid solution (2.0 mL) was stirred overnight at room temperature. Toluene (20 mL) was added thereto, and the mixture was concentrated under reduced pressure. The residue was diluted with EtOAc, and the insoluble substance was collected by filtration, and washed with EtOAc to give a solid. To a mixture of the obtained solid, THF (2.0 mL) and 2M aqueous sodium hydroxide solution (0.50 mL) was added cyclopropanecarbonyl chloride (81 mg) at room temperature. The mixture was stirred at room temperature for 30 min. To the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (88.2 mg).

MS: [M+H]⁺ 407.0.

B) N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide (22.0 mg), (3,5-difluorophenyl)boronic acid (17.3 mg), 1M aqueous potassium phosphate solution (0.130 mL) and DME (0.500 mL) was added XPhos Pd G3 (5.0 mg) at room temperature. The mixture was stirred overnight under nitrogen atmosphere at 90° C. The mixture was purified by silica gel column chromatography (NH₄ EtOAc/hexane), and recrystallized from EtOAc/hexane to give the title compound (13.5 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.05-1.48 (7H, m), 2.68-4.30 (8H, m), 4.60-5.41 (3H, m), 6.76-7.54 (6H, m).

Example 239

N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide (22.0 mg), phenylboronic acid (17.0 mg), 1 M aqueous potassium phosphate solution (0.130 mL) and DME (0.500 mL) was added XPhos Pd G3 (6.0 mg) at room temperature. The mixture was stirred overnight under nitrogen atmosphere at 90° C. The mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane), then recrystallization (EtOAc/heptane) to give the title compound (13.2 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.09-1.47 (7H, m), 2.60-4.31 (8H, m), 4.60-5.38 (3H, m), 6.98-7.57 (8H, m).

Example 245

N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (141 mg) and cyclopropanecarboxylic acid (0.044 mL) in DMF (4 mL) were added HATU (212 mg) and DIPEA (0.260 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (134 mg).

MS: [M+H]⁺ 411.0.

B) N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-

3-yl]methanesulfonamide (134 mg), 4,4,4',4',5,5,5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (124 mg), XPhos Pd G3 (55.2 mg) and potassium acetate (64.0 mg) in toluene (4 mL) was stirred at 80° C. under nitrogen atmosphere for 8 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The mixture of the obtained residue (0.041 g), cesium carbonate (0.080 g), 1-bromo-3-chloro-5-fluorobenzene (0.051 g), PdCl₂(dppf) (0.012 g) and DME (1 mL)/water (0.300 mL) was stirred in a sealed flask at 80° C. under nitrogen atmosphere for 1 h. The solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (13.8 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.05-1.47 (5H, m), 2.83-5.42 (10H, m), 7.07-7.47 (6H, m).

Example 248

N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (144 mg), 4,4,4',4',5,5,5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (129 mg), XPhos Pd G3 (57.4 mg) and potassium acetate (66.5 mg) in toluene (4 mL) was stirred at 80° C. under nitrogen atmosphere for 36 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The mixture of the obtained residue (0.043 g), cesium carbonate (0.081 g), 1-bromo-3-chloro-5-fluorobenzene (0.052 g), PdCl₂(dppf) (0.012 g) and DME (1 mL)/water (0.300 mL) was stirred in a sealed flask at 80° C. under nitrogen atmosphere for 1 h. The solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (6.00 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.06-1.51 (8H, m), 2.84-5.44 (9H, m), 7.05-7.48 (6H, m).

Example 249

N-[(2S,3R)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (144 mg), 4,4,4',4',5,5,5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (129 mg), XPhos Pd G3 (57.4 mg) and potassium acetate (66.5 mg) in toluene (4 mL) was stirred at 80° C. under nitrogen atmosphere for 36 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The mixture of the obtained residue (0.043 g), cesium carbonate (0.081 g), 1-bromo-3-chloro-2-fluorobenzene (0.052 g) and PdCl₂(dppf) (0.012 g) in DME (1 mL)/water (0.300 mL) was stirred in a sealed flask at 80° C. under nitrogen atmosphere for 1 h. The solvent was removed under reduced pressure, and the residue was purified

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fied by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (9.30 mg).

¹H NMR (400 MHz, CDCl₃) δ -0.11-0.22 (1H, m), 0.47-1.48 (7H, m), 2.79-4.98 (9H, m), 6.95-7.61 (6H, m).

Example 250

N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide (134 mg), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (124 mg), XPhos Pd G3 (55.2 mg) and potassium acetate (64.0 mg) in toluene (4 mL) was stirred at 80° C. under nitrogen atmosphere for 8 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The mixture of the obtained residue (0.041 g), cesium carbonate (0.080 g), 1-chloro-3-iodobenzene (0.058 g) and PdCl₂(dppf) (0.012 g) in DME (1 mL)/water (0.300 mL) was stirred in a sealed flask at 80° C. under nitrogen atmosphere for 1 h. The solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (0.013 g).

¹H NMR (400 MHz, CDCl₃) δ 0.07-1.48 (5H, m), 2.76-5.27 (10H, m), 7.12-7.51 (7H, m).

Example 251

N-[(2S,3R)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-1-(cyclopropanecarbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide (144 mg), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (129 mg), XPhos Pd G3 (57.4 mg) and potassium acetate (66.5 mg) in toluene (4 mL) was stirred at 80° C. under nitrogen atmosphere for 36 h. The reaction mixture was diluted with water and extracted with EtOAc. The extract was washed with saturated brine, dried over magnesium sulfate, filtered and concentrated under reduced pressure. The mixture of the obtained residue (0.043 g), cesium carbonate (0.081 g), 2-bromo-4-chloro-1-fluorobenzene (0.052 g), PdCl₂(dppf) (0.012 g) and DME (1 mL)/water (0.300 mL) was stirred in a sealed flask at 80° C. under nitrogen atmosphere for 1 h. The solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (10.6 mg).

¹H NMR (400 MHz, CDCl₃) δ 0.11-1.47 (8H, m), 2.77-5.39 (9H, m), 6.97-7.61 (6H, m).

Example 257

N-[(2S,3R)-1-butanoyl-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide

To a solution of N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfo-

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namide hydrochloride (0.020 g) and butyric acid (6.08 μL) in DMF (1 mL) were added HATU (0.025 g) and DIPEA (0.031 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane), and then purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (9.90 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 0.44-0.69 (3H, m), 0.70-1.33 (6H, m), 1.81-2.31 (1H, m), 2.56-3.24 (4H, m), 3.64-4.82 (4H, m), 7.08-7.58 (7H, m), 8.11-8.41 (1H, m).

Example 267

N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A) tert-butyl (2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (51 mg), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (42.5 mg), XPhos Pd G3 (18.9 mg), potassium acetate (21.9 mg) and toluene (2 mL) was stirred under nitrogen atmosphere at 80° C. for 36 hr. The reaction mixture was diluted with water, and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. A mixture of the obtained residue, cesium carbonate (108 mg), 1-chloro-3-iodobenzene (39.3 mg), PdCl₂(dppf) (16.1 mg) and DME (1 mL)/water (0.3 mL) was stirred under sealed condition, under nitrogen atmosphere at 80° C. for 1 hr. The solvent was evaporated under reduced pressure, and the residue was purified by silica gel column chromatography (NH, EtOAc/hexane) to give the title compound (22 mg).

MS: [M-H]⁻ 531.1.

B) N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide Hydrochloride

A mixture of tert-butyl (2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (22 mg) and 4M hydrogen chloride/CPME solution (15 mL) was stirred overnight at room temperature. The insoluble substance was collected by filtration to give the title compound (4.5 mg).

MS: [M-H]⁻ 431.0.

C) N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

To a mixture of N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (4.5 mg), oxetane-2-

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carboxylic acid (1.47 mg) and DMF (1 mL) were added HATU (5.47 mg) and DIPEA (4.96 mg) at room temperature. The mixture was stirred overnight at room temperature, and concentrated under reduced pressure. The residue was purified by HPLC (YMC-Actus Triant C18 mobile phase: water/CH₃CN (containing 10 mM ammonium bicarbonate)) to give the title compound (0.4 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.18-1.32 (3H, m), 2.23-2.40 (1H, m), 2.61-3.16 (5H, m), 3.78-5.11 (7H, m), 7.12-7.61 (7H, m), 8.12-8.42 (1H, m).

Example 272

N-[(2S,3R)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

To a solution of N-[(2S,3R)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide hydrochloride (0.030 g) and 1-hydroxycyclobutane-1-carboxylic acid (10.2 μL) in DMF (1 mL) were added HATU (0.036 g) and DIPEA (0.045 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (0.016 g).

¹H NMR (400 MHz, CDCl₃) δ 1.36 (3H, t, J=7.3 Hz), 1.77-1.92 (1H, m), 1.96-2.11 (3H, m), 2.45-2.62 (2H, m), 2.90-3.19 (4H, m), 3.35-3.67 (1H, m), 3.98-4.39 (3H, m), 4.86-5.08 (2H, m), 6.78-6.86 (1H, m), 7.02-7.12 (2H, m), 7.15-7.23 (1H, m), 7.24-7.30 (1H, m), 7.35-7.45 (1H, m).

Example 302

N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

A) benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

A mixture of benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (129 mg), phenylboronic acid (66.5 mg), XPhos Pd G3 (23.1 mg) and 1 M aqueous potassium phosphate solution (0.818 mL) in DME (4 mL) was stirred at 80° C. under nitrogen atmosphere for 1 h. The mixture was diluted with water and extracted with EtOAc. The combined organic layers were dried over magnesium sulfate, filtered and concentrated under reduced pressure. The residue obtained was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (92 mg).

MS: [M+H]⁺ 515.1.

B) N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide Hydrobromide

To benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (90 mg) was added 30% HBr/acetic acid

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solution (1 mL) at room temperature. The mixture was stirred at room temperature for 1.5 h, and the solvent was evaporated with toluene. The solid obtained was suspended in IPE and collected by filtration to give the title compound (80.9 mg).

MS: [M+H]⁺ 381.0.

C) N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

To a solution of N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide hydrobromide (35 mg) and oxetane-2-carboxylic acid (23.2 mg) in DMF (2 mL) were added HATU (43.3 mg) and DIPEA (0.053 mL) at 0° C. The mixture was stirred at room temperature for 1.5 h. The mixture was quenched with water and extracted with EtOAc. The organic layer was dried over magnesium sulfate, filtered and concentrated under reduced pressure. The residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (11.5 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.19-1.31 (3H, m), 2.30-2.46 (1H, m), 2.60-2.86 (2H, m), 3.02-3.20 (3H, m), 3.46-3.68 (1H, m), 3.77-4.28 (4H, m), 4.39-5.40 (3H, m), 7.09-7.61 (8H, m), 7.81 (1H, br s).

Example 304

N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A) benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

To a mixture of benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (129 mg), (3,5-difluorophenyl)boronic acid (94.6 mg), XPhos Pd G3 (21.9 mg) and DME (2.0 mL) was added 1 M aqueous potassium phosphate solution (0.60 mL) at room temperature. The mixture was stirred at 90° C. under nitrogen atmosphere for 4 h. To the mixture was added water, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (30.4 mg).

MS: [M+H]⁺ 551.2.

B) N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with Shorter Retention Time

A mixture of benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (30.4 mg) and 10% Pd—C (3.0 mg) in MeOH (1.0 mL) was hydrogenated under balloon pressure overnight at room temperature. The catalyst was removed by filtration, and the filtrate was concentrated under reduced pressure. To a mixture of the obtained residue, 2-oxetanecarboxylic acid (10.2 mg), DIPEA (0.030

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mL) and DMF (1.0 mL) was added HATU (36.1 mg) at room temperature. The mixture was stirred at room temperature for 1 h. To the mixture was added saturated aqueous sodium hydrogen carbonate solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (5.2 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.17-1.33 (3H, m), 2.22-2.45 (1H, m), 2.58-3.20 (5H, m), 3.48-5.42 (8H, m), 7.13-7.51 (6H, m), 7.73-7.91 (1H, m).

Example 305

N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with Longer Retention Time

A mixture of benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (30.4 mg) and 10% Pd—C (3.0 mg) in MeOH (1.0 mL) was hydrogenated under balloon pressure overnight at room temperature. The catalyst was removed by filtration, and the filtrate was concentrated under reduced pressure. To a mixture of the obtained residue, 2-oxetanecarboxylic acid (10.2 mg), DIPEA (0.030 mL) and DMF (1.0 mL) was added HATU (36.1 mg) at room temperature. The mixture was stirred at room temperature for 1 h. To the mixture was added saturated aqueous sodium hydrogen carbonate solution, and the mixture was extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (4.1 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.19-1.33 (3H, m), 2.22-2.44 (1H, m), 2.61-3.22 (4H, m), 3.42-5.41 (9H, m), 7.12-7.56 (6H, m), 7.73-8.00 (1H, m).

Example 307

N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with Shorter Retention Time

A) benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

To a mixture of benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (129 mg), (2,3-difluorophenyl)boronic acid (91.9 mg), XPhos Pd G3 (21.9 mg) and DME (2.0 mL) was added 1 M aqueous potassium phosphate solution (0.60 mL) at room temperature. After being stirred at 90° C. under nitrogen atmosphere for 4 h, the mixture was concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (32.7 mg).

MS: [M+H]⁺ 551.2.

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B) N-{(2S,3R,4S)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A mixture of benzyl (2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (32.7 mg) and 10% Pd—C (3.0 mg) in MeOH (1.0 mL) was hydrogenated under balloon pressure overnight at room temperature. The catalyst was removed by filtration, and the filtrate was concentrated under reduced pressure to give the title compound (23.6 mg).

MS: [M+H]⁺ 417.1.

C) N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with Shorter Retention Time

To a mixture of N-{(2S,3R,4S)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide (23.6 mg), 2-oxetanecarboxylic acid (16.1 mg), DIPEA (0.050 mL) and DMF (1.0 mL) was added HATU (26.0 mg) at room temperature. After being stirred at room temperature for 1 h, the mixture was concentrated under reduced pressure. The residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (8.7 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.24 (3H, t, J=7.3 Hz), 2.23-2.45 (1H, m), 2.58-2.89 (2H, m), 3.00-3.18 (3H, m), 3.41-5.39 (8H, m), 7.07-7.59 (6H, m), 7.82 (1H, br s).

Example 346

N-{(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl}ethanesulfonamide

A) N-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide Hydrobromide

To benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoropyrrolidine-1-carboxylate (2.18 g) was added 30% HBr/acetic acid solution (20 mL) at 0° C. After being stirred at room temperature for 30 min, the mixture was concentrated under reduced pressure with heptane, and the residue was diluted with IPE. The insoluble material was collected by filtration, and washed with IPE to give the title compound (1.83 g).

MS: [M+H]⁺ 339.0.

B) N-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl}ethanesulfonamide

To a mixture of N-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide hydrobromide (195 mg) and DIPEA (0.402 mL) in THF (4 mL) was added alpha-acetoxy-isobutryl chloride (0.081 mL) at 0° C., and the mixture was stirred at same temperature for 10 min. To the mixture were added water (1 mL) and 4 M aqueous lithium hydroxide solution (1.16 mL), and the mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution, and extracted with EtOAc. The

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extract was washed with saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (180 mg).

MS: [M+H]⁺ 425.0.

C) N-[(2S,3S,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide (45 mg), (3-chlorophenyl)boronic acid (21.5 mg), XPhos Pd G3 (8.96 mg) and 1 M aqueous potassium phosphate solution (0.318 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) and then column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (11.4 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.10-1.22 (3H, m), 1.40 (6H, s), 2.73-3.27 (5H, m), 3.94-4.31 (3H, m), 4.76-5.33 (3H, m), 7.13-7.55 (7H, m).

Example 374

N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A) N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-(3-chloro-2-fluorobenzyl)-4-fluoropyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]methanesulfonamide hydrobromide (100 mg) and bicyclo[1.1.1]pentane-1-carboxylic acid (0.044 mL) in DMF (3 mL) were added HATU (141 mg) and DIPEA (0.129 mL) at 0° C. The mixture was stirred at room temperature for 1.5 h, then it was quenched with a saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The combined organic layers were washed with saturated brine, dried over magnesium sulfate, filtered and concentrated. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (102 mg).

MS: [M+H]⁺ 419.0.

B) N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-(3-chloro-2-fluorobenzyl)-4-fluoropyrrolidin-3-yl]methanesulfonamide (20 mg), (3,5-difluorophenyl)boronic acid (15.1 mg), XPhos Pd G3 (4.04 mg) and 1 M aqueous potassium phosphate solution (0.143 mL) in DME (0.8 mL) was stirred at 80° C. under nitrogen atmosphere for 2 h. The mixture was purified by column chromatography (silica gel, eluted with EtOAc/hexane) followed by preparative HPLC (Column: L-Column 2 ODS, mobile phase: H₂O 5 mM NH₄HCO₃/CH₃CN) to give the title compound (15.5 mg).

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¹H NMR (400 MHz, CDCl₃) δ 1.75-1.94 (1H, m), 2.07-2.38 (5H, m), 2.53 (3H, d, J=26.2 Hz), 2.88-3.31 (3H, m), 3.62-4.47 (3H, m), 4.65-5.35 (3H, m), 6.78-6.88 (1H, m), 7.02-7.12 (2H, m), 7.16-7.26 (2H, m), 7.46-7.55 (1H, m).

Example 375

N-[(2S,3R)-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (28 mg), (3,5-difluorophenyl)boronic acid (19.4 mg), XPhos Pd G3 (5.19 mg) and 1 M aqueous potassium phosphate solution (0.184 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (28.3 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.21-1.73 (4H, m), 2.00-2.42 (3H, m), 2.60-3.26 (5H, m), 3.68-5.13 (5H, m), 7.11-7.55 (6H, m), 8.09-8.40 (1H, m).

Example 376

N-[(2S,3R)-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (28 mg), (2,5-difluorophenyl)boronic acid (19.4 mg), XPhos Pd G3 (5.19 mg) and 1 M aqueous potassium phosphate solution (0.184 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (18.2 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.19-1.74 (4H, m), 2.01-2.42 (3H, m), 2.57-3.22 (5H, m), 3.66-5.17 (5H, m), 7.13-7.48 (6H, m), 8.11-8.37 (1H, m).

Example 378

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1r,3S)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (28 mg), phenylboronic acid (14.9 mg), XPhos Pd G3 (5.19 mg) and 1 M aqueous potassium phosphate solution (0.184 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (26.9 mg).

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¹H NMR (300 MHz, DMSO-d₆) δ 1.20-1.71 (4H, m), 1.99-2.43 (3H, m), 2.60-3.23 (5H, m), 3.68-5.14 (5H, m), 7.09-7.56 (8H, m), 8.11-8.38 (1H, m).

Example 380

N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1S,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (17 mg), (3,5-difluorophenyl)boronic acid (11.8 mg), XPhos Pd G3 (3.15 mg) and 1 M aqueous potassium phosphate solution (0.112 mL) in DME (1 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (15.8 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.11-3.20 (12H, m), 3.73-5.00 (5H, m), 7.11-7.52 (6H, m), 8.14-8.35 (1H, m).

Example 381

N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1S,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (17 mg), (2,5-difluorophenyl)boronic acid (7.64 mg), XPhos Pd G3 (3.15 mg) and 1 M aqueous potassium phosphate solution (0.112 mL) in DME (1 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (12.0 mg).

¹H NMR (300 MHz, DMSO-d₆) δ 1.19-1.30 (3H, m), 1.70-2.44 (5H, m), 2.54-3.22 (4H, m), 3.70-5.12 (5H, m), 7.10-7.49 (6H, m), 8.10-8.38 (1H, m).

Example 383

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(1S,3R)-3-fluorocyclobutane-1-carbonyl]pyrrolidin-3-yl]ethanesulfonamide (17 mg), phenylboronic acid (9.07 mg), XPhos Pd G3 (3.15 mg) and 1 M aqueous potassium phosphate solution (0.112 mL) in DME (1 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) and preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (8.00 mg).

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¹H NMR (300 MHz, DMSO-d₆) δ 1.12-3.24 (12H, m), 3.68-5.03 (5H, m), 7.06-7.59 (8H, m), 8.10-8.40 (1H, m).

Example 388

N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A) benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate

Methanesulfonic anhydride (225 mg) was added to a stirred solution of benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate (328 mg) and TEA (0.360 mL) in THF (10 mL) at room temperature. After 0.5 h, the reaction mixture was concentrated under reduced pressure, and the residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (336 mg).

MS: [M+H]⁺ 459.0.

B) N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]methanesulfonamide Hydrobromide

To benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-3-[(methanesulfonyl)amino]pyrrolidine-1-carboxylate (1.17 g) was added 30% HBr/acetic acid solution (10 mL) at room temperature. The mixture was stirred at room temperature for 30 min. The mixture was evaporated with toluene, and the residue was suspended in IPE, and the insoluble substance was collected by filtration and washed with IPE to give the title compound (0.974 g).

MS, found: 325.1.

C) N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

To a mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]methanesulfonamide hydrobromide (100 mg) and DIPEA (0.299 mL) in THF (3 mL) was added alpha-acetoxy-isobutyryl chloride (0.071 mL) at 0° C., and the mixture was stirred at the same temperature for 10 min. To the mixture were added water (1 mL) and 4 M aqueous lithium hydroxide solution (1 mL), and the mixture was stirred at room temperature for 4 h. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The extract was washed with saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (96 mg).

MS: [M+H]⁺ 411.1.

D) N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide (23 mg), (3-chlorophenyl)boronic acid (17.5 mg), XPhos Pd G3 (4.74 mg) and 1 M aqueous potassium phosphate solution (0.168 mL) in DME

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(2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA) and column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (5.40 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.42 (6H, s), 2.60 (6H, brs), 4.01-4.30 (3H, m), 4.77-5.34 (3H, m), 7.16-7.23 (1H, m), 7.27-7.55 (6H, m).

Example 392

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Longer Retention Time

A) N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide

To a solution of N-[(2S,3R)-2-(3-chloro-2-fluorobenzyl)-4,4-difluoropyrrolidin-3-yl]methanesulfonamide hydrochloride (0.310 g) and 3-fluorocyclobutane-1-carboxylic acid (0.133 mL) in DMF (10 mL) were added HATU (0.466 g) and DIPEA (0.571 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane, and then silica gel, eluted with EtOAc/hexane) to give the title compound (0.196 g).

MS: [M+H]⁺ 443.0.

B) N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Longer Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), (3-fluorophenyl)boronic acid (18.3 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (11.3 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.56-1.72 (1H, m), 1.95-3.14 (9H, m), 3.36-3.92 (2H, m), 4.26-4.53 (2H, m), 4.88-5.24 (2H, m), 7.26 (7H, s).

Example 393

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)

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pyrrolidin-3-yl]methanesulfonamide (29 mg), (3-fluorophenyl)boronic acid (18.3 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (10.2 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.84-3.14 (10H, m), 3.32-3.94 (2H, m), 4.27-5.13 (4H, m), 7.03-7.46 (7H, m).

Example 431

N-[(2S,3R)-4,4-difluoro-1-((1R,3S)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), (3,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (9.90 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.56-1.70 (1H, m), 1.98-3.14 (9H, m), 3.27-3.98 (2H, m), 4.25-4.52 (2H, m), 4.88-5.23 (2H, m), 6.78-6.88 (1H, m), 6.99-7.09 (2H, m), 7.16-7.45 (3H, m).

Example 433

N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), (3,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (12.3 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.29-3.22 (10H, m), 3.30-5.24 (6H, m), 6.77-7.47 (6H, m).

Example 435

N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide with Longer Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)

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pyrrolidin-3-yl]methanesulfonamide (29 mg), (3,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (9 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.60-3.18 (10H, m), 3.36-3.94 (2H, m), 4.28-4.53 (2H, m), 4.92-5.24 (2H, m), 7.05-7.46 (6H, m).

Example 436

N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), (3,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IB, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (7.6 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.54-3.23 (10H, m), 3.27-5.08 (6H, m), 7.04-7.47 (6H, m).

Example 437

N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide with Longer Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), (2,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK IA, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (6.1 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.54-3.36 (10H, m), 3.33-3.94 (2H, m), 4.27-4.53 (2H, m), 4.91-5.24 (2H, m), 7.02-7.46 (6H, m).

Example 439

N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)

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pyrrolidin-3-yl]methanesulfonamide (29 mg), (2,5-difluorophenyl)boronic acid (20.7 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK TA, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (8.4 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.30-3.21 (10H, m), 3.25-5.07 (6H, m), 6.97-7.47 (6H, m).

Example 440

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Longer Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), phenylboronic acid (16.0 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by preparative HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK OJ-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (11.8 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.54-3.15 (10H, m), 3.32-3.93 (2H, m), 4.27-4.52 (2H, m), 4.91-5.23 (2H, m), 7.09-7.53 (8H, m).

Example 441

N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide with Shorter Retention Time

A mixture of N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide (29 mg), phenylboronic acid (16.0 mg), XPhos Pd G3 (5.54 mg) and 1 M aqueous potassium phosphate solution (0.196 mL) in DME (2 mL) was stirred at 80° C. for 2 h. After cooling back to room temperature, the solvent was removed under reduced pressure, and the residue was purified by HPLC (Column: YMC-Actus Triant C18, mobile phase: H₂O 0.1% TFA/CH₃CN 0.1% TFA, and then Column: CHIRALPAK OJ-H, mobile phase: carbon dioxide/(MeOH/diethylamine=1000/3)=900/100 (v/v)) to give the title compound (8 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.29-3.20 (10H, m), 3.35-5.19 (6H, m), 7.10-7.53 (8H, m).

Example 442

(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide

A tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate

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(300 mg), dimethylsulfamoyl chloride (800 mg) and DMAP (201 mg) was stirred overnight under nitrogen atmosphere at 50° C. The reaction mixture was diluted with saturated aqueous sodium hydrogen carbonate solution, and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (225 mg).

MS: [M-H]⁻ 470.1.

B) tert-butyl (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (200 mg), (3-methylphenyl)boronic acid (86 mg), XPhos Pd G3 (35.9 mg), 1M aqueous potassium phosphate solution (0.636 mL) and THF (3 mL) was irradiated with microwave at 80° C. for 1 hr. The mixture was neutralized with saturated aqueous sodium hydrogen carbonate solution at 0° C., and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH₄ EtOAc/hexane) to give the title compound (150 mg).

MS: [M-H]⁻ 526.2.

C) N'-{(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide Hydrochloride

A mixture of tert-butyl (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxylate (145 mg) and 4M hydrogen chloride/CPME solution (5 mL) was stirred at 60° C. for 4 hr. The reaction mixture was concentrated under reduced pressure to give the title compound (125 mg).

MS, found: 428.1.

D) (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carbonyl Chloride

To a mixture of N'-{(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide hydrochloride (50 mg), DIPEA (27.9 mg) and THF (4.7 mL) was added bis(trichloromethyl) carbonate (25.6 mg) at 0° C. The reaction mixture was stirred at 0° C. for 10 min, and then stirred at room temperature for 10 min. The mixture was neutralized with saturated aqueous sodium hydrogen carbonate solution at 0° C., and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure to give the title compound (48 mg).

MS: [M+H]⁺ 490.1.

E) (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide

To a mixture of (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carbonyl chloride (48 mg) and THF

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(0.5 mL) was added 2M N-methylmethanamine/THF solution (0.5 mL) at 0° C. The mixture was stirred at room temperature for 2 hr, and purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (35 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.40 (3H, s), 2.60 (6H, s), 2.76-2.88 (7H, m), 2.98-3.05 (1H, m), 3.66-3.78 (1H, m), 3.88-4.00 (1H, m), 4.14-4.27 (1H, m), 4.82-4.90 (1H, m), 5.05 (1H, d, J=9.7 Hz), 7.09-7.34 (7H, m).

Example 443

N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

To a mixture of (2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carbonyl chloride (50 mg) and THF (0.5 mL) was added azetidine (17.5 mg) at 0° C. The mixture was stirred at room temperature for 2 hr, and purified by silica gel column chromatography (EtOAc/hexane) to give the title compound (20 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.99-2.11 (2H, m), 2.41 (3H, s), 2.78 (6H, s), 2.83-2.93 (1H, m), 2.96-3.09 (1H, m), 3.58-3.91 (6H, m), 4.10-4.26 (1H, m), 4.69-4.87 (1H, m), 4.87-5.01 (1H, m), 7.12-7.23 (2H, m), 7.24-7.36 (5H, m).

Example 450

N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A) N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide

To a mixture of N-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide hydrobromide (60 mg), bis(trichloromethyl)carbonate (25.5 mg) and THF (1.4 mL) was added DIPEA (0.075 mL) at 0° C. After being stirred at 0° C. for 30 min, the mixture was poured into water and extracted with EtOAc. The combined organic layers were dried over sodium sulfate, filtered and concentrated under reduced pressure. The residue was diluted with THF (1.4 mL), and azetidine (0.029 mL) was added thereto at room temperature. After being stirred at room temperature for 2 h, the mixture was poured into aqueous saturated ammonium chloride solution and extracted with EtOAc. The combined organic layers were dried over sodium sulfate, filtered and concentrated under reduced pressure. The residue was purified by preparative HPLC (Column: L-Column 2 ODS, mobile phase: H₂O 10 mM NH₄HCO₃/CH₃CN) to give the title compound (49.0 mg).

MS: [M+H]⁺ 422.1.

B) N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A mixture of N-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide (23 mg), (3,5-difluorophenyl)boronic acid (17.2 mg), XPhos Pd G3 (4.61 mg) and 1 M aqueous potassium phosphate solution (0.164 mL) in DME (0.8 mL)

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was stirred at 80° C. under nitrogen atmosphere for 2 h. The mixture was directly purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (24.5 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.28 (3H, t, J=7.4 Hz), 2.13 (2H, quin, J=8.0 Hz), 2.95 (3H, q, J=7.3 Hz), 3.04-3.14 (1H, m), 3.61-3.79 (4H, m), 3.89-4.08 (3H, m), 4.66-4.75 (1H, m), 4.86 (1H, d, J=10.3 Hz), 5.06-5.27 (1H, m), 6.82 (1H, t, J=8.9 Hz), 7.08 (2H, d, J=7.0 Hz), 7.15-7.22 (1H, m), 7.27-7.31 (1H, m), 7.36 (1H, t, J=7.1 Hz).

Example 451

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide

A mixture of N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}ethanesulfonamide (23 mg), m-tolylboronic acid (14.8 mg), XPhos Pd G3 (4.61 mg) and 1 M aqueous potassium phosphate solution (0.164 mL) in DME (0.8 mL) was stirred at 80° C. under nitrogen atmosphere for 2 h. The mixture was directly purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (23.5 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.22 (3H, t, J=7.3 Hz), 2.06-2.18 (2H, m), 2.38-2.45 (3H, m), 2.87 (2H, q, J=7.2 Hz), 2.92-3.01 (1H, m), 3.05-3.14 (1H, m), 3.60-3.82 (4H, m), 3.88-4.09 (3H, m), 4.68 (1H, q, J=7.3 Hz), 4.88 (1H, d, J=10.3 Hz), 5.04-5.27 (1H, m), 7.13-7.21 (2H, m), 7.27-7.37 (5H, m).

Example 458

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

A) benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoropyrrolidine-1-carboxylate

A mixture of benzyl (2S,3R,4S)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidine-1-carboxylate (100 mg), DMAP (64.2 mg) and dimethylsulfamoyl chloride (3 mL) was stirred overnight under nitrogen atmosphere at 50° C. The reaction mixture was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (114 mg).

MS: [M+H]⁺ 488.1.

B) N'-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide Hydrobromide

To benzyl (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoropyrrolidine-1-carboxylate (225 mg) was added 30% HBr/acetic acid solution (10 mL) at room temperature. The mixture was stirred at room temperature 16 h then the solvent was evaporated with toluene. The solid obtained was suspended in IPE and collected by filtration to give the title compound (180 mg).

MS: [M+H]⁺ 354.0.

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C) (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl methyl)-3-[(dimethylsulfamoyl)amino]-4-fluoropyrrolidine-1-carbonyl Chloride

Bis(trichloromethyl) carbonate (76 mg) was added to a solution of N'-{(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide hydrobromide (140 mg) and DIPEA (0.112 mL) in THF (4.7 mL) at 0° C. After being stirred at 0° C. for 10 min and at room temperature for 10 min, the mixture was neutralized with saturated aqueous sodium hydrogen carbonate solution at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure to give the title compound (120 mg).

MS: [M-H]⁻ 414.0.

D) N'-{(2S,3S,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

Azetidine (300 μL) was added to a solution of (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoropyrrolidine-1-carbonyl chloride (80 mg) in THF (3 mL) at 0° C. The mixture was stirred at room temperature under a dry atmosphere for 2 h. The mixture was neutralized with 0.05 M hydrochloric acid at 0° C. and extracted with EtOAc. The organic layer was separated, washed with saturated aqueous sodium hydrogen carbonate solution, water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure to give the title compound (70.0 mg).

MS: [M+H]⁺ 437.1.

E) N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide

A mixture of N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide (35 mg), (3,5-difluorophenyl)boronic acid (19.0 mg), XPhos Pd G3 (6.78 mg) and 1 M aqueous potassium phosphate solution (0.120 mL) in THF (3 mL) was heated at 80° C. for 1 h under microwave irradiation. The mixture was neutralized with aqueous saturated ammonium chloride solution at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The crude was purified by preparative TLC (silica gel, eluted with EtOAc/hexane) to give the title compound (20.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.99-2.21 (2H, m), 2.71 (6H, s), 2.84-3.00 (1H, m), 3.00-3.18 (1H, m), 3.63-3.80 (4H, m), 3.80-4.01 (3H, m), 4.58-4.81 (1H, m), 4.77-4.97 (1H, m), 5.05-5.37 (1H, m), 6.70-6.90 (1H, m), 7.02-7.21 (3H, m), 7.25-7.30 (1H, m), 7.32-7.45 (1H, m).

Example 459

(2S,3R,4S)-3-[(dimethylsulfamoyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide

A) (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide

Dimethylamine (366 μL) was added to a solution of (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dim-

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ethylsulfamoyl)amino]-4-fluoropyrrolidine-1-carbonyl chloride (40 mg) in THF (2 mL) at 0° C. The mixture was stirred at room temperature under a dry atmosphere for 2 h. The mixture was neutralized with 0.05 M hydrochloric acid at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure to give the title compound (35.0 mg).

MS: [M+H]⁺ 425.0.

B) (2S,3R,4S)-3-[(dimethylsulfamoyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide

A mixture of (2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide (35 mg), m-tolylboronic acid (16.8 mg), XPhos Pd G3 (6.97 mg) and 1 M potassium phosphate, aqueous solution (0.247 mL) in THF (3 mL) was heated at 85° C. for 3 h under microwave irradiation. The mixture was neutralized with saturated aqueous sodium hydrogen carbonate solution at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The crude was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (13.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.41 (3H, s), 2.65 (6H, s), 2.80 (6H, s), 2.82-2.90 (1H, m), 3.00-3.08 (1H, m), 3.58-3.76 (1H, m), 3.78-4.04 (2H, m), 4.75-4.89 (2H, m), 5.01-5.39 (1H, m), 7.07-7.15 (1H, m), 7.15-7.23 (2H, m), 7.24-7.27 (1H, m), 7.28-7.36 (3H, m).

Example 462

N'-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric Diamide

A) N'-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric Diamide

To a solution of N'-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl]-N,N-dimethylsulfuric diamide hydrobromide (40 mg) and 1-hydroxycyclobutane-1-carboxylic acid (12.8 mg) in DMF (1 mL) were added HATU (52.5 mg) and DIPEA (0.048 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (40.0 mg).

MS: [M+H]⁺ 452.0.

B) N'-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric Diamide

A mixture of N'-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide (40 mg),

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m-tolylboronic acid (18.1 mg), XPhos Pd G3 (7.49 mg) and 1 M aqueous potassium phosphate solution (0.266 mL) in THF (3 mL) was heated at 85° C. for 3 h under microwave irradiation. The mixture was neutralized with aqueous saturated ammonium chloride solution at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (32.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 1.59-1.70 (1H, m), 1.82-1.97 (1H, m), 1.97-2.14 (2H, m), 2.15-2.20 (1H, m), 2.41 (3H, s), 2.57 (8H, s), 2.98-3.11 (1H, m), 3.11-3.27 (1H, m), 3.74-4.30 (3H, m), 4.72-4.84 (1H, m), 4.85-5.01 (1H, m), 5.09-5.34 (1H, m), 7.10-7.23 (2H, m), 7.26-7.30 (1H, m), 7.31 (3H, s), 7.36-7.47 (1H, m).

Example 463

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

A mixture of N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide (35 mg), m-tolylboronic acid (16.3 mg), XPhos Pd G3 (6.78 mg) and 1 M aqueous potassium phosphate solution (0.240 mL) in THF (3 mL) was heated at 85° C. for 3 h under microwave irradiation. The mixture was neutralized with aqueous saturated ammonium chloride solution at 0° C. and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over magnesium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (13.0 mg).

¹H NMR (400 MHz, CDCl₃) δ 2.03-2.16 (2H, m), 2.41 (3H, s), 2.63-2.69 (6H, m), 2.90-2.99 (1H, m), 3.03-3.15 (1H, m), 3.59-4.01 (7H, m), 4.63-4.76 (1H, m), 4.76-4.91 (1H, m), 5.03-5.41 (1H, m), 7.10-7.21 (2H, m), 7.26-7.37 (5H, m).

Example 505

N-[(2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Longer Retention Time

A) tert-butyl (2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-2-(3-chloro-2-fluorophenyl)-3-(ethylsulfonamido)-4,4-difluoropyrrolidine-1-carboxylate (270 mg), (3-(difluoromethyl)phenyl)boronic acid (203 mg), XPhos Pd G3 (50.0 mg) and 1 M aqueous potassium phosphate solution (1.77 mL) in THF (7 mL) was stirred overnight at 90° C. in sealed tube. The mixture was quenched with saturated aqueous sodium hydrogen carbonate solution and extracted with EtOAc. The organic layer was separated, washed with saturated brine, dried over sodium sulfate and concentrated under reduced pressure.

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The mixture was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (231 mg).

MS: $[M-H]^-$ 547.2.

B) N-[(2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide Hydrochloride

The mixture of tert-butyl (2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl]-4,4-difluoropyrrolidine-1-carboxylate (231 mg) and 4 M HCl/CPME solution (5 mL) was stirred at room temperature for 2 h. White solid was collected by filtration to give the title compound (195 mg).

MS: $[M+H]^+$ 449.1.

C) N-[(2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with Longer Retention Time

To a mixture of 3-fluorocyclobutanecarboxylic acid (29.2 mg), N-[(2S,3R)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide hydrochloride (60 mg), DIPEA (0.130 mL) and DMF (1 mL) was added HATU (75 mg) at room temperature. The mixture was stirred at room temperature for 3 h. The mixture was quenched with aqueous saturated ammonium chloride solution at room temperature and extracted with EtOAc. The organic layer was separated, washed with saturated aqueous sodium hydrogen carbonate solution and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (silica gel, eluted with EtOAc/hexane) and then preparative HPLC (Column: CHIRALCEL OJ-H, mobile phase: carbon dioxide/MeOH=900/100 (v/v)) to give the title compound (25.6 mg).

1H NMR (400 MHz, $CDCl_3$) δ 1.25-1.48 (3H, m), 1.63-3.30 (9H, m), 3.70-5.30 (6H, m), 6.53-6.88 (1H, m), 7.14-7.71 (7H, m).

Example 542

N'-{(2S,3R)-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

A) (2R)-oxetane-2-carboxylic Acid

Iodobenzene diacetate (8.04 g), (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl (0.532 g) and (R)-oxetan-2-ylmethanol (1.00 g) were combined, and to this mixture were added CH_3CN (50 mL) and water (50.0 mL) and then the mixture was stirred for 3 h at room temperature. The mixture was neutralized with 1 M NaOH aqueous solution at 0° C. and the aqueous layer was washed with IPE. The aqueous layer was acidified with 2 M hydrochloric acid at 0° C. To the mixture was added NaCl and the mixture was extracted with THF/EtOAc. The organic layer was separated, dried over magnesium sulfate and concentrated under reduced pressure. To the residue was added toluene and the mixture was concentrated under reduced pressure to give the title compound (0.510 g).

1H NMR (400 MHz, $CDCl_3$) δ 2.69-2.90 (1H, m), 3.00-3.22 (1H, m), 4.68-4.88 (2H, m), 5.07-5.33 (1H, m).

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B) tert-butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoropyrrolidine-1-carboxylate

A mixture of tert-butyl (2S,3R)-3-amino-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidine-1-carboxylate (500 mg) and DMAP (335 mg) in dimethylsulfamoyl chloride (4 mL) was stirred at 65° C. for 5 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature and purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (541 mg).

MS, found: 416.0.

C) N'-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide Hydrochloride

tert-Butyl (2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoropyrrolidine-1-carboxylate (541 mg) was stirred in 4 M HCl/CPME solution (10 mL) at 60° C. for 1 h. The mixture was concentrated under reduced pressure, triturated with IPE and collected by filtration to give the title compound (439 mg).

MS: $[M+H]^+$ 372.0.

D) N'-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

To a solution of N'-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide hydrochloride (100 mg) and (2R)-oxetane-2-carboxylic acid (0.034 mL) in DMF (5 mL) were added HATU (140 mg) and DIPEA (0.171 mL) at room temperature. The mixture was stirred overnight at room temperature. The mixture was quenched with aqueous saturated ammonium chloride solution and extracted with EtOAc. The organic layer was separated, washed with water and saturated brine, dried over sodium sulfate and concentrated under reduced pressure. The residue was purified by column chromatography (NH silica gel, eluted with EtOAc/hexane) to give the title compound (109 mg).

MS: $[M+H]^+$ 456.0.

E) N'-{(2S,3R)-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric Diamide

A mixture of N'-{(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide (22 mg), (3,5-difluorophenyl)boronic acid (15.2 mg), XPhos Pd G3 (4.08 mg) and 1M aqueous potassium phosphate solution (0.145 mL) in DME (0.8 mL) was stirred at 80° C. under nitrogen atmosphere for 2 h. The mixture was purified by column chromatography (silica gel, eluted with EtOAc/hexane) to give the title compound (16.4 mg).

1H NMR (400 MHz, $CDCl_3$) δ 2.71-3.06 (9H, m), 3.06-3.25 (1H, m), 3.69-3.90 (1H, m), 4.05-5.17 (7H, m), 6.77-6.87 (1H, m), 7.06 (2H, d, J=7.0 Hz), 7.21 (1H, q, J=7.3 Hz), 7.27-7.48 (2H, m).

The compounds of Examples are shown in the following tables. MS in the tables means actual measured value. The compounds of Examples 6, 9-12, 14-20, 22-24, 27-34,

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36-43, 47-49, 53-55, 57-65, 69-72, 74, 75, 80, 82-86, 89, 90, 93, 95, 96, 99-105, 107-115, 117-120, 123, 125-128, 130, 132, 134-143, 148-170, 172-201, 203, 204, 206-209, 213-219, 221-224, 227-235, 237, 238, 240-244, 246, 247, 252-256, 258-266, 268-271, 273-301, 303, 306, 308-345, 347-

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373, 377, 379, 382, 384-387, 389-391, 394-430, 432, 434, 438, 444-449, 452-457, 460, 461, 464-504, 506-541 and 543-616 in the following tables were produced according to the methods described in the above-mentioned Examples, or methods analogous thereto.

TABLE 1-1

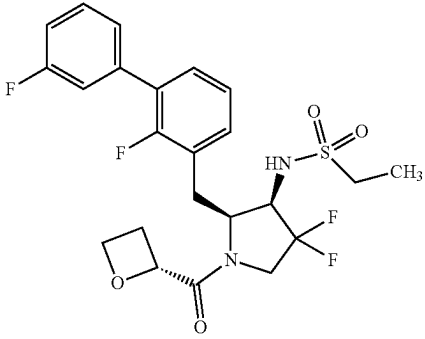
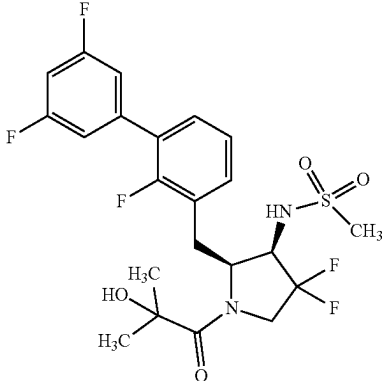
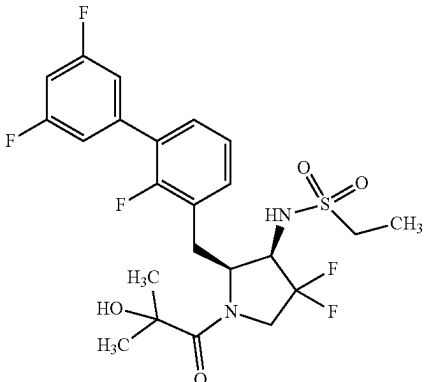
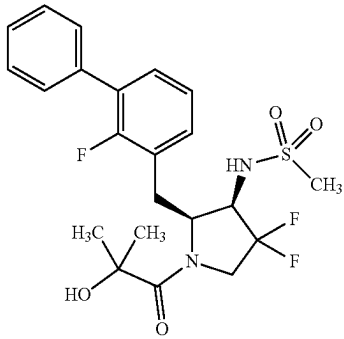
EX.	IUPAC NAME	Structure	ADDITIVE	MS
1	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			501.4
2	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			507.1
3	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			521.1
4	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			471.2

TABLE 1-1-continued

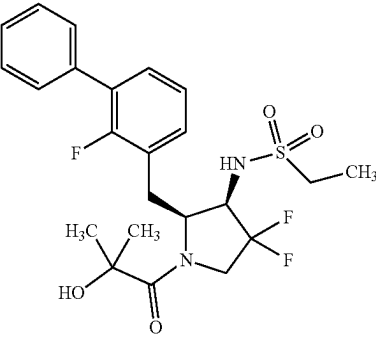
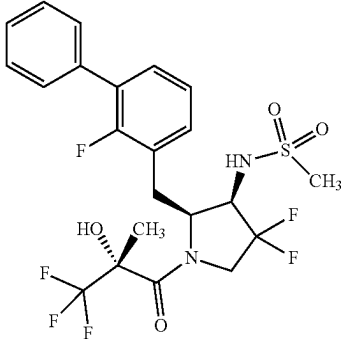
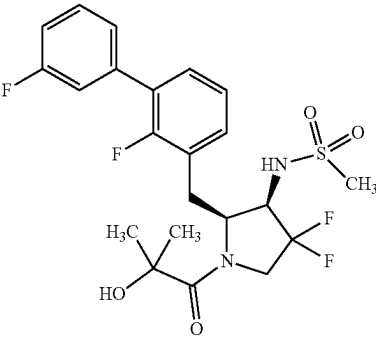
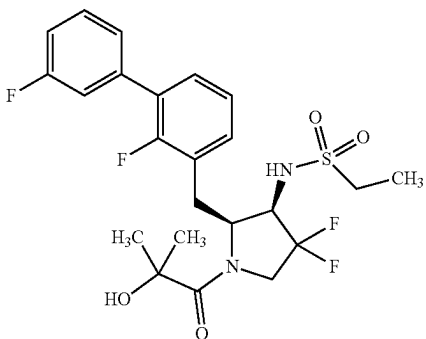
EX.	IUPAC NAME	Structure	ADDITIVE	MS
5	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			485.2
6	N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-3,3-trifluoro-2-hydroxy-2-methylpropanoyl]-pyrrolidin-3-yl}methanesulfonamide			523.3
7	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			487.2
8	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			503.1

TABLE 1-1-continued

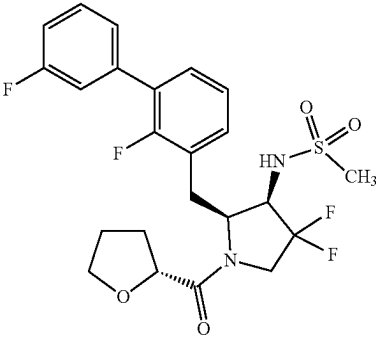
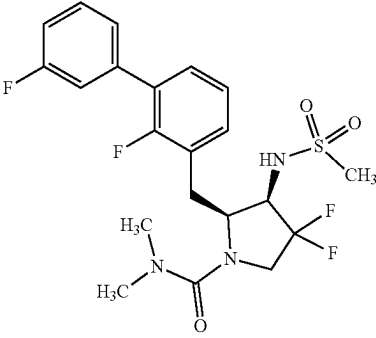
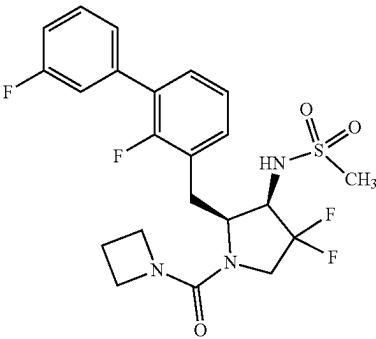
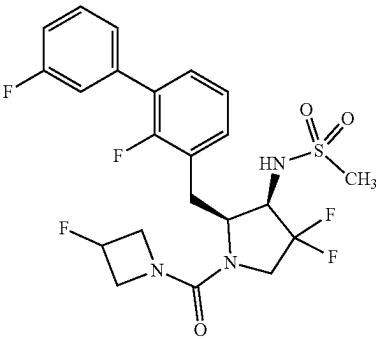
EX.	IUPAC NAME	Structure	ADDITIVE	MS
9	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			501.1
10	(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			474.1
11	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			486.1
12	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluoroazetidine-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide			504.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
13	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxy-cyclobutane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide			501.1
14	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			503.1
15	rac-N-[(2S,3R,4R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			471.2
16	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylbutanoyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			503.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
17	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylbutanoyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			503.2
18	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			481.2
19	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			503.2
20	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			485.2

TABLE 1-1-continued

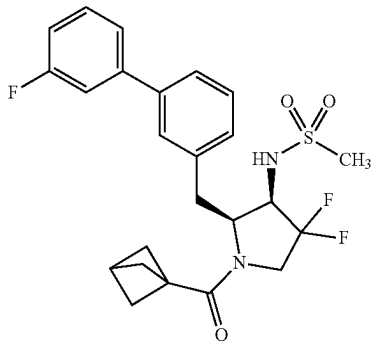
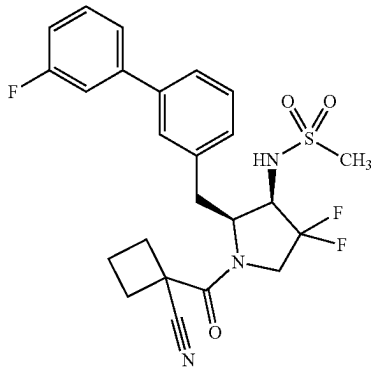
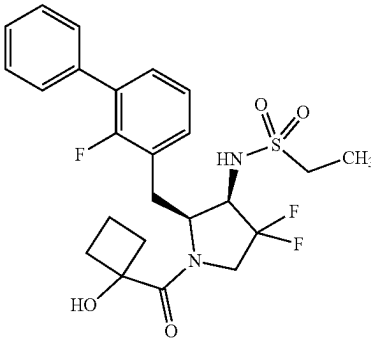
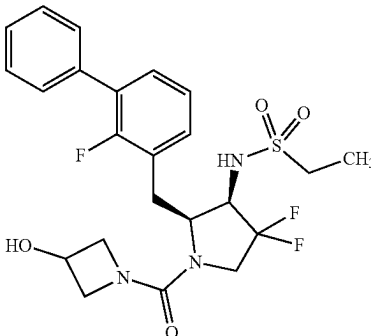
EX.	IUPAC NAME	Structure	ADDITIVE	MS
21	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			479.2
22	N-[(2S,3R)-1-(1-cyanocyclobutane-1-carbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			490.2
23	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide			497.2
24	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-hydroxyazetidine-1-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide			498.2

TABLE 1-1-continued

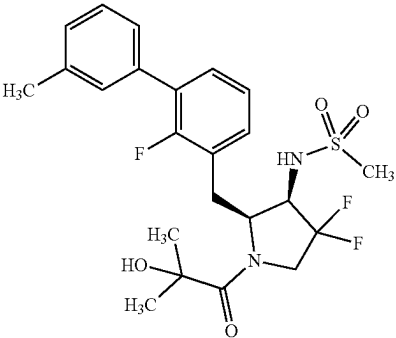
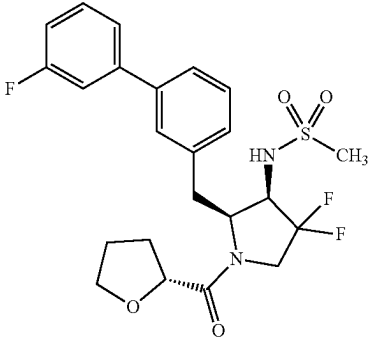
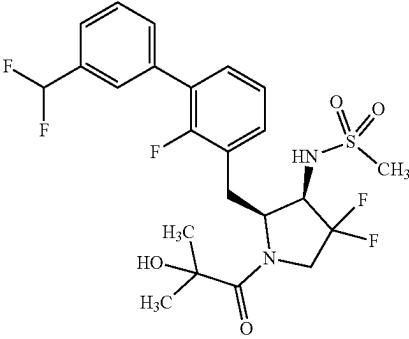
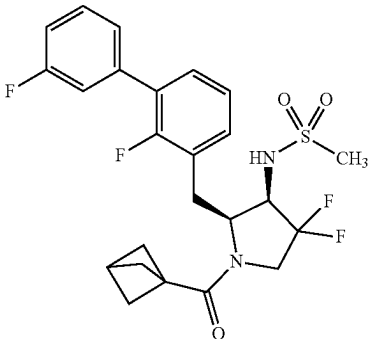
EX.	IUPAC NAME	Structure	ADDITIVE	MS
25	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			485.2
26	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			483.2
27	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-pyrrolidin-3-yl]methanesulfonamide			521.2
28	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			497.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
29	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)-methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl)methanesulfonamide			465.2
30	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl)methanesulfonamide			461.2
31	N-[(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl)methanesulfonamide			503.2
32	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(3-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)methanesulfonamide			475.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
33	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			501.2
34	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-methylcyclopropane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide			483.2
35	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			473.2
36	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			501.2

TABLE 1-1-continued

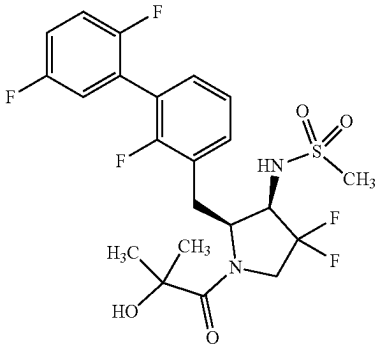
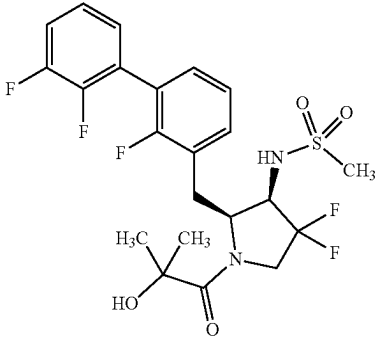
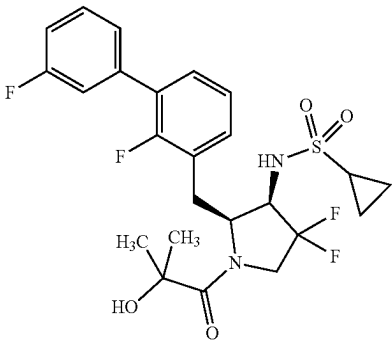
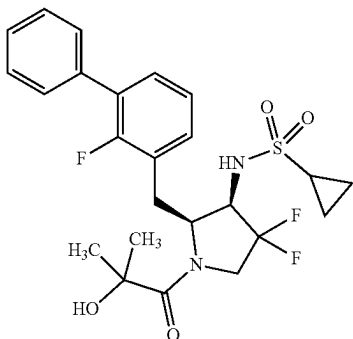
EX.	IUPAC NAME	Structure	ADDITIVE	MS
37	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			507.1
38	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			507.2
39	N-[(2S,3R)-2-[(2,3,-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]cyclopropanesulfonamide			515.2
40	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]cyclopropanesulfonamide			497.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
41	N-[(2S,3R)-2-{[3'-(difluoromethoxy)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			537.1
42	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(3'-chloro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			495.1
43	N-[(2S,3R)-2-[(3'-chloro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			499.1
44	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			505.1

TABLE 1-1-continued

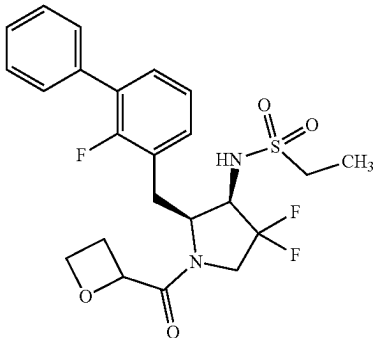
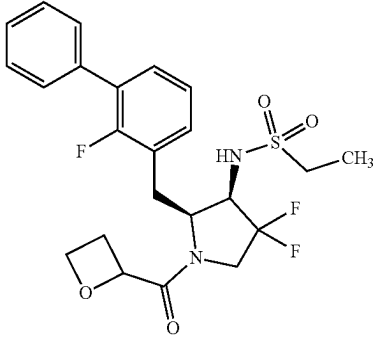
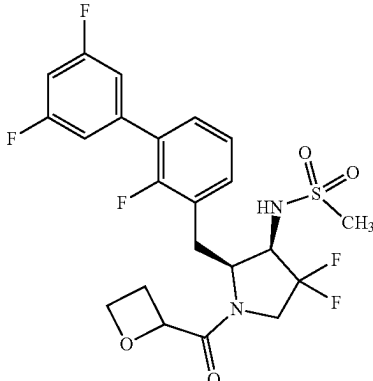
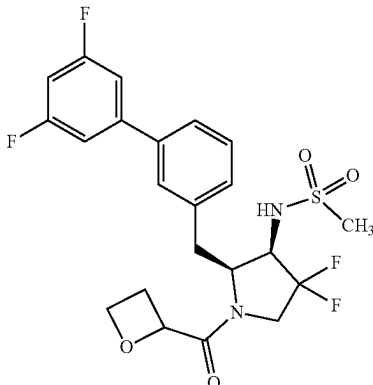
EX.	IUPAC NAME	Structure	ADDITIVE	MS
45	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			483.2
46	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			483.1
47	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			487.2
48	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			487.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
49	N-[(2S,3R)-2-[(2,3,-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			501.1
50	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			465.2
51	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			465.2
52	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			483.2

TABLE 1-1-continued

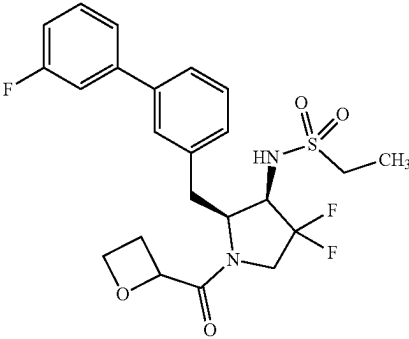
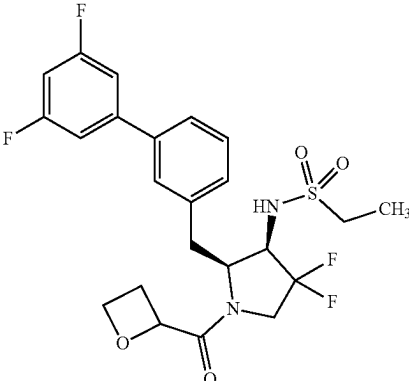
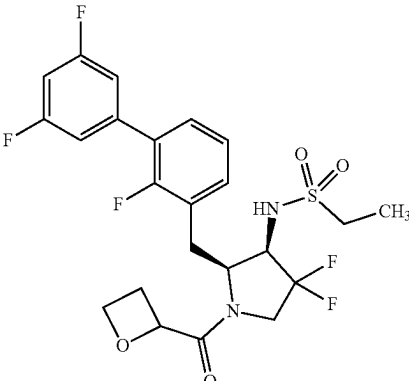
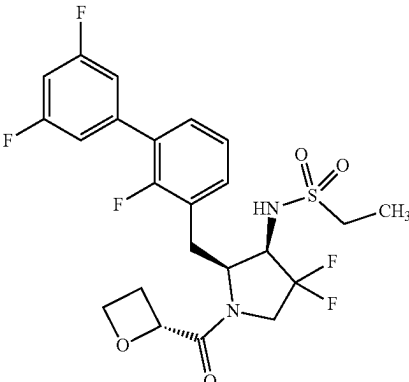
EX.	IUPAC NAME	Structure	ADDITIVE	MS
53	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			483.2
54	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			501.2
55	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			501.1
56	N-{(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-ethanesulfonamide			519.1

TABLE 1-1-continued

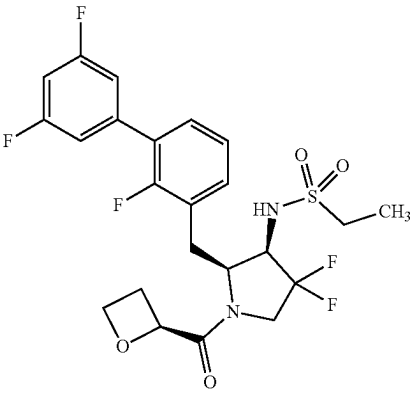
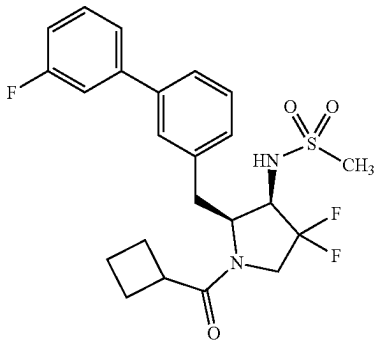
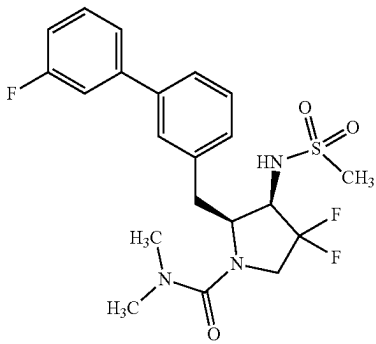
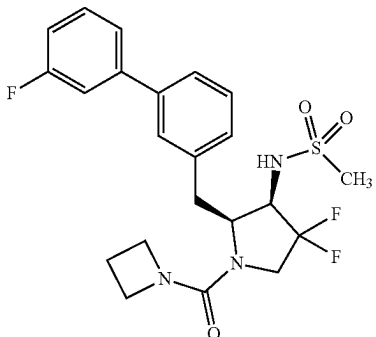
EX.	IUPAC NAME	Structure	ADDITIVE	MS
57	N-[(2S,3R)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			519.1
58	N-[(2S,3R)-1-(cyclobutanecarbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			467.2
59	(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			456.2
60	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			468.2

TABLE 1-1-continued

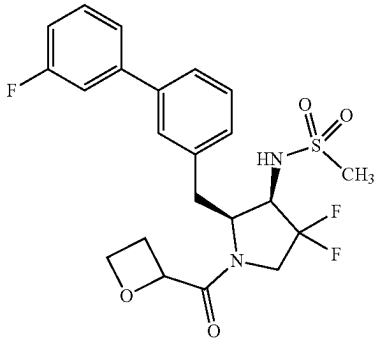
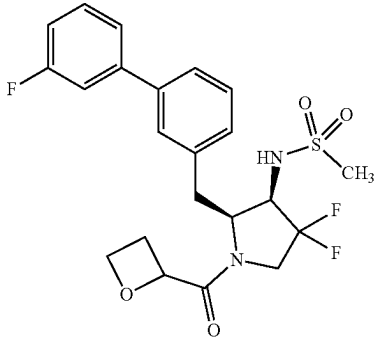
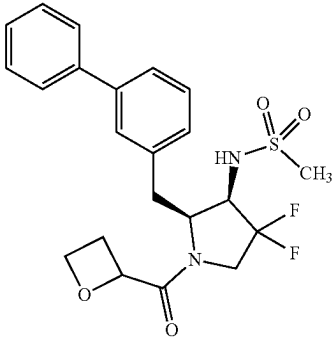
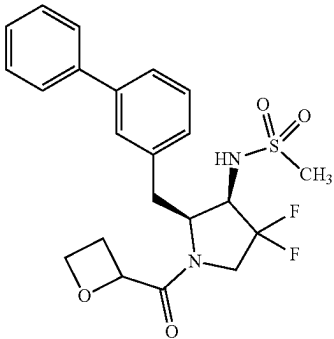
EX.	IUPAC NAME	Structure	ADDITIVE	MS
61	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			469.2
62	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			469.2
63	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			451.2
64	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			451.2

TABLE 1-1-continued

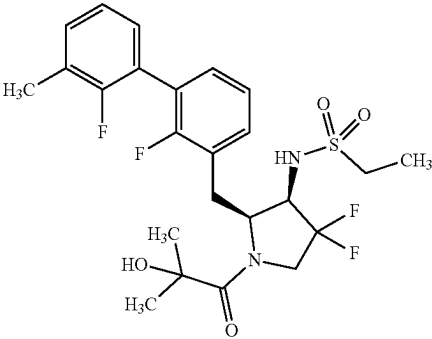
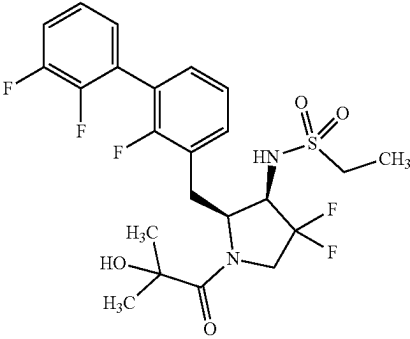
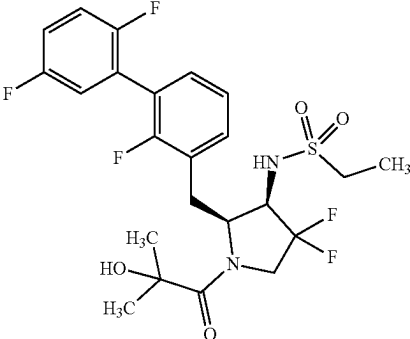
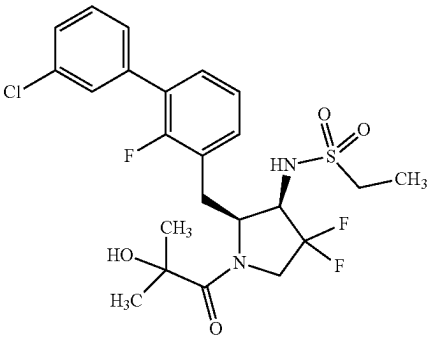
EX.	IUPAC NAME	Structure	ADDITIVE	MS
65	N-[(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			517.2
66	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			521.1
67	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			521.1
68	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			519.1

TABLE 1-1-continued

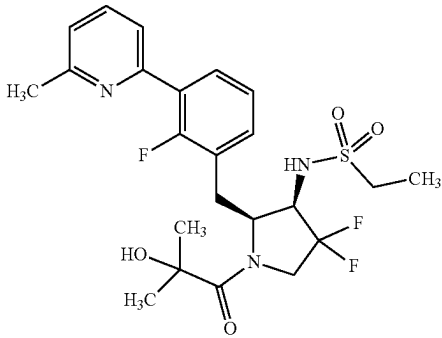
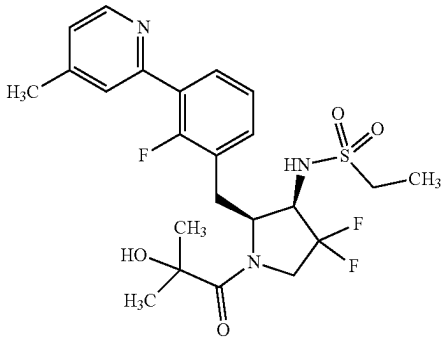
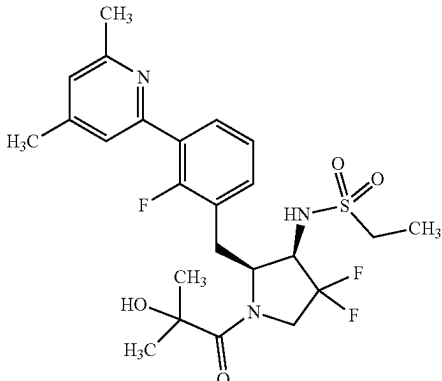
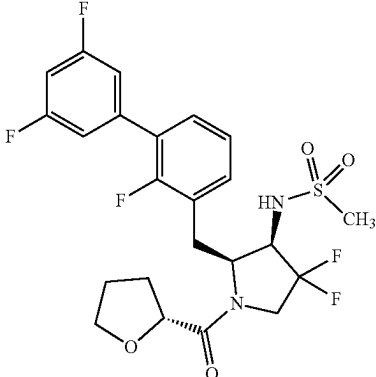
EX.	IUPAC NAME	Structure	ADDITIVE	MS
69	N-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			500.2
70	N-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(4-methylpyridin-2-yl)phenyl]methyl}-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			500.2
71	N-[(2S,3R)-2-{{[3-(4,6-dimethylpyridin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			514.2
72	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			519.1

TABLE 1-1-continued

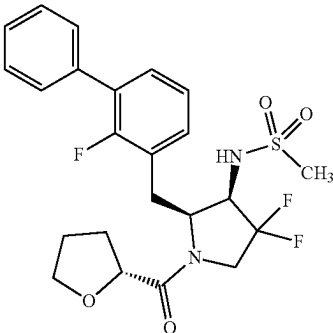
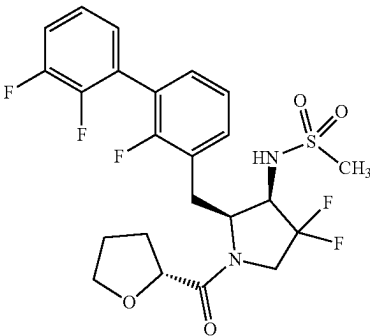
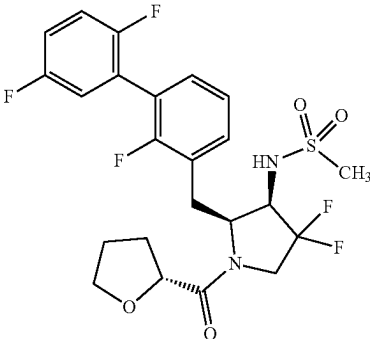
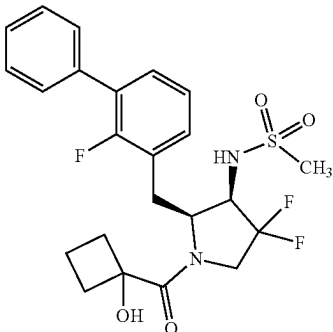
EX.	IUPAC NAME	Structure	ADDITIVE	MS
73	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl)methanesulfonamide			483.2
74	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',3',-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			519.1
75	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			519.1
76	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide			483.2

TABLE 1-1-continued

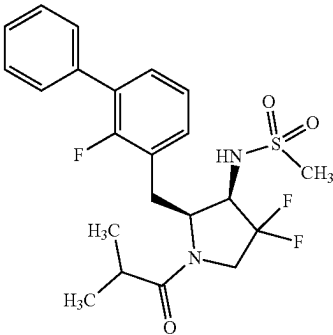
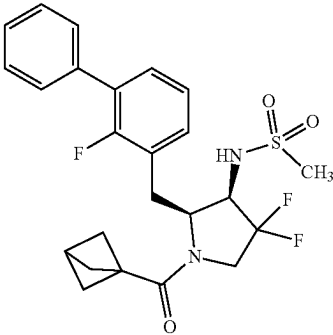
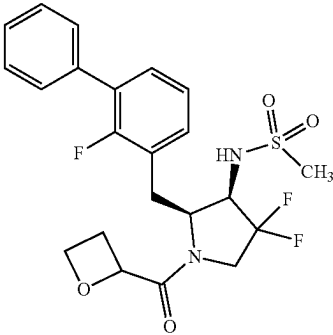
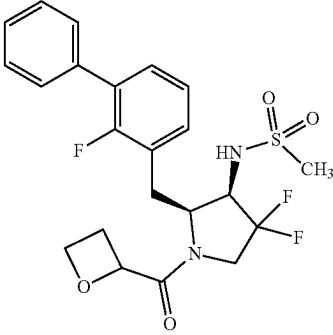
EX.	IUPAC NAME	Structure	ADDITIVE	MS
77	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			455.2
78	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			479.2
79	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			469.2
80	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			469.2

TABLE 1-1-continued

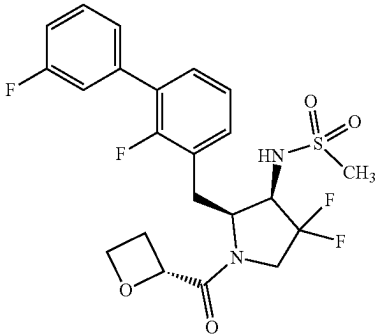
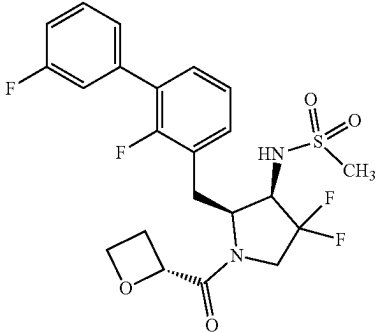
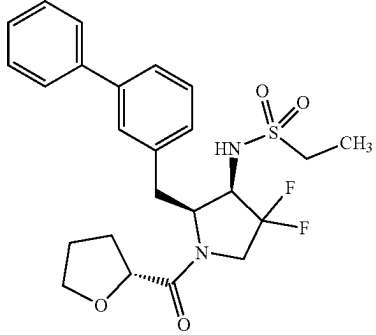
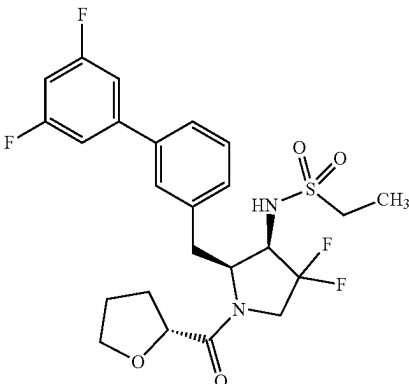
EX.	IUPAC NAME	Structure	ADDITIVE	MS
81	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			487.2
82	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			487.2
83	N-{(2S,3R)-2-[(1,1'-biphenyl)-3-yl]-methyl}-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl}ethanesulfonamide			479.2
84	N-{(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl}ethanesulfonamide			515.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
85	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			515.2
86	N-[(2S,3R)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			515.2
87	N-[(2S,3R)-1-(bicyclo[1.1.1.]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			515.3
88	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			497.2

TABLE 1-1-continued

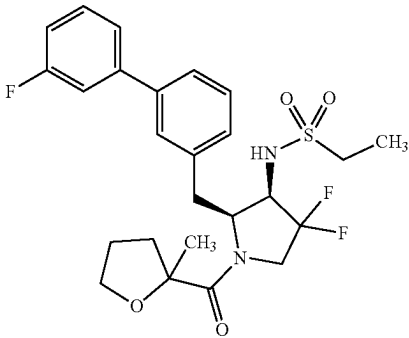
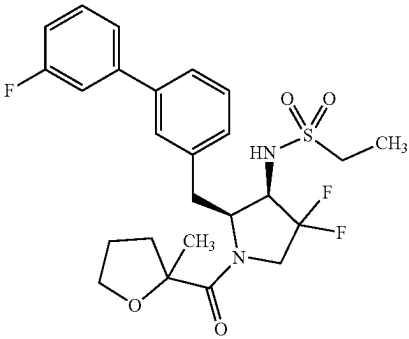
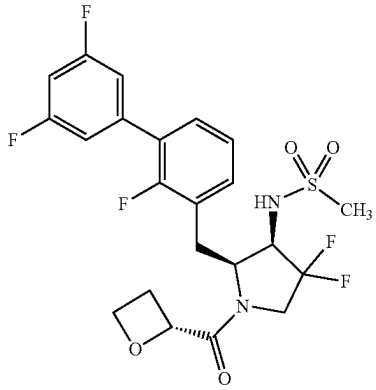
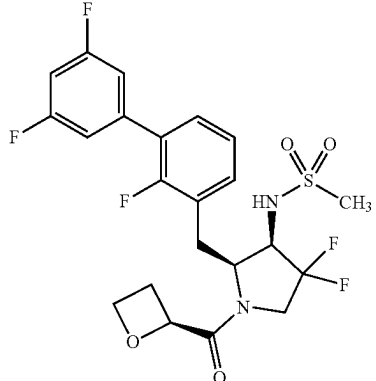
EX.	IUPAC NAME	Structure	ADDITIVE	MS
89	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyloxolane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention			511.2
90	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyloxolane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			511.2
91	N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			505.1
92	N-[(2S,3R)-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			505.1

TABLE 1-1-continued

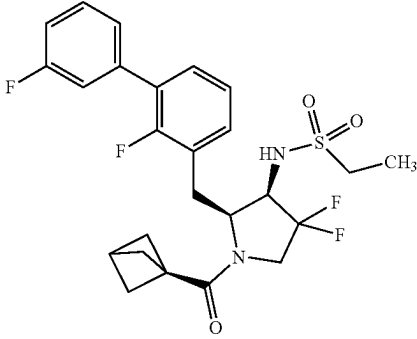
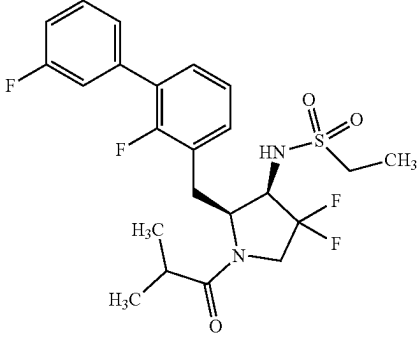
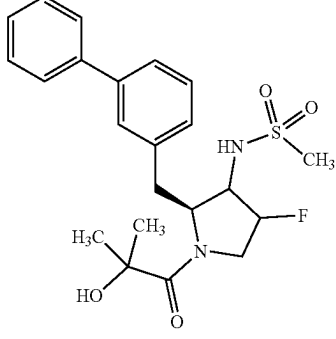
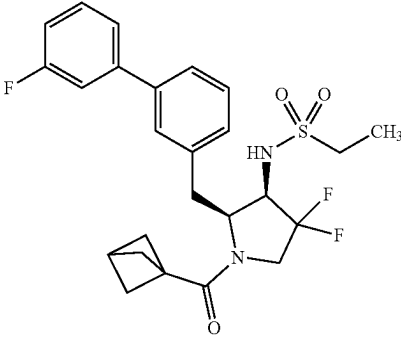
EX.	IUPAC NAME	Structure	ADDITIVE	MS
93	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			511.2
94	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			487.2
95	rac-N-[(2S,3R,4S)-2-[(1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			435.2
96	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			493.2

TABLE 1-1-continued

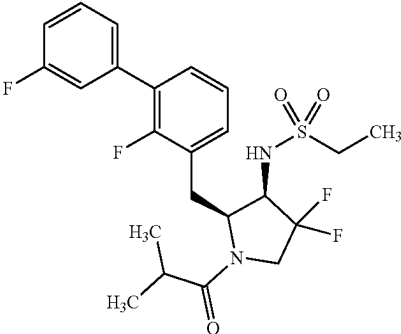
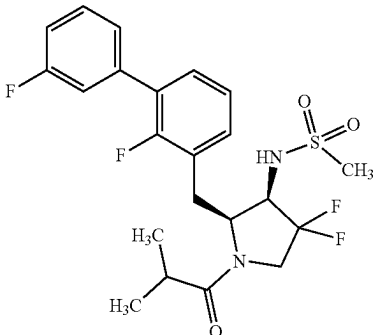
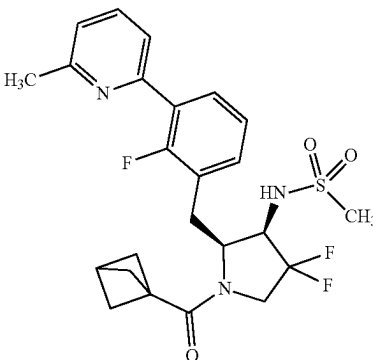
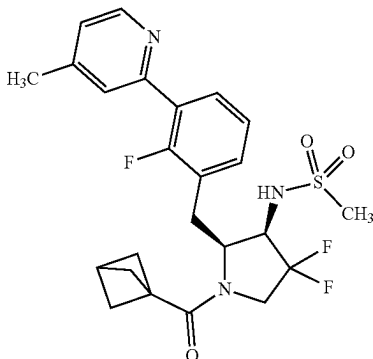
EX.	IUPAC NAME	Structure	ADDITIVE	MS
97	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			469.2
98	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			473.2
99	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl]-pyrrolidin-3-yl]methanesulfonamide			494.2
100	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[[2-fluoro-3-(4-methylpyridin-2-yl)phenyl]methyl]-pyrrolidin-3-yl]methanesulfonamide			494.2

TABLE 1-1-continued

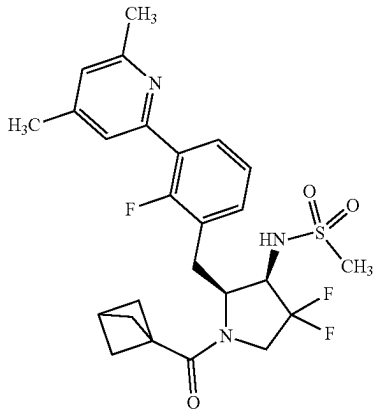
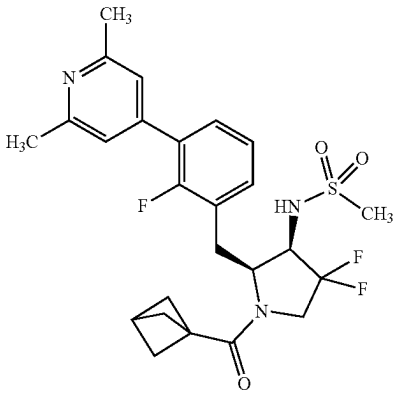
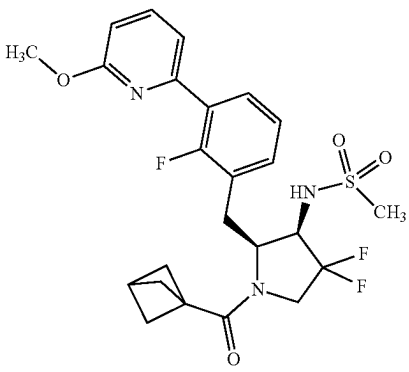
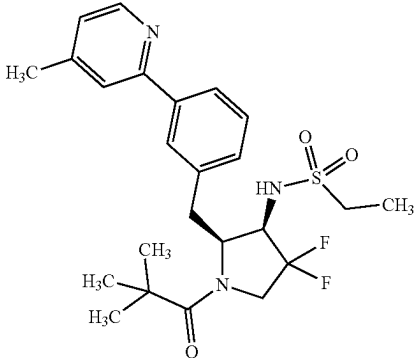
EX.	IUPAC NAME	Structure	ADDITIVE	MS
101	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{{[3-(4,6-dimethylpyridin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			508.2
102	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{{[3-(2,6-dimethylpyridin-4-yl)-2-fluorophenyl]methyl}-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			508.2
103	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-{{[2-fluoro-3-(6-methoxypyridin-2-yl)phenyl]methyl}-pyrrolidin-3-yl]methanesulfonamide			510.2
104	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-4,4-difluoro-2-{{[3-(4-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			480.2

TABLE 1-1-continued

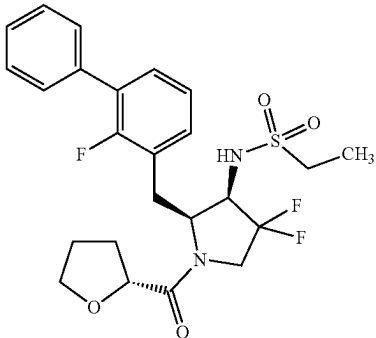
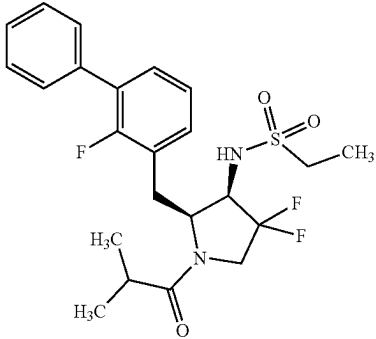
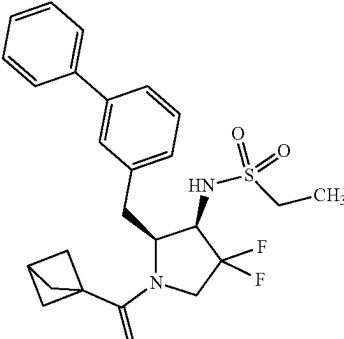
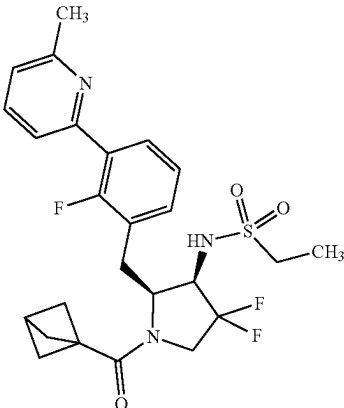
EX.	IUPAC NAME	Structure	ADDITIVE	MS
105	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			497.1
106	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			469.2
107	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			475.2
108	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-{[3-(6-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			490.2

TABLE 1-1-continued

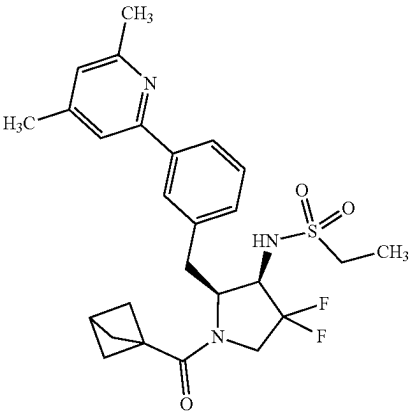
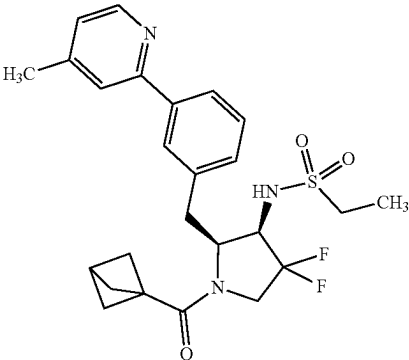
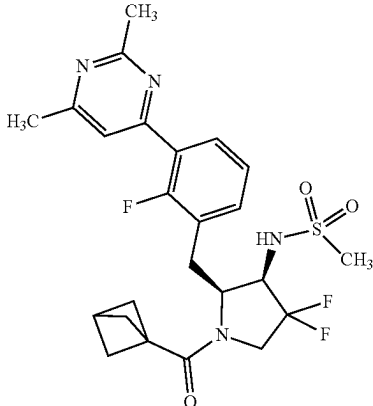
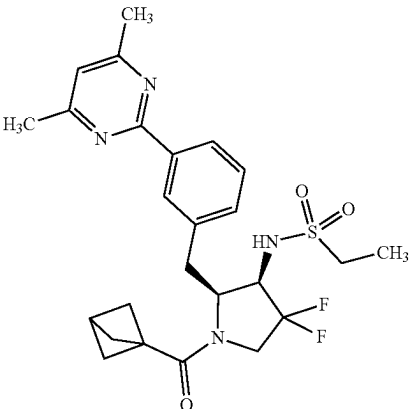
EX.	IUPAC NAME	Structure	ADDITIVE	MS
109	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{{[3-(4,6-dimethylpyridin-2-yl)phenyl]methyl}-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			504.2
110	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-{{[3-(4-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			490.2
111	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{{[3-(2,6-dimethylpyrimidin-4-yl)-2-fluorophenyl]methyl}-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			509.1
112	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{{[3-(4,6-dimethylpyrimidin-2-yl)phenyl]methyl}-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			505.2

TABLE 1-1-continued

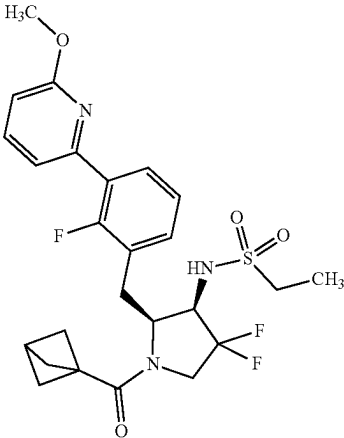
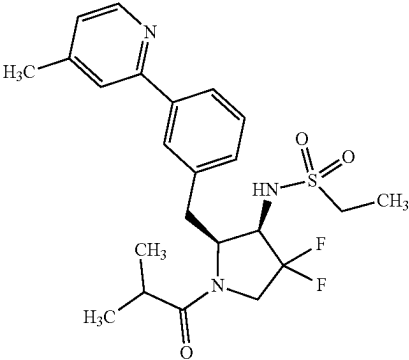
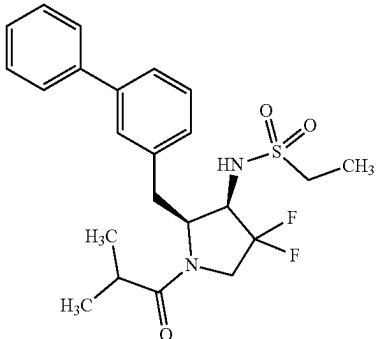
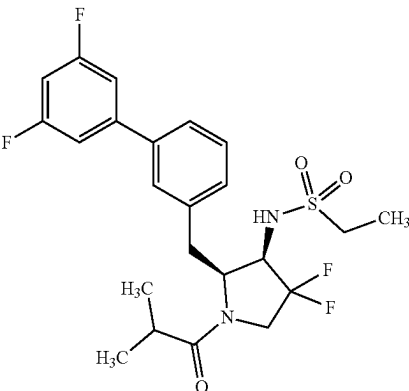
EX.	IUPAC NAME	Structure	ADDITIVE	MS
113	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-{[3-(6-methoxypyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			506.2
114	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-{[3-(4-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			466.2
115	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl]-methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			451.2
116	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			487.2

TABLE 1-1-continued

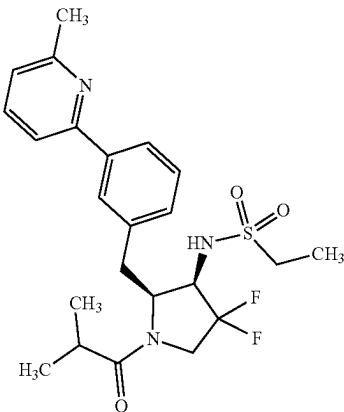
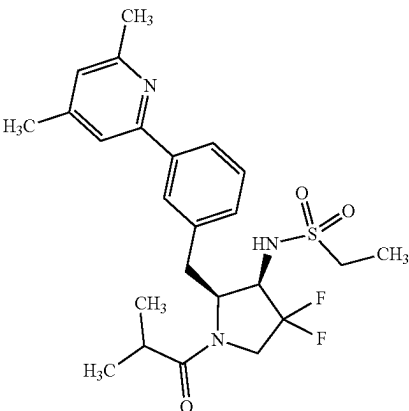
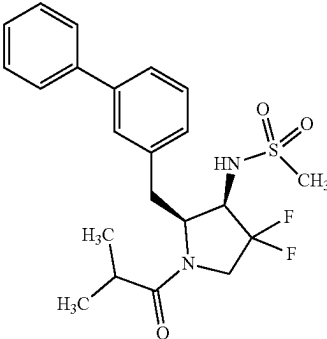
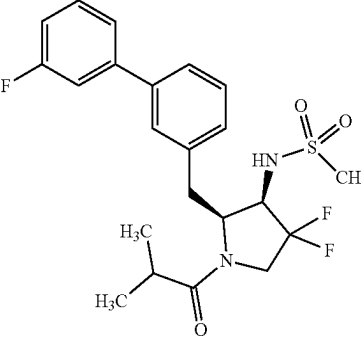
EX.	IUPAC NAME	Structure	ADDITIVE	MS
117	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-{[3-(6-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]ethanesulfonamide			466.2
118	N-[(2S,3R)-2-{[3-(4,6-dimethylpyridin-2-yl)phenyl]methyl}-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			480.2
119	N-[(2S,3R)-2-[(1,1'-biphenyl)-3-yl]-methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			437.2
120	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			453.2

TABLE 1-1-continued

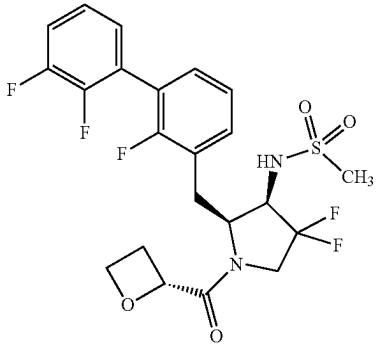
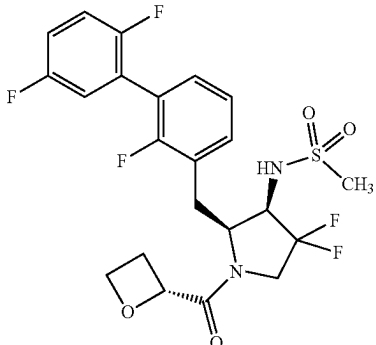
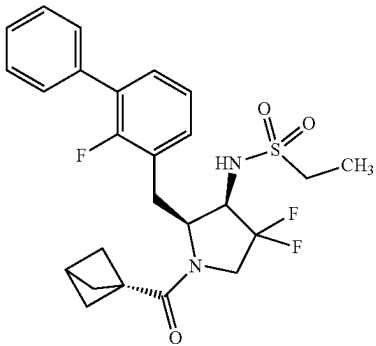
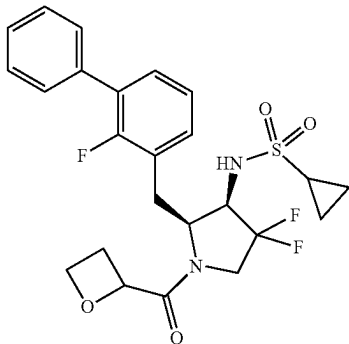
EX.	IUPAC NAME	Structure	ADDITIVE	MS
121	N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			505.1
122	N-[(2S,3R)-4,4-difluoro-1-((2R)-oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			505.1
123	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			493.3
124	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time			495.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
125	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			495.2
126	N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-2-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl}cyclopropane-sulfonamide			509.2
127	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyl-oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time by HPLC (column: CHIRALPAK IC (4.6 mmI · D. × 250 mmL., 5 μm), mobile phase: hexane/EtOH/diethylamine = 550/450/1 (v/v/v))			483.2
128	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyl-oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time by HPLC (column: CHIRALPAK IC (4.6 mmI · D. × 250 mmL., 5 μm), mobile phase: hexane/EtOH/diethylamine = 550/450/1 (v/v/v))			483.2

TABLE 1-1-continued

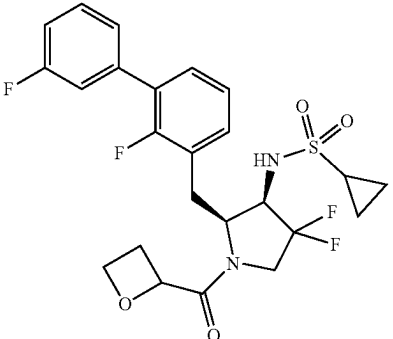
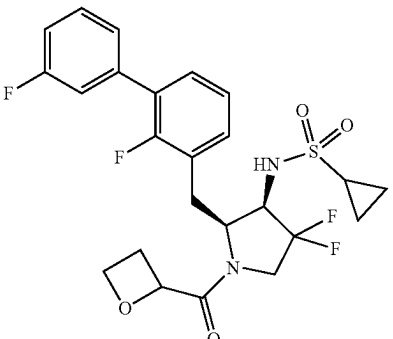
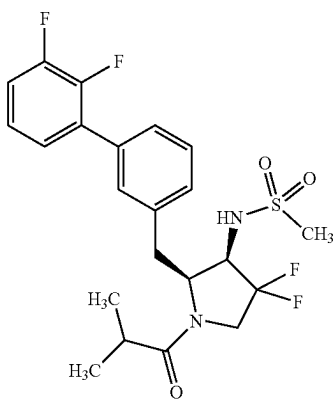
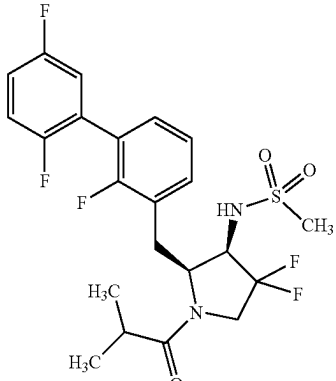
EX.	IUPAC NAME	Structure	ADDITIVE	MS
129	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time			513.2
130	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			513.1
131	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			473.2
132	N-[(2S,3R)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			473.2

TABLE 1-1-continued

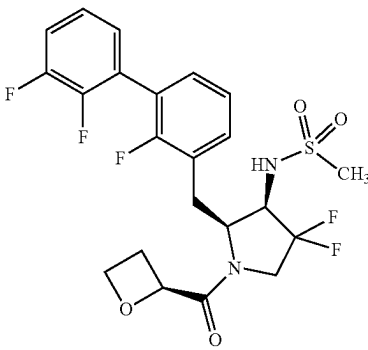
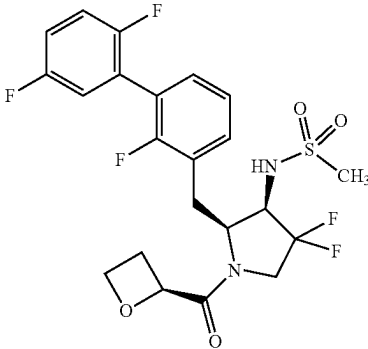
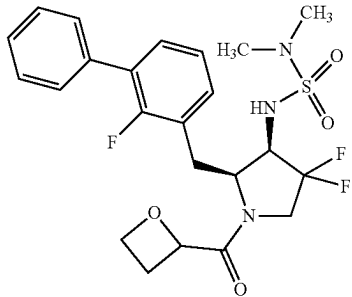
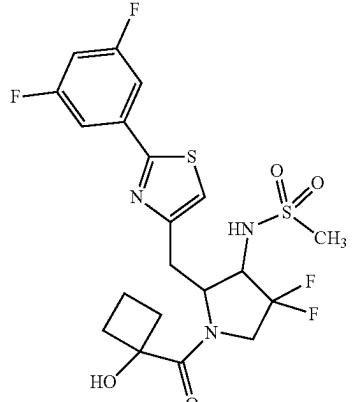
EX.	IUPAC NAME	Structure	ADDITIVE	MS
133	N-[(2S,3R)-4,4-difluoro-1-({2S}-oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			505.1
134	N-[(2S,3R)-4,4-difluoro-1-({2S}-oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			505.1
135	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide with shorter retention time			498.1
136	rac-N-[(2S,3R)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide			508.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
137	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide with longer retention time			498.1
138	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			485.2
139	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			497.2
140	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			500.3
141	N'-{(2S,3R)-1-(cyclobutanecarbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			496.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
142	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-N-methoxy-N-methylpyrrolidine-1-carboxamide			501.2
143	N'-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			508.2
144	N'-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethane-sulfonamide			500.8
145	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			512.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
146	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			505.2
147	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			505.2
148	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			505.2
149	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			484.2

TABLE 1-1-continued

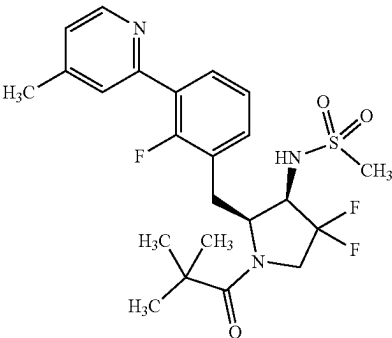
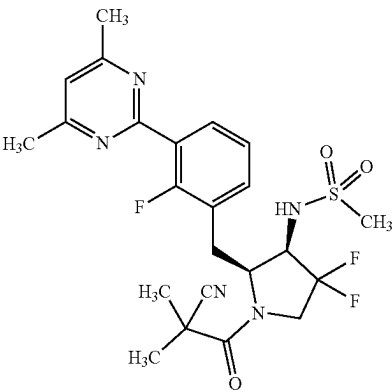
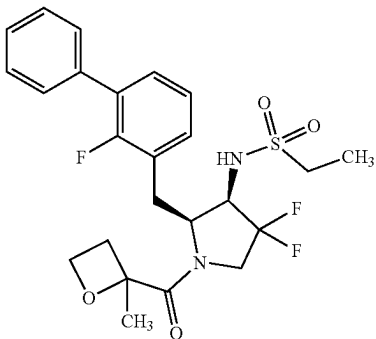
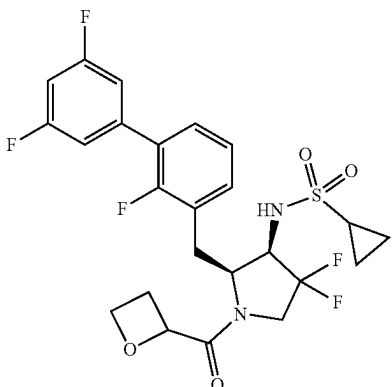
EX.	IUPAC NAME	Structure	ADDITIVE	MS
150	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-4,4-difluoro-2-{{[2-fluoro-3-(4-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]methanesulfonamide			484.2
151	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-2-{{[3-(4,6-dimethylpyrimidin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-pyrrolidin-3-yl]methanesulfonamide			499.3
152	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyloxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by SFC (column: CHIRALPAK IC (4.6 mmI · D. × 150 mmL, 5 μm), mobile phase: carbon dioxide/(MeOH/diethylamine) = 1000/1) = 750/250 (v/v))			497.2
153	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropanesulfonamide with longer retention time			531.1

TABLE 1-1-continued

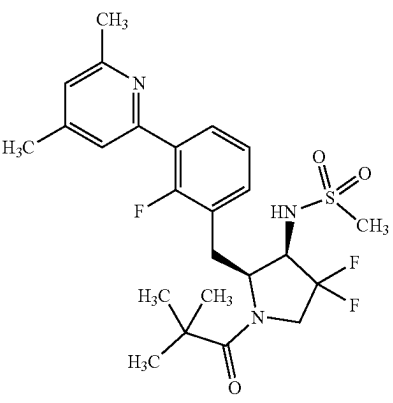
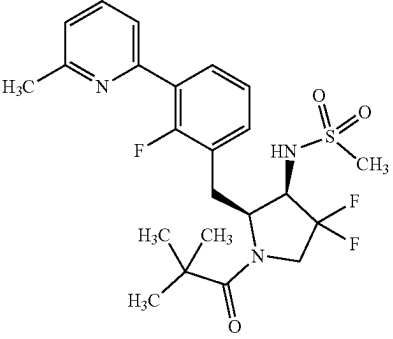
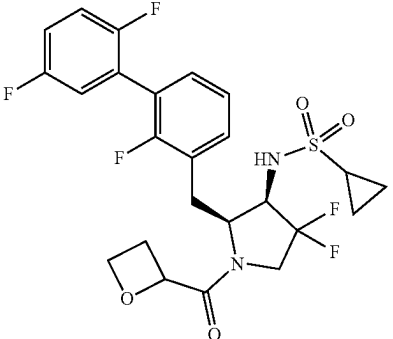
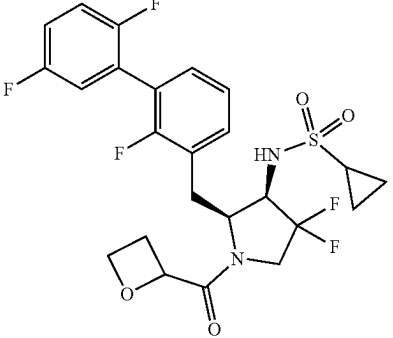
EX.	IUPAC NAME	Structure	ADDITIVE	MS
154	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-2-[[3-(4,6-dimethylpyridin-2-yl)-2-fluorophenyl]methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			498.2
155	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-4,4-difluoro-2-[[3-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl]pyrrolidin-3-yl]methanesulfonamide			484.2
156	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropanesulfonamide with longer retention time			531.1
157	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropanesulfonamide with shorter retention time			531.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
158	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methyloxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			501.2
159	N-{(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}cyclopropanesulfonamide with longer retention time			531.2
160	N-[(2S,3R)-2-[(3',4'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]-methanesulfonamide			471.2
161	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methyloxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time by SFC (column: CHIRALPAK IC (4.6 mmL · D. × 150 mmL, 5 μm), mobile phase: carbon dioxide/(MeOH/diethylamine) = 1000/1) = 750/250 (v/v))			497.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
162	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide with shorter retention time			531.2
163	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]cyclopropane-sulfonamide			499.2
164	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methyloxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			501.1
165	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide with shorter retention time			531.1

TABLE 1-1-continued

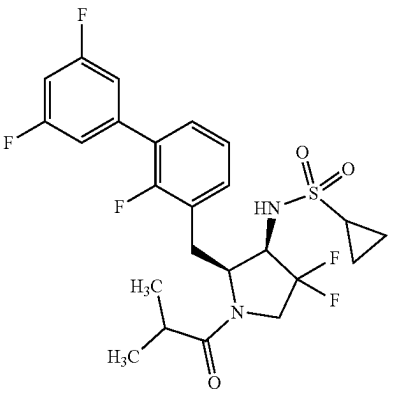
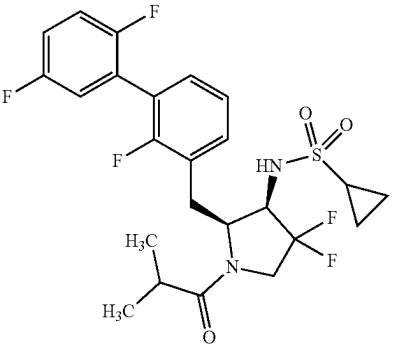
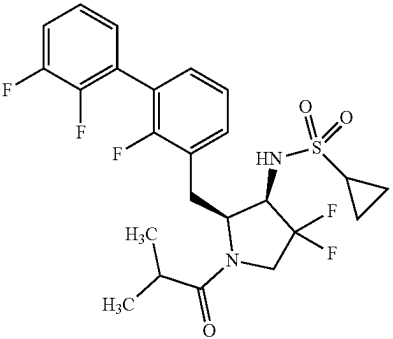
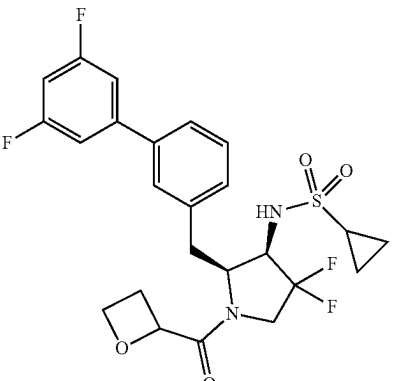
EX.	IUPAC NAME	Structure	ADDITIVE	MS
166	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			517.2
167	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			517.1
168	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			517.2
169	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with shorter retention time			513.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
170	N-[(2S,3R)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			513.2
171	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]cyclopropane-sulfonamide			481.2
172	N-[(2S,3R)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention			501.1
173	N-[(2S,3R)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			501.1

TABLE 1-1-continued

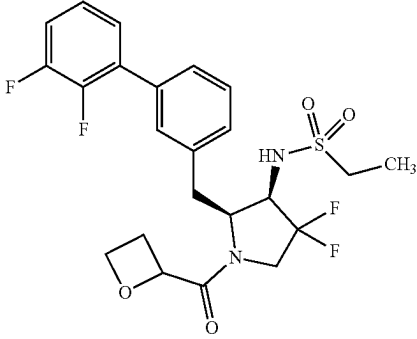
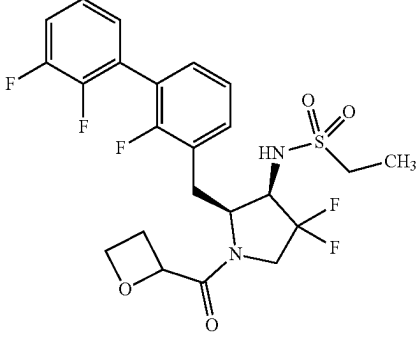
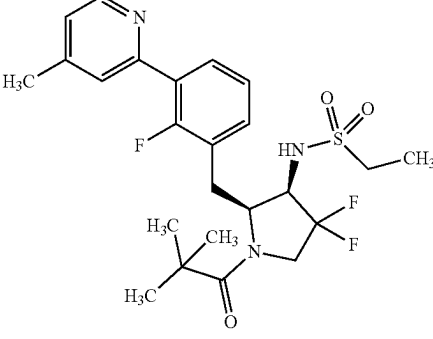
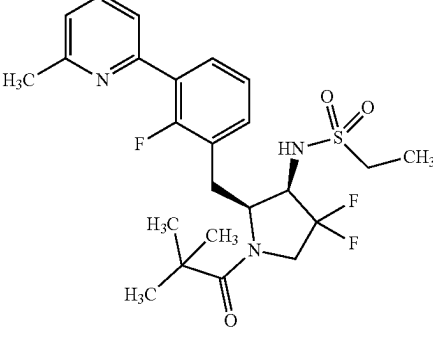
EX.	IUPAC NAME	Structure	ADDITIVE	MS
174	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			501.2
175	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			501.2
176	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-4,4-difluoro-2-{{[2-fluoro-3-(4-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl}ethanesulfonamide			498.2
177	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-4,4-difluoro-2-{{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl}ethanesulfonamide			498.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
178	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-2-[[3-(4,6-dimethylpyridin-2-yl)-2-fluorophenyl]methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			512.2
179	N-[(2S,3R)-1-(2,2-dimethylpropanoyl)-2-[[3-(4,6-dimethylpyrimidin-2-yl)-2-fluorophenyl]methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			513.3
180	N-[(2S,3R)-2-[[[1,1'-biphenyl]-3-yl]-methyl]-4,4-difluoro-1-(oxetane-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with retention time			477.2
181	N-[(2S,3R)-2-[[[1,1'-biphenyl]-3-yl]-methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			477.2

TABLE 1-1-continued

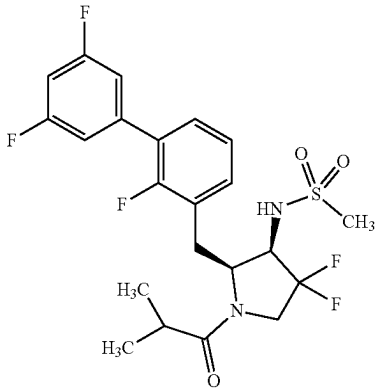
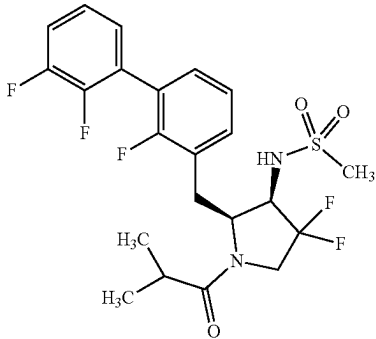
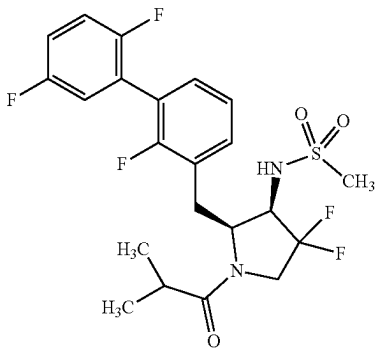
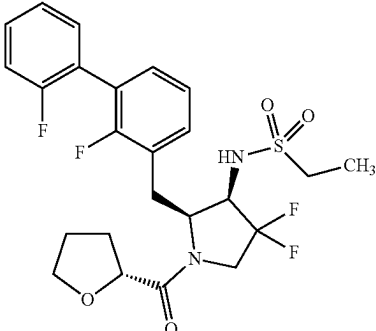
EX.	IUPAC NAME	Structure	ADDITIVE	MS
182	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			489.2
183	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			491.1
184	N-[(2S,3R)-4,4-difluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			491.2
185	N-[(2S,3R)-2-[(2,2',-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			515.2

TABLE 1-1-continued

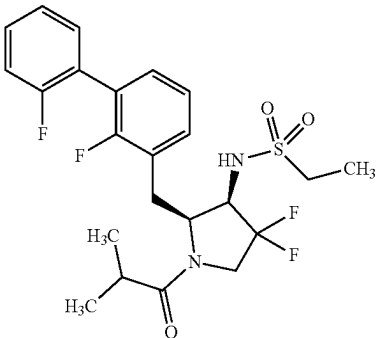
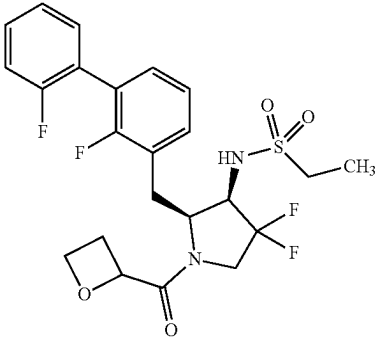
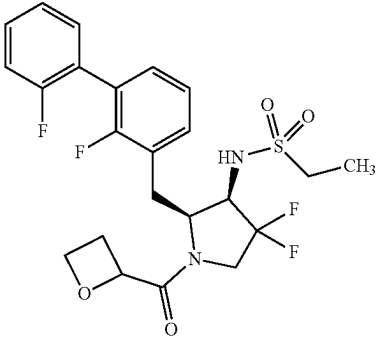
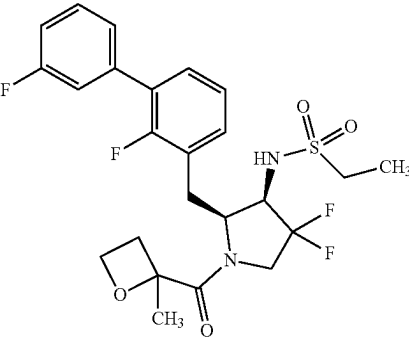
EX.	IUPAC NAME	Structure	ADDITIVE	MS
186	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			487.2
187	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			501.2
188	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			501.1
189	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methyloxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			515.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
190	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]cyclopropane-sulfonamide			499.2
191	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			513.2
192	N-[(2S,3R)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time			513.2
193	N-{(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2S)-oxolane-2-carbonyl]pyrrolidin-3-yl}methanesulfonamide			483.2

TABLE 1-1-continued

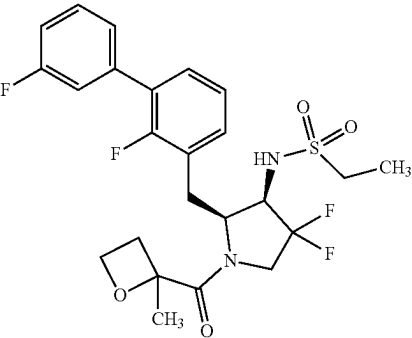
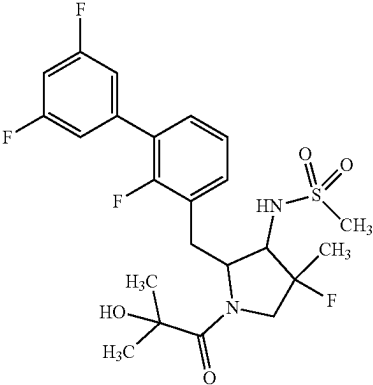
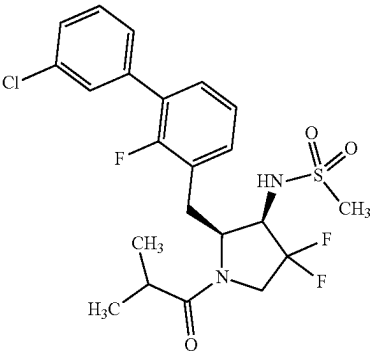
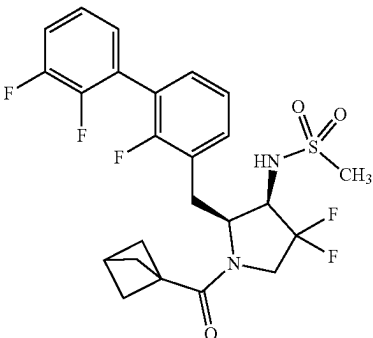
EX.	IUPAC NAME	Structure	ADDITIVE	MS
194	N-[(2S,3R)-2-[(2,3,-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methyl-oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			515.1
195	rac-N-[(2S,3R,4R)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-4-methyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl]methanesulfonamide			503.2
196	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			489.2
197	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			515.2

TABLE 1-1-continued

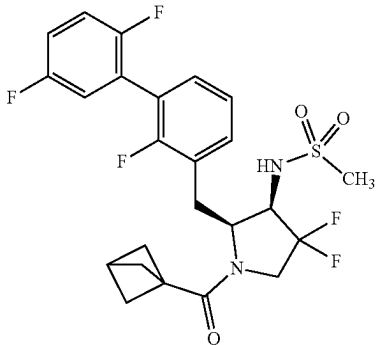
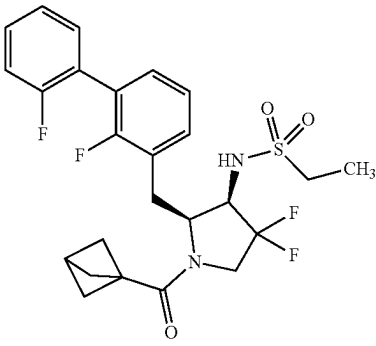
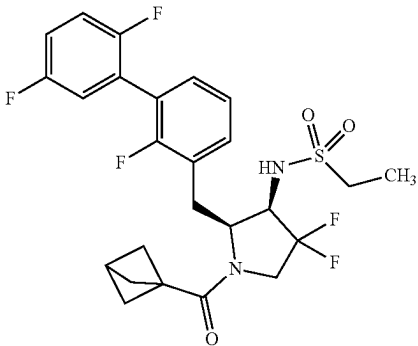
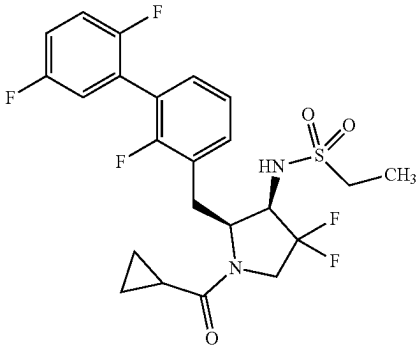
EX.	IUPAC NAME	Structure	ADDITIVE	MS
198	N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide			515.2
199	N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}ethanesulfonamide			511.2
200	N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide			529.0
201	N-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide			503.1

TABLE 1-1-continued

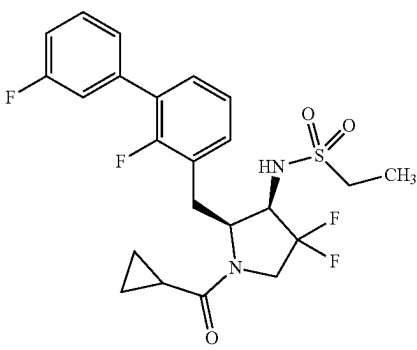
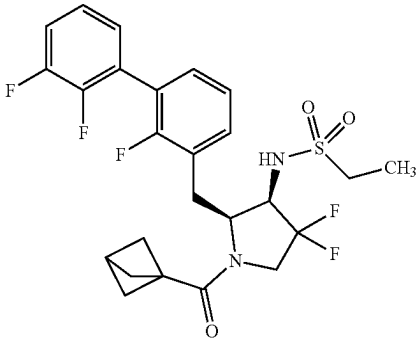
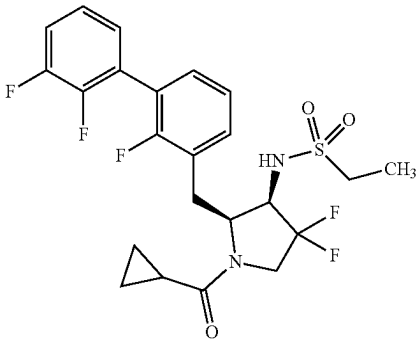
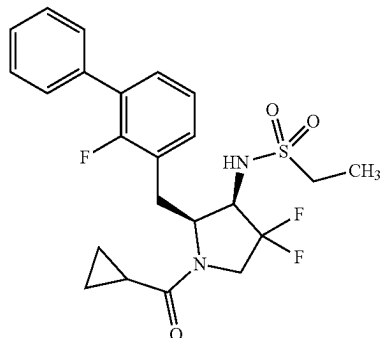
EX.	IUPAC NAME	Structure	ADDITIVE	MS
202	N-[(2S,3R)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			485.1
203	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			529.1
204	N-[(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			503.2
205	N-[(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			467.2

TABLE 1-1-continued

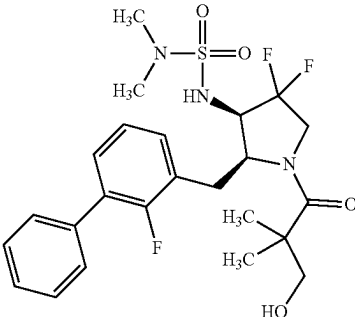
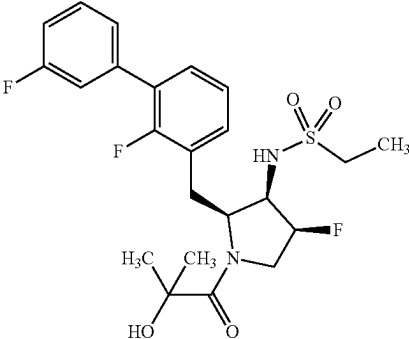
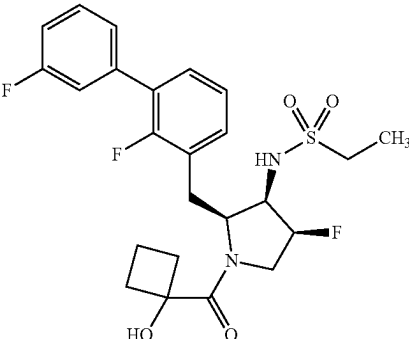
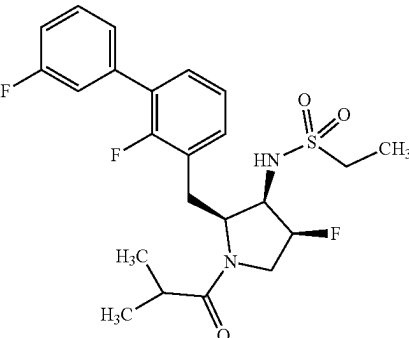
EX.	IUPAC NAME	Structure	ADDITIVE	MS
206	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-hydroxy-2,2-dimethylpropanoyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			514.2
207	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			485.2
208	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			497.1
209	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			469.2

TABLE 1-1-continued

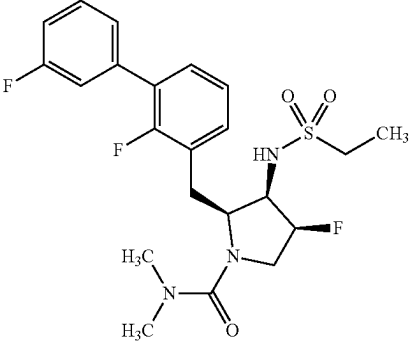
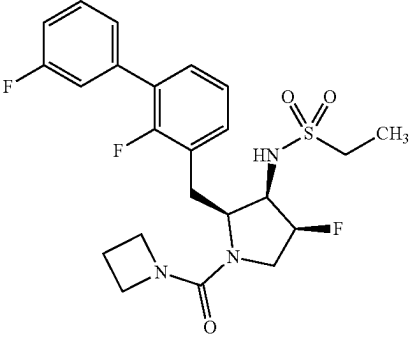
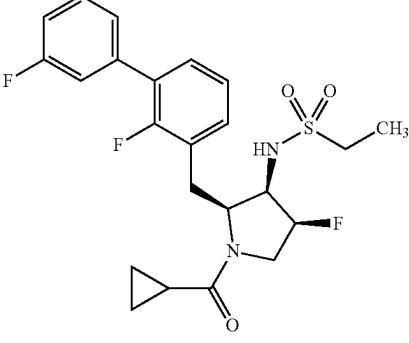
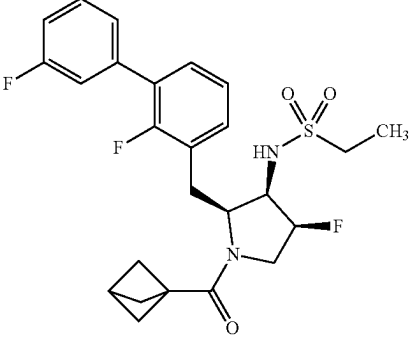
EX.	IUPAC NAME	Structure	ADDITIVE	MS
210	(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			470.2
211	N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			482.2
212	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			467.2
213	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			493.2

TABLE 1-1-continued

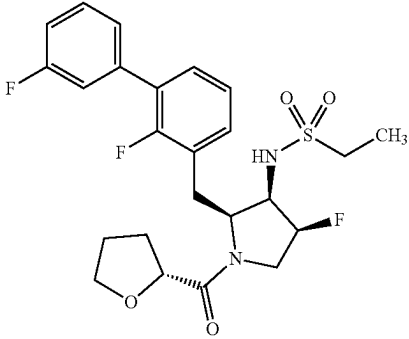
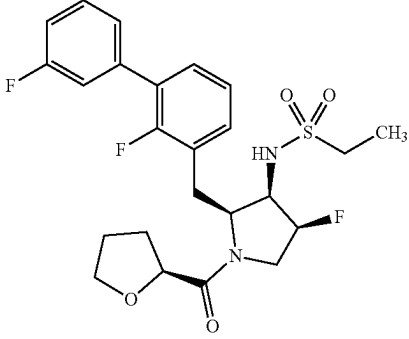
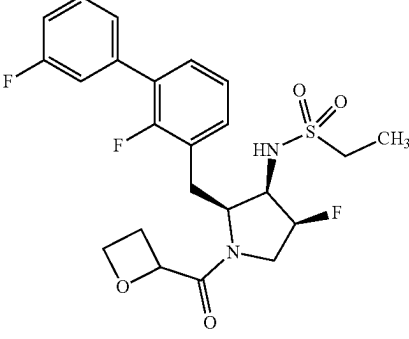
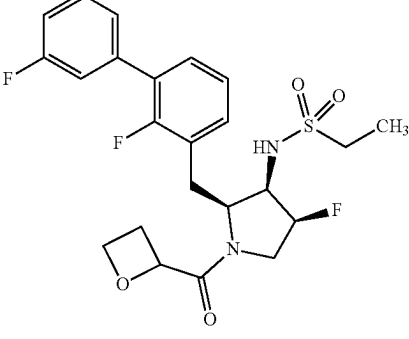
EX.	IUPAC NAME	Structure	ADDITIVE	MS
214	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]-ethanesulfonamide			497.3
215	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-[(2S)-oxolane-2-carbonyl]pyrrolidin-3-yl]-ethanesulfonamide			497.3
216	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			483.2
217	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			483.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
218	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3,3-difluorocyclobutane-1-carbonyl)-4,4-difluoropyrrolidin-3-yl]-ethanesulfonamide			533.3
219	N-[(2S,3R)-1-(3,3-difluoroazetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			536.4
220	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			503.4
221	N-[(2S,3R,4S)-4-fluoro-1-{2-hydroxy-2-methylpropanoyl}-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			503.4

TABLE 1-1-continued

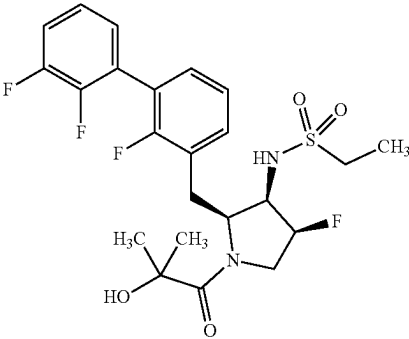
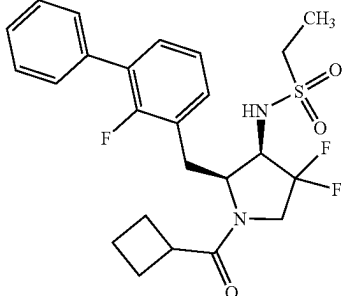
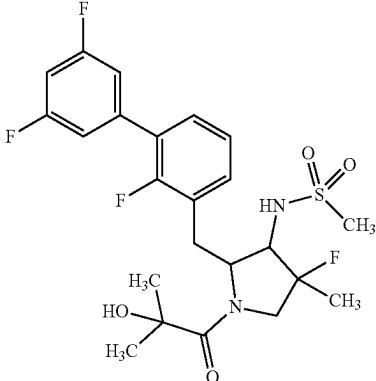
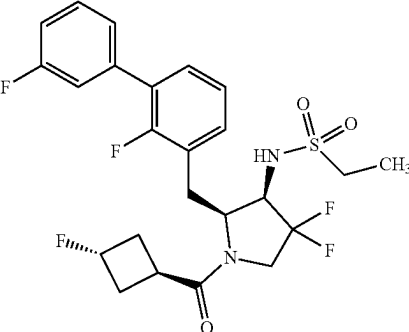
EX.	IUPAC NAME	Structure	ADDITIVE	MS
222	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			503.4
223	N-[(2S,3R)-1-(cyclobutanecarbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			481.4
224	rac-N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-4-methyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			503.4
225	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((1r,3S)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			515.2

TABLE 1-1-continued

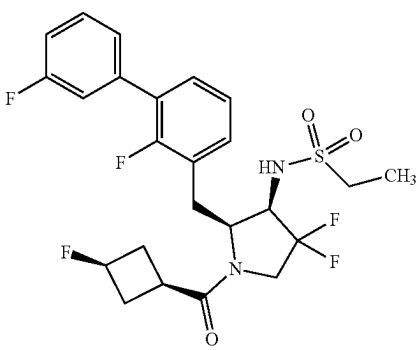
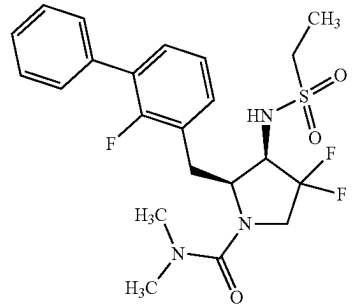
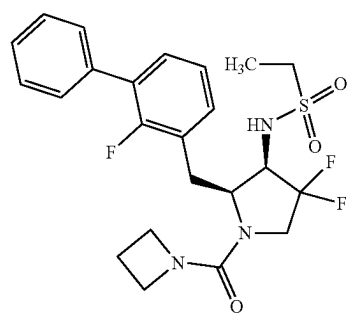
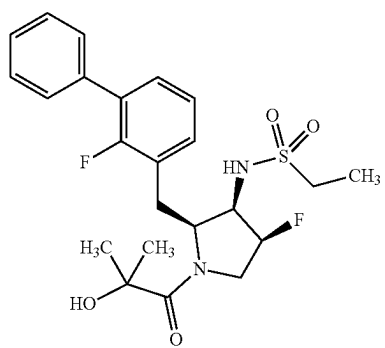
EX.	IUPAC NAME	Structure	ADDITIVE	MS
226	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			515.2
227	(2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			470.4
228	N-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide			482.4
229	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			467.4

TABLE 1-1-continued

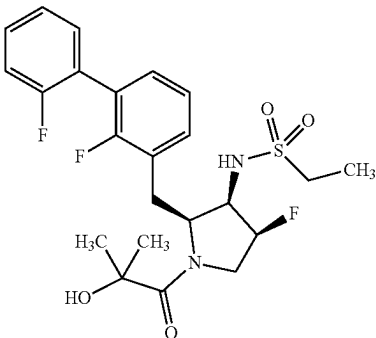
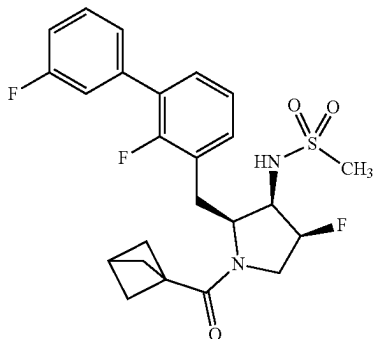
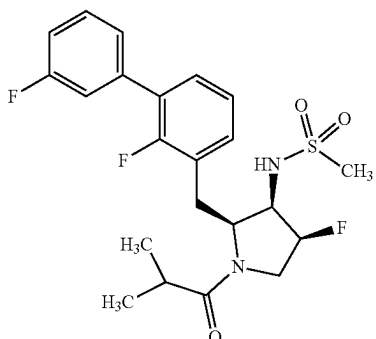
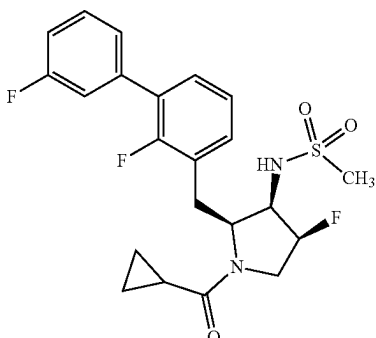
EX.	IUPAC NAME	Structure	ADDITIVE	MS
230	N-[(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			485.5
231	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]-methanesulfonamide			479.4
232	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]-methanesulfonamide			455.4
233	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)-methyl]-4-fluoropyrrolidin-3-yl}methanesulfonamide			453.4

TABLE 1-1-continued

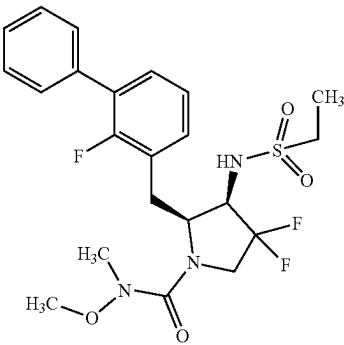
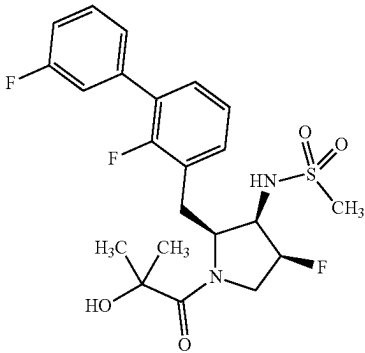
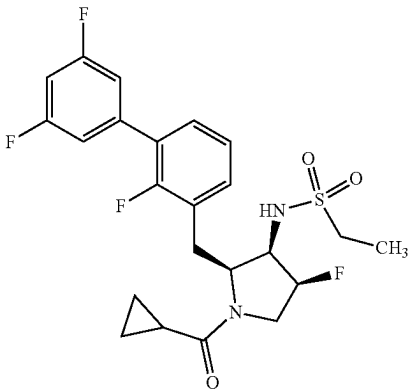
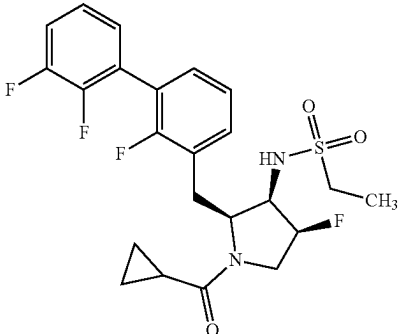
EX.	IUPAC NAME	Structure	ADDITIVE	MS
234	(2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-N-methoxy-N-methylpyrrolidin-1-carboxamide			486.4
235	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			471.4
236	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			485.4
237	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			485.4

TABLE 1-1-continued

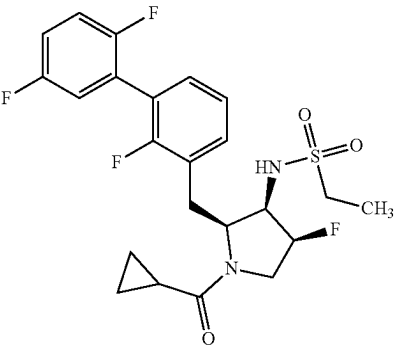
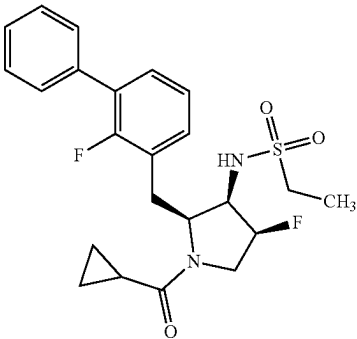
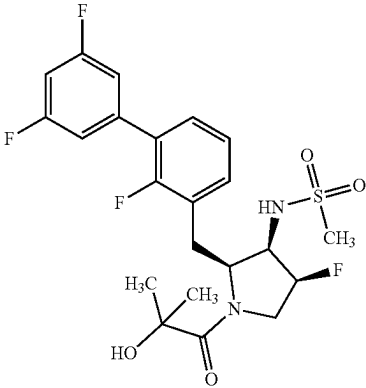
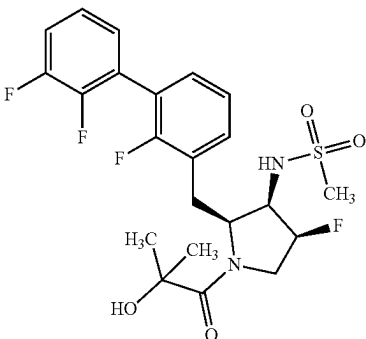
EX.	IUPAC NAME	Structure	ADDITIVE	MS
238	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			485.4
239	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			449.4
240	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			489.3
241	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			489.4

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
242	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			489.4
243	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]-methanesulfonamide			453.4
244	N-[(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]-methanesulfonamide			471.4
245	N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-methanesulfonamide			503.1

TABLE 1-1-continued

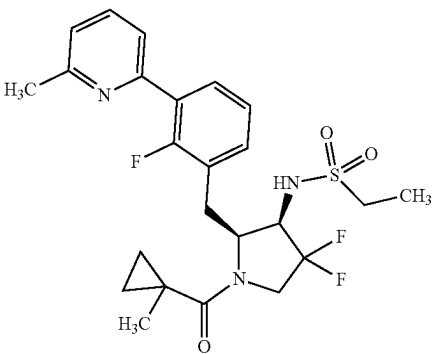
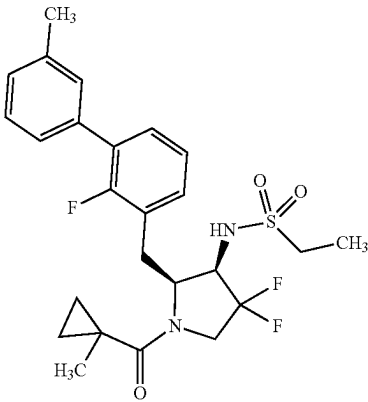
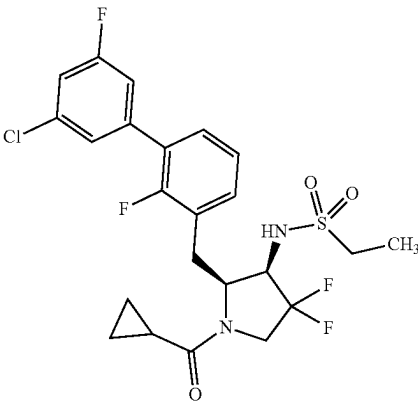
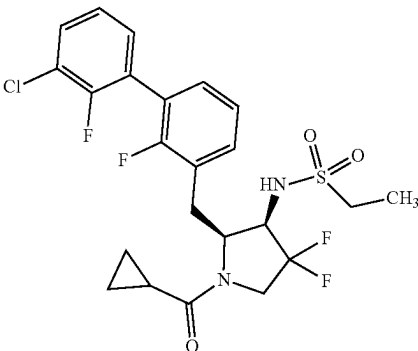
EX.	IUPAC NAME	Structure	ADDITIVE	MS
246	N-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-1-(1-methylcyclopropane-1-carbonyl)pyrrolidin-3-yl}ethanesulfonamide		496.2	
247	N-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(4-methylphenyl)phenyl]methyl}-1-(1-methylcyclopropane-1-carbonyl)pyrrolidin-3-yl}ethanesulfonamide		496.1	
248	N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide		519.1	
249	N-[(2S,3R)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide		519.1	

TABLE 1-1-continued

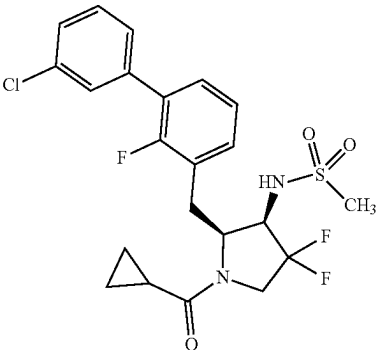
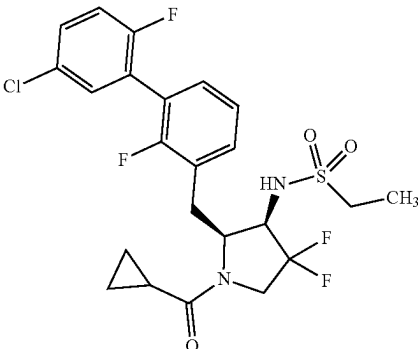
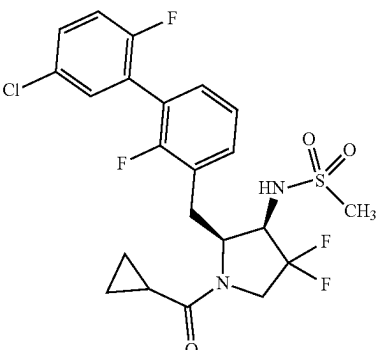
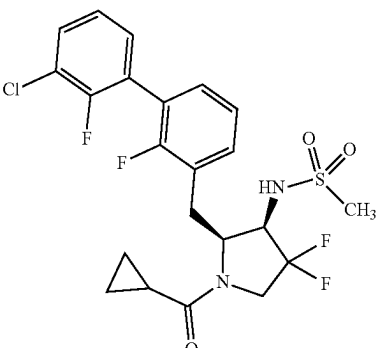
EX.	IUPAC NAME	Structure	ADDITIVE	MS
250	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-methanesulfonamide			487.1
251	N-[(2S,3R)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-ethanesulfonamide			519.1
252	N-[(2S,3R)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-methanesulfonamide			505.1
253	N-[(2S,3R)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-methanesulfonamide			505.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
254	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropane-carbonyl)-4,4-difluoropyrrolidin-3-yl]-ethanesulfonamide			501.1
255	N-[(2S,3R)-2-{[3-(4,6-dimethylpyrimidin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-methylcyclopropane-1-carbonyl)-pyrrolidin-3-yl]ethanesulfonamide			511.2
256	ethyl(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)-amino]-4,4-difluoropyrrolidine-1-carboxylate			487.2
257	N-[(2S,3R)-1-butanoyl-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			487.4

TABLE 1-1-continued

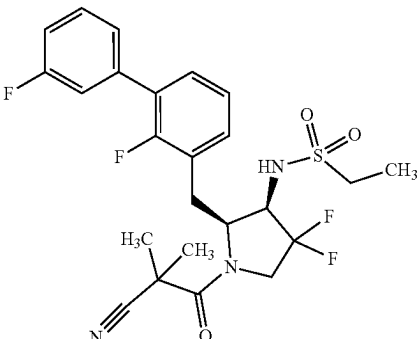
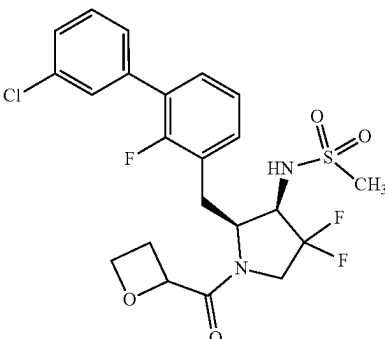
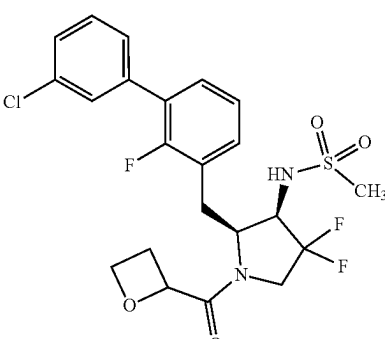
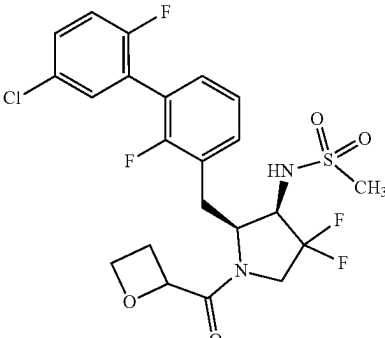
EX.	IUPAC NAME	Structure	ADDITIVE	MS
258	N-[(2S,3R)-1-(2-cyano-2-methylpropanoyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			510.2
259	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			503.4
260	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with longer retention time			503.4
261	N-[(2S,3R)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide with shorter retention time			521.3

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
262	N-[(2S,3R)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			521.3
263	N-[(2S,3R)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			521.3
264	N-[(2S,3R)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			521.3
265	N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			521.3

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
266	N-[(2S,3R)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			521.3
267	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			517.4
268	N-[(2S,3R)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			517.4
269	rac-N-[(2S,3R)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			510.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
270	rac-N-[(2S,3R)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4,4-difluoro-1-{1-hydroxycyclobutane-1-carbonyl}pyrrolidin-3-yl]ethanesulfonamide			522.1
271	rac-N-[(2S,3R,4R)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-4-methyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			517.2
272	N-[(2S,3R)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			531.2
273	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			515.2

EX.	IUPAC NAME	Structure	ADDITIVE	MS
274	N-[(2S,3R)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide		533.2	
275	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide		511.2	
276	N-[(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide		529.2	
277	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time		497.2	

TABLE 1-1-continued

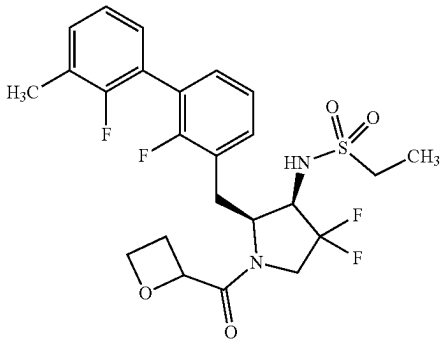
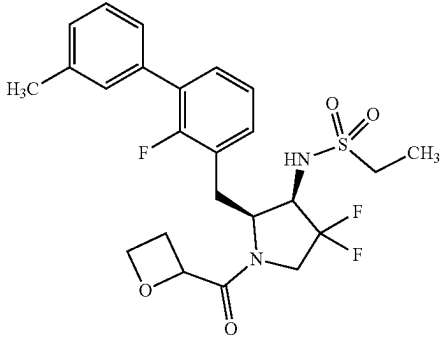
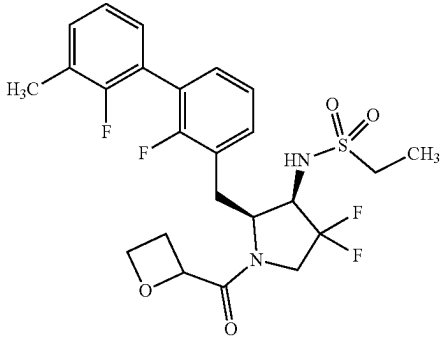
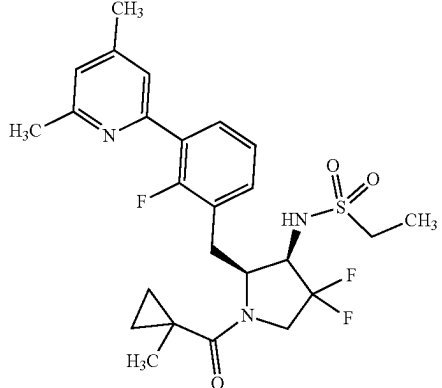
EX.	IUPAC NAME	Structure	ADDITIVE	MS
278	N-[(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			515.2
279	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			497.2
280	N-[(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			515.2
281	N-[(2S,3R)-2-{[3-(4,6-dimethylpyridin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-methylcyclopropane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			510.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
282	(2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-N,N-dimethyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			506.2
283	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			518.2
284	(2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			484.2
285	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			496.2

TABLE 1-1-continued

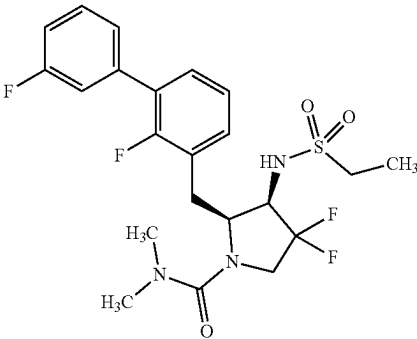
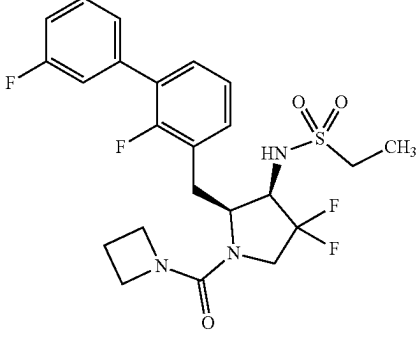
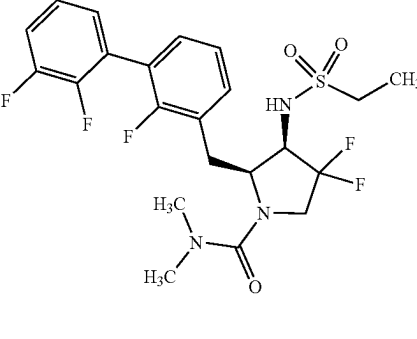
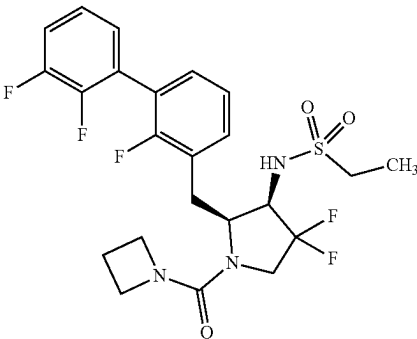
EX.	IUPAC NAME	Structure	ADDITIVE	MS
286	(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			488.2
287	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			500.2
288	(2S,3R)-3-[(ethanesulfonyl)amino]-4,4-difluoro-N,N-dimethyl-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			506.2
289	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			518.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
290	(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			502.2
291	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]-ethanesulfonamide			514.1
292	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			487.2
293	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			451.3

TABLE 1-1-continued

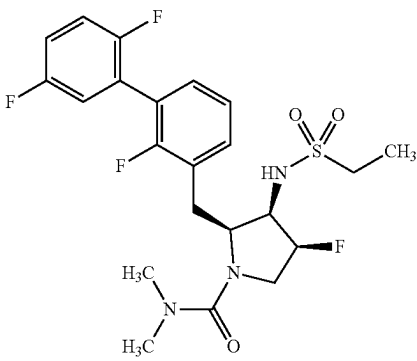
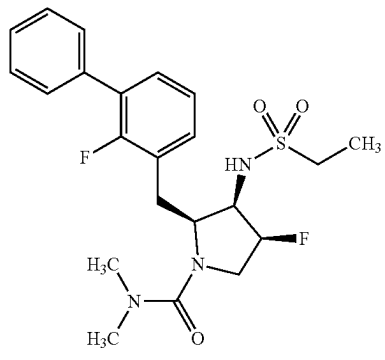
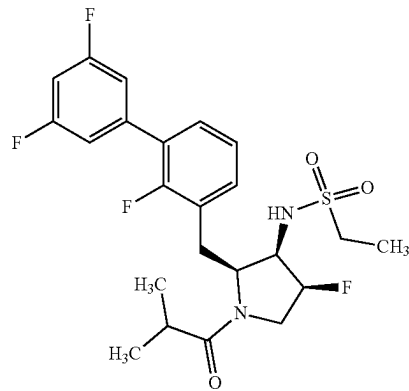
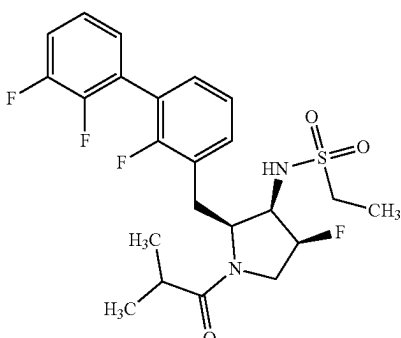
EX.	IUPAC NAME	Structure	ADDITIVE	MS
294	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethyl-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			488.2
295	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			452.2
296	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			487.2
297	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			487.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
298	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			485.2
299	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			467.2
300	N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with shorter retention time			501.2
301	N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with longer retention time			501.1

TABLE 1-1-continued

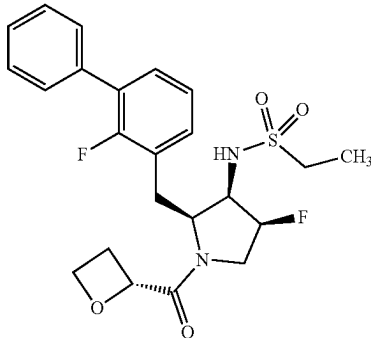
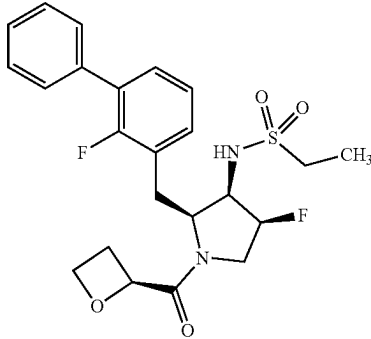
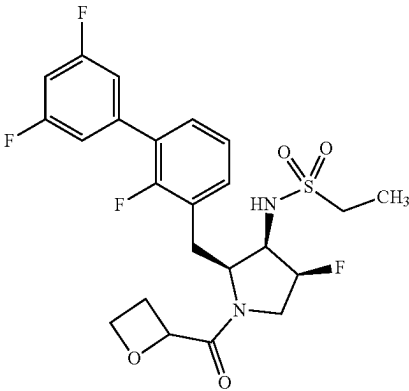
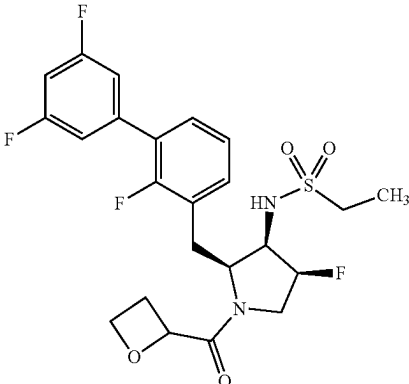
EX.	IUPAC NAME	Structure	ADDITIVE	MS
302	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((2R)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			465.2
303	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((2S)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			465.2
304	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			501.2
305	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with longer retention time			501.1

TABLE 1-1-continued

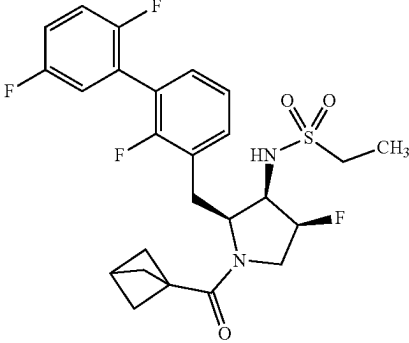
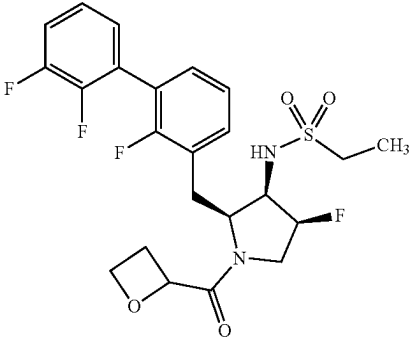
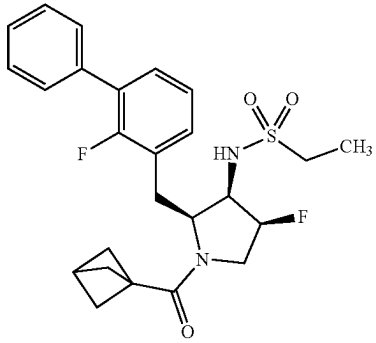
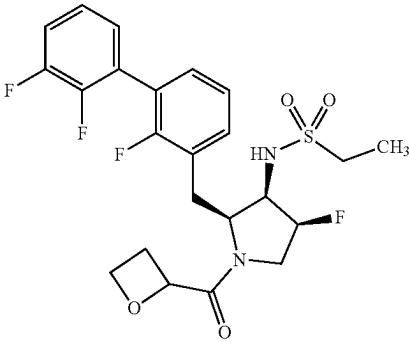
EX.	IUPAC NAME	Structure	ADDITIVE	MS
306	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			511.2
307	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			501.1
308	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			475.3
309	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			501.1

TABLE 1-1-continued

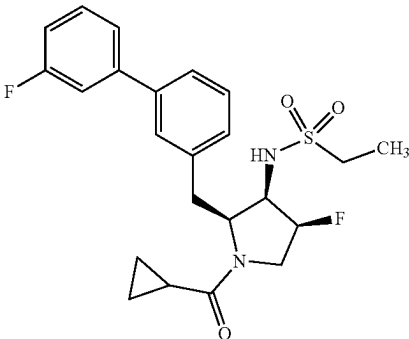
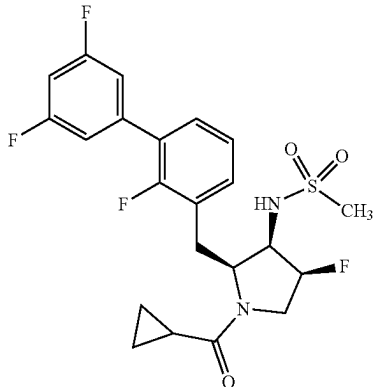
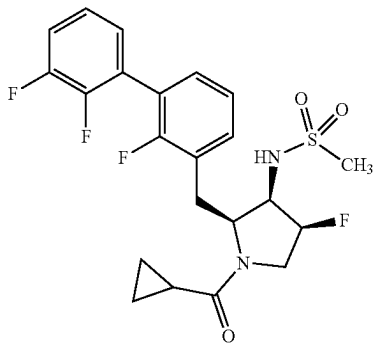
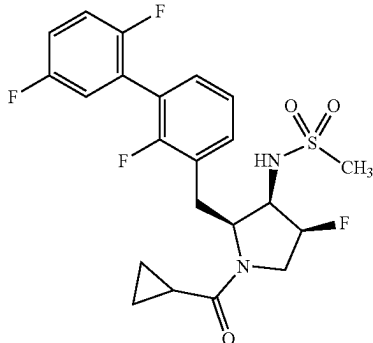
EX.	IUPAC NAME	Structure	ADDITIVE	MS
310	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			449.0
311	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			468.9
312	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			471.0
313	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			471.0

TABLE 1-1-continued

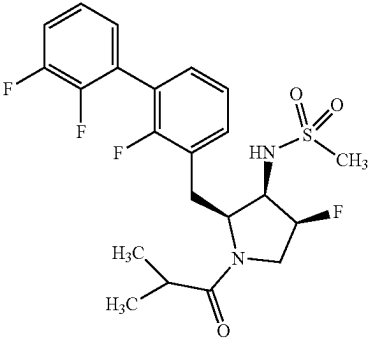
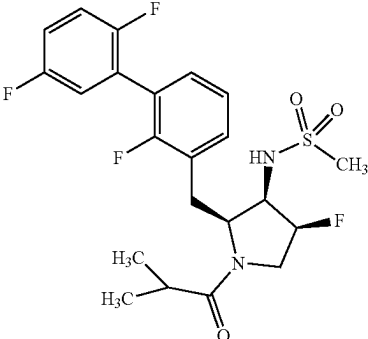
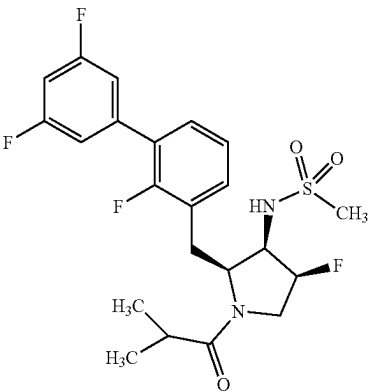
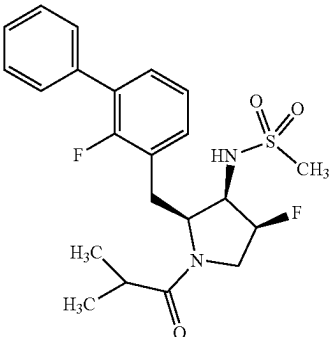
EX.	IUPAC NAME	Structure	ADDITIVE	MS
314	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			473.0
315	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			473.0
316	N-[(2S,3R,4S)-4-fluoro-1-(2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			473.0
317	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			437.1

TABLE 1-1-continued

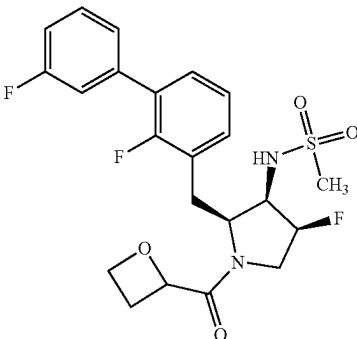
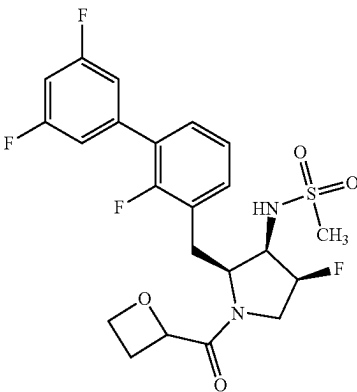
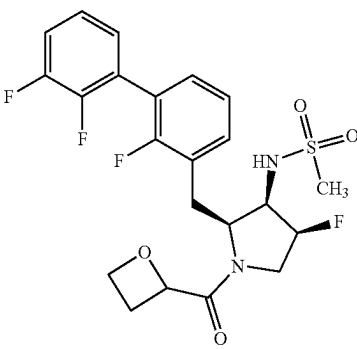
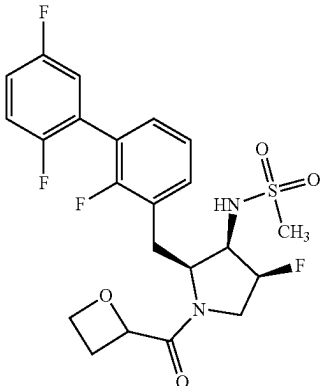
EX.	IUPAC NAME	Structure	ADDITIVE	MS
318	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			469.0
319	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide with longer retention time			484.8
320	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide with longer retention time			486.9
321	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide with longer retention time			486.9

TABLE 1-1-continued

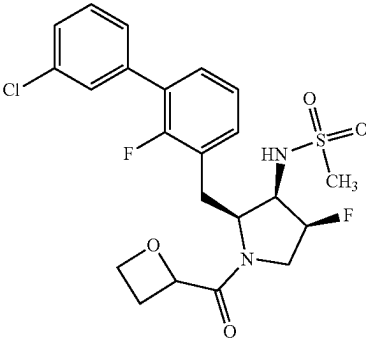
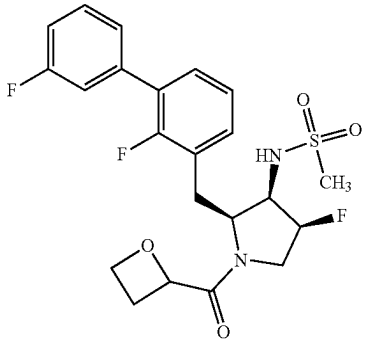
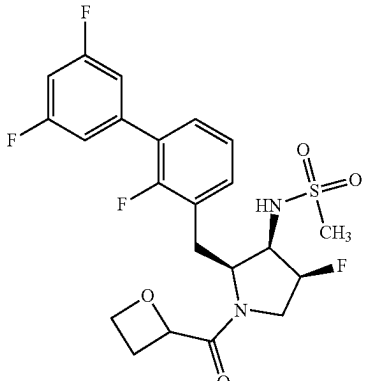
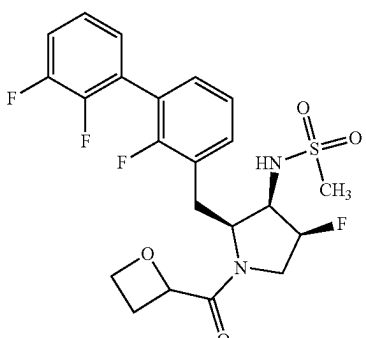
EX.	IUPAC NAME	Structure	ADDITIVE	MS
322	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			484.9
323	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			469.0
324	N-{(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-methanesulfonamide with shorter retention time			486.9
325	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			486.9

TABLE 1-1-continued

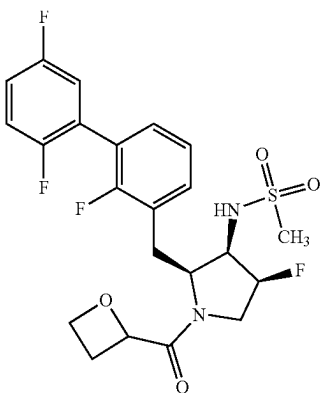
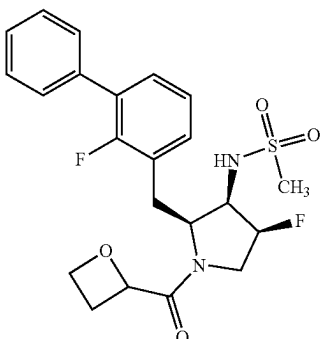
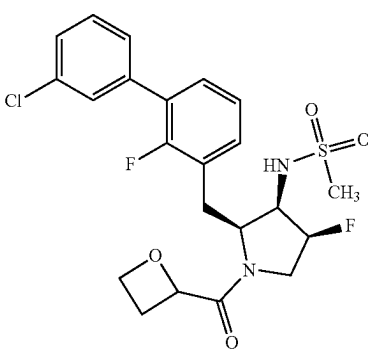
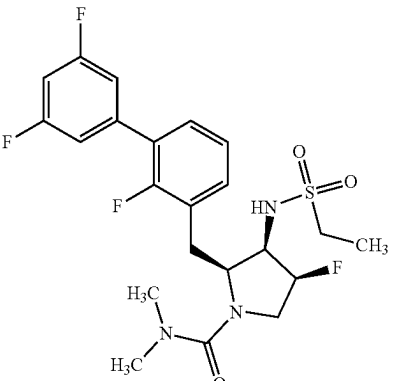
EX.	IUPAC NAME	Structure	ADDITIVE	MS
326	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			486.9
327	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]methanesulfonamide derived from N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA))			451.0
328	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			484.9
329	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-fluoro-N,N-dimethyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidine-1-carboxamide			488.0

TABLE 1-1-continued

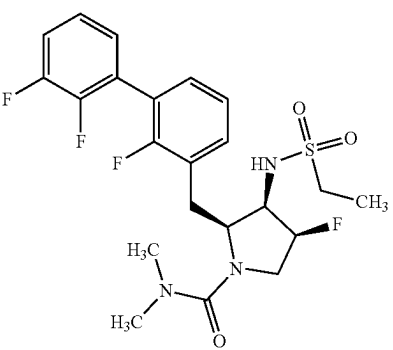
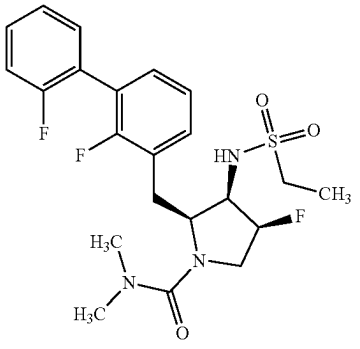
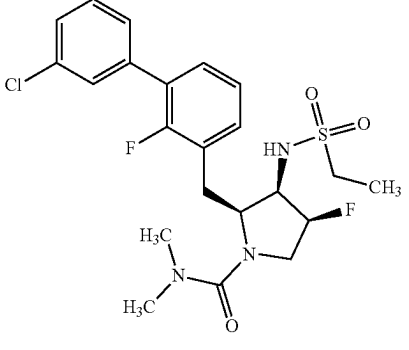
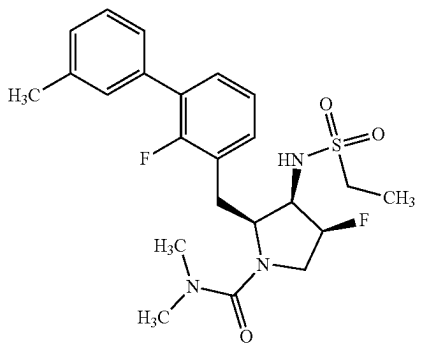
EX.	IUPAC NAME	Structure	ADDITIVE	MS
330	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-fluoro-N,N-dimethyl-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			488.0
331	(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			470.0
332	(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			485.9
333	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			466.0

TABLE 1-1-continued

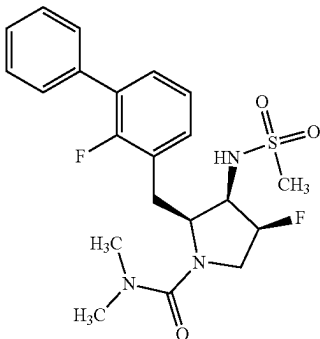
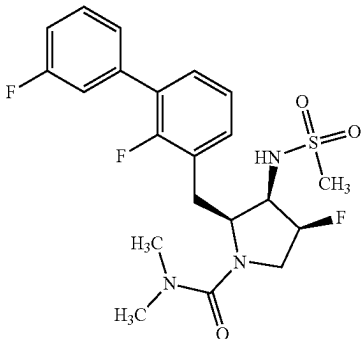
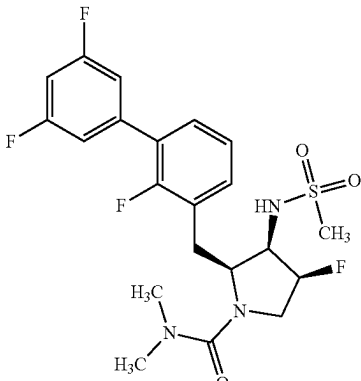
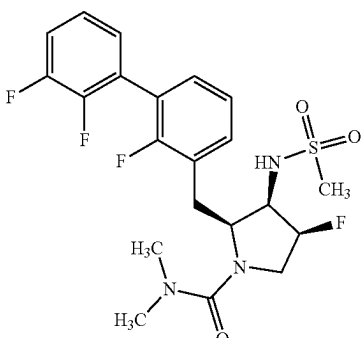
EX.	IUPAC NAME	Structure	ADDITIVE	MS
334	(2S,3R,4S)-4-fluoro-2-[2-fluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			438.1
335	(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			456.0
336	(2S,3R,4S)-4-fluoro-3-[(methanesulfonyl)amino]-N,N-dimethyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			474.0
337	(2S,3R,4S)-4-fluoro-3-[(methanesulfonyl)amino]-N,N-dimethyl-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			474.0

TABLE 1-1-continued

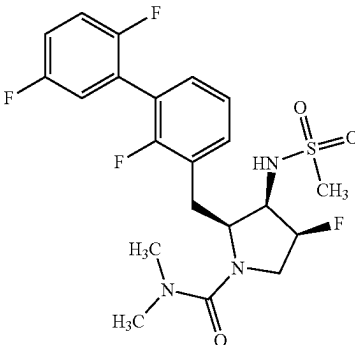
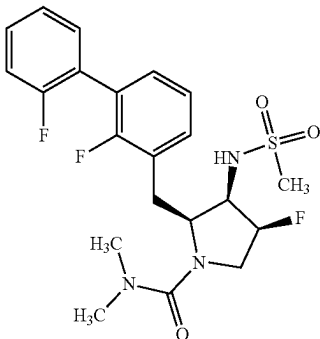
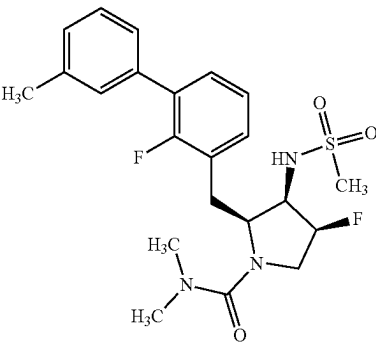
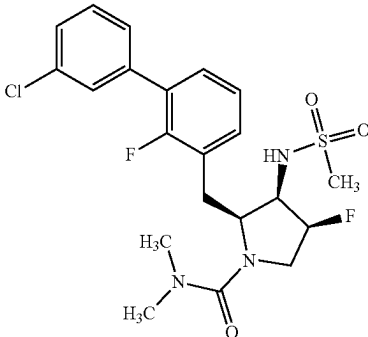
EX.	IUPAC NAME	Structure	ADDITIVE	MS
338	(2S,3R,4S)-4-fluoro-3-[(methanesulfonyl)-amino]-N,N-dimethyl-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidine-1-carboxamide			474.0
339	(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-3-[(methanesulfonyl)-amino]-N,N-dimethylpyrrolidine-1-carboxamide			456.0
340	(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			452.0
341	(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			471.9

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
342	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]4-fluoropyrrolidin-3-yl)methanesulfonamide			452.9
343	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)methanesulfonamide			435.1
344	(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(dimethylsulfamoyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			485.0
345	N-[(2S,3R,4S)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			516.8

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
346	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			501.0
347	N-[(2S,3R,4S)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			518.9
348	N-[(2S,3R,S)-2-[(5'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-2-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			518.9
349	N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with longer retention time			532.9

TABLE 1-1-continued

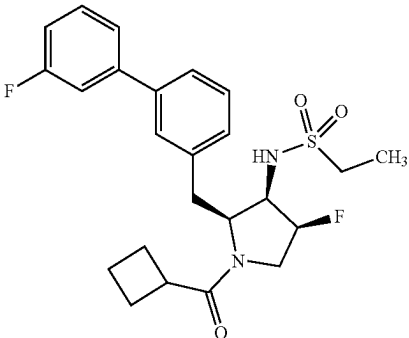
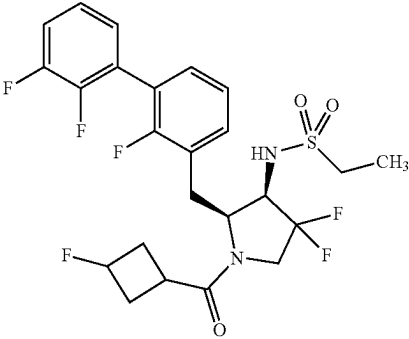
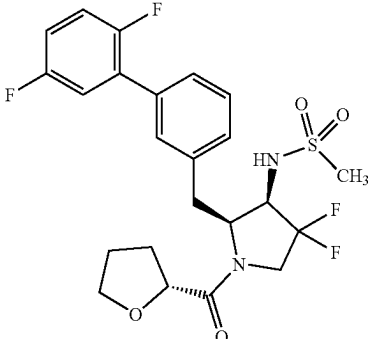
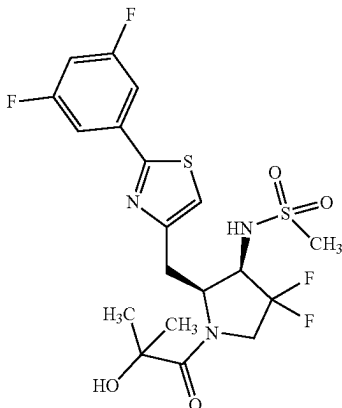
EX.	IUPAC NAME	Structure	ADDITIVE	MS
350	N-[(2S,3R,4S)-1-(cyclobutanecarbonyl)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			463.0
351	N-[(2S,3R)-4,4-difluoro-1-(3-fluoro-cyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			532.9
352	N-[(2S,3R)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			501.2
353	rac-N-[(2S,3R)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			496.1

TABLE 1-1-continued

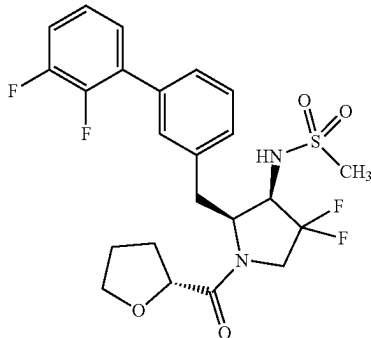
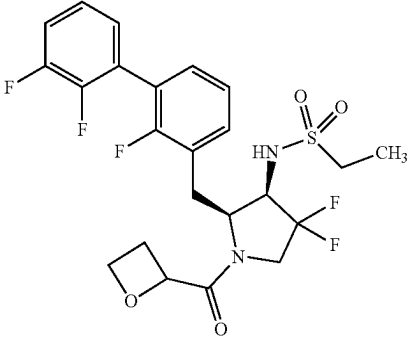
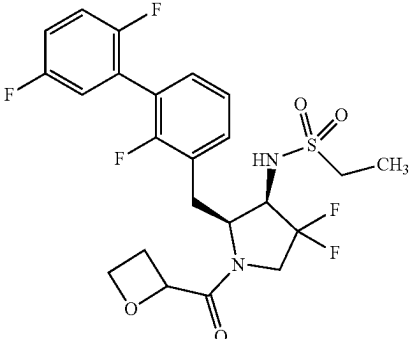
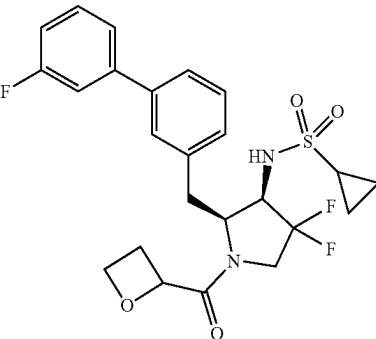
EX.	IUPAC NAME	Structure	ADDITIVE	MS
354	N-[(2S,3R)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]methanesulfonamide			501.2
355	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide derived from N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50 mmL, 3μm), mobile phase: water/acetonitrile (including 0.05% TFA))			519.1
356	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide derived from N-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA))			519.2
357	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time			495.2

TABLE 1-1-continued

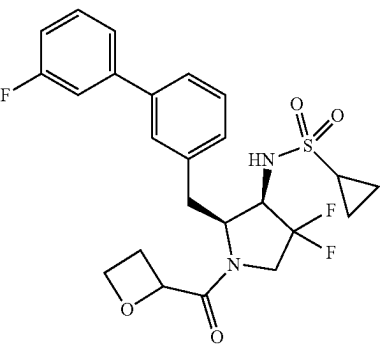
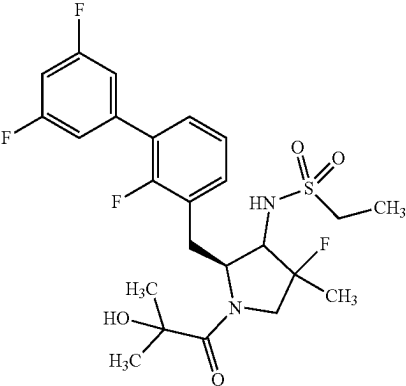
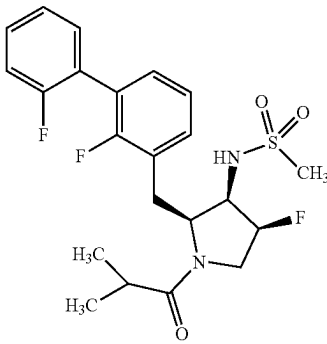
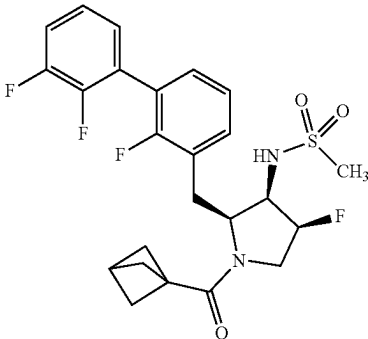
EX.	IUPAC NAME	Structure	ADDITIVE	MS
358	N-[(2S,3R)-4,4-difluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			495.2
359	rac-N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-4-methyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl]ethanesulfonamide			517.3
360	N-[(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]-methanesulfonamide			455.2
361	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)-methanesulfonamide			497.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
362	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			461.2
363	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			497.2
364	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			479.2
365	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			511.2

TABLE 1-1-continued

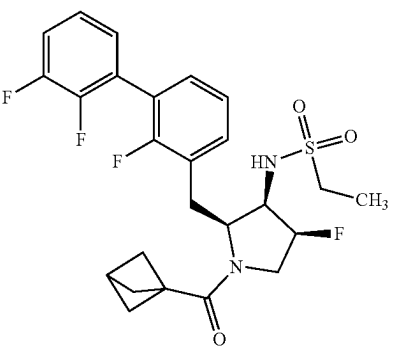
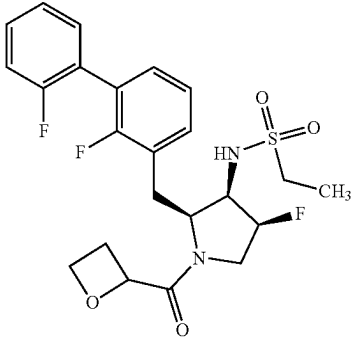
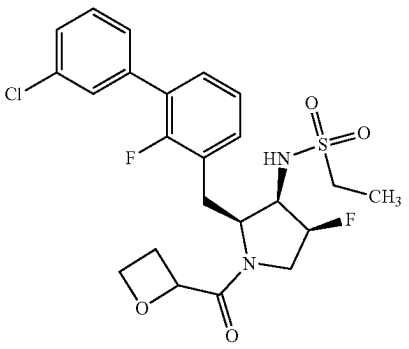
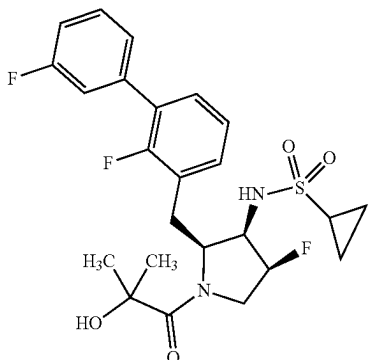
EX.	IUPAC NAME	Structure	ADDITIVE	MS
366	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			511.1
367	N-[(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			483.2
368	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			499.2
369	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]cyclopropanesulfonamide			497.2

TABLE 1-1-continued

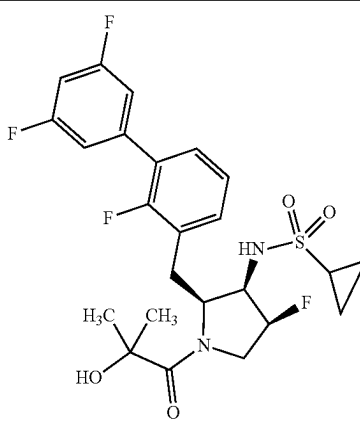
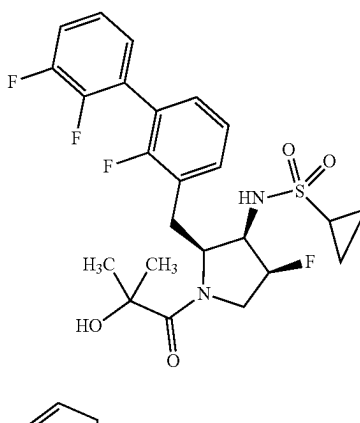
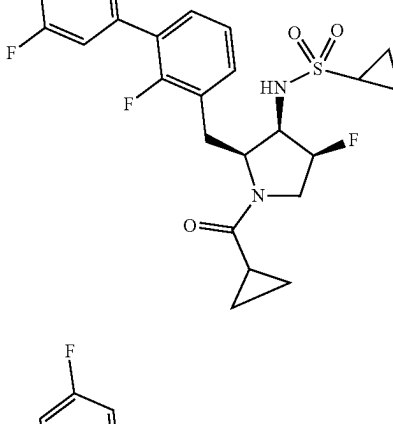
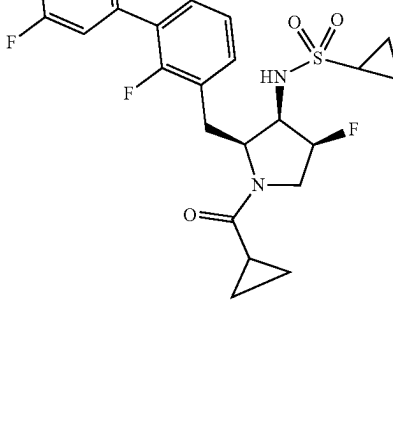
EX.	IUPAC NAME	Structure	ADDITIVE	MS
370	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			515.2
371	N-[(2S,3R,4S)-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			515.1
372	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]cyclopropane-sulfonamide			479.2
373	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropane-sulfonamide			497.2

TABLE 1-1-continued

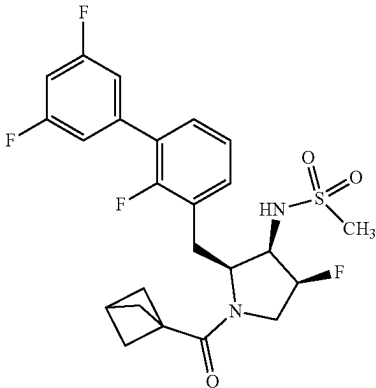
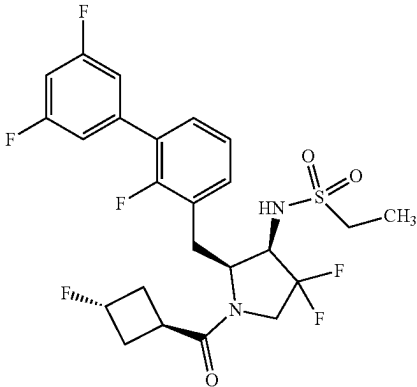
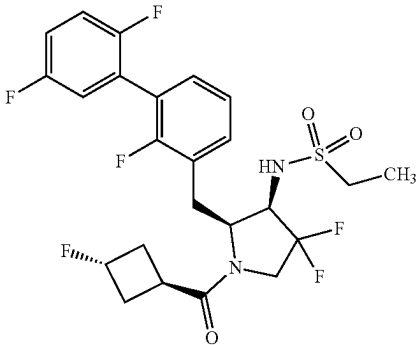
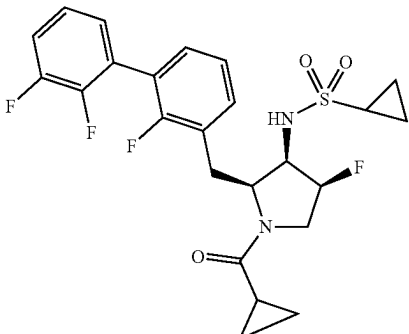
EX.	IUPAC NAME	Structure	ADDITIVE	MS
374	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-methanesulfonamide			497.2
375	N-[(2S,3R)-4,4-difluoro-1-((1r,3S)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			533.2
376	N-[(2S,3R)-4,4-difluoro-1-((1r,3S)-3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			533.2
377	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropanesulfonamide			497.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
378	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((1R,3S)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			497.2
379	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropane-sulfonamide			497.2
380	N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			533.2
381	N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			533.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
382	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]-cyclopropanesulfonamide			481.2
383	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			499.2
384	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]cyclopropanesulfonamide			505.2
385	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with shorter retention time			495.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
386	N-[(2S,3R,4S)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			495.2
387	(2S,3R,4S)-3-[(cyclopropanesulfonyl)-amino]-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			482.2
388	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			487.1
389	N-[(2S,3R,4S)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			505.1

TABLE 1-1-continued

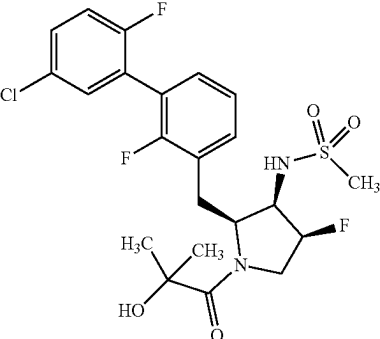
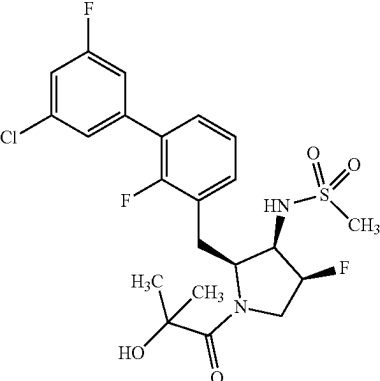
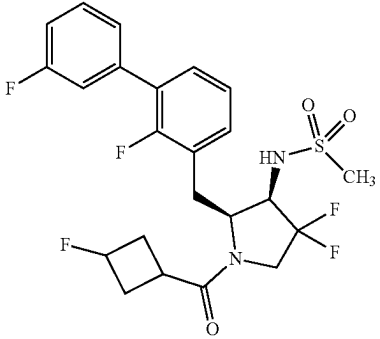
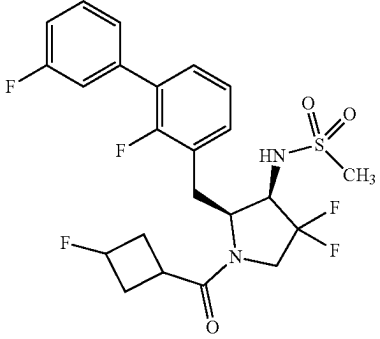
EX.	IUPAC NAME	Structure	ADDITIVE	MS
390	N-[(2S,3R,4S)-2-[(5'-chloro-2,2'-difluoro-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			505.1
391	N-[(2S,3R,4S)-2-[(3'-chloro-2,5'-difluoro-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			505.1
392	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			501.2
393	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			501.2

TABLE 1-1-continued

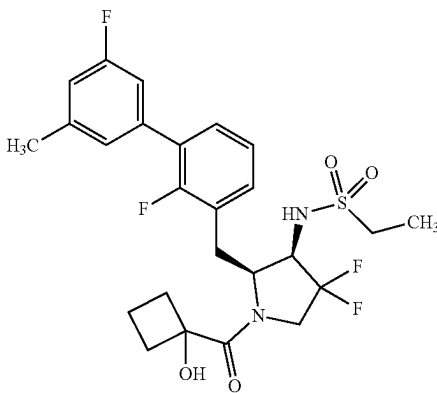
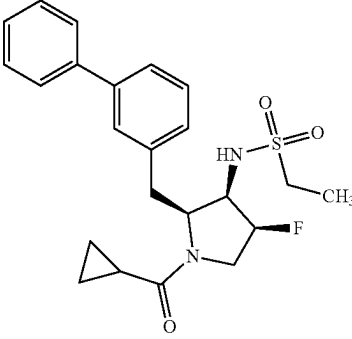
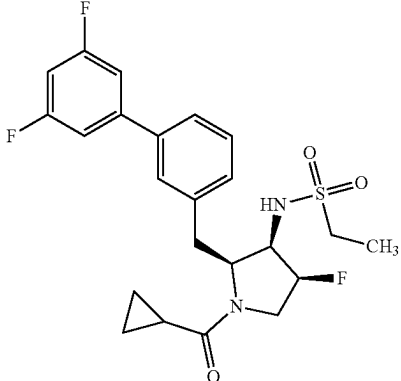
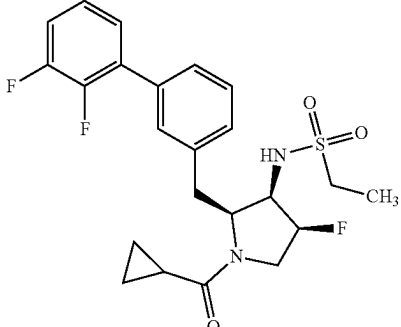
EX.	IUPAC NAME	Structure	ADDITIVE	MS
394	N-[(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			529.1
395	N-[(2S,3R,4S)-2-[(1,1'-biphenyl]-3-yl)-methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide			431.1
396	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(3',5'-difluoro[1,1'-biphenyl]-3-yl)-methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			467.0
397	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2',3'-difluoro[1,1'-biphenyl]-3-yl)-methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			467.0

TABLE 1-1-continued

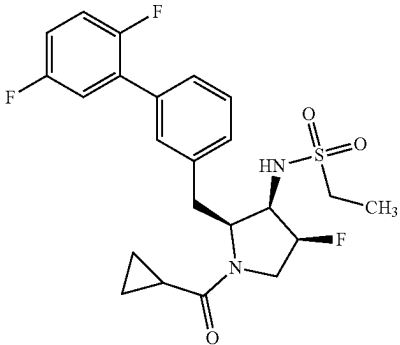
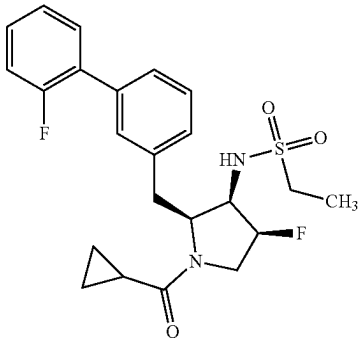
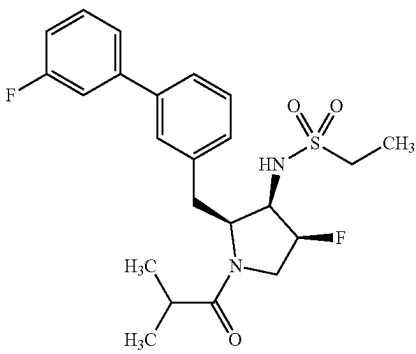
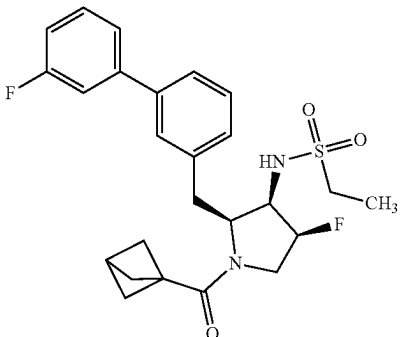
EX.	IUPAC NAME	Structure	ADDITIVE	MS
398	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2',5'-difluoro[1,1'-biphenyl]-3-yl)-methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			467.0
399	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			449.1
400	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			451.1
401	N-[(2S,3R,4S)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			475.0

TABLE 1-1-continued

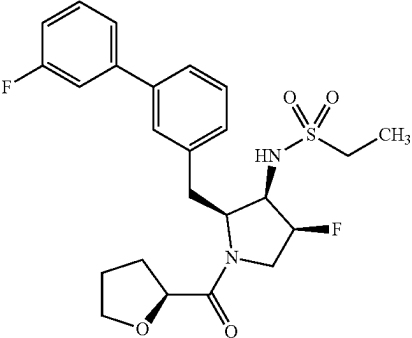
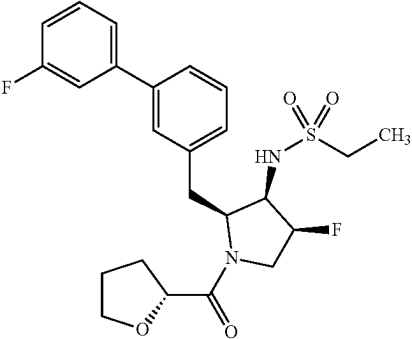
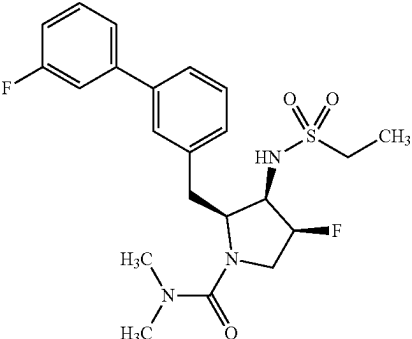
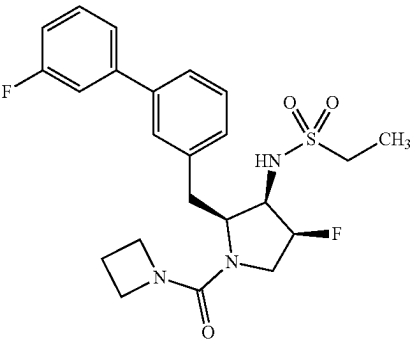
EX.	IUPAC NAME	Structure	ADDITIVE	MS
402	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2S)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			479.0
403	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			479.0
404	(2S,3R,4S)-3-[(ethanesulfonyl)amino]-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			452.0
405	N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			464.0

TABLE 1-1-continued

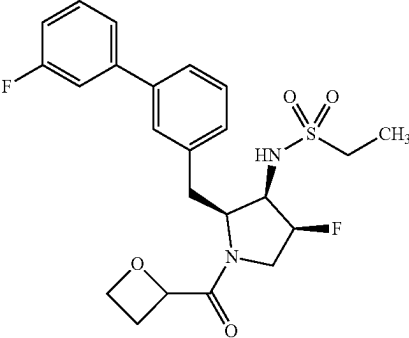
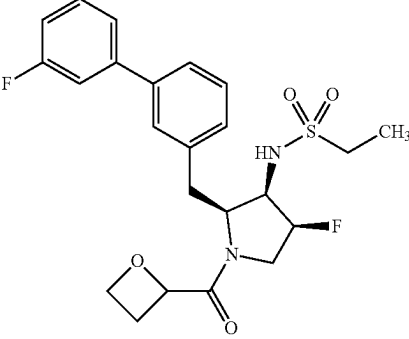
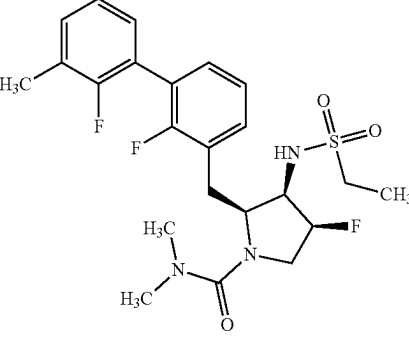
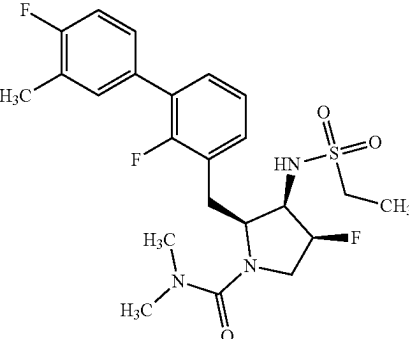
EX.	IUPAC NAME	Structure	ADDITIVE	MS
406	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			465.0
407	N-[(2S,3R,4S)-4-fluoro-2-[(3'-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			465.0
408	(2S,3R,4S)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)-amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			484.0
409	(2S,3R,4S)-2-[(2,4'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)-amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			484.0

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
410	(2S,3R,4S)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			484.0
411	(2S,3R,4S)-2-[(2,2'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			484.0
412	(2S,3R,4S)-2-[(2,4'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)amino]-4-fluoro-N,N-dimethylpyrrolidine-1-carboxamide			470.0
413	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			479.0

TABLE 1-1-continued

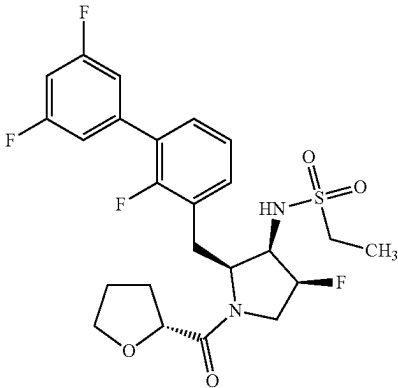
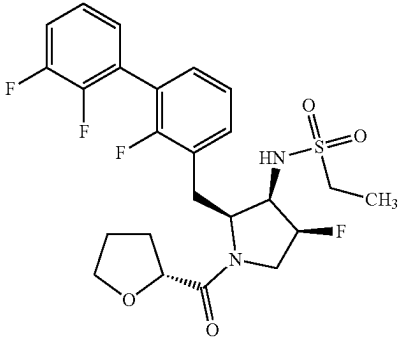
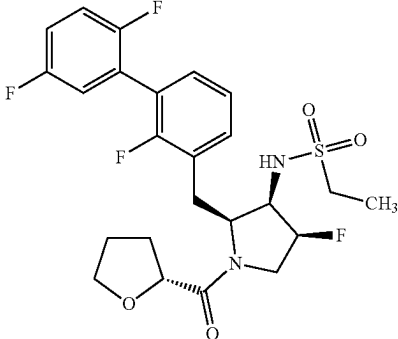
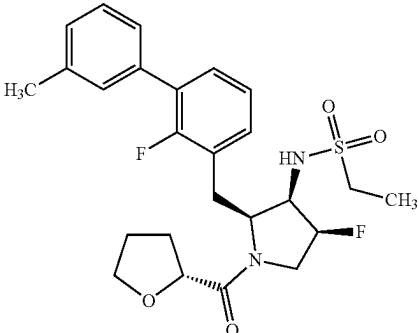
EX.	IUPAC NAME	Structure	ADDITIVE	MS
414	N-[(2S,3R,4S)-4-fluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			515.0
415	N-[(2S,3R,4S)-4-fluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			515.0
416	N-[(2S,3R,4S)-4-fluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			515.0
417	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			493.0

TABLE 1-1-continued

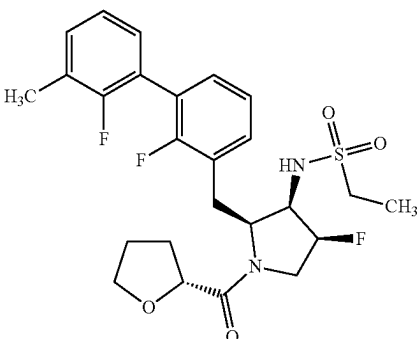
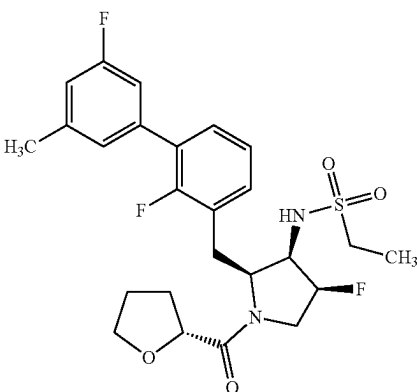
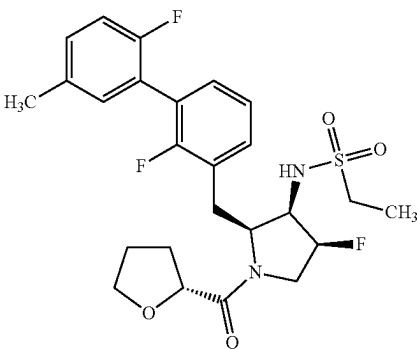
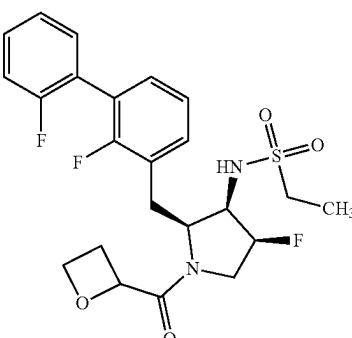
EX.	IUPAC NAME	Structure	ADDITIVE	MS
418	N-((2S,3R,4S)-2-[(2,2'-difluoro-3'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl)ethanesulfonamide			511.0
419	N-((2S,3R,4S)-2-[(2,3'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl)ethanesulfonamide			511.0
420	N-((2S,3R,4S)-2-[(2,2'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl)ethanesulfonamide			511.0
421	N-[2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			483.0

TABLE 1-1-continued

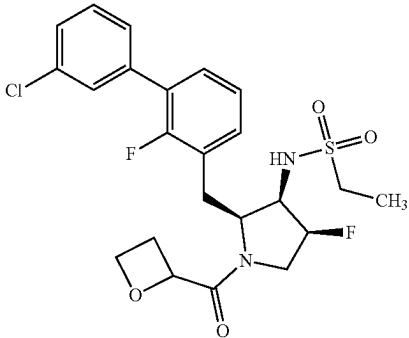
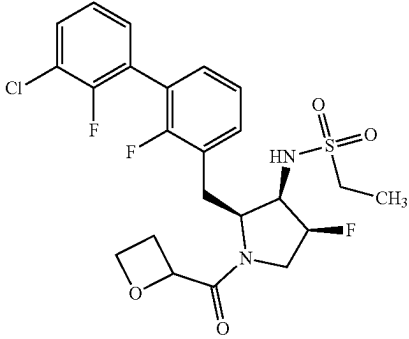
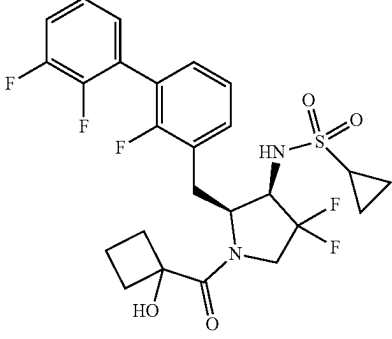
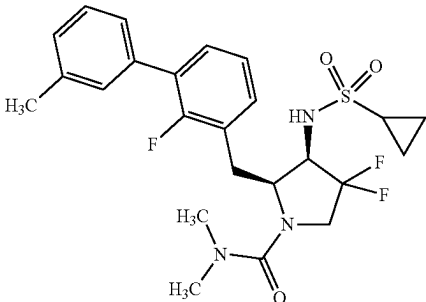
EX.	IUPAC NAME	Structure	ADDITIVE	MS
422	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			498.9
423	N-[(2S,3R,4S)-2-[(3'-chloro-2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide derived from N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50mmL, 3 μm) mobile phase: water/acetonitrile (including 0.05% TFA)			516.9
424	N-{(2S,3R)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}cyclopropanesulfonamide			544.9
425	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			496.0

TABLE 1-1-continued

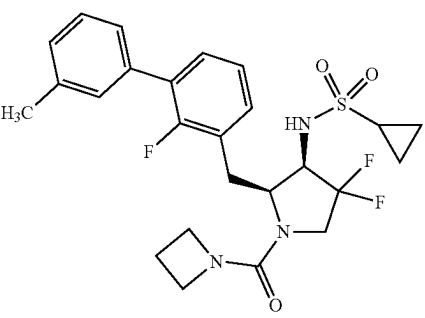
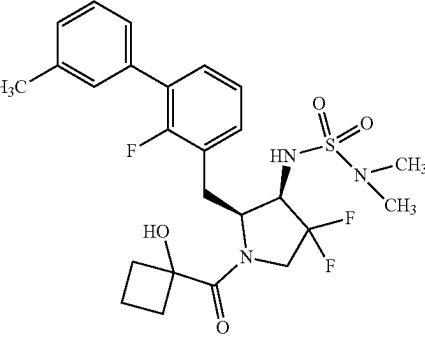
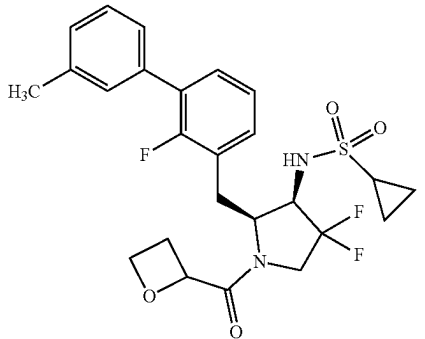
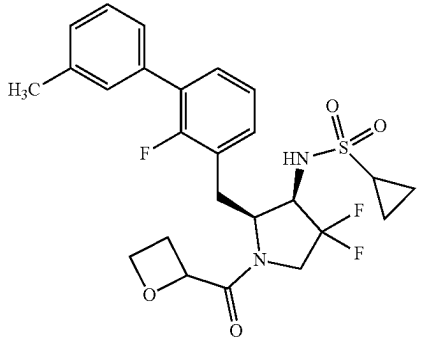
EX.	IUPAC NAME	Structure	ADDITIVE	MS
426	N'-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide			508.0
427	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			526.0
428	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with shorter retention time			509.0
429	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			509.0

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
430	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide			523.0
431	N-[(2S,3R)-4,4-difluoro-1-((1R,3S)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			518.8
432	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide with shorter retention time			512.0
433	N-[(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]methanesulfonamide			518.9

TABLE 1-1-continued

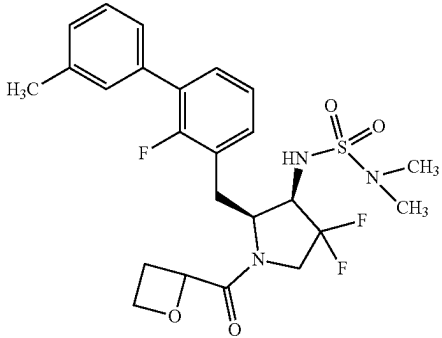
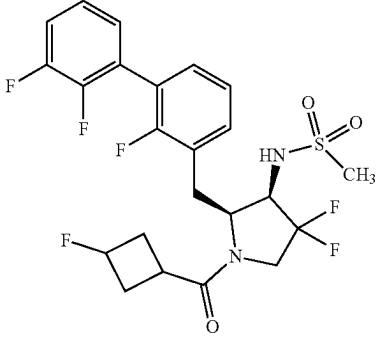
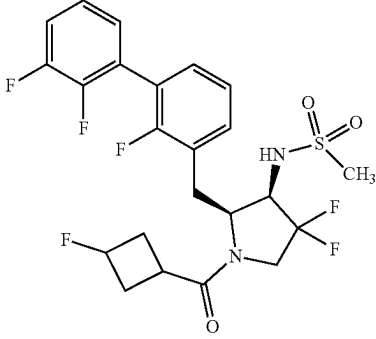
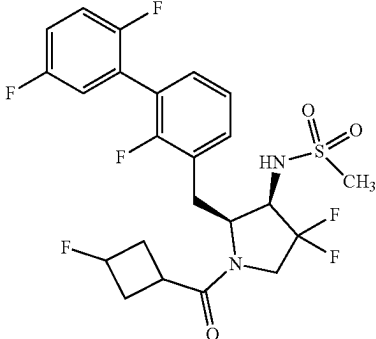
EX.	IUPAC NAME	Structure	ADDITIVE	MS
434	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide with longer retention time			512.0
435	N'-{(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl}methanesulfonamide with longer retention time			518.8
436	N'-{(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl}methanesulfonamide with shorter retention time			518.9
437	N'-{(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl}methanesulfonamide with longer retention time			518.8

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
438	N'-{(2S,3R)-4,4-difluoro-1-(1-hydroxy-cyclobutane-1-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			547.9
439	N'-{(2S,3R)-4,4-difluoro-1-(3-fluoro-cyclobutane-1-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl}methanesulfonamide with shorter retention time			518.9
440	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-fluoro-cyclobutane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with longer retention time			482.9
441	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(3-fluoro-cyclobutane-1-carbonyl)pyrrolidin-3-yl]-methanesulfonamide with shorter retention time			482.9

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
442	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			499.0
443	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			511.0
444	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}-ethanesulfonamide			514.0
445	N'-[(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			515.0

TABLE 1-1-continued

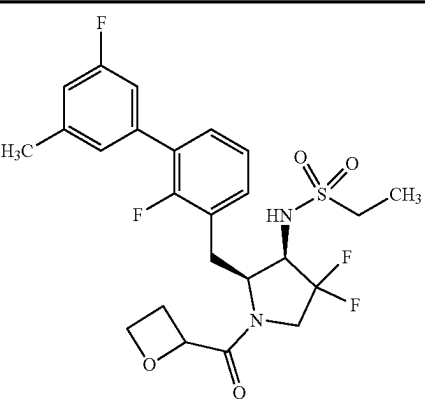
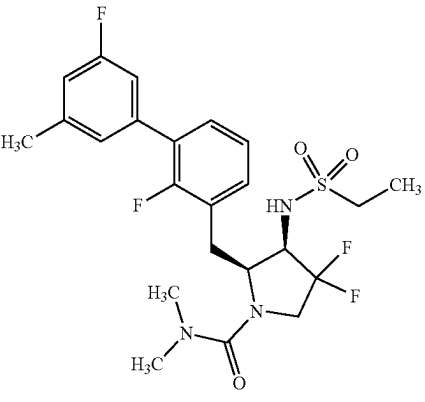
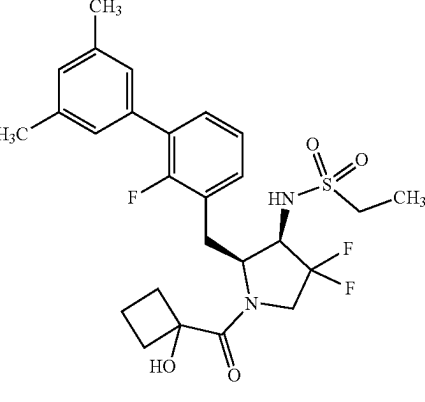
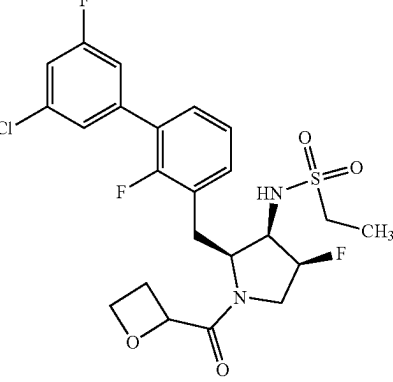
EX.	IUPAC NAME	Structure	ADDITIVE	MS
446	N-[(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			515.0
447	(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(ethanesulfonyl)-amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			501.9
448	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3',5'-dimethyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			525.0
449	N-[(2S,3R,4S)-2-[(3'-chloro-2,5'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide derived from N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA))			517.0

TABLE 1-1-continued

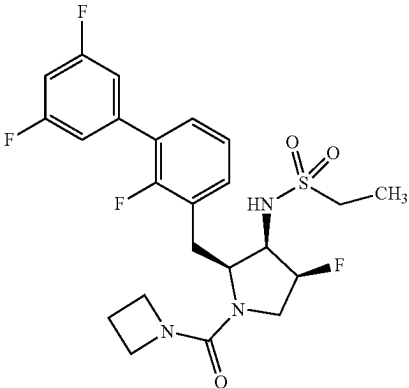
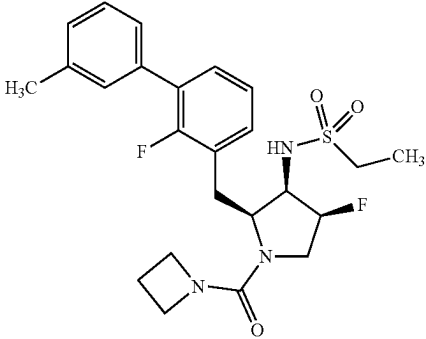
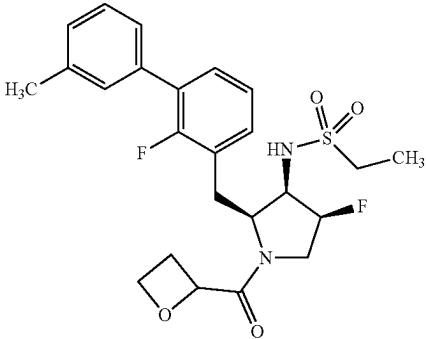
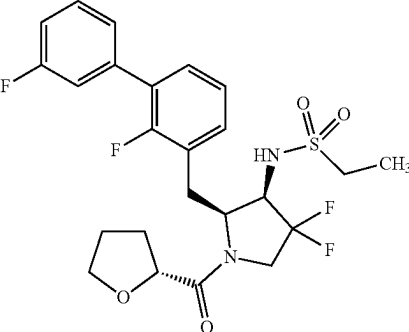
EX.	IUPAC NAME	Structure	ADDITIVE	MS
450	N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			500.2
451	N-[(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			478.3
452	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			479.2
453	N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			515.2

TABLE 1-1-continued

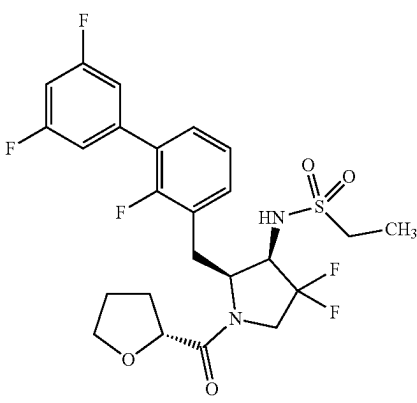
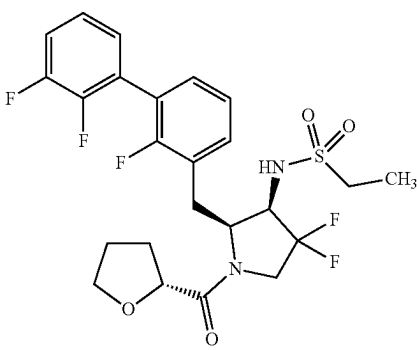
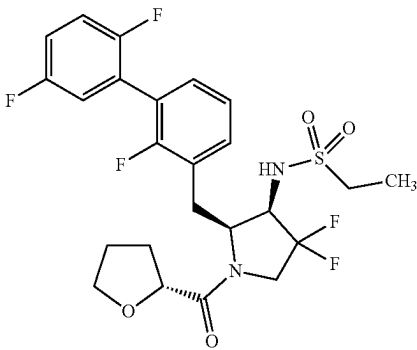
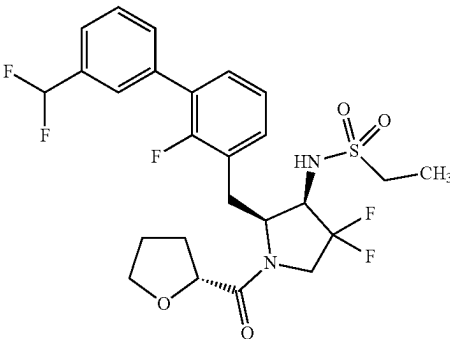
EX.	IUPAC NAME	Structure	ADDITIVE	MS
454	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			533.2
455	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			533.2
456	N-[(2S,3R)-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			533.2
457	N-[(2S,3R)-2-[(3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxolane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			547.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
458	N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethyl-sulfuric diamide			515.2
459	(2S,3R,4S)-3-[(dimethylsulfamoyl)amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			481.3
460	N'-[(2S,3R,4S)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			497.2
461	N'-[(2S,3R,4S)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			497.2

TABLE 1-1-continued

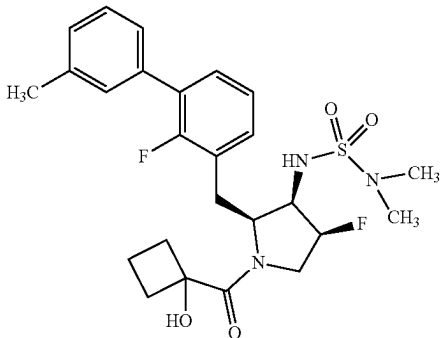
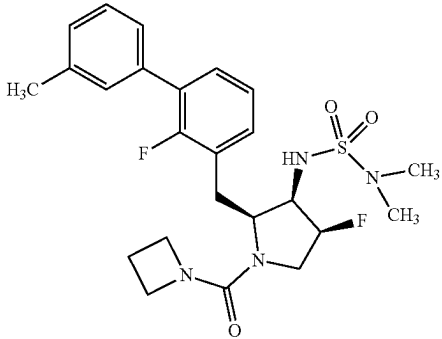
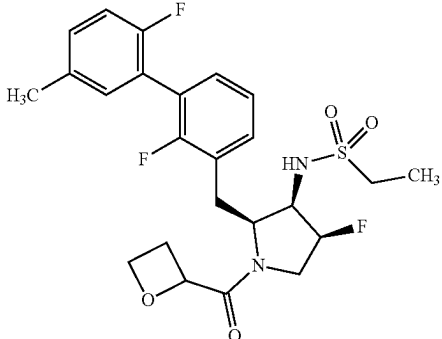
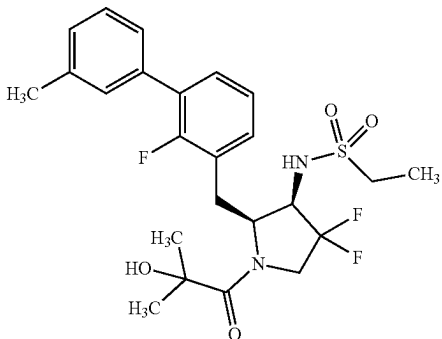
EX.	IUPAC NAME	Structure	ADDITIVE	MS
462	N'-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			508.2
463	N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			493.3
464	N'-[(2S,3R,4S)-2-[(2,2'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with shorter retention time			497.2
465	N'-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			499.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
466	N-[(2S,3R)-2-[(2,4'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			517.1
467	N-{(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',4'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide with longer retention time			517.2
468	N-{(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',4'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-ethanesulfonamide with shorter retention time			519.2
469	N-{(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4',5'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-ethanesulfonamide with longer retention time			535.1

TABLE 1-1-continued

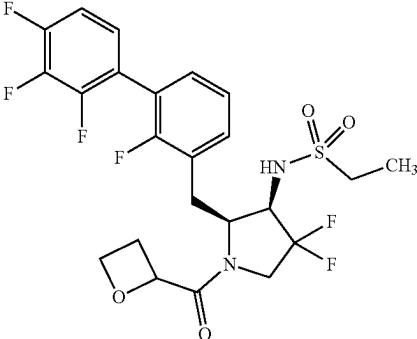
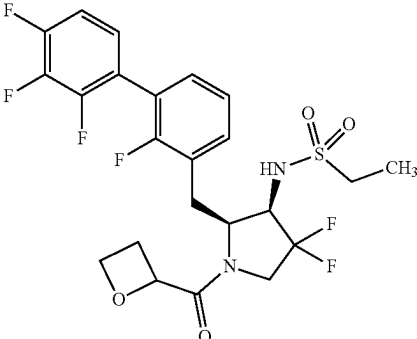
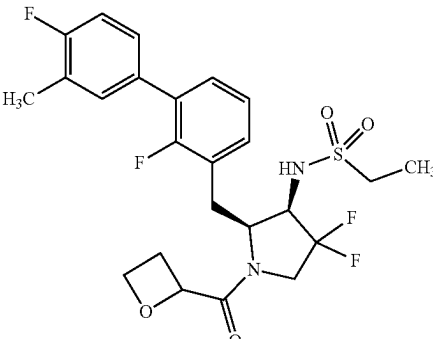
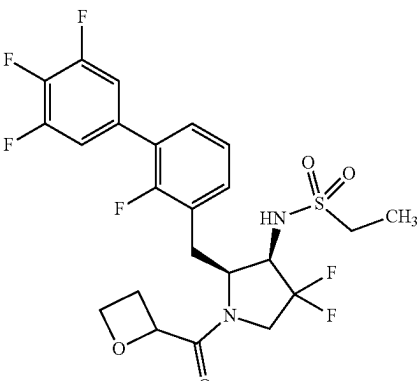
EX.	IUPAC NAME	Structure	ADDITIVE	MS
470	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3',4'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide with shorter retention			537.1
471	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3',4'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			537.1
472	N-[(2S,3R)-2-[(2,4'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			515.2
473	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',4',5'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			535.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
474	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4'-trifluoro-3'-methyl-[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with longer retention time			533.2
475	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,3',4',5'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			535.2
476	N-[(2S,3R)-2-[(2,4'-difluoro-3'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			515.2
477	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4',5'-tetrafluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			537.2

TABLE 1-1-continued

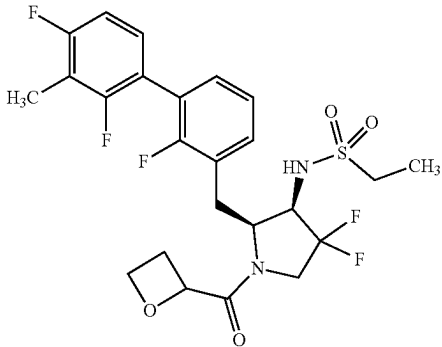
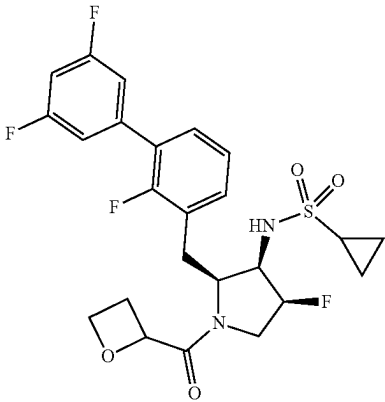
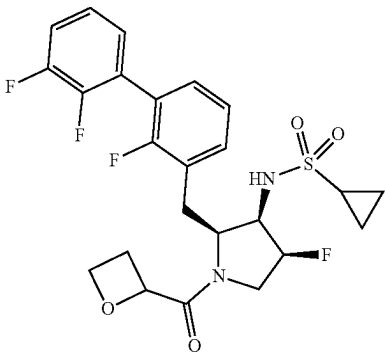
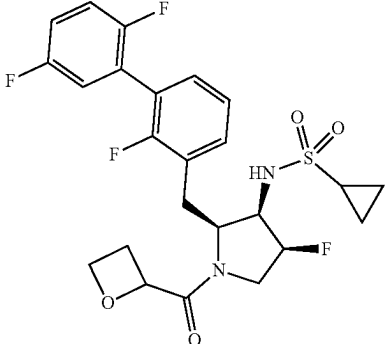
EX.	IUPAC NAME	Structure	ADDITIVE	MS
478	N-((2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4'-trifluoro-3'-methyl-[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)ethanesulfonamide with shorter retention time			533.2
479	N-((2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)-cyclopropanesulfonamide with shorter retention time			513.2
480	N-((2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)-cyclopropanesulfonamide with shorter retention time			513.2
481	N-((2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)-cyclopropanesulfonamide with shorter retention time			513.2

TABLE 1-1-continued

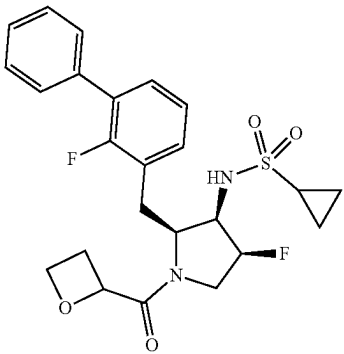
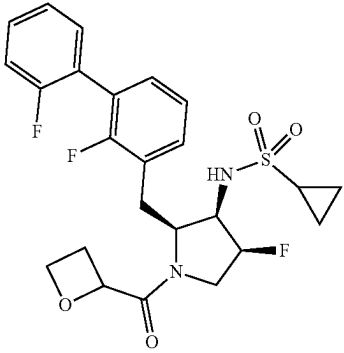
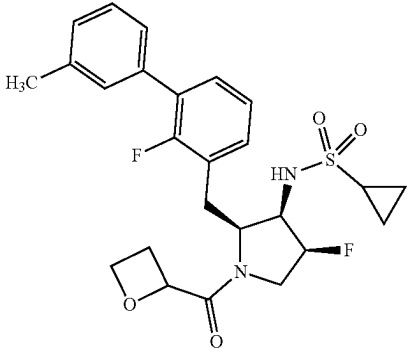
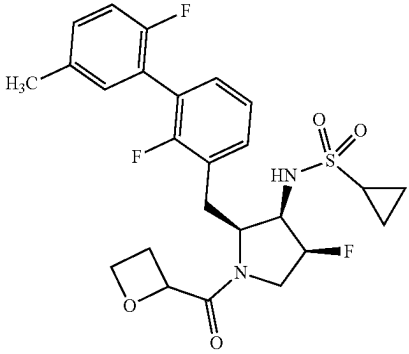
EX.	IUPAC NAME	Structure	ADDITIVE	MS
482	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time			477.2
483	N-[(2S,3R,4S)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide derived from N-[(2S,3R,4S)-2-[(3-chloro-2-fluorophenyl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmI · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA))			495.2
484	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with shorter retention time			491.2
492	N-[(2S,3R,4S)-2-[(2,2'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide with longer retention time			509.1

TABLE 1-1-continued

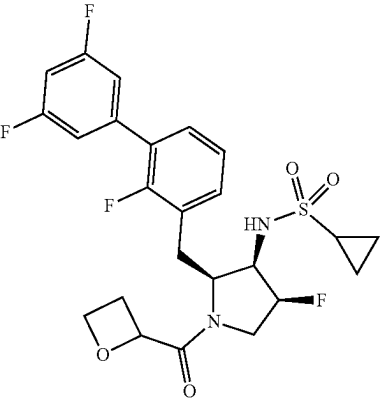
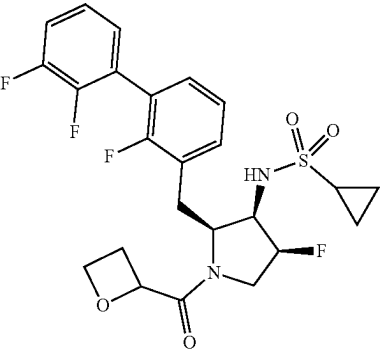
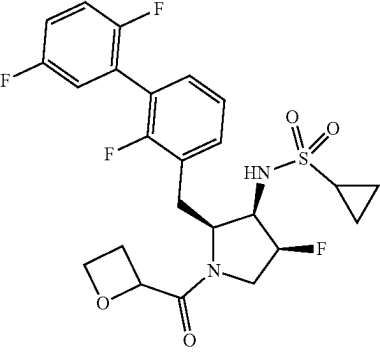
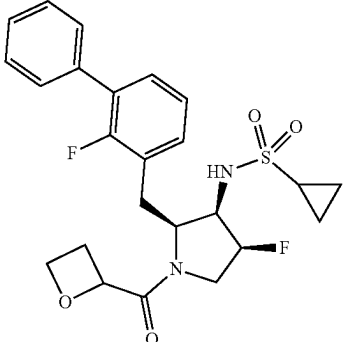
EX.	IUPAC NAME	Structure	ADDITIVE	MS
493	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropanesulfonamide with longer retention time			513.1
494	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',3'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			513.1
495	N-[(2S,3R,4S)-4-fluoro-1-(oxetane-2-carbonyl)-2-[(2,2',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			513.2
496	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]cyclopropane-sulfonamide with longer retention time			477.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
497	N-[(2S,3R,4S)-2-[(3-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-cyclopropanesulfonamide with longer retention time			511.1
498	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			479.2
499	N-[(2S,3R,4S)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			497.2
500	N-[(2S,3R,4S)-2-[(2,4'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide derived from N-[(2S,3R,4S)-2-[3-chloro-2-fluorophenyl]-methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time by HPLC (column: L-Column2 ODS (3.0 mmI · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA))			497.2

TABLE 1-1-continued

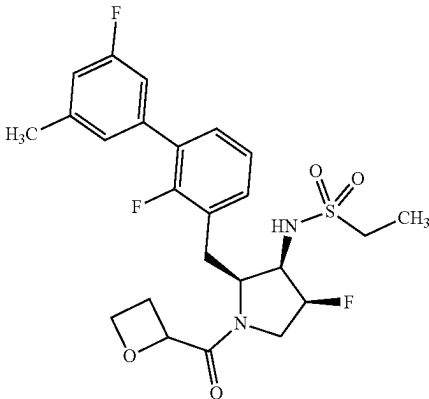
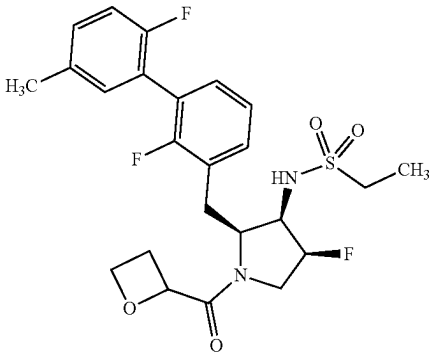
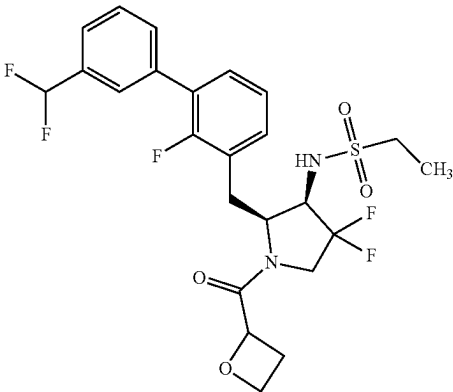
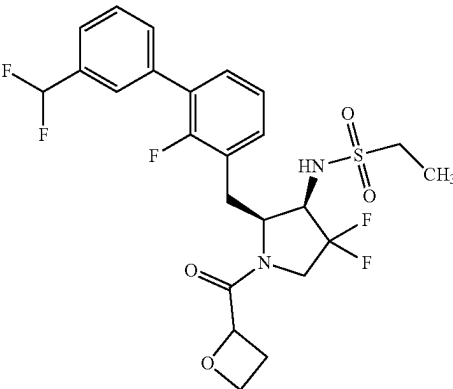
EX.	IUPAC NAME	Structure	ADDITIVE	MS
501	N-[(2S,3R,4S)-2-[(2,3'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide with longer retention time			497.2
502	N-[(2S,3R,4S)-2-[(2,2'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			497.2
503	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with longer retention time			533.1
504	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			533.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
505	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			547.2
506	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			547.2
507	N-[(2S,3R)-2-{[3'-(difluoromethyl)-2-fluoro(1,1'-biphenyl)-3-yl]methyl}-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			519.2
508	N-[(2S,3R)-1-(cyclopropanecarbonyl)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-pyrrolidin-3-yl]ethanesulfonamide			517.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
509	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-[[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl]-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			543.2
510	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			523.2
511	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			499.1
512	N-[(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[[2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			497.2

TABLE 1-1-continued

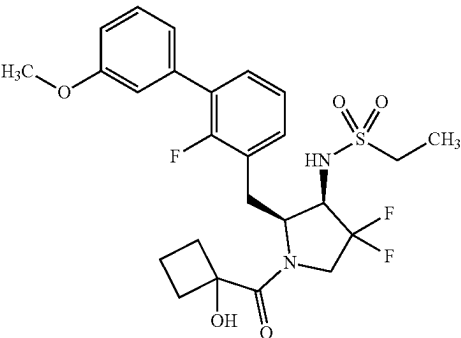
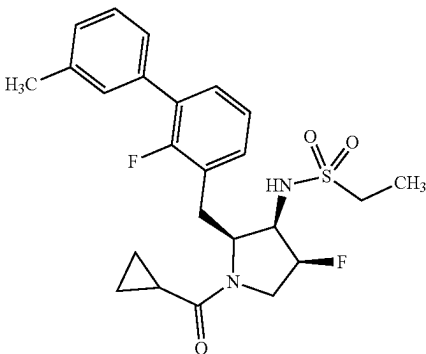
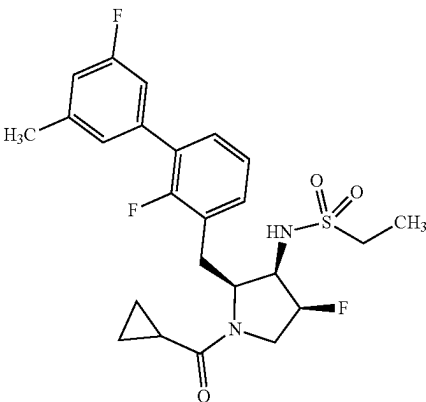
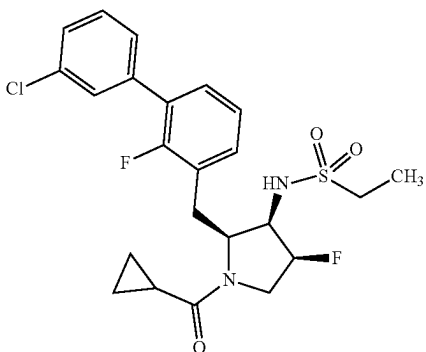
EX.	IUPAC NAME	Structure	ADDITIVE	MS
513	N-[(2S,3R)-4,4-difluoro-2-[(2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)-pyrrolidin-3-yl]ethanesulfonamide			527.2
514	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]-ethanesulfonamide			463.2
515	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]-ethanesulfonamide			481.2
516	N-[(2S,3R,4S)-2-[(3'-chloro-2-fluoro[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide			483.2

TABLE 1-1-continued

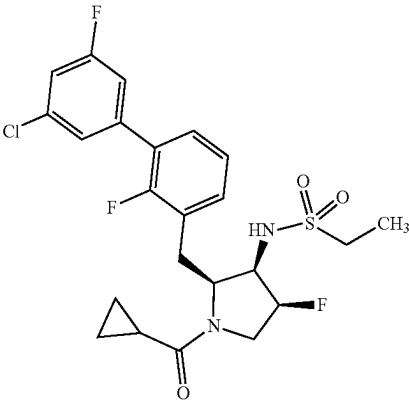
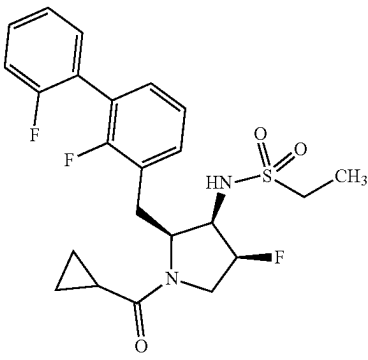
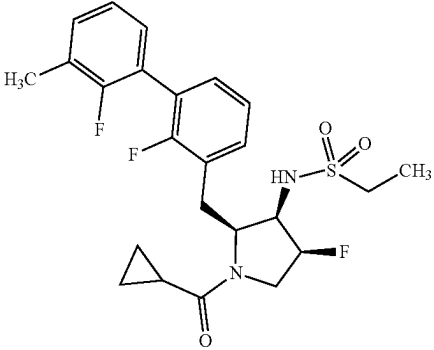
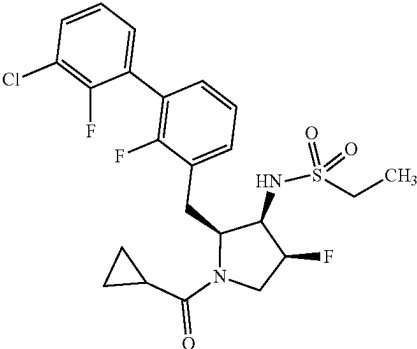
EX.	IUPAC NAME	Structure	ADDITIVE	MS
517	N-[(2S,3R,4S)-2-[(3'-chloro-2,5'-difluoro-[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide			501.1
518	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,2'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			467.1
519	N-[(2S,3R,4S)-1-(cyclopropanecarbonyl)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4-fluoropyrrolidin-3-yl]ethanesulfonamide			481.1
520	N-[(2S,3R,4S)-2-[(3'-chloro-2,2'-difluoro-[1,1'-biphenyl]-3-yl)methyl]-1-(cyclopropanecarbonyl)-4-fluoropyrrolidin-3-yl]ethanesulfonamide			501.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
521	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			514.2
522	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			514.2
523	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			500.2
524	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-4,4-difluoro-N,N-dimethyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-pyrrolidine-1-carboxamide			518.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
525	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]cyclopropane-sulfonamide			530.1
526	(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			517.2
527	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]-cyclopropanesulfonamide			526.1
528	(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			503.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
529	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]-cyclopropanesulfonamide			526.2
530	N-[(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]cyclopropanesulfonamide			527.2
531	(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-N,N-dimethylpyrrolidine-1-carboxamide			517.2
532	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-N,N-dimethyl-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidine-1-carboxamide			521.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
533	N-[(2S,3R)-2-[(2,3,-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]cyclopropanesulfonamide			527.1
534	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]cyclopropane-sulfonamide			512.2
535	N-[(2S,3R,4S)-2-{[3'-(difluoromethyl)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4-fluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide derived from N-[(2S,3R,4S)-2-{[3-chloro-2-fluorophenyl]-methyl}-4-fluoro-1-(oxetane-2-carbonyl)-pyrrolidin-3-yl]ethanesulfonamide with shorter retention time by HPLC (column: L-Column2 ODS (3.0 mmL · D. × 50 mmL, 3 μm), mobile phase: water/acetonitrile (including 0.05% TFA)			515.1
536	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,r-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			493.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
537	N-[(2S,3R,4S)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide			505.3
538	(2S,3R,4S)-3-[(cyclopropanesulfonyl)-amino]-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-N,N-dimethylpyrrolidine-1-carboxamide			478.2
539	N'-{(2S,3R)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			530.2
540	N'-{(2S,3R)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			530.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
541	N'-{(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			516.2
542	N'-{(2S,3R)-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl)-N,N-dimethylsulfuric diamide			534.1
543	N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-cyclopropanesulfonamide			490.1
544	N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-cyclopropane-sulfonamide			512.2

TABLE 1-1-continued

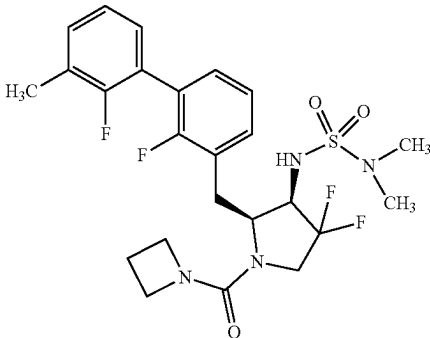
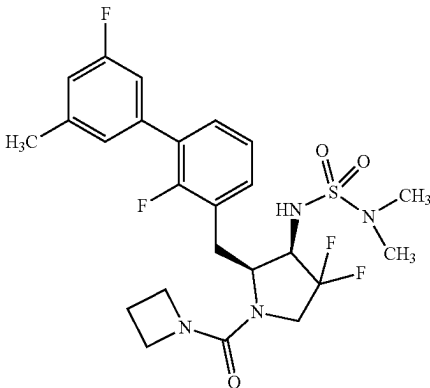
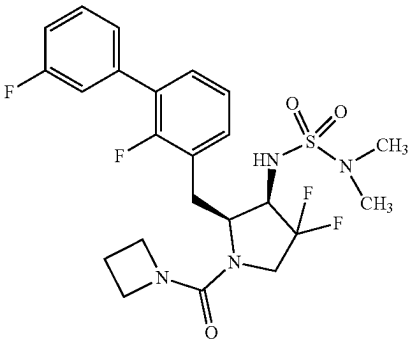
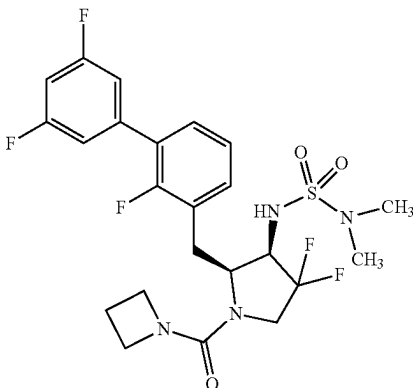
EX.	IUPAC NAME	Structure	ADDITIVE	MS
545	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,2'-difluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			529.2
546	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro-5'-methyl[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			529.2
547	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			515.2
548	N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			533.2

TABLE 1-1-continued

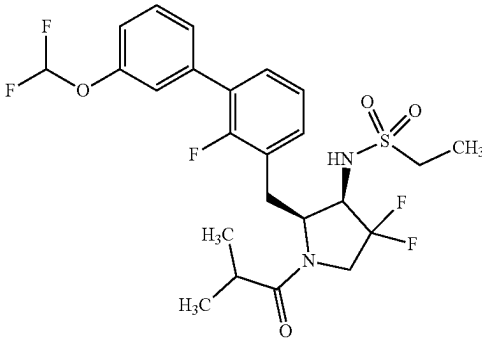
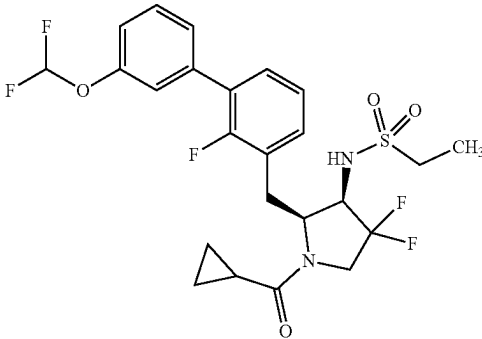
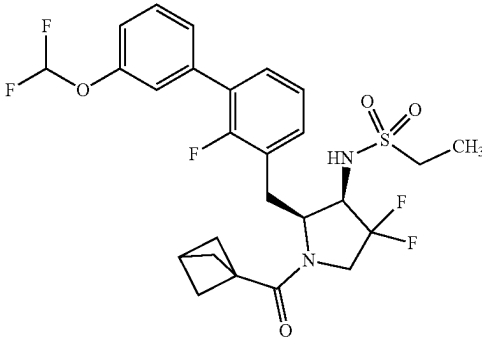
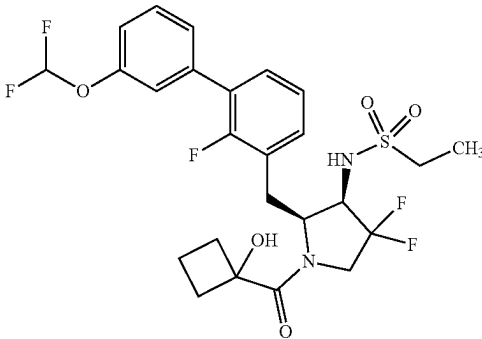
EX.	IUPAC NAME	Structure	ADDITIVE	MS
549	N-[(2S,3R)-2-{[3'-(difluoromethoxy)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			535.1
550	N-[(2S,3R)-1-(cyclopropanecarbonyl)-2-{[3'-(difluoromethoxy)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			533.2
551	N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{[3'-(difluoromethoxy)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoropyrrolidin-3-yl]ethanesulfonamide			559.2
552	N-[(2S,3R)-2-{[3'-(difluoromethoxy)-2-fluoro[1,1'-biphenyl]-3-yl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			563.1

TABLE 1-1-continued

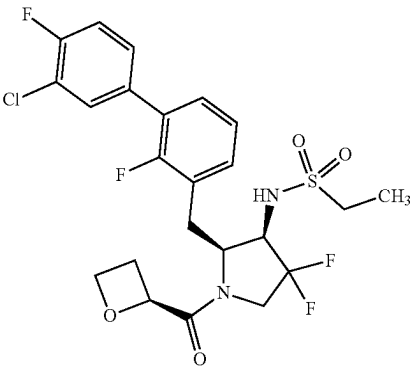
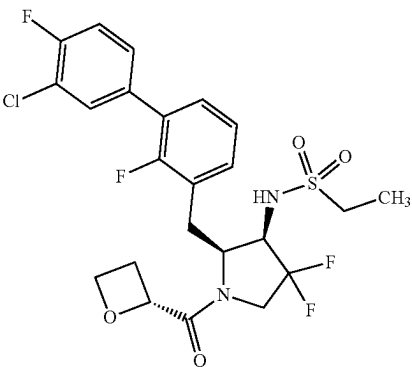
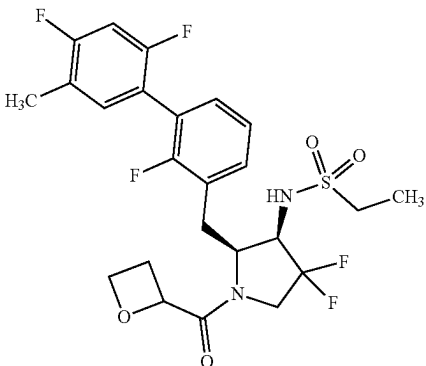
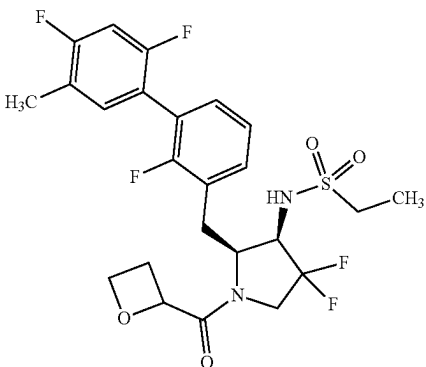
EX.	IUPAC NAME	Structure	ADDITIVE	MS
553	N-[(2S,3R)-2-[(3'-chloro-2,4'-difluoro-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-((2S)-oxetane-2-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			535.0
554	N-[(2S,3R)-2-[(3'-chloro-2,4'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl]ethanesulfonamide			535.1
555	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4'-trifluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with longer retention time			533.1
556	N-[(2S,3R)-4,4-difluoro-1-(oxetane-2-carbonyl)-2-[(2,2',4'-trifluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			533.1

TABLE 1-1-continued

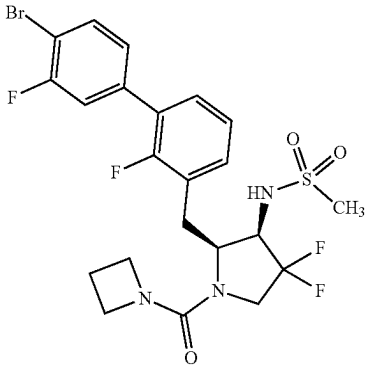
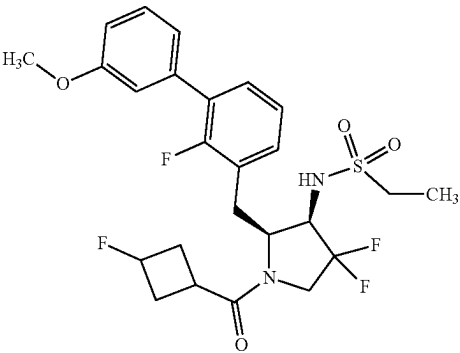
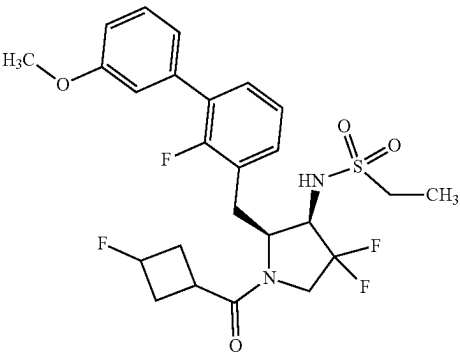
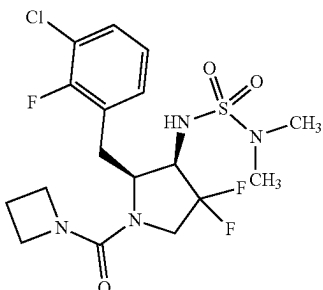
EX.	IUPAC NAME	Structure	ADDITIVE	MS
557	N-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(4'-bromo-2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoropyrrolidin-3-yl]methanesulfonamide			564.0
558	N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2-fluoro-3'-methoxy-[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl]ethanesulfonamide with longer retention time			529.1
559	N-[(2S,3R)-4,4-difluoro-1-(3-fluorocyclobutane-1-carbonyl)-2-[(2-fluoro-3'-methoxy[1,1'-biphenyl]-3-yl)methyl]-pyrrolidin-3-yl]ethanesulfonamide with shorter retention time			529.1
560	N'-[(2S,3R)-1-(azetidine-1-carbonyl)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoropyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			455.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
561	rac-N-[(2S,3R)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			490.1
562	rac-N-[(2S,3R)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]methanesulfonamide			502.1
563	N-{(2S,3R)-2-[(2,2'-difluoro-3-methyl-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl}-cyclopropanesulfonamide			527.1
564	N-{(2S,3R)-2-[(2,3'-difluoro-5'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl}-cyclopropanesulfonamide			527.1

TABLE 1-1-continued

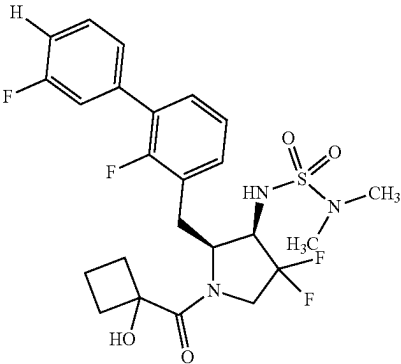
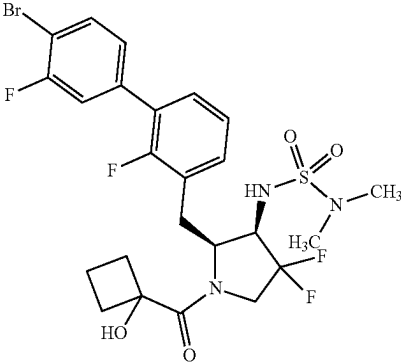
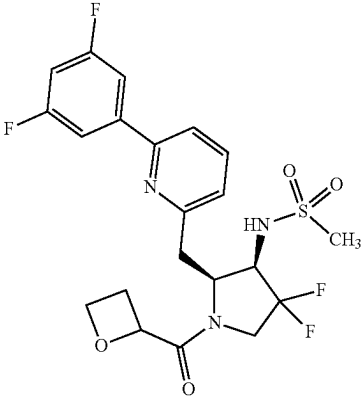
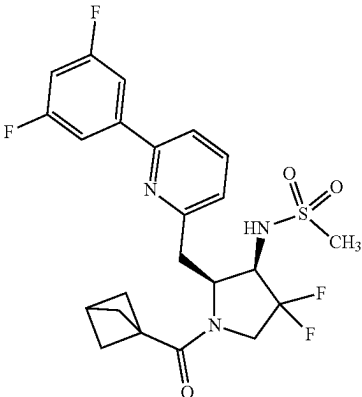
EX.	IUPAC NAME	Structure	ADDITIVE	MS
565	N'-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			530.1
566	N'-[(2S,3R)-2-[(4'-bromo-2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			608.0
567	N'-[(2SR,3RS)-2-{[6-(3,5-difluorophenyl)pyridin-2-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide			488.1
568	rac-N'-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{[6-(3,5-difluorophenyl)pyridin-2-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-methanesulfonamide			498.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
569	[(2SR,3RS)-2-{{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl}-methanesulfonamide			488.1
570	N-{{(2S,3R)-2-{{(5'-chloro-2,2',4'-trifluoro-[1,1'-biphenyl]-3-yl)methyl}-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}ethanesulfonamide			553.1
571	N-{{(2S,3R)-2-{{(3'-chloro-2,2',4'-trifluoro-[1,1'-biphenyl]-3-yl)methyl}-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}ethanesulfonamide			553.0
572	rac-N-[(2S,3R)-2-{{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl}ethanesulfonamide			504.1

TABLE 1-1-continued

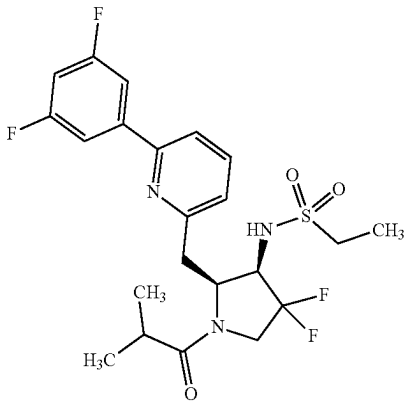
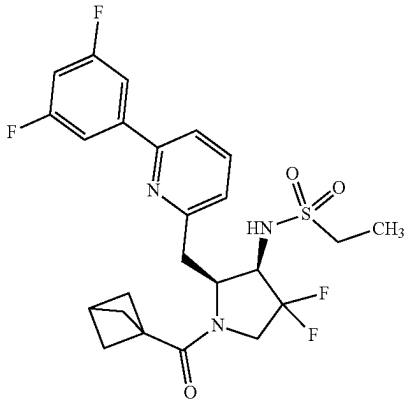
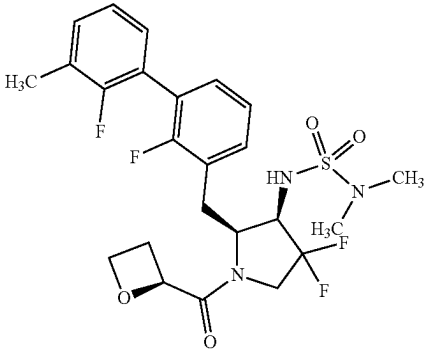
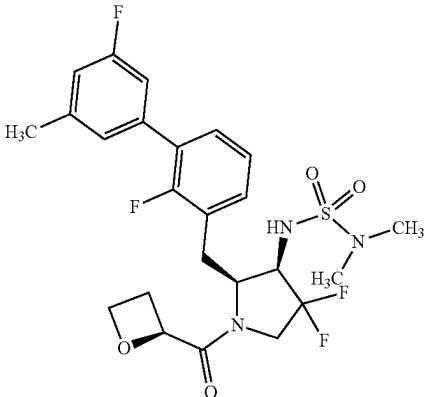
EX.	IUPAC NAME	Structure	ADDITIVE	MS
573	rac-N-[(2S,3R)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide			488.1
574	rac-N-[(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-pyrrolidin-3-yl]ethanesulfonamide			512.1
575	N'-{(2S,3R)-2-[(2,2'-difluoro-3'-methyl-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			530.2
576	N'-{(2S,3R)-2-[(2,3-difluoro-5-methyl-[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			530.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
577	N'-{(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			516.1
578	N'-{(2S,3R)-4,4-difluoro-1-[(2S)-oxetane-2-carbonyl]-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide			534.1
579	N-[(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,2',4'-trifluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl]ethanesulfonamide			535.1
580	rac-N-[(2S,3R)-2-{[6-(3,5-difluorophenyl)pyridin-2-yl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]ethanesulfonamide			516.1

TABLE 1-1-continued

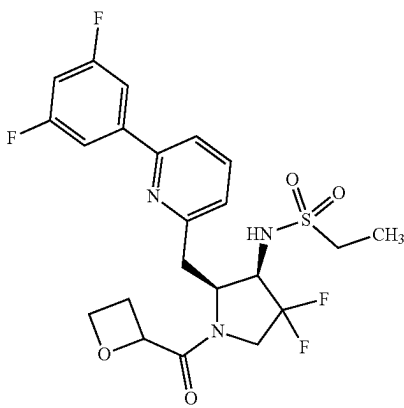
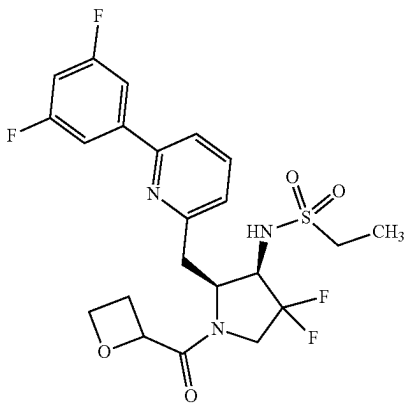
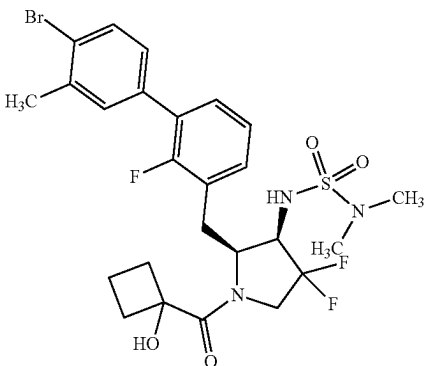
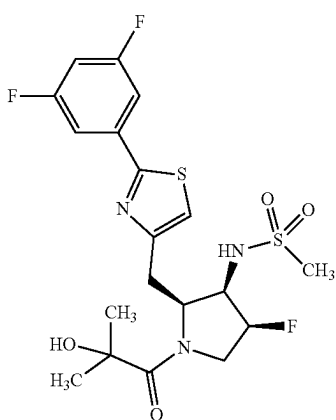
EX.	IUPAC NAME	Structure	ADDITIVE	MS
581	N-[(2SR,3RS)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide			502.1
582	N-[(2SR,3RS)-2-{[6-(3,5-difluorophenyl)-pyridin-2-yl]methyl}-4,4-difluoro-1-(oxetane-2-carbonyl)pyrrolidin-3-yl]-ethanesulfonamide			502.1
583	N'-[(2S,3R)-2-{(4'-bromo-2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			602.1
584	rac-N-[(2S,3R,4S)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4-fluoro-1-(2-hydroxy-2-methylpropanoyl)pyrrolidin-3-yl]methanesulfonamide			478.1

TABLE 1-1-continued

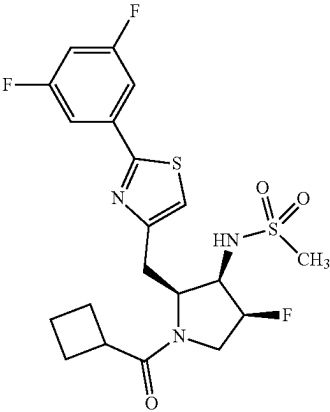
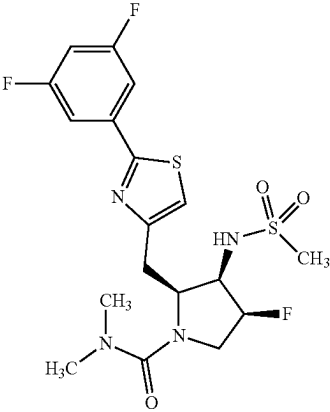
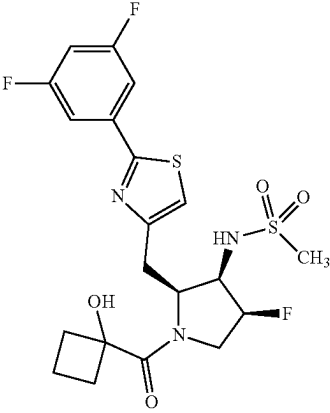
EX.	IUPAC NAME	Structure	ADDITIVE	MS
585	rac-N-[(2S,3R,4S)-1-(cyclobutanecarbonyl)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4-fluoropyrrolidin-3-yl]-methanesulfonamide			474.1
586	rac-(2S,3R,4S)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4-fluoro-3-[(methanesulfonyl)amino]-N,N-dimethylpyrrolidine-1-carboxamide			463.1
587	rac-N-((2S,3R,4S)-2-{[2-(3,5-difluorophenyl)-1,3-thiazol-4-yl]methyl}-4-fluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl)methanesulfonamide			490.0

TABLE 1-1-continued

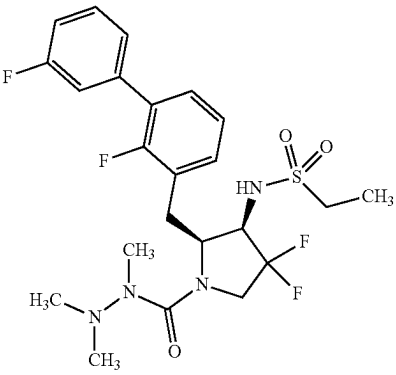
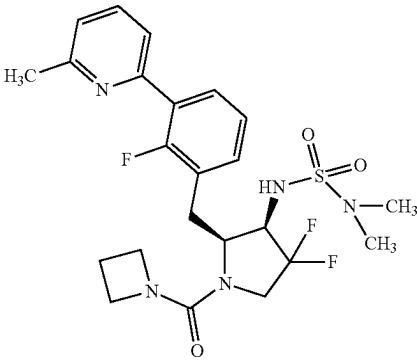
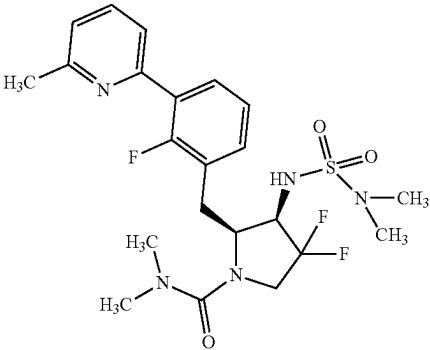
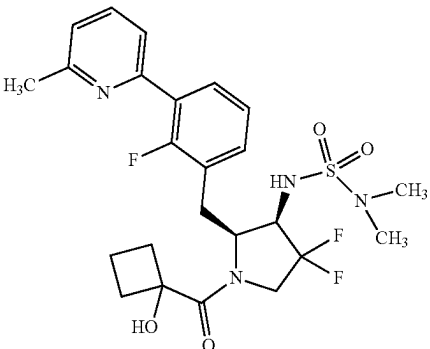
EX.	IUPAC NAME	Structure	ADDITIVE	MS
588	N'-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(trimethylhydrazinecarbonyl)pyrrolidin-3-yl]-ethanesulfonamide			517.0
589	N'-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			512.0
590	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-N,N-dimethylpyrrolidine-1-carboxamide			500.0
591	N'-[(2S,3R)-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			527.0

TABLE 1-1-continued

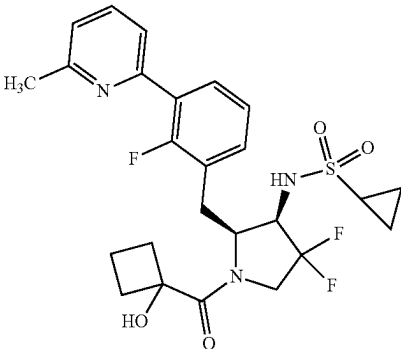
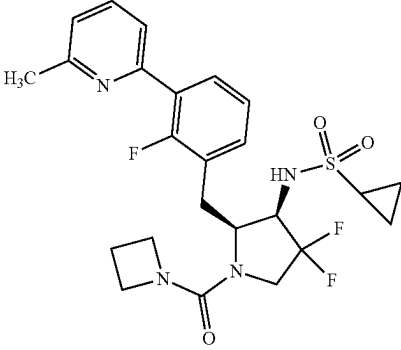
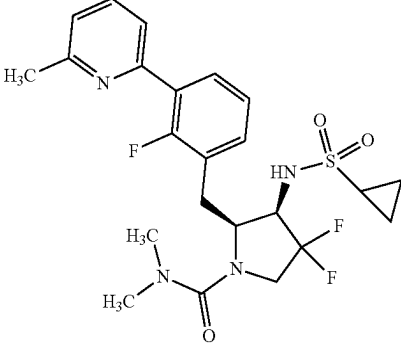
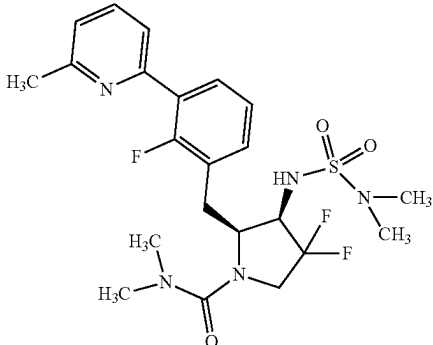
EX.	IUPAC NAME	Structure	ADDITIVE	MS
592	N-[(2S,3R)-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-2-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]cyclopropanesulfonamide			523.9
593	N-[(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}pyrrolidin-3-yl]cyclopropanesulfonamide			509.0
594	(2S,3R)-3-[(cyclopropanesulfonyl)amino]-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-N,N-dimethylpyrrolidine-1-carboxamide			497.0
595	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-{[2-fluoro-3-(6-methylpyridin-2-yl)phenyl]methyl}-N,N-dimethylpyrrolidine-1-carboxamide			500.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
596	(2S,3R)-3-[(dimethylsulfamoyl)amino]-4,4-difluoro-2-{{[2-fluoro-3-(5-methylpyridazin-3-yl)phenyl]methyl}-N,N-dimethylpyrrolidine-1-carboxamide			501.2
597	N'-[(2S,3R)-1-(acetidine-1-carbonyl)-4,4-difluoro-2-{{[2-fluoro-3-(5-methylpyridazin-3-yl)phenyl]methyl}pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			513.2
598	N'-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(5-methylpyridazin-3-yl)phenyl]methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			528.2
599	N-((2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(trimethylhydrazinecarbonyl)pyrrolidin-3-yl)-ethanesulfonamide			517.2

TABLE 1-1-continued

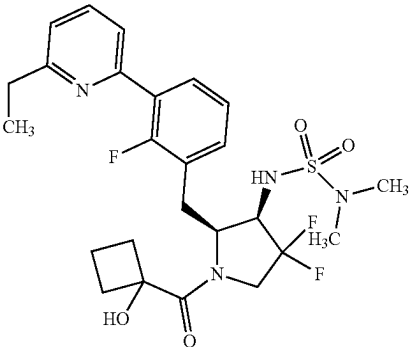
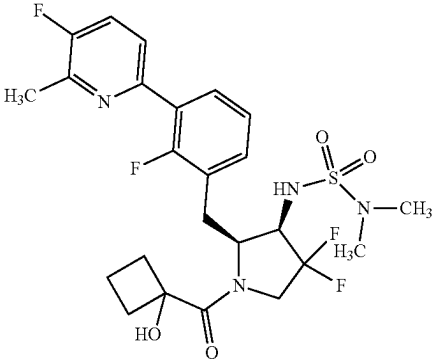
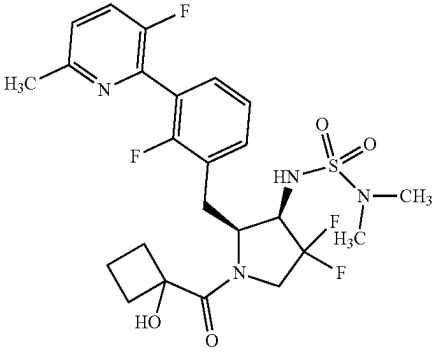
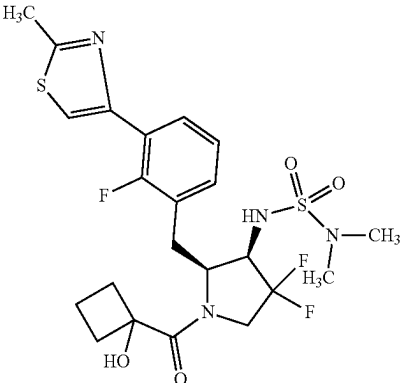
EX.	IUPAC NAME	Structure	ADDITIVE	MS
600	N'-[(2S,3R)-2-{[3-(6-ethylpyridin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			540.9
601	N'-[(2S,3R)-4,4-difluoro-2-{[2-fluoro-3-(5-fluoro-6-methylpyridin-2-yl)phenyl]-methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			544.9
602	N'-[(2S,3R)-4,4-difluoro-2-{[2-fluoro-3-(3-fluoro-6-methylpyridin-2-yl)phenyl]-methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			544.9
603	N'-[(2S,3R)-4,4-difluoro-2-{[2-fluoro-3-(2-methyl-1,3-thiazol-4-yl)phenyl]methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			532.9

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
604	N'-[(2S,3R)-4,4-difluoro-2-({2-fluoro-3-[6-(trifluoromethyl)pyridin-2-yl]phenyl}-methyl)-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			580.9
605	N'-[(2S,3R)-2-({3-[6-(difluoromethyl)pyridin-2-yl]-2-fluorophenyl}methyl)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			562.9
606	N'-[(2S,3R)-2-({3-[4,6-dimethylpyridin-2-yl]-2-fluorophenyl}methyl)-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			540.9
607	ethyl 2-[3-({(2S,3R)-1-(tert-butoxycarbonyl)-3-[(dimethylsulfamoyl)amino]-4,4-difluoropyrrolidin-2-yl}methyl)-2-fluorophenyl]-1,3-oxazole-4-carboxylate			577.2

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
608	N'-[(2S,3R)-2-[(3-chloro-2-fluorophenyl)methyl]-4,4-difluoro-2-(2-hydroxycyclobutane-2-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			468.1
609	N'-[(2S,3R)-4,4-difluoro-2-({2-fluoro-3-[4-(hydroxymethyl)-1,3-oxazol-2-yl]phenyl)methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			532.9
610	N'-[(2S,3R)-4,4-difluoro-2-({[2-fluoro-3-(4-methyl-1,3-oxazol-2-yl)phenyl]methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			517.2
611	N'-[(2S,3R)-2-({[3-(5-chloro-2-methyl-1,3-thiazol-4-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)-pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			567.1

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
612	N'-[(2S,3R)-4,4-difluoro-2-{{[2-fluoro-3-(2-methyl-1,3-oxazol-4-yl)phenyl]methyl}-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			517.2
613	N'-[(2S,3R)-2-{{[3-(5-chloro-2-methyl-1,3-oxazol-4-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			551.1
614	N'-[(2S,3R)-2-{{[3-(4-ethyl-6-methylpyridin-2-yl)-2-fluorophenyl]methyl}-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-3-yl]-N,N-dimethylsulfuric diamide			555.1
615	N'-{(2S,3R)-2-[(3'-chloro-2,4',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-[(2R)-oxetane-2-carbonyl]pyrrolidin-3-yl}ethanesulfonamide			553.0

TABLE 1-1-continued

EX.	IUPAC NAME	Structure	ADDITIVE	MS
616	[3-({(2S,3R)-3-[(N,N-dimethylsulfamoyl)-amino]-4,4-difluoro-1-(1-hydroxycyclobutane-1-carbonyl)pyrrolidin-2-yl}-methyl)-2-fluorophenyl]boronic acid			480.16

Experimental Example 1: Obtainment of Cell
Stably Expressing Human Orexin Type 2 Receptor
(hOX2R)

To obtain a cell clone stably expressing human orexin type 2 receptor, human orexin type 2 receptor cDNA was inserted into pcDNA3.1(+) plasmid vector (Invitrogen), and a plasmid DNA for expression of human orexin type 2 receptor (pcDNA3.1(+)/hOX2R) was cloned. The plasmid DNA was introduced into CHO-dhfr cell by an electroporation method, and human orexin type 2 receptor expressing clone cells were obtained by limiting dilution method by using G418 drug resistance as a selection marker.

Experimental Example 2: Measurement of Orexin
Type 2 Receptor Agonist Activity

CHO cells forcibly expressing human OX2 receptor were seeded in each well of 384 well black transparent bottom plate (BD Falcon) at 7,500 cells/well, and cultured for one day in a 5% CO₂ incubator at 37° C. After removal of the medium in the cell plate, assay buffer A containing a calcium indicator (HBSS (Thermo Fisher Scientific), 20 mM HEPES (Thermo Fisher Scientific), 0.1% BSA (Sigma-Aldrich), 2.5 µg/mL Fluo-4 AM (DOJINDO Chemical), 0.08% Pluronic F127 (DOJINDO Chemical), 1.25 mM probenecid (DOJINDO Chemical)) was added at 30 µL/well. The plate was stood for 30 min in a 5% CO₂ incubator at 37° C., and further stood at room temperature for 30 min. A test compound prepared by diluting with assay buffer B (HBSS, 20 mM HEPES, 0.1% BSA) was added at 10 µL/well, and the fluorescence value was measured by FDSSµCELL (Hamamatsu Photonics K. K.) every one sec for 1 min, and thereafter every two sec for 1 min 40 sec. The activity (%) of the test compound was calculated assuming that variation in the fluorescence value when DMSO was added instead of the test compound was 0%, and variation in the fluorescence value when orexin A (human) (PEPTIDE INSTITUTE, INC.) was added at the final concentration of 10 nM was 100%. The activity of each compound at the concentration of 3 µM was shown in Table 2. As is clear from the results, the compound of the present invention was shown to have an agonist activity on human orexin type 2 receptor.

TABLE 2

Test compound	OX2R agonist activity (3 µM, %)
1	100
2	106

TABLE 2-continued

Test compound	OX2R agonist activity (3 µM, %)
3	106
4	101
5	117
7	101
8	99
11	85
13	93
21	102
23	100
25	108
26	100
35	96
44	98
45	98
46	102
50	99
51	99
52	100
56	95
65	100
66	104
67	111
68	106
73	104
76	98
77	95
78	93
79	102
81	95
87	93
88	98
91	94
92	99
93	98
94	99
97	90
98	91
105	99
106	97
116	86
121	94
122	89
124	89
129	92
131	94
133	96
136	87
138	97
139	93
144	88
145	99
146	92
147	91
162	92

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TABLE 2-continued

Test compound	OX2R agonist activity (3 μ M, %)	
163	105	
166	98	5
171	99	
174	90	
175	95	
202	94	
204	89	
205	91	10
207	88	
210	95	
211	94	
212	96	
216	95	
217	96	15
220	90	
222	97	
225	96	
226	100	
228	99	
231	95	20
236	100	
238	98	
239	102	
245	94	
248	90	
249	90	
250	96	25
251	91	
257	96	
259	96	
263	103	
264	98	
265	96	30
266	83	
267	92	
268	94	
272	88	
273	83	
274	93	35
275	85	
276	88	
277	94	
282	96	
283	93	
284	92	
285	98	40
287	94	
289	80	
291	90	
292	87	
293	83	
294	96	45
295	95	
296	96	
297	95	
298	96	
302	85	
304	99	50
305	103	
306	89	
307	100	
308	94	
314	109	
316	101	55
323	92	
324	91	
328	86	
329	88	
330	98	
332	95	
333	78	60
344	102	
345	92	
346	93	
347	102	
348	103	
349	98	65
356	97	

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TABLE 2-continued

Test compound	OX2R agonist activity (3 μ M, %)	
365	93	
366	96	
368	95	
369	85	
370	91	
371	94	
372	89	
373	88	
374	93	
375	91	
376	99	
377	95	
378	106	
379	93	
380	93	
381	95	
382	102	
383	94	
384	98	
385	95	
387	85	
388	89	
389	89	
391	92	
392	97	
393	94	
394	86	
408	88	
409	92	
413	99	
414	95	
415	107	
416	101	
422	102	
423	90	
424	95	
425	95	
426	90	
427	99	
428	100	
430	96	
431	106	
432	95	
433	101	
434	101	
435	88	
436	92	
437	84	
438	95	
439	93	
440	94	
441	93	
442	96	
443	90	
444	98	
447	95	
450	86	
451	82	
453	85	
454	84	
455	85	
456	77	
457	80	
458	84	
459	78	
462	78	
463	79	
465	106	
466	103	
467	105	
468	102	
469	91	
470	103	
471	94	
472	95	
473	97	
474	96	
475	100	

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TABLE 2-continued

Test compound	OX2R agonist activity (3 μ M, %)
476	100
477	96
478	103
479	93
480	101
481	104
482	100
483	108
484	90
485	93
486	88
487	95
488	95
489	92
490	93
491	93
492	90
493	95
494	96
495	92
496	95
497	91
498	99
499	96
500	98
501	96
502	96
503	89
504	90
505	88
506	100
507	97
508	96
509	85
510	97
511	98
512	95
513	97
514	95
515	98
516	102
517	100
518	98
519	98
520	94
521	97
522	97
523	86
524	87
525	85
526	83
527	95
528	83
529	95
530	94
531	95
532	96
533	101
534	98
535	102
536	98
537	100
538	106
539	101
540	100
541	100
542	100
543	99
544	90
545	93
546	96
547	98
548	94
549	94
550	93
551	93
552	94

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Experimental Example 3: Evaluation of Microsome Stability in Human

Human liver microsomes were purchased from Xenotech, LLC (Lenexa, KS). An incubation mixture consisted of microsomes in 50 mmol/L KH_2PO_4 — K_2HPO_4 phosphate buffer (pH 7.4) and 1 μ mol/L test compound. The concentration of microsomal protein was 0.2 mg/mL. An NADPH-generating system (5 mmol/L MgCl_2 , 5 mmol/L glucose-6-phosphate, 0.5 mmol/L beta-NADP+ and 1.5 unit/mL glucose-6-phosphate dehydrogenase) was added to the incubation mixture with a half volume of the reaction mixture to initiate the enzyme reaction. The reaction was terminated 15 and 30 minutes after the initiation of the reaction by mixing the reaction mixture with acetonitrile, followed by centrifugation at 2500 rpm for 10 min. The supernatant was subjected to LC/MS/MS analysis. The metabolic rate constant was calculated as the slope of the remaining rate-time plot. The in vitro intrinsic metabolic clearance was calculated by dividing initial metabolic rate constant by microsomal protein in the incubation mixture. The results were shown in Table 3.

TABLE 3

Test compound	Clearance (μ L/min/mg)
1	75
2	40
3	-3
5	12
56	135
66	61
67	4
87	9
91	65
94	44
144	13
146	83
225	60
236	67
302	3
375	35
380	39
433	16

Experimental Example 4: Evaluation of Wake-Promoting Effects in Cynomolgus Monkeys

The wake-promoting effects were evaluated by measuring the electroencephalogram (EEG) and electromyogram (EMG) in cynomolgus monkeys. Under isoflurane anesthesia (1-5%, Pfizer Japan Inc., Tokyo, Japan), male cynomolgus monkeys (3-5 years old, Hamri Co., Ltd., Ibaraki, Japan) were surgically implanted with radio-telemetry transmitters (TL10M3-D70-EEE, Data Sciences International Inc., MN, USA). EEG electrodes were screwed into the skull at the parietal area. EMG electrodes were implanted on the cervical muscles. After the surgery, each monkey was given penicillin (100,000 units/head, i.m., Meiji Seika Pharma Co., Ltd., Tokyo, Japan), buprenorphine (0.02 mg/kg, i.m., Otsuka Pharmaceutical Co., Ltd., Tokyo, Japan) and prednisolone (1 mg/kg, s.c., Kyoritsu Seiyaku Co., Ltd., Tokyo, Japan) daily for one week. After at least a 1-month recovery period in home cages, the monkeys were habituated to the recording chamber placed in a soundproof room. EEG and EMG signals were recorded using the telemetry system (Dataquest ART software, Data Sciences International Inc., MN, USA) and the signals were analyzed using SleepSign

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software (Kissei Comtec Co., Ltd., Nagano, Japan). After confirming long sleep in dark phase in the experimental room, we used animals to examine the wake-promoting effect of compounds.

Oral test compounds (3 or 10 mg/kg) suspended in 0.5% methylcellulose aqueous solution, or vehicle (i.e., 0.5% methylcellulose aqueous solution) was administered orally (p.o.) to monkeys at zeitgeber time 12 (ZT12) in a volume of 5 mL/kg body weight in cross-over design or pre-post design (n=1-4). EEG and EMG recordings were performed for 4 h after the compound administration. The time spent in wakefulness for 4 h after administration (% of vehicle treatment) was calculated by using SleepSign. The results are shown in Table 4.

Parenteral test compounds (0.1 or 0.3 mg/kg) dissolved in a mixed solution comprising 5% DMSO, 5% Cremophor EL, 20% PEG400 and 70% soluplus (1% (w/v)), or vehicle (i.e., a mixed solution comprising 5% DMSO, 5% Cremophor EL, 20% PEG400 and 70% soluplus (1% (w/v))) was administered subcutaneously (s.c.) to monkeys at ZT12 in a volume of 0.5 mL/kg body weight in a pre-post design (n=1-2). EEG and EMG recordings were performed for 4 h after the compound administration. The time spent in wakefulness for 4 h after administration (% of vehicle treatment) was calculated by using SleepSign. The results are shown in Table 5.

TABLE 4

Example No	Dose (mg/kg)	Wakefulness time (% of vehicle treatment) (Mean)
1	3	233.91
2	3	599.68
3	3	612.62
5	10	411.70
56	3	417.22
66	3	463.00
67	3	355.56
87	3	832.34
91	10	548.00
94	3	441.88
144	3	649.41
146	3	426.26
225	3	317.65
236	10	716.00
302	10	287.88
375	3	396.49
380	3	483.53
433	3	727.27

TABLE 5

Example No	Dose (mg/kg)	Wakefulness time (% of vehicle treatment) (Mean)
443	0.1	532.00
450	0.3	813.51
451	0.1	225.10
459	0.3	431.39
462	0.3	365.46
463	0.1	483.15
542	0.3	505.45

As is clear from Table 4 and Table 5, the test compounds of the present invention increased the wakefulness time compared to the vehicle treatment group in cynomolgus monkeys. That is, these compounds were suggested to be potential therapeutics for narcolepsy.

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Formulation Example 1 (Production of Capsule)

1) compound of Example 1	30 mg
2) crystalline cellulose	10 mg
3) lactose	19 mg
4) magnesium stearate	1 mg
total	60 mg

1), 2), 3) and 4) are mixed and filled in a gelatin capsule.

Formulation Example 2 (Production of Tablet)

1) compound of Example 1	30 g
2) lactose	50 g
3) cornstarch	15 g
4) calcium carboxymethylcellulose	44 g
5) magnesium stearate	1 g
1000 tablets	140 g in total

The total amount of 1), 2), 3) and 30 g of 4) are kneaded with water, vacuum dried and sieved. The sieved powder is mixed with 14 g of 4) and 1 g of 5), and the mixture is punched by a tableting machine. In this way, 1000 tablets containing 30 mg of the compound of Example 1 per tablet are obtained.

INDUSTRIAL APPLICABILITY

The compound of the present invention has an orexin type 2 receptor agonist activity, and is useful as an agent for the prophylaxis or treatment of narcolepsy.

This application is based on patent application No. 2019-015488 filed on Jan. 31, 2019 in Japan, the contents of which are encompassed in full herein.

The invention claimed is:

1. A method of treating excessive sleepiness, cataplexy, or a combination thereof, in a patient in need thereof, the method comprising administering to the patient a compound selected from the group consisting of:

N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide;

N-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide;

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide;

N-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide;

N-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide;

N-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide;

N-{(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide;

N-{(2S,3R)-4,4-difluoro-1-((1S,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide; and

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N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide; or a salt thereof.

2. The method of claim 1, wherein the compound is N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof.

3. The method of claim 1, wherein the compound is N'-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide or a salt thereof.

4. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof.

5. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof.

6. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof.

7. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof.

8. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide or a salt thereof.

9. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide or a salt thereof.

10. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-

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fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide or a salt thereof.

11. The method of claim 1, wherein the compound is N'-{(2S,3R,4S)-1-(azetidine-1-carbonyl)-4-fluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide.

12. The method of claim 1, wherein the compound is N'-[(2S,3R)-2-[(2,3'-difluoro[1,1'-biphenyl]-3-yl)methyl]-4,4-difluoro-1-(2-methylpropanoyl)pyrrolidin-3-yl]ethanesulfonamide.

13. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide.

14. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-(2-hydroxy-2-methylpropanoyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide.

15. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(bicyclo[1.1.1]pentane-1-carbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide.

16. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(cyclopropanecarbonyl)-4,4-difluoro-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide.

17. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}ethanesulfonamide.

18. The method of claim 1, wherein the compound is N'-{(2S,3R)-4,4-difluoro-1-((1s,3R)-3-fluorocyclobutane-1-carbonyl)-2-[(2,3',5'-trifluoro[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}methanesulfonamide.

19. The method of claim 1, wherein the compound is N'-{(2S,3R)-1-(azetidine-1-carbonyl)-4,4-difluoro-2-[(2-fluoro-3'-methyl[1,1'-biphenyl]-3-yl)methyl]pyrrolidin-3-yl}-N,N-dimethylsulfuric diamide.

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